



SpaceOrbitLAB

Professional Training Program in Space Sciences

MASTER THE PHYSICS OF MOTION: From first principles to real missions

Mastering the Math and Physics of
Space Orbits Simulation

Part II – Foundations of Astrodynamics

Richard L. Lachance, Ph.D.

SpaceOrbitLAB.com

RLLac@ULaval.ca



(The link)



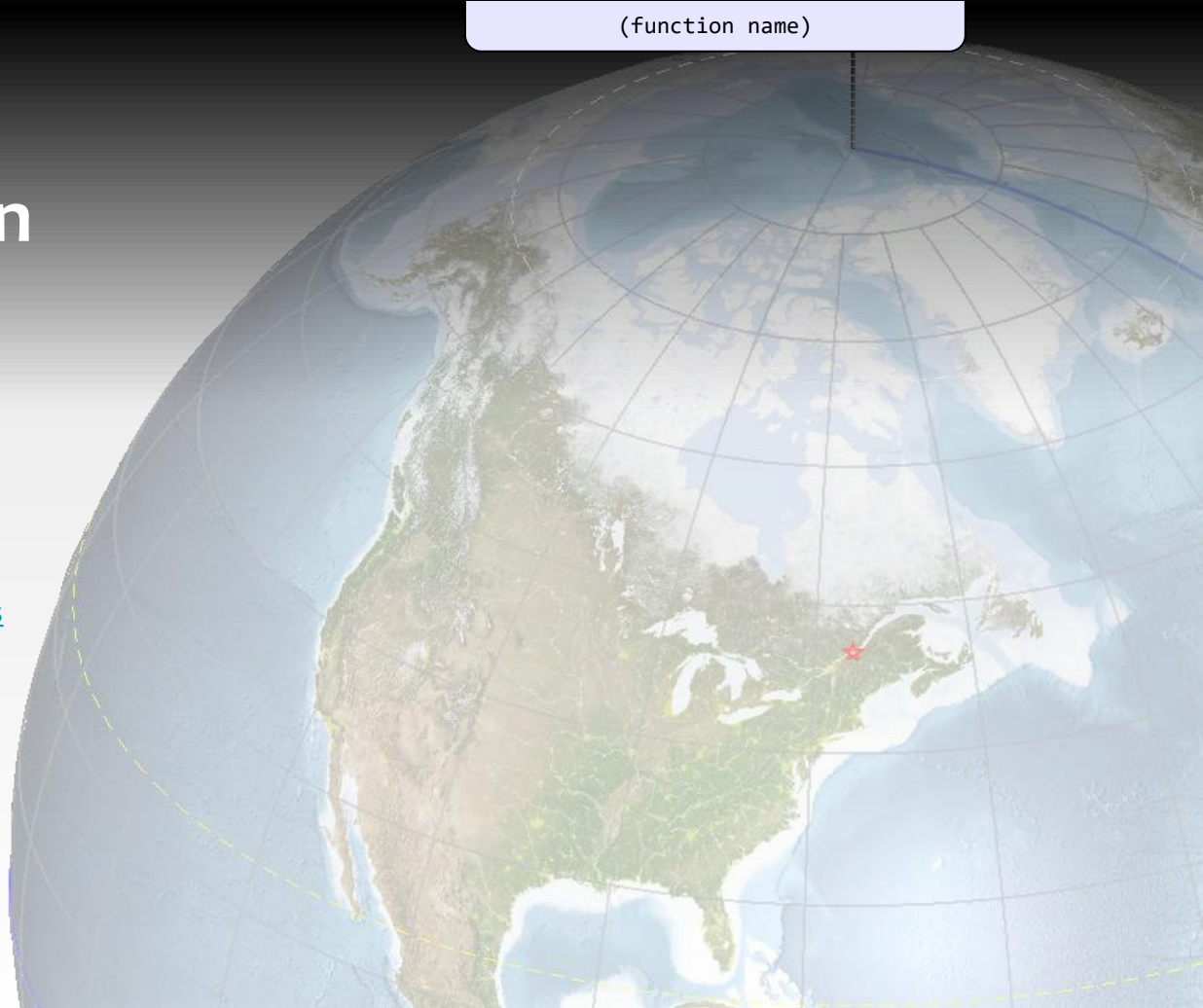
In collaboration with
ABB Space and Defence
Systems

Presentation plan

Course outline

Part II – Foundations of Astrodynamics

1. [Astrodynamics laws](#)
 2. [Orbital motion & Trajectory equation](#)
 3. [Kepler problem](#)
 4. [Reference Frames & Coordinate systems](#)
 5. [Orbital elements](#)
 6. [Satellite Ground Tracks](#)
 7. [Orbit types](#)
 8. [Launch into Orbit](#)
 9. [Maneuvering in Space](#)
 10. [Rockets & Launch vehicles](#)
- A. [N-Body problem](#)



II– Foundations of Astrodynamics – Course outline



1. Astrodynamics laws

Sellers Ch. 4

1.1 Kepler's laws

1.1.1 Orbital period

1.2 Newton's laws

1.2.1 Preliminary notions

1.2.2 3 laws + Universal gravitation

1.2.3 Constants of motion (ε & h)

1.2.4 Equation of relative motion ($\ddot{r}$)

2. Orbital motion

Sellers Ch. 4

2.1 Simplified 2-Body Keplerian orbit (2BK)

2.2 Trajectory equation & Conic sections

2.2.1 Escape velocity

2.3 Geometry of the Ellipse

2.4 Parabolas & Hyperbolas

2.5 Reaching orbit

2.6 Orbit propagation using Euler & Newton's methods

(→ now in Part III)

3. Kepler problem

Sellers Ch. 8

3.1 Anomalies (M , E , v)

3.2 Kepler equation

3.3 Numerical solution by Newton's method

4. Reference Frames & Coordinate systems

Sellers Ch. 4

4.1 Heliocentric-Ecliptic (ECL)

4.2 Perifocal (PQW)

4.3 ECI – Earth-Centered Inertial (IJK)

4.4 ECEF – Earth-Centered, Earth-Fixed (XYZ)

4.5 Topocentric (SEZ, ENU, NED)

4.6 Frame Instances

4.7 Transformation chains

II– Foundations of Astrodynamics – Course outline



5. [Orbital elements](#)

Sellers Ch. 5

- 5.1 [Classical orbital elements \(COEs\)](#)
- 5.2 [Two-line orbital parameters \(TLEs\)](#)
 - 5.2.1 [Space-Track & CelesTrak](#)

6. [Satellite Ground Tracks](#)

Sellers Ch. 5

- 6.1 [Day/Night Terminator](#)
- 6.2 [Side Track Paths](#)
- 6.3 [Satellite footprint](#)

7. [Orbit types](#)

- 7.1 [LEO – Low Earth Orbit](#)
 - 7.1.1 [SSO – Sun Synchronous Orbit](#)
 - 7.1.2 [LTAN/LTDN](#)
- 7.2 [MEO – GPS, Molniya, Tundra](#)
- 7.3 [HEO/GEO – Geosynchronous & Geostationary](#)
- 7.4 [Lagrange Points](#)
- 7.5 [Satellite database study](#)

8. [Launch into Orbit](#)

Sellers Ch. 9

- 8.1 [Orbit Insertion](#)
- 8.2 [Launch Windows](#)

9. [Maneuvering in Space](#)

Sellers Ch. 6

- 9.1 [Hyperbolic trajectories](#)
- 9.2 [Oberth effect](#)
- 9.3 [Hohmann transfer](#)
- 9.4 [Orbital plane change](#)
- 9.5 [Orbital rendezvous](#)
- 9.6 [Gravity assist maneuvers](#)
 - Gravitational slingshot maneuvers
- 9.7 [Interplanetary travel](#)
 - 9.7.1 Mission to Mars
 - 9.7.2 [Constant-thrust travel](#) (Sci-Fi concepts)
- 9.8 [Porkchop plots](#)

Sellers Ch. 7

Sellers Ch. 7

II– Foundations of Astrodynamics – Course outline



10. Rockets & Launch vehicles

Sellers Ch. 14

10.1 The “The Rocket equation”

Specific impulse, ΔV , etc.

10.2 Propulsion: Rocket fuels & propellants

10.3 Staging

10.4 Modeling rocket launch Rocket launch simulation

A. N-Body problem

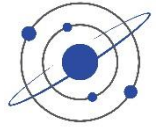
A.1 Circular-Restricted 3-Body problem

A.2 3-Body problem

B. Use of AI in Astrodynamics

B.1 TBC

Greek letters & special symbols



α alpha

β beta

γ gamma

δ delta

ε epsilon

θ theta

λ lambda

μ mu

ν nu

π pi ($\approx 3.1416\dots$)

ρ rho

σ sigma

Alternate forms → ϕ phi
 Φ Phi (uppercase)

$\angle$ angle

$\emptyset$ diameter

ψ psi (lowercase)

Ψ Psi (uppercase)

ω omega (lowercase)

Ω Omega (uppercase)

$\odot$ Sun

$\oplus$ Earth

$\lrcorner$ Moon

$\star$ Sidereal

Υ First Point of Aries (**1stA**) – *Vernal Equinox* (Υ - Υ)

$\ Cancer$ Summer Solstice (Cancer)

Ω Ascending / Descending node

$\diamond$ Epoch

ℓ Ellipse semi-parameter; focal parameter (p).

$\equiv$ Defined as / Identity, e.g.: $\sin^2(x) + \cos^2(x) \equiv 1$; $\sin(90^\circ) \equiv 1$
 (also congruence in modular arithmetic: $22 \equiv 2 \pmod{10}$)



Astrodynamics notation



$\vec{r}$	Position vector, $ \vec{r} = \rho$
$\vec{v}$	Velocity vector, $ \vec{v} \equiv \text{speed}$
$\vec{h}$	Specific Angular Momentum vector, $ \vec{h} = h$
$\vec{n}$	Ascending Node vector, $ \vec{n} = r_p$
$\vec{e}$	Eccentricity vector, $ \vec{e} = e$
μ	Gravitational parameter / constant
ε	Energy
e	Eccentricity
a	Semi-Major Axis
i	Inclination
ω	Argument of Perigee / angular rotation speed
ϖ	Longitude of Perihelion (used in planetary astronomy)
Ω	RA of Ascending Node (RAAN) / also Solid Angle
ν	True Anomaly (sometimes θ)
ρ	Orbit radius length
ϵ	Earth obliquity of the ecliptic (23.44°)

α	Right Ascension (RA) / Azimuth (Azim)
δ	Declination (Dec) / Elevation (Elev)
λ	Longitude
ϕ	Geocentric latitude
φ	Geodetic latitude (sub-satellite)

ϕ	Flight Path Angle (FPA)
γ	Zenith Angle
β	Launch Azimuth (LA) angle
$\beta_{\odot}$	Sun Beta Angle
ψ	Orbit heading angle

σ	Wavenumber
λ	Wavelength
β	Fraction of diffuse light
ψ	phase angle
Φ	phase integral / radiant flux
Ψ	reflected radiance factor

Radiometry

Space Mission Architecture



- Part II-A Trajectories and Orbits
- Part II-B Launch Vehicles

Further reading:

Sellers 2004 [Understanding Space](#)

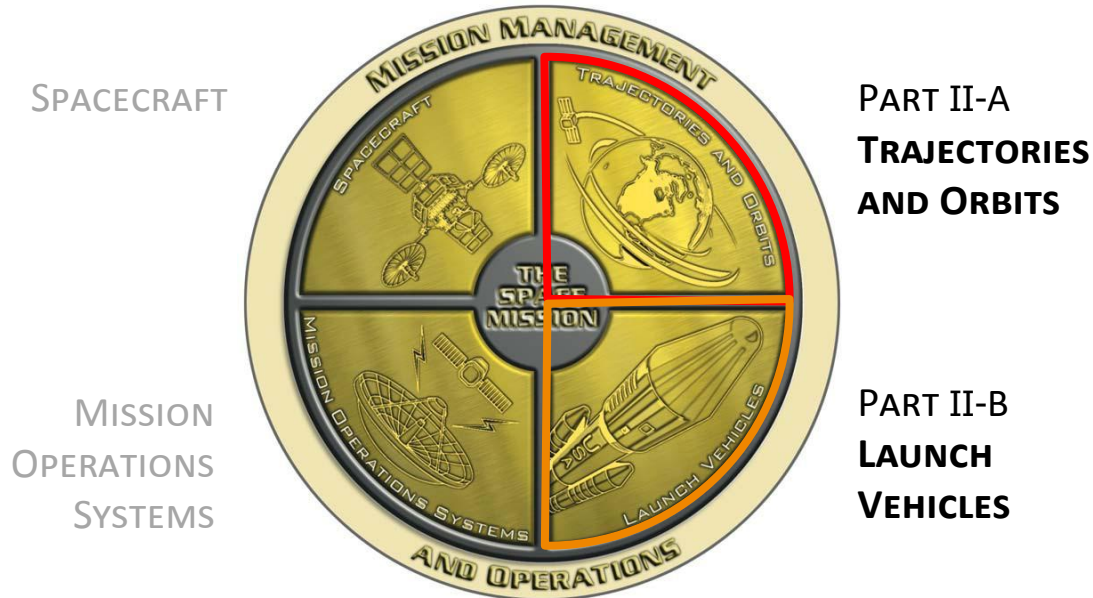
Chapter 1 – Elements of a Space Mission

Introduction



Space Mission Architecture

– Sellers 2004, *Understanding Space; An Introduction to Astronautics* (2nd Ed)



Space Mission Architecture



- The **PAYLOAD** consists of the hardware and software that sense or interact with the subject. Typically, we trade off and combine several sensors and experiments to form the payload, which largely determines the mission's cost, complexity, and effectiveness. The subsystems of the **spacecraft bus** support the payload by providing orbit and attitude maintenance, power, command, telemetry and data handling, structure and rigidity, and temperature control. The payload and spacecraft bus together are called the spacecraft, **space segment**, or **launch vehicle payload**
- The **LAUNCH SYSTEM** includes the launch facility, launch vehicle and any upper stage required to place the spacecraft in orbit, as well as interfaces, payload fairing, and associated ground-support equipment and facilities. The selected launch system constrains the size, shape, and mass of the spacecraft
- The **ORBIT** is the spacecraft's path or trajectory. Typically, there is a separate initial parking orbit, transfer orbit, and final mission orbit. There may also be an end-of-life or disposal orbit. The mission orbit significantly influences every element of the mission and provides many options for trades in mission architecture
- The **COMMUNICATIONS ARCHITECTURE** is the arrangement of components which satisfy the mission's communication, command, and control requirements. It depends strongly on the amount and timing requirements of data to be transferred, as well as the number, location, availability, and communicating ability of the space and ground assets

Space Mission Architecture (continued)



- The **GROUND SYSTEM** consists of fixed and mobile ground stations around the globe connected by various data links. They allow us to command and track the spacecraft, receive and process telemetry and mission data, and distribute the information to the operators and users. Mission operations consist of the people, hardware, and software that execute the mission, the mission operations concept, and attendant policies, procedures, and data flows.
- Finally, the **COMMAND, CONTROL, AND COMMUNICATIONS (C3)** architecture contains the spacecraft, communications architecture, ground segment, and mission operations elements.



ASTRODYNAMICS:

The application of Newtonian mechanics to human-made objects in space, such as rockets and spacecraft.

It is a subfield of celestial mechanics and ballistics

→ The mathematical analysis of orbits and their properties

“The study of the motion of bodies moving in space,
subjected to natural* and artificial[†] forces”

What is Astrodynamics?



- Astrodynamics, subfield of orbital mechanics, is a critical engineering field that encompasses the design and control of spacecraft missions
- It involves applying mathematical and physical principles to understand and predict satellite motion, calculate spacecraft trajectories, determine orbits, and control spacecraft
- The study of astrodynamics is also crucial for various aspects of space exploration, including the design of interplanetary missions and the maintenance of satellites in orbit for communication, navigation, and other purposes
- Astrodynamics principles are used to determine the types of propulsion systems required, propellant needs, and control strategies necessary to place a satellite into orbit, maintain it there, and maneuver it to achieve the mission objectives

Définition de termes



- **Astronomie**

- Science des astres, des corps célestes (y compris la Terre) et de la structure de l'univers.

- **Aérospatiale**

- Discipline scientifique qui rassemble les techniques de **l'aéronautique** (déplacement dans l'atmosphère, utilisant des avions ou des hélicoptères par exemple) et de **l'astronautique** (déplacements spatiaux, c'est-à-dire trajets hors atmosphère et interplanétaires, en utilisant des fusées)

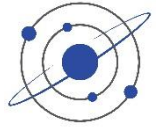
- **Astronautique**

- Science qui a pour objet l'étude de la navigation spatiale ; cette navigation. Relatif aux astronautes.
 - Pratique consistant à envoyer des engins spatiaux au-delà de l'atmosphère terrestre, dans l'espace. Les vols spatiaux constituent l'une de ses principales applications et les sciences spatiales en sont le domaine général.

- **Astrodynamique (ou mécanique spatiale)**

- Science qui étudie le mouvement des corps dans l'espace, comme les satellites, les fusées et les sondes, sous l'effet de la gravité et d'autres forces. C'est une branche de l'astronomie et de l'ingénierie aérospatiale qui calcule les trajectoires orbitales, planifie les manœuvres (transferts interplanétaires, changements d'orbite) et détermine l'attitude des engins spatiaux. Elle repose sur les lois de Newton et utilise des outils mathématiques avancés pour comprendre et prédire les mouvements célestes et artificiels.

Interesting introduction

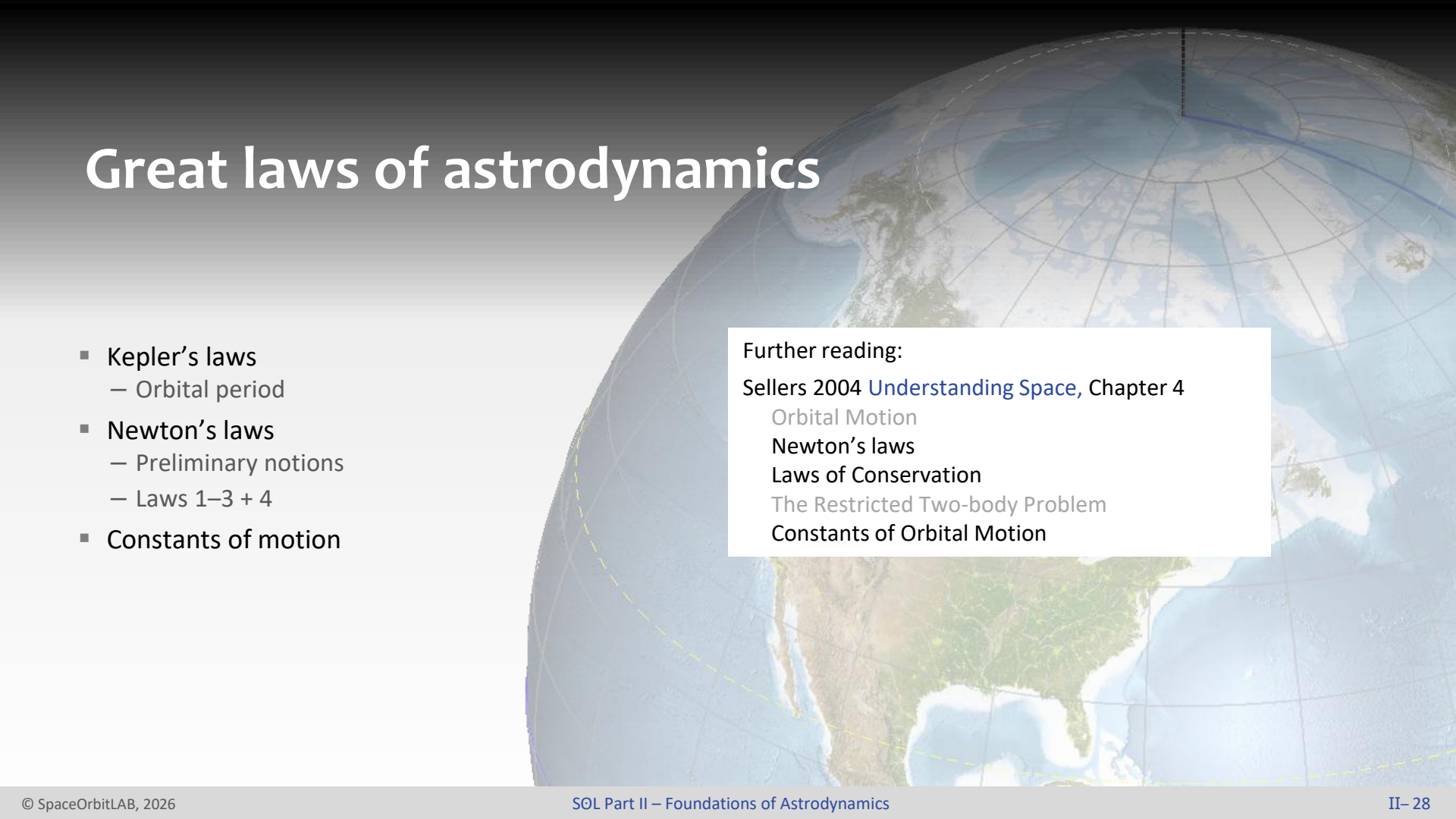


- Mark Rober: “I Tried Building My Own Space Satellite”

https://youtu.be/6KcV1C1Ui5s?si=oFO_08-tzOmcRIBM



Great laws of astrodynamics



- Kepler's laws
 - Orbital period
- Newton's laws
 - Preliminary notions
 - Laws 1–3 + 4
- Constants of motion

Further reading:

Sellers 2004 [Understanding Space](#), Chapter 4

Orbital Motion

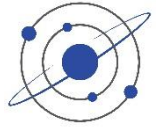
Newton's laws

Laws of Conservation

The Restricted Two-body Problem

Constants of Orbital Motion

Orbital mechanics



Kepler's three laws of planetary motion, Newton's three laws of motion, and Newton's law of universal gravitation form the foundations of orbital mechanics

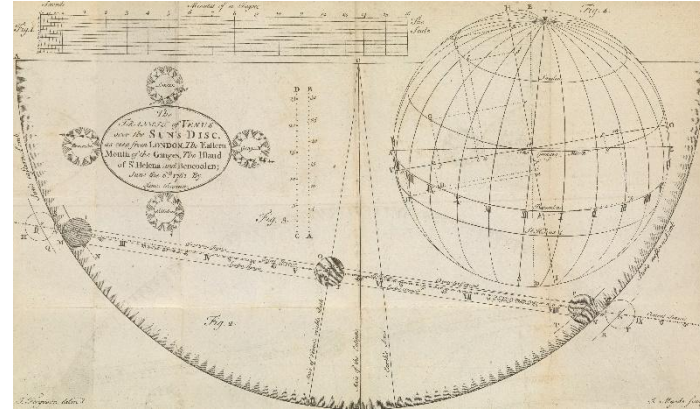
Let's start with the basics: Kepler and Newton gave theoretical explanations based on the laws of physics: *motion in space can be:*

1. **Described (visualized)** using **KEPLER'S LAWS**

Empirical laws of Planetary motion
Doesn't explain underlying forces

2. **Explained (understood)** using **NEWTON'S LAWS**

Fundamental laws of Gravitational Attraction
and dynamics of motion



We will use these principles to develop the equations and parameters we need to describe orbital motion, and to see how satellites work and the different types of orbits available

Style guide for naming conventions in astrodynamics



- “Newton’s laws” (with apostrophe) is the standard form
 - For Isaac Newton, the phrase “Newton’s laws of motion” is treated as a **named set of principles historically attributed to a person**, not just a descriptive label
 - This differs from terms like **Kepler problem**, which describe a class of mathematical problems rather than a canonical named set
 - Established usage: In physics and engineering, the overwhelmingly standard forms are:
 - **Newton’s laws of motion** and **Newton’s law of gravitation**
 - These are deeply entrenched in textbooks, education, and literature
- *The distinction is largely historical convention + linguistic inertia, not strict grammar*
- Capitalize proper names, but **do not capitalize generic terms**: equations, problem, method, model

Case	Form	Reason
Newton’s laws Kepler’s law	possessive	named canonical laws
Kepler problem Kepler orbits Laplace & Euler equation Euler angles Gauss method	no apostrophe	descriptive classification
Navier-Stokes equations Lagrange points	no apostrophe	attributive naming convention
Maxwell’s equations	often possessive (exception)	historically fixed phrasing

Kepler's laws

- 1) Law of orbits
- 2) Law of areas
- 3) Law of periods



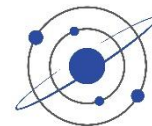
Tycho Brahe's key observations



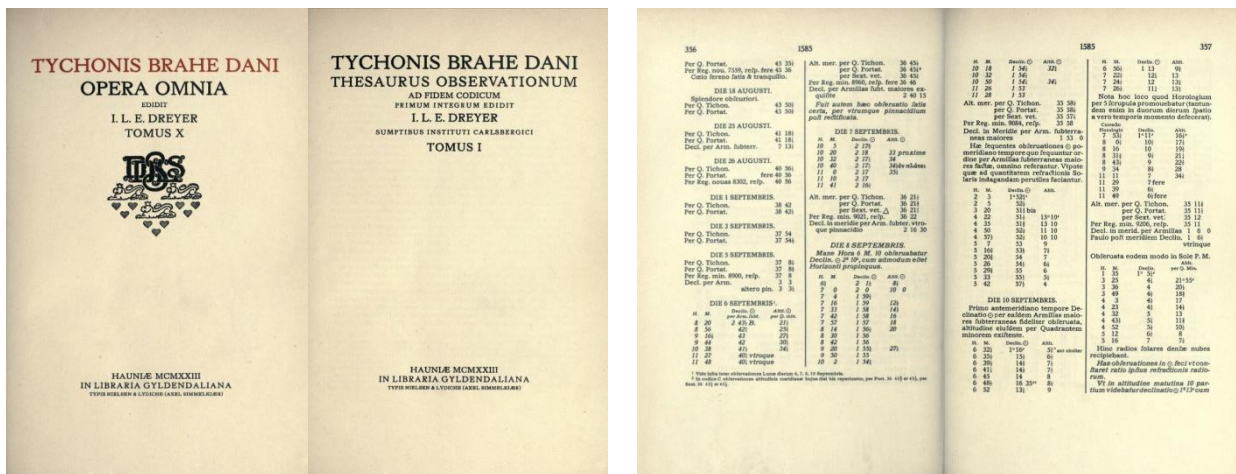
- Tycho Brahe was convinced of the need to undertake measurements of planetary orbits over very long periods of time (*i.e.*, over several years) in order to uncover the laws governing the motion of the planets
- He thus meticulously measured the orbit of Mars, as well as other planets (i.e. Mercury, Venus, Jupiter, Saturn), over a period of about twenty years
- His observations have been collected in notebooks that have been reproduced in *Tychonis Brahe Dani Opera Omnia*, a series of books from which all the data presented here could be extracted
- More specifically, Tycho Brahe accurately measured the declination of Mars, δ , expressed in degrees [°] as a function of time, over the period extending from 1582 to 1600

It was these very data that allowed Johannes Kepler to establish his laws of planetary motion

Opera omnia, by Tycho Brahe



- **Tycho Brahe** published his foundational Mars observations and other data in his posthumously released masterpiece, *Historia Coelestis*, but detailed Mars data also appeared in his earlier book on instruments, *Astronomiae instauratae mechanica*, and his *Opera Omnia*, with his *Tabulae Rudolphinae* (published by Kepler) being the ultimate catalog
- His detailed, raw data for Mars (1582–1600) was compiled in volumes of his Opera Omnia, particularly Tychonis Brahe Dani Opera Omnia, TOMUS X–XIII.



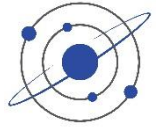
1913 edition, by Dreyer <https://archive.org/details/operaomniaedit10brahuoft/page/n9/mod/2up>

Kepler's analysis of Brahe's observations



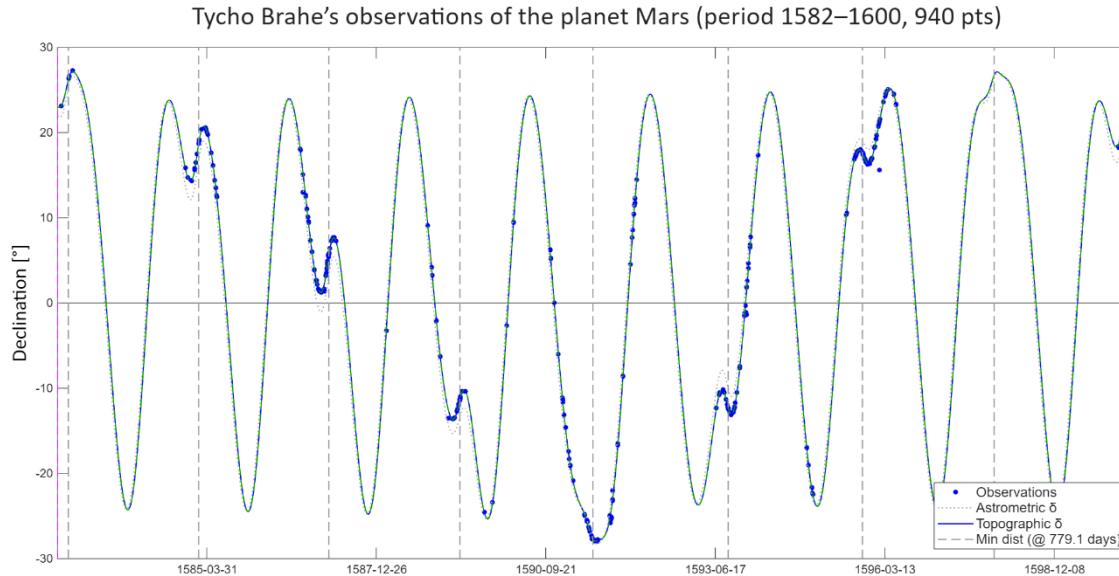
→ Before his death in 1601, [Brahe challenged Kepler to calculate the orbit of Mars](#)

- Brahe's choice of planets was fortunate, for of the six planets then known, Mars had the second most eccentric orbit (Mercury being the most eccentric)
 - Eccentric means “off center,” and [eccentricity](#) e describes the deviation of a shape from a perfect circle
A circle has an $e = 0$, and an ellipse has $0 < e < 1$
- As Kepler examined Tycho Brahe's meticulous observations of Mars, [he encountered a troubling inconsistency: the planet's motion could not be reconciled with a circular orbit](#). His calculations repeatedly revealed a discrepancy of $\approx 8'$ (arcminutes) between the predictions of circular motion and the observed positions. While many astronomers might have dismissed such a small deviation, Kepler did not. He regarded Brahe's measurements as authoritative and refused to set them aside
- Rather than forcing the data to fit a preconceived model, [Kepler explored alternative orbital shapes](#). After briefly experimenting with oval paths, he arrived at a decisive insight: planets move around the Sun in elliptical orbits, with the Sun located not at the center but at one *focus* (from the Latin term meaning “hearth” or “fireplace”)
- Trusting both his mathematical reasoning and the precision of Brahe's observations, Kepler formalized this result as a law of planetary motion
 - Although it was the second law he discovered chronologically, it is now known as [Kepler's first law](#)



“Mars is a star who defies observation” – Pliny the Elder (23–79 AD)

- How would you fit the observations?



Raw data (•) collected by Tycho Brahe from his observations of the declination of Mars

The solid line curve (—) is calculated using the ephemeris data retrieved from the NASA Horizons database

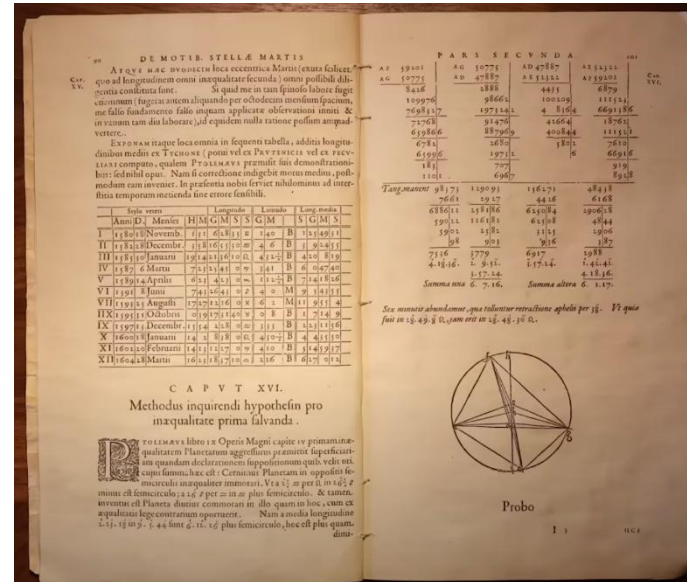
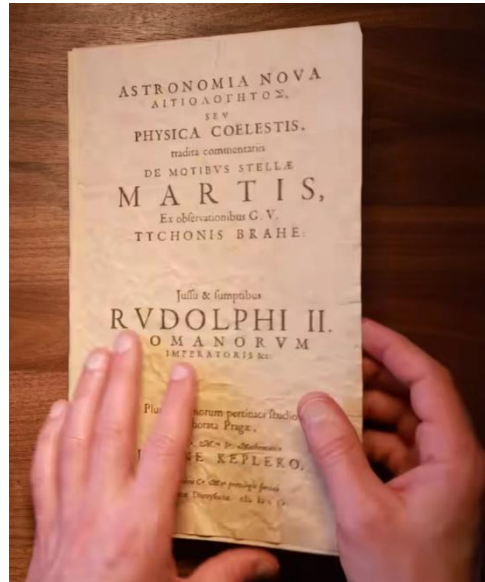


Part III – Mars's retrograde motion analysis

Astronomia Nova, by Johannes Kepler (1609)



- Kepler published the detailed data and his derivations concerning the orbit of Mars, in *Astronomia Nova* in 1609
 - The full Latin title translates to: “*New Astronomy Based upon Causes, or Celestial Physics Treated by Means of Commentaries on the Motions of the Star Mars, from the Observations of Tycho Brahe, Gent*”
 - This title highlights the central role that his extensive, meticulous study of Mars’s motion played in the book



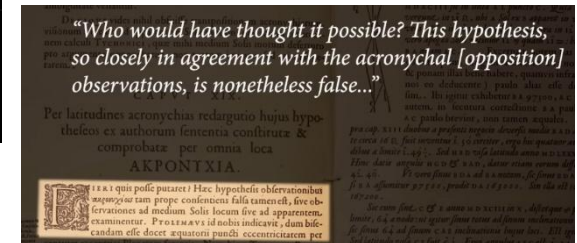
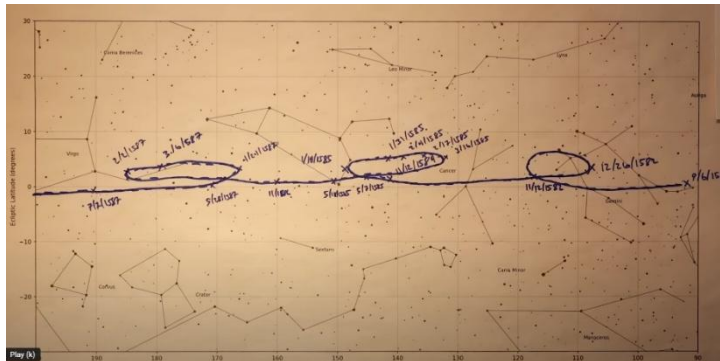
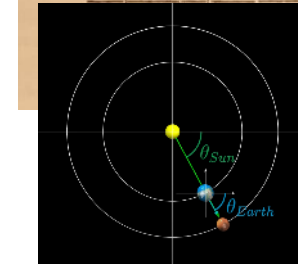
How Kepler actually discovered his laws



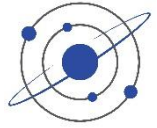
Welch Labs: How the Bizarre Path of Mars Reshaped Astronomy

- Kepler's laws Part 1
www.youtube.com/watch?v=Phscjl0u6TI [16:39]
 - Kepler's laws Part 2
www.youtube.com/watch?v=MprJN5teQxc [15:13]
 - Kepler's impossible equation
www.youtube.com/watch?v=hBkmyJ3TE0g&t=4s [22:41]
- See SOL Part II – 1.1) Kepler's laws for more details

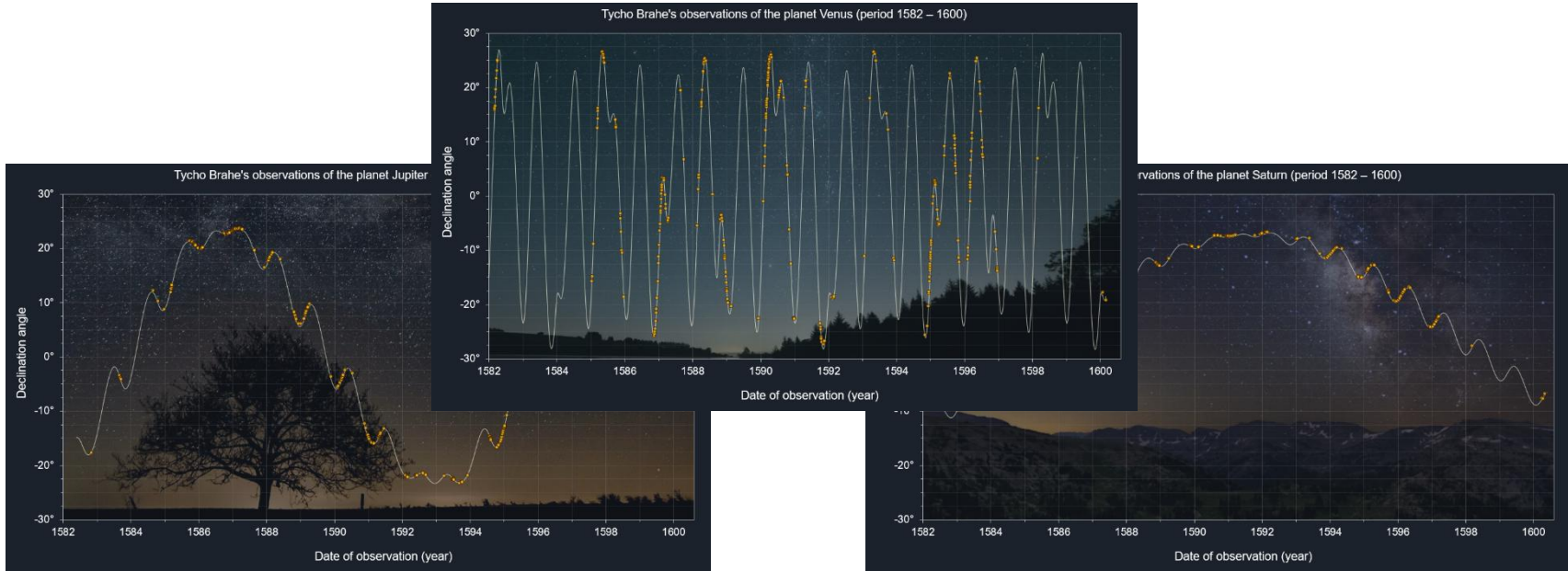
	Stylo veteri				Longitudo				Latitudo		Long. media					
	Anni	D.	Menfes	H	M	G	M	S	S	G	M		S	G	M	S
I	1580	18	Novemb.	1	31	6	28	35	11	1	40	B	1	25	49	31
II	1582	28	Decembr.	3	58	16	55	30	5	4	6	B	3	9	24	55
III	1585	30	Januarii	19	14	21	36	10	8	4	3 2 ¹ / ₂	B	4	20	8	19
IV	1587	6	Martii	7	23	25	43	0	17	3	41	B	6	0	47	40
V	1589	14	Aprilis	6	23	4	23	0	11	1	2 1 ¹ / ₂	B	7	14	18	26
VI	1591	8	Junii	7	43	26	43	0	2	4	0	M	9	5	43	55
VII	1593	25	Augusti	17	27	12	16	0	11	6	2	M	11	9	55	4
IIIX	1595	31	Octobris	0	39	17	31	40	8	0	8	B	1	7	14	9
IX	1597	13	Decembr.	15	54	2	28	0	5	3	3	B	2	23	11	56
				14	2	8	38	0	8	4	3 0 ¹ / ₂	B	4	4	35	50
				14	13	12	27	0	17	4	10	B	5	14	59	37
				16	23	18	37	10	5	2	26	B	6	27	0	12



Tycho Brahe's observations of other planets



- Brahe meticulously measured the orbit of Mars, as well as other planets (i.e., Mercury, [Venus](#), [Jupiter](#), [Saturn](#)), over a period of eighteen years (1582 –1600)

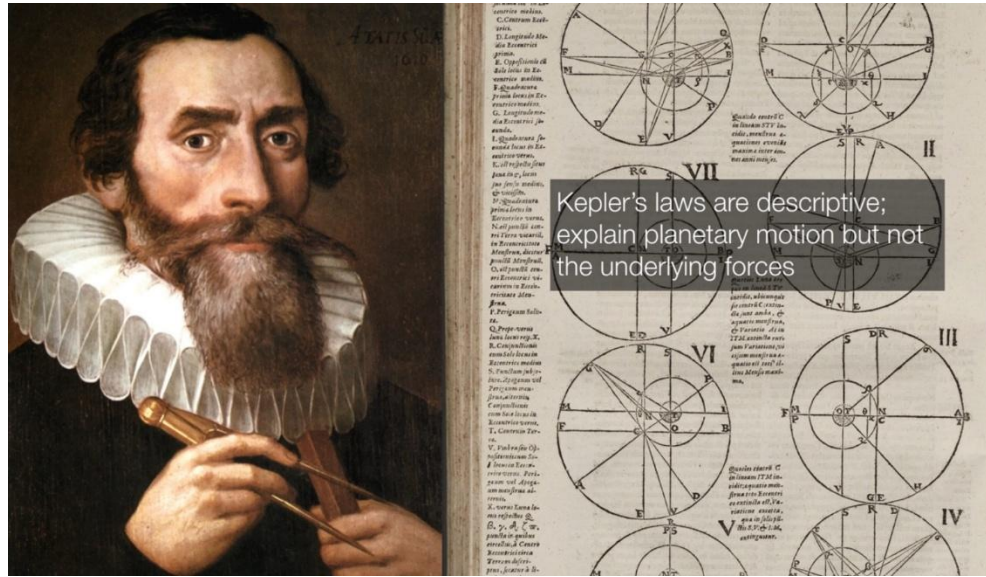


Kepler's laws are descriptive

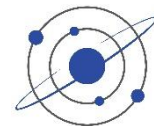


Kepler's laws of planetary motion

Launch Pad Astronomy: https://youtu.be/lcDMv_bs4LY?si=1_r5sb4S1pQckpEF [10:40]

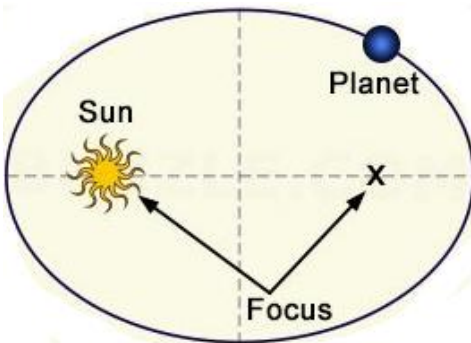


Kepler's first law (law of orbits)

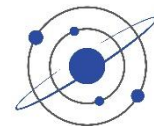


- Its first law was the result of a great intuition: **the planets don't necessarily move in perfect circular orbits**, as had long been assumed
- Using Tycho's measurements of Mars' motion, **Kepler discovered that planets move in elliptical orbits** where the Sun is not at the center of the ellipse, but at one of its foci (the other focus is empty)
 - It's a deceptively simple law that required amazing insight and 5 years of hard work

The planets of the solar system **describe elliptical trajectories** of which the **Sun occupies one of the foci**



Kepler's second law (law of areas)

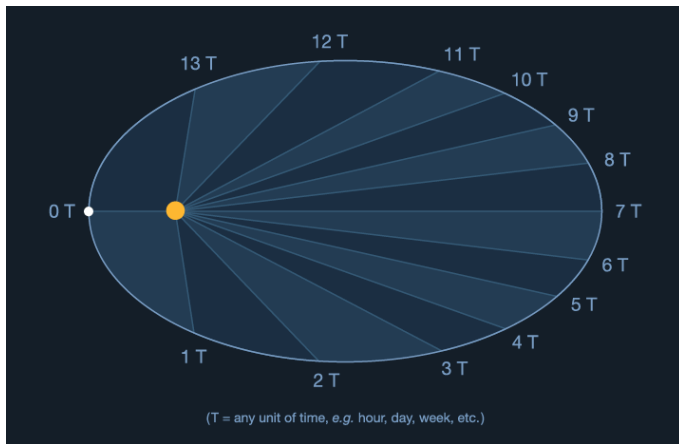


- Kepler found another law by studying the orbit of Mars (published in 1609 in: *De Motu Stellae Mars*)

The radius vector joining the Sun to a planet sweeps
equal areas in equal time intervals

Fundamentally:
Angular momentum is
conserved along the orbit:

$$L = m(\vec{R} \times \vec{V}) = \text{constant}$$



→ This law implies that the
velocity of a planet in an
elliptical orbit varies,
→ and that it is greater at
perihelion than at *aphelion*

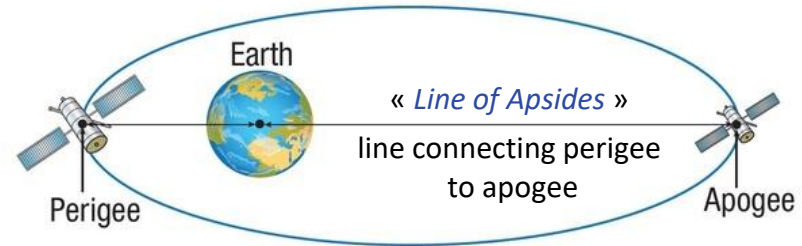
→ This 2nd law implies a very important fact: orbital motion is caused by a force directed towards the Sun.
This is a discovery that Kepler made and that guided Newton to obtain the law of universal gravitation

Apsides

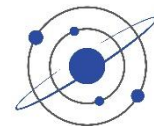


- The **apsides** (plural of apsis) are the two extreme points of the orbit of a celestial body, for which the distance to the attracting body (more exactly, to the center of mass of the two bodies, to the main focus of the ellipse) is:
 - either **minimal**: inferior apsis, or **periapsis** (general term)
 - or **maximal**: upper apsis, or **apoapsis**.

Body	Nearest	Farthest
General	periapsis	apoapsis
Earth ⊕	perigee	apogee
Sun ☉	perihelion	aphelion
Moon ☾	perilune	apolune



Kepler's third law (law of periods)



- Kepler needed another 10 years of research to find the third empirical law:

The **square of a planet's period** of revolution is **proportional** to the **cube of the semi-major axis** of its elliptical orbit

- This law derives from the fact that the gravitational force exerted by the Sun on the planet is proportional to the inverse of the square of the distance. For circular orbits, the gravitational force plays the role of the centripetal force:

$$\sum F_c = \frac{GM_{\odot}m}{r^2} = m \frac{v^2}{r}$$

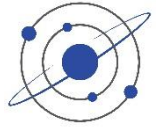
Newton' law of universal gravitation

- Inserting the period of revolution for a circular orbit $P = 2\pi r/v$ with r = semi-major axis a we find:

$$P^2 = \left(\frac{4\pi^2}{G(M+m)} \right) a^3$$

For artificial satellites, $m \ll M$
The same constant of proportionality applies to all objects orbiting the Sun ($G \cdot M_{\odot} = \mu_{\odot}$)

- The farther the satellite is from the central mass, the **slower its velocity** and the **longer its period**
 - The period of revolution **does not depend on the mass** of the orbiting object (m)
- A very useful law for astrophysicists, which allows to calculate the mass of a celestial body if only the period and radius of the orbit of one of its satellites are known



Empirical law

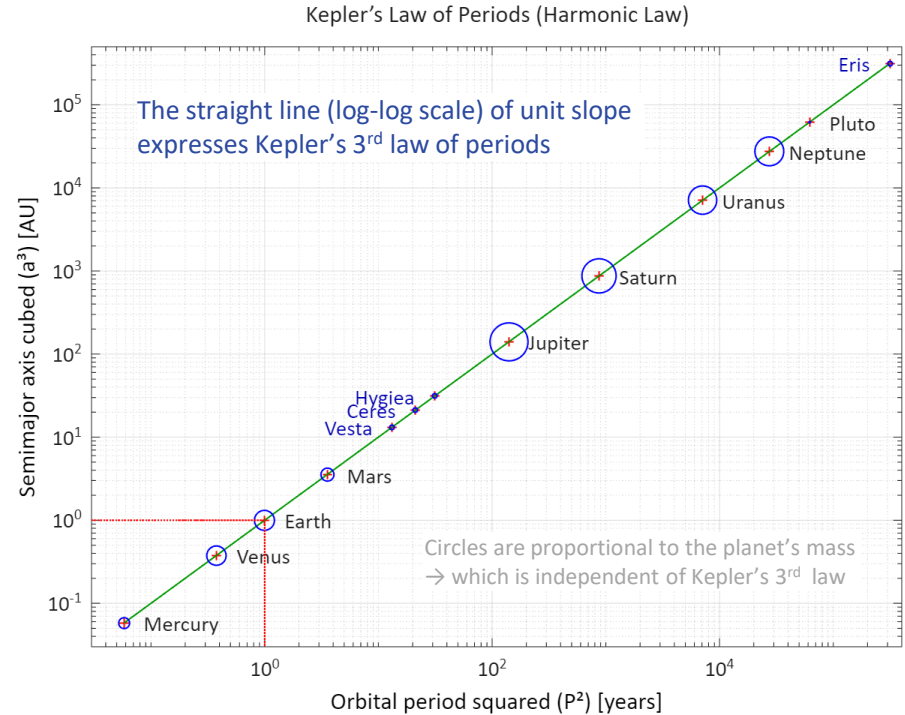
Planet	e [-]	a [AU]	P [y]	P ² /a ³
MERCURY	0.211	0.40	0.25	0.999
VENUS	0.005	0.71	0.60	0.999
EARTH	0.023	0.99	0.99	0.999
MARS	0.092	1.54	1.90	0.999
Ceres	0.077	2.75	4.57	0.999
Vesta	0.089	2.37	3.66	0.999
Hygiea	0.113	3.13	5.53	0.999
JUPITER	0.049	5.20	11.85	1.002
SATURN	0.055	9.54	29.45	1.000
URANUS	0.047	19.19	83.99	0.999
NEPTUNE	0.009	30.08	164.90	0.999
PLUTO	0.248	39.66	249.61	0.999
Eris	0.438	67.84	558.35	0.999

mean = 0.999

The square of the period **in years** is equal to the cube of the semi-major axis in **astronomical units**:

$$P^2 = a^3$$

$$\text{because: } AU \equiv \left(\mu_{\odot} \left(\frac{24 \times 60^2 \times 365.25}{2\pi} \right)^2 \right)^{1/3}$$



Exercise



- Determine the semi-major axis of Mars if by observation we know that its orbital period is 1.88 years
- Introducing unit values for the Earth and using the approximate solution provided by equation 4.6 we have:

```
[Earth, Moon, Sun, G, Planets] = AstroConstants();  
k = find(ismember(Planets.Name, 'Mars'));  
Mars = structfun(@(x) x(k), Planets, 'UniformOutput', false);
```

```
Mars.P % = 1.88 years
```

```
Mars.P^(2/3) % = 1.523 [A.U.] == Mars.a
```

```
Mars.P^(2/3) * Earth.AU % = 2.27875e8 [km]
```

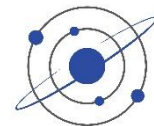
```
[ Mars.a , Earth.a / Earth.AU * (Mars.P / 1.0)^(2/3) ]
```

$$\frac{P_{\oplus}^2}{P_{\text{Mars}}^2} = \frac{a_{\oplus}^3}{a_{\text{Mars}}^3}$$

$$a_{\text{Mars}} = a_{\oplus} \left(\frac{P_{\text{Mars}}}{P_{\oplus}} \right)^{2/3}$$

$$1 \text{ AU} \left(\frac{1.88 \text{ years}}{1 \text{ year}} \right)^{2/3} = 1.523 \text{ AU}$$

Example: Calculating the mass of the Sun and Earth



- Calculating the mass of the Sun's using Earth's orbital data:

The Earth revolves around the Sun with a period of 1 year: $T = 3.156 \times 10^7$ s

in an orbit with an average radius of $r = 1.496 \times 10^{11}$ m $\equiv$ 1 AU

- By reformulating Kepler's 3rd law: $T^2 = \frac{4\pi^2}{GM} r^3 \rightarrow M_{\odot} = \frac{4\pi^2 r^3}{GT^2}$

→ Inserting the numeric values we find $M_{\odot} = 1.99 \times 10^{30}$ kg (=333 000 $\times M_{\oplus}$)

```
[ Sun.mass_kg , 4*pi^2 * (Earth.a*1000)^3 / (G * (365.2425 * 24*60^2)^2) ]
```

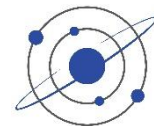
- For calculating the mass of the Earth, we can use the orbital data of the Moon

```
[ Earth.mass_kg , 4*pi^2 * (Moon.a*1000)^3 / (G * (365.2425 * 24*60^2)^2) ]
```

– This simple calculation is accurate to better than 0.3%, which is remarkable given the approximations

→ This method works because **orbital motion directly encodes gravitational mass** – one of the most powerful results linking celestial mechanics and Newtonian gravity

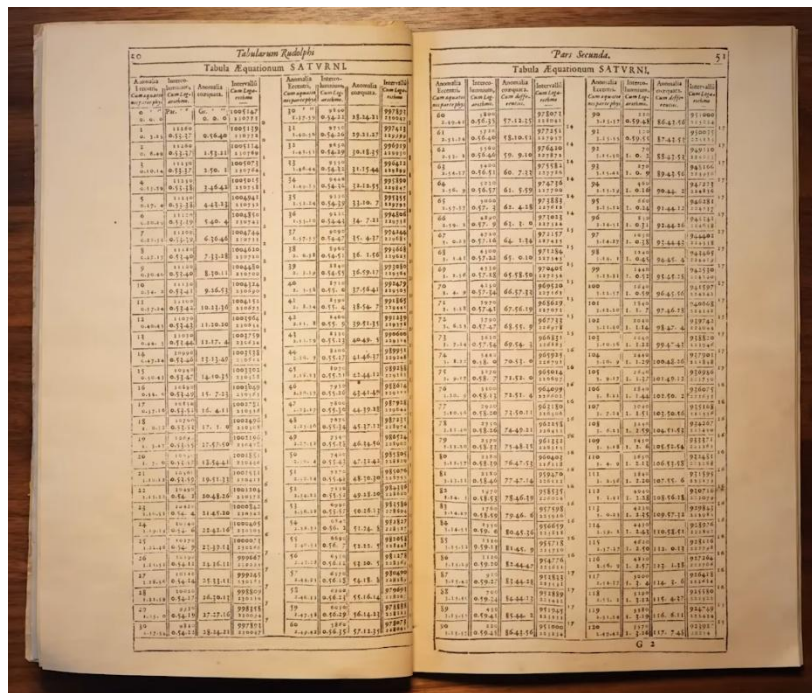
Rudolphine Tables



- The *Rudolphine Tables* consist of a star catalogue and planetary tables published by Johannes Kepler in 1627, using observational data collected by Tycho Brahe (1546–1601)

– The tables are named in memory of Rudolf II, Holy Roman Emperor, in whose employ Brahe and Kepler had begun work on the tables

- The main purpose of the Rudolphine tables was to allow the computation of the positions of the then known planets of the Solar System, and they were considerably more precise than earlier such tables



Orbital period

- Sidereal period / Keplerian period
- Synodic period
- Anomalistic period
- Nodal period (or “Draconic” or “Draconitic” period)
- Tropical year & Leap years





Orbital period

Orbital period is the time it takes for a satellite to complete one orbit around its primary body. Different types of period exist. (usually measured in seconds)

1) Sidereal period

- Determined by a 360° revolution of one body around its primary **relative to the fixed stars projected in the sky**
- This is the orbital period in an inertial (non-rotating) frame of reference; the “absolute” period in the Newtonian sense, as often used in astronomy

KEPLERIAN PERIOD:

- The time it takes for a satellite to complete one orbit around its primary, as described by **Kepler’s third law of planetary motion**
(Note: second mass might or might not be not negligible wrt to primary)

$$P_K = 2\pi \sqrt{\frac{a^3}{G(M+m)}} = 2\pi \sqrt{\frac{a^3}{\mu}} = \frac{2\pi}{n'}$$

$$P_{\oplus-\odot} = 2\pi * \text{sqrt}(\text{Earth}.a^3 / (G/1000^3 * (\text{Sun}.mass_kg + \text{Earth}.mass_kg))) / (24*60^2) \% = 365.2563 \text{ days}$$

$$\text{Earth}.Sidereal_year \% = 365.2564 \text{ days}$$

$$P_{\zeta-\oplus} = 2\pi * \text{sqrt}(\text{Moon}.a^3 / (G/1000^3 * (\text{Earth}.mass_kg + \text{Moon}.mass_kg))) / (24*60^2) \% = 27.29 \text{ days}$$

$$\text{Moon}.Sidereal_period \% = 27.32 \text{ days}$$



Orbital period (continued)

2) Synodic period

- The time it takes for a celestial object to return to the same position relative to another object, often as observed from a specific vantage point. For example:
 - For a planet, it's the time between successive conjunctions or oppositions as seen from Earth
 - For the Moon, it's the time between successive full moons or new moons
 - For an **artificial satellite**, the time to return to the **same position relative to the Sun** as seen by an Earth observer

IMP: the Keplerian orbital period of an artificial satellite *using TLE data* is the **Synodic period** (wrt to the Sun, not the stars)

- Frame of Reference: Observational alignment (often relative to Earth and the Sun)
- This depends on both the sidereal periods of the object in question and the observer's frame of reference. For planets, it accounts for the fact that Earth is also orbiting the Sun, altering how frequently we observe alignments

$$\frac{1}{P_{syn}} = \frac{1}{P_{sid}} - \frac{1}{P_{\oplus}}$$

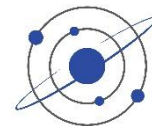
[Moon.Synodic_period , 1/(1/Moon.Sidereal_period - 1/Earth.Sidereal_year)] == 29.53059 days

Sidereal vs Synodic periods



- Key differences:

Aspect	Keplerian (sidereal) period	Synodic period
Definition	True orbital period around the primary	Time to align relative to another object
Reference Frame	Fixed stars (inertial frame)	Observer-dependent (e.g., Earth–Sun)
Example	Earth’s sidereal year ≈ 365.25 days	Earth’s synodic “year” for Venus is ≈ 584 days (time between Venus’s appearances at the same position in the sky)
Mathematics	Independent of observer motion	Depends on relative orbital speeds



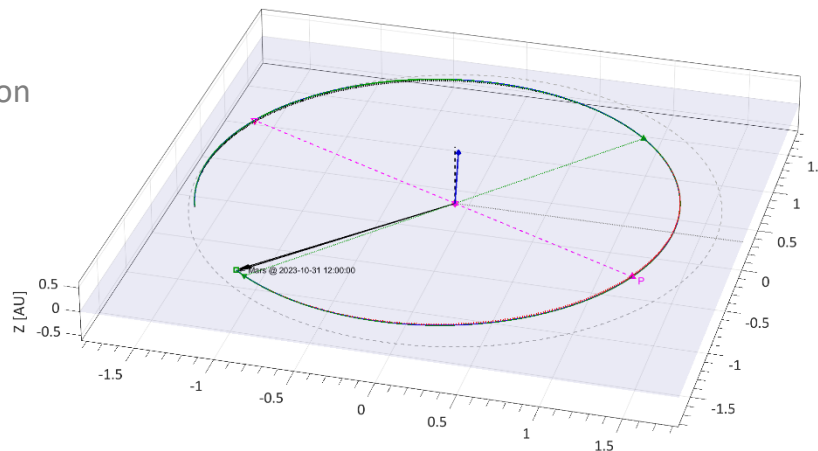
Orbital period (continued)

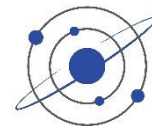
3) Anomalistic period:

- Fundamental quantity used to specify the orbital period.
- The time required for the satellite to complete **one revolution with respect to perigee**, including the drift rate for argument of perigee (the time to advance 360° in mean anomaly)
- In the absence of perturbations, the anomalistic period is equal to the Keplerian period
- The anomalistic period is a perturbed or osculating Keplerian period P_K

→ Computed & Assigned in `OrbitPropagation.m`

→ Used when computing planet's orbit, to generate a fraction or a number of orbits:





Orbital period (continued)

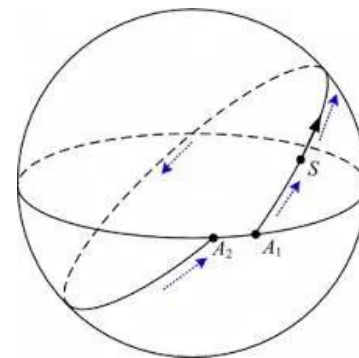
4) Nodal period (or “Draconic” or “Draconitic” period)

- The time interval it takes for the satellite to make **two successive equator crossings** (ascending or descending)
- Function of perturbation term J_2 , e , a , i , and ω : → see [non-spherical Earth perturbation](#)

$$P_{\text{nodal}} = 2\pi \sqrt{\frac{a^3}{\mu}} \left(1 - \frac{3J_2(4 - 5\sin^2(i))}{4(a/R)^2\sqrt{1-e^2}(1+e\cos(\omega))^2} - \frac{3J_2(1+e\cos(\omega))^3}{2(a/R)^2(1-e^2)^3} \right)$$

$$= P_K \times (\text{correction terms})$$

- For the Moon wrt to Earth’s ecliptic, this period is 18.61 years, also called the “lunar nodal cycle” or Bradley’s period
- As with the Anomalistic period, in the absence of perturbations the nodal periods is equal to the Keplerian period



→ Nodal period is used for computing [Ground track](#) shift and [Repeat-Ground Track](#) orbits

Newton's laws



Preliminary notions



Mass, inertia, weight, momentum



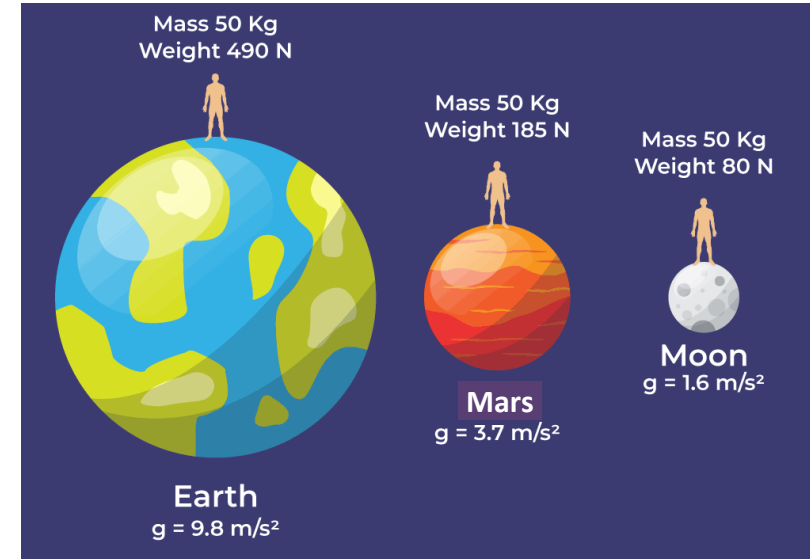
- **Mass – Inertia** m [kg] → amount of matter in an object
 - Mass is usually defined in terms of inertia: *mass is the amount or measure of an object's inertia*
 - The greater the mass of an object, the greater the inertia it contains and the more it tends to resist change (scalar quantity)
 - I have a mass of **50 kg**; it remains the same anywhere: on Earth, on the Moon, even in orbit
- **Weight** W [N] → The force of gravity exerted on an object ($F = m a$)
 - With this mass of 50 kg; my weight is = $50 \text{ kg} \times 9.8 \text{ m/s}^2 = \mathbf{490 \text{ N on Earth}}$
 - = $50 \text{ kg} \times 3.72 \text{ m/s}^2 = \mathbf{185 \text{ N on Mars}}$,
 - = $50 \text{ kg} \times 1.62 \text{ m/s}^2 = \mathbf{80 \text{ N on the Moon}}$,
 - and my weight is **NULL** in the ISS
- **Moment of inertia** I [kg/m²] → resistance of an object to acceleration due to a force
- **Momentum** p [kg·m/s] → product of mass by velocity
 - Property of a moving mass to resist change in its velocity (vector quantity)
 - It is momentum, not inertia, that drives an object on a ballistic trajectory, because momentum incorporates speed

Mass, inertia, weight, momentum

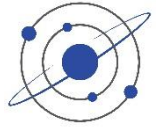


Observations:

- An object of constant mass has a constant inertia, regardless of its velocity,
i.e.: a very massive object at zero speed always retains a lot of inertia
- Inertia is the reason why, even in a zero-gravity environment, an astronaut pushing on a massive free-floating object will still feel that object push his hand back, even if it has no weight



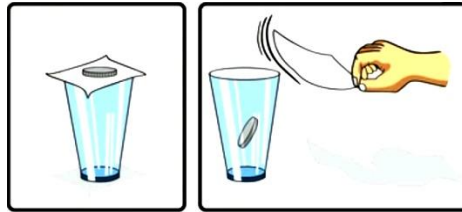
Inertia, momentum



- **Inertia** is the resistance offered by an object to change its state of rest or state of motion without the application of an external force:

$$I \text{ [kg} \cdot \text{m}^2]$$

- **Newton's first law** is known as the *Law of inertia*
 - The law states that if a body is at rest or moving at a constant speed in a straight line, it will remain at rest or keep moving in a straight line at constant speed unless it is acted upon by a force.



When the cardboard is pulled quickly, the coin falls into the glass. This is because the inertia of the piece maintains its state of rest, so that it then falls into the glass under the effect of gravity

- **Momentum** is the tendency of an object to continue to move
- Equal to the product of mass by the velocity of the object (vectorial quantity):

$$\vec{p} = m \cdot \vec{v} \text{ [kg} \cdot \text{m/s]}$$

Newton's laws

1-3+4



Newton's laws



Nearly a century after Kepler, Isaac Newton formulated his **THREE LAWS OF MOTION**:

- 1st Law:** An object at rest remains at rest, and an object in motion remains in motion at constant speed and in a straight line, unless acted on by an unbalanced force
- 2nd Law:** The acceleration of an object depends on the mass of the object and the amount (and direction) of force applied
- 3rd Law:** Whenever one object exerts a force on another object, the second object exerts an equal and opposite on the first

After these 3 laws, to complete Kepler's laws, Newton then asked the following question: *what must be the force that produces the elliptical orbits observed by Kepler?* With his answer, Newton revealed to the world the reason why celestial bodies have orbital motion: the **LAW OF UNIVERSAL GRAVITATION**

The force of gravity between two bodies is directly proportional to the product of their two masses, and inversely proportional to the square of the distance between them

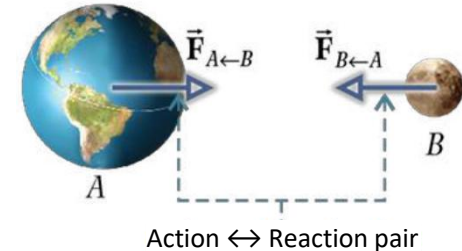
Newton's laws



- The principle of equivalence: **Mass is the measure of inertia** and is the source of gravitational force F_g ; An object of mass m placed in a gravitational field g experiences a force equal to:

$$F_g = m \cdot g \quad \text{Gravitational force (weight)}$$

- What makes Newton's dynamics great is that it starts from elementary axioms to arrive at **universal gravitation**. Its laws unify mechanics associated with movements produced by human actions and physics associated with natural movements (such as the fall of bodies)
- The **third law (action-reaction) is the most original**, since the first two were indebted to Galileo, Descartes, and Huygens
 - It was also crucial in developing the concept of mutual attraction between celestial bodies, which allowed Newton to demonstrate that the operation of this force between the Earth and the Moon had the same characteristics as gravity, responsible for the fall of bodies on Earth



→ *Gravitation is therefore universal and the Earth's dynamics now extend to celestial phenomena*

1st Newton's law



“Every object perseveres in its state of rest, or of uniform motion in a right line, except insofar as it is compelled to change that state by forces impressed thereon”

(What is normally called the definition of inertia)

- The essential point is that if no net force acts on an object (if all external forces cancel each other out), the object will maintain a constant speed:

$$\sum \vec{F} = 0 \leftrightarrow \frac{d\vec{V}}{dt} = 0$$

- If this speed is zero, the object remains at rest
- If an outside force is applied, the speed will change due to that force
- Objects with higher masses have more inertia: it takes a greater force to change their motion

→ *Contrary to what Aristotle initially thought*

2nd Newton's law



“The time rate of change of the momentum of a body
is equal in both magnitude and direction to the force imposed on it”

- This law defines a force to be equal to a change in momentum (mass times velocity) per change in time

$$\text{Linear Momentum: } \vec{p} = m \cdot \vec{v}$$

$$\text{Angular Momentum: } h = I \cdot \omega = r \times (m v) = m(r \times v) = L$$

- This is the famous equation that **Force = mass times acceleration** (*assuming the mass stays constant*)

$$\vec{F} = \frac{\Delta \vec{p}}{\Delta t} = \frac{d(m\vec{v})}{dt}$$

$$\vec{F} = m \frac{d\vec{v}}{dt} + \vec{v} \frac{dm}{dt}$$

$$\vec{F} = m\dot{v} \rightarrow \vec{F} = m \mathbf{a}$$

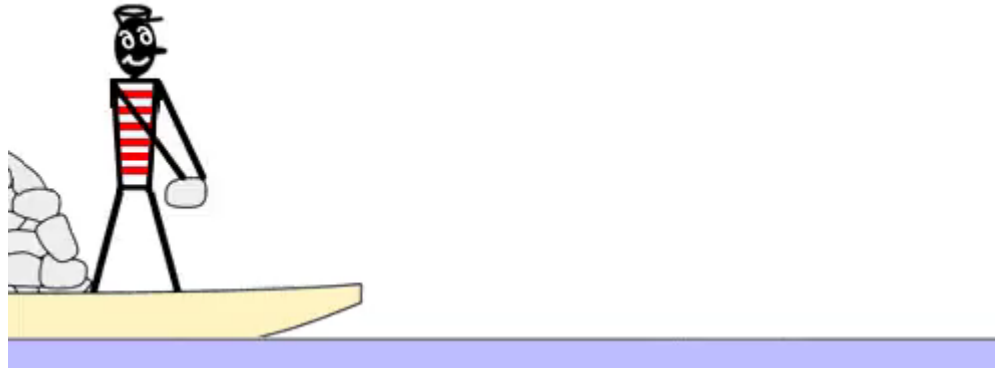
3rd Newton's law



« To every action, there is always opposed an equal reaction; or,
the mutual actions of two bodies upon each other are always equal, and directed to contrary parts »

$$\vec{F}_{1-2} = -\vec{F}_{2-1}$$

→ This law is the basis of rocket propulsion:



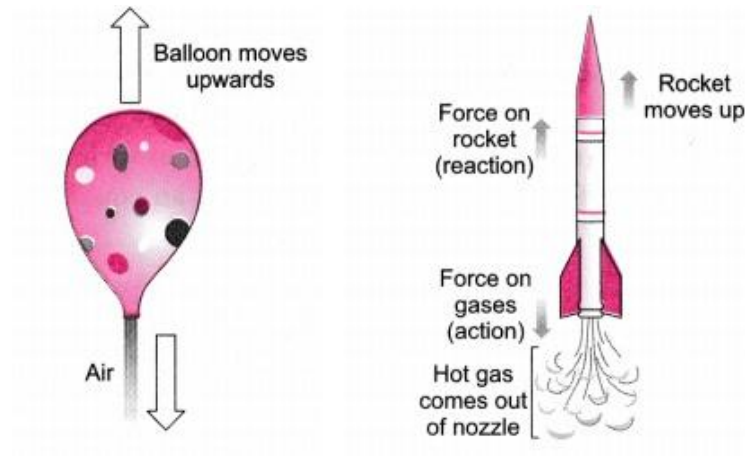
3rd Newton's law



« To every action, there is always opposed an equal reaction; or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts »

$$\vec{F}_{1-2} = -\vec{F}_{2-1}$$

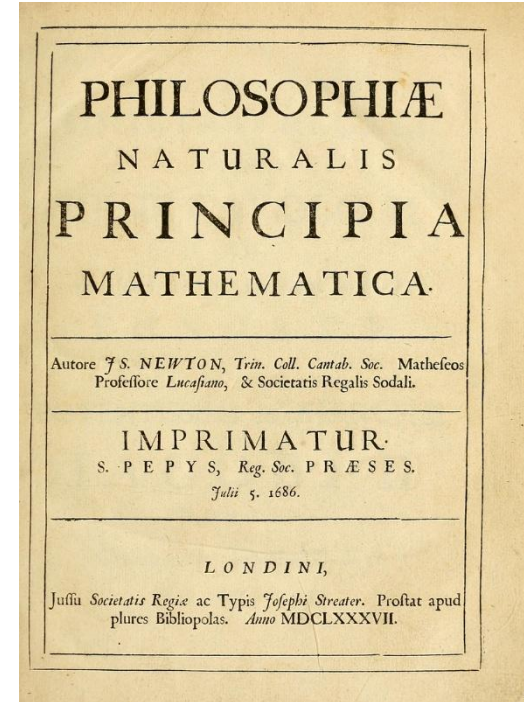
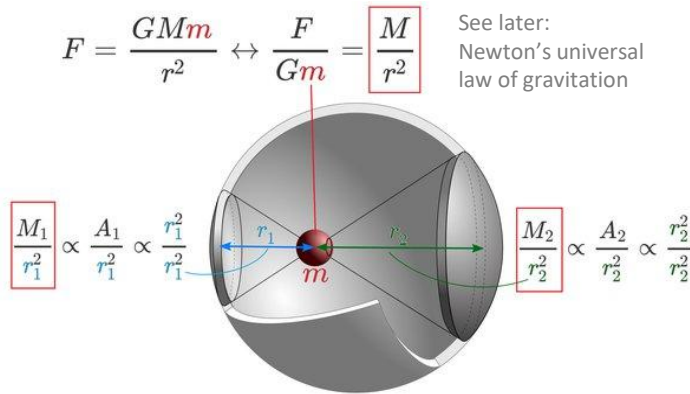
→ This law is the basis of rocket propulsion:



22 years delay...



- However, Newton did not publish the detailed theory of his laws until 1687 in *Philosophiæ Naturalis Principia Mathematica*
- One of the main causes of this 22-year delay was Newton's inability (at that time) to demonstrate that spherically symmetrical objects (e.g. the Earth) behaved gravitationally as if all mass were concentrated at the center, which required the development of **differential and integral calculus**



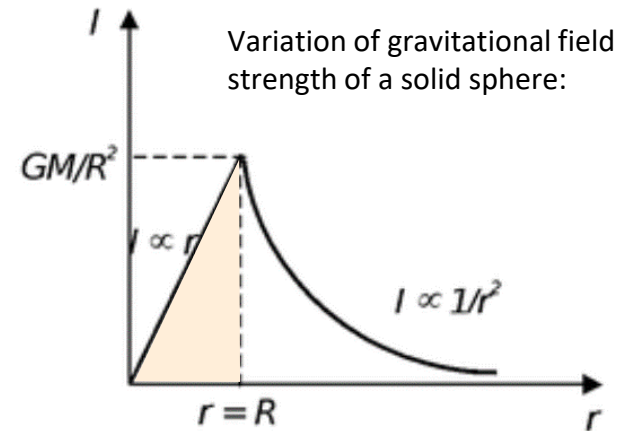
Newton Shell Theorem



- For all calculations, the bodies can be considered as **points of spherical symmetry**

The gravitational field of a mass of spherical symmetry can be calculated by treating the entire mass as a point particle at the center of the sphere

- This theorem is a consequence of the law of gravitation:
Each surface element of a homogeneous spherical shell produces a force on an object, and the sum of these forces provides a simple result that is expressed in two parts:
 - The force on an object *inside* the shell is zero
 - The force on an object *outside* the shell is equal to that of a mass equal to the shell that would be located in the center of that shell



Gravity in function of altitude

Law of universal gravitation



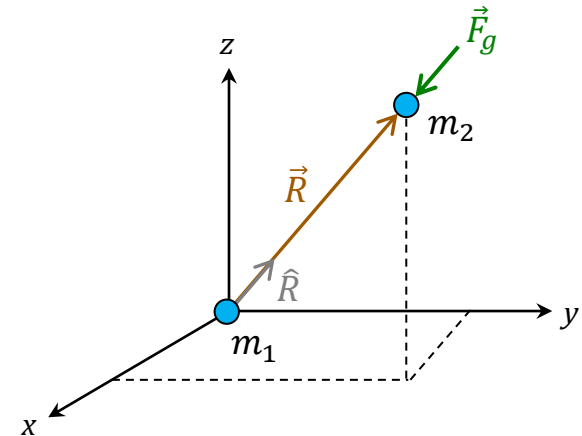
“The force of gravity between two bodies is directly proportional to the product of their masses and inversely proportional to the square of the distance between them”

Equation of the acceleration vector of the relative motion of the **2-Body problem** (in 3D)

$$\vec{F}_g = -\frac{G m_1 m_2}{r^2} \hat{R}$$

$\vec{F}_g$	force-vector between masses [N]
m_1, m_2	masses of both bodies [kg]
$\vec{R}$	position-vector connecting the 2 masses [m]
$r = \vec{R} $	magnitude of the position-vector (distance) [m]
$\hat{R} = \frac{\vec{R}}{r}$	unit vector in the direction of $\vec{R}$ [n.u.]
G	Universal Gravitational Constant

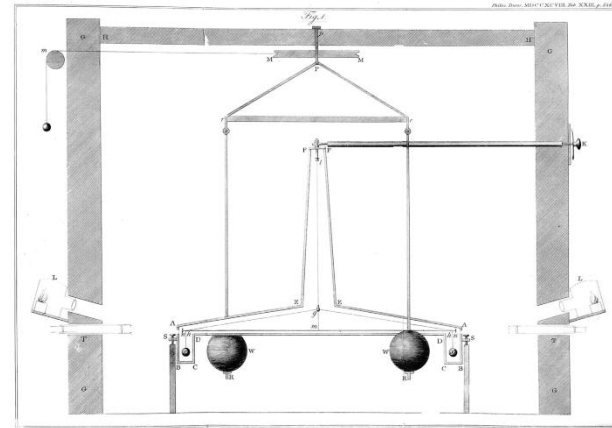
whose value ($6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$) was determined by Henry Cavendish in 1798



Measuring Gravity Constant G



- In 1798, a physicist named [Henry Cavendish](#) performed an experiment that helped him determine Newton gravitational constant, G
- He took a rod with a tiny, solid sphere at each end and hung it from a [torsion balance](#), using a thin metal thread. Then he placed two big, solid spheres close to and level with the small ones, but not attached. [Slowly, the small spheres moved toward the big ones, due to the gravitational pull.](#) By choosing an appropriate material for the thread, he simulated a *free-fall condition* between the big and small spheres
- His torsion balance experiment was remarkably precise; using precise optical deflection measurement techniques, he was able to evaluate the numerical value of G [with an accuracy of \$\approx 1\%\$](#)
- Although he never intended to determine the gravitational constant, because his goal was to find the mass of Earth, we recognize him as the first to compute a value for G



Standard Gravitational Parameter μ



- Although the procedure used finds a representative value for G , **its accuracy is limited** because the gravitational attraction between small masses **is very small and thus difficult to measure accurately**
 - Modern evaluation of the Newtonian gravitational constant is done via ultra-precise, laboratory-based experiments – primarily using sophisticated torsion balances or cold-atom interferometry
 - Today, we commonly measure the quantity $\mu = GM_{\oplus}$ from satellite observations
- The quantity μ is called the **Gravitational Parameter**. The EGM-96 values for m and G are:

$$\mu = 3.986004415 \times 10^5 \text{ [km}^3/\text{s}^2] = G (M_{\oplus} + m_{sat}) \approx G M_{\oplus}$$

- Although the precise definition includes the masses of the Earth and the satellite, we neglect the satellite's mass because it's so small relative to the Earth's. Whenever the objects of interest are more nearly the same size (the Sun and Jupiter, for instance), then both masses contribute to μ

$$G = 6.67430(15) \times 10^{-11} \text{ [N} \cdot \text{m}^2/\text{kg}^2 \equiv \text{m}^3/(\text{kg} \cdot \text{s}^2)]$$

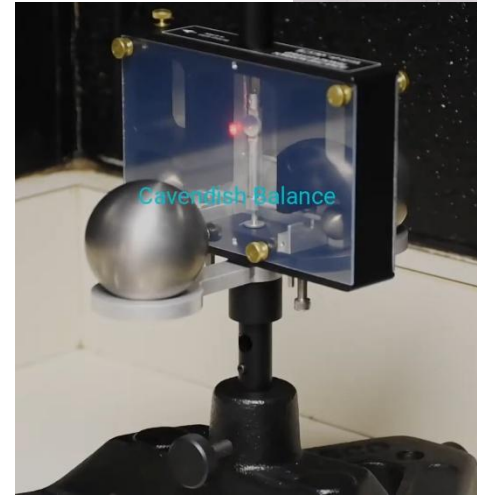
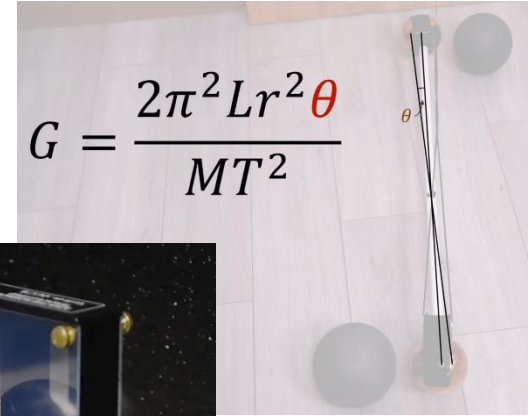
$$M_{\oplus} = 5.9733320 \times 10^{24} \text{ [kg]}$$

→ Although μ and G are experimentally measured, the mass of the Earth $M_{\oplus}$ is a *derived quantity*

Cavendish experiment



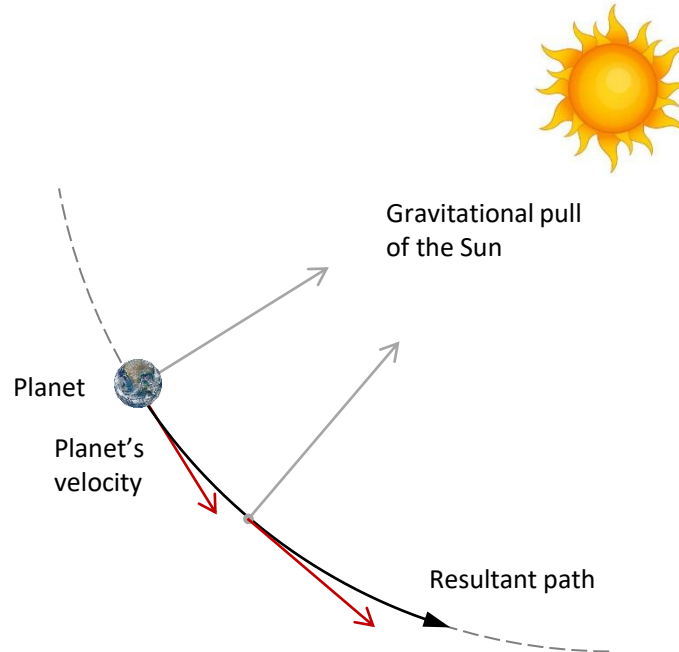
- See SOL Part I – [The Amazing World of Gravity](#) video (31:00 – 35:20 min)
- “Watch gravity pull two metal balls together”
Steve Mould: www.youtube.com/watch?v=70-GBymrck&t=12s [12:46]



Universal law of gravitation



- Illustration for the Earth-Sun system:



Exam question



Describe Kepler and Newton's contributions to our understanding of celestial mechanics

- **Johannes Kepler** introduced 3 *empirical* laws of planetary motion
 - He transformed astronomy by discovering that planetary motions follow precise mathematical rules derived from Tycho Brahe's observational data. His three laws describe *how* planets move
 - Kepler provided the first accurate dynamical description of planetary motion, replacing the Ptolemaic and Copernican circular models. However, he did not identify the physical cause of these motions
 - Kepler discovered *what* planetary orbits look like (ellipses, variable speeds, harmonic relation)
- **Isaac Newton** provided physical explanation through universal gravitation
 - He supplied the *underlying physical laws* explaining Kepler's empirical findings
 - Newton created *classical celestial mechanics*, enabling prediction of planetary perturbations, comet trajectories, tides, and eventually spaceflight trajectories
 - Newton explained *why* they behave that way, deriving Kepler's laws from universal physical principles
- Together, their work laid the foundation for all modern orbital mechanics, from planetary dynamics to spacecraft navigation

Constants of motion

- Conservation of specific mechanical energy
- Conservation of specific angular momentum

Because satellites operate in a conservative gravitational field, they **conserve mechanical energy** (ϵ) as well as **angular momentum** (h)

These two fundamental principles allow us to determine and predict their motion in time and space



Specific mechanical energy



- Total mechanical energy combines potential energy U to kinetic energy K , forming a relationship between a spacecraft's mass m , its position R , its velocity V , and the local gravitational parameter μ

$$E = \frac{1}{2} m V^2 - \frac{\mu m}{R}$$

- To generalize the equation and eliminate the dependence on mass, **divide both sides by m** . This leads to the definition of **specific mechanical energy**, ε , which is independent of mass
- As a result, the energy of an orbit can be described without regard to whether the object is a golf ball or the International Space Station
- In essence, the specific mechanical energy ε is the total mechanical energy normalized by the spacecraft's mass

$$\varepsilon \equiv \frac{E}{m} \left[\frac{\text{km}^2}{\text{s}^2} \right]$$

Conservation of mechanical energy

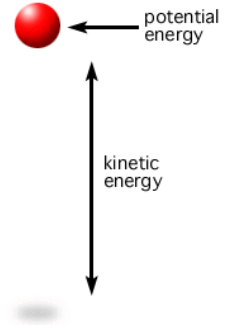


- Since the gravitational field is conservative, an object moving only under its influence does not lose or gain total mechanical energy
- Although mechanical energy remains constant, it *exchanges* kinetic energy for potential energy
→ A satellite moves fastest at perigee, and slowest at apogee

- Total mechanical energy per unit mass, the *specific mechanical energy* is

$$\varepsilon \equiv \frac{E}{m} = \text{kinetic} + \text{potential} \left[\frac{\text{km}^2}{\text{s}^2} \right]$$

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = \text{constant} \quad \text{“Vis-viva” equation}$$



- Note: Potential energy is zero at an infinite altitude ($v = 0, r \rightarrow \infty$), and increasingly negative towards the origin
- Since ε remains constant, it is related to the constant size of the orbit, a , or to the semi-major axis by:

$$\varepsilon = -\frac{\mu}{2a} \quad (\text{true for all conics})$$

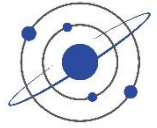


Orbital energy and eccentricity



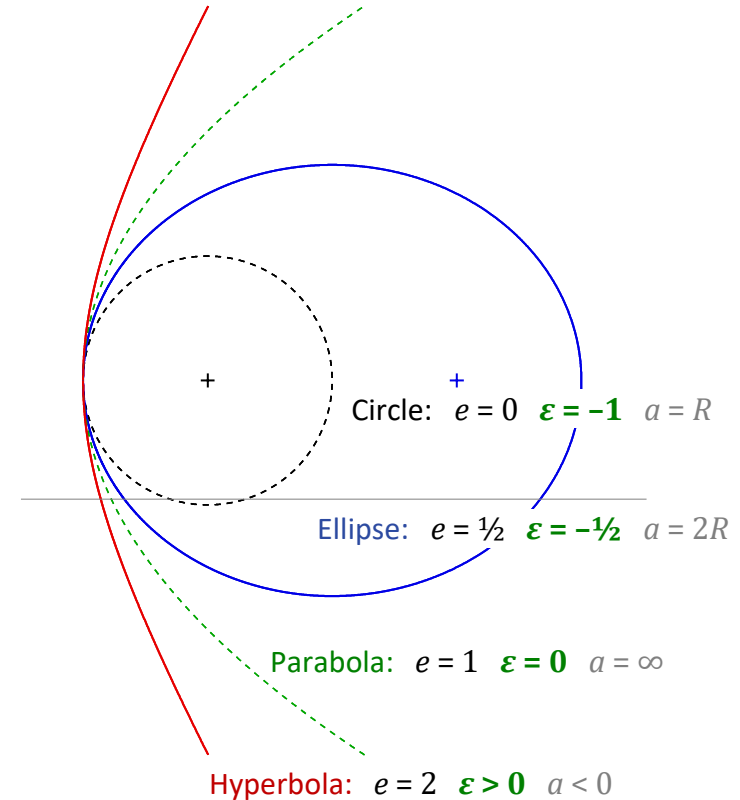
- We must bear in mind that the circular and elliptic orbits have their semi-major axes defined as a positive quantity, whereas it is considered as infinite in the parabolic orbits, and negative in the hyperbolic orbits
- This means, based on $\varepsilon = -\mu/(2a)$, the specific mechanic energy of a body is:
 - *negative in elliptic orbit (or circular),*
 - *zero for parabolic orbits,* and
 - *positive for hyperbolic orbits*
- Also, for any conic: $e = \sqrt{1 + 2\varepsilon(h/\mu)^2}$; we can conclude that:
 - if the energy is negative then the eccentricity is positive and < 1 , i.e. an ellipse
 - if the energy is zero, then the eccentricity is $= 1$, i.e. a parabola
 - if the energy is positive, then the eccentricity will also be positive, but > 1 and therefore the orbit will be hyperbolic

Orbital energy



- Example of Specific Mechanical Energy:
 - Satellite in GEO orbit ($H = 35786$ km): $\varepsilon = -4.7 \text{ km}^2/\text{s}^2$
 - At a LEO altitude (ISS @ 430 km): $\varepsilon = -29.3 \text{ km}^2/\text{s}^2$
→ Lower Energy
- Energy is related to the eccentricity of the orbit e
- The Vis-viva equation also tells us that
$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r \rightarrow \infty}$$

outside the Earth's sphere of influence ($H > 1 \text{ M km}$), since the potential energy term is zero, only the positive kinetic energy remains, so the overall specific mechanical energy is positive
 - In this case, the semi-major axis becomes negative ($a < 0$), which is the case for hyperbolic trajectories describing interplanetary orbits relative to the Earth



→ see table in Geometry of the ellipse

“Slow down to speed up” paradox



- The height of the orbit, or distance between the satellite and Earth’s surface, determines how quickly the satellite moves around the Earth. An Earth-orbiting satellite’s motion is mostly controlled by Earth’s gravity. As satellites get closer to Earth, the pull of gravity gets stronger, and the satellite moves more quickly
- Changing a satellite’s height changes its orbital speed. *This introduces a strange paradox*
- If we want to increase the satellite’s orbital speed, we can’t simply fire the thrusters to accelerate the satellite. Doing so would boost the orbit (increase the altitude), which would slow the orbital speed
- Instead, we must fire the thrusters in a direction opposite to the satellite’s forward motion, an action that on the ground would slow a moving vehicle. This change will push the satellite into a lower orbit, which will increase its forward velocity

“Slow down to speed up”

The Only Video Needed to Understand Orbital Mechanics

www.youtube.com/watch?v=bcvnfQlz1x4 [7:37]

www.youtube.com/watch?v=hKdpzagTUxk [1:15]



Getting to orbit

Paradoxes of Space Navigation



- Two Paradoxes of Space Navigation

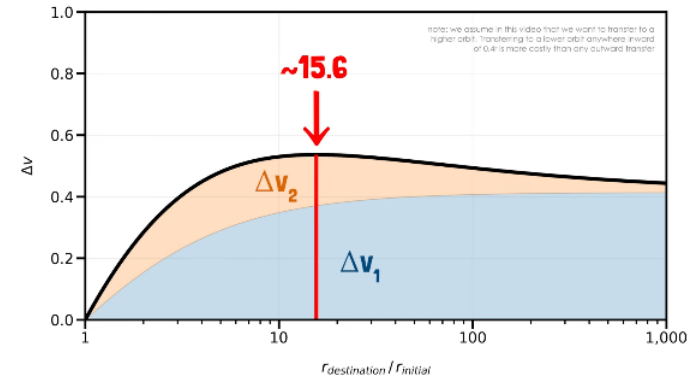
minutephysics: www.youtube.com/watch?v=dLtRtdg67PU [03:17]

- This video looks at the counter-intuitive weirdness of navigating two “simple” space manoeuvres

- This Orbit is the WORST

minutephysics: www.youtube.com/watch?v=R2G8Xy6hiyg [07:31]

→ *More about orbital transfer and space navigation later in the course*

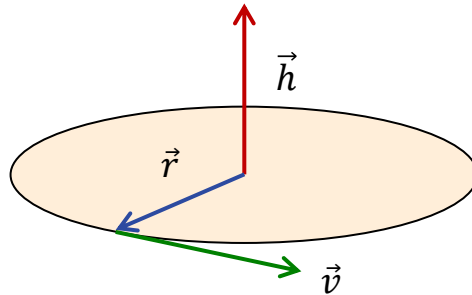
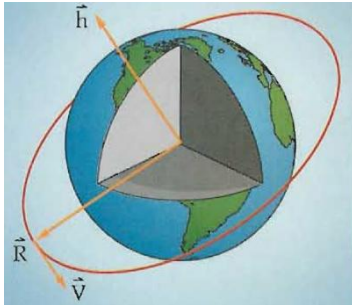


Conservation of angular momentum



- Angular momentum is the angular quantity analogous to linear momentum
- Since the gravitational field is always directed radially towards the center of the large mass, the angular momentum of the object around the large mass **does not change**
- Angular momentum per unit mass*, is called **specific angular momentum** (or reduced angular momentum)

$$\vec{h} = \vec{r} \times \vec{v} = \text{constant} \quad \left[\frac{\text{km}^2}{\text{s}^2} \right]$$

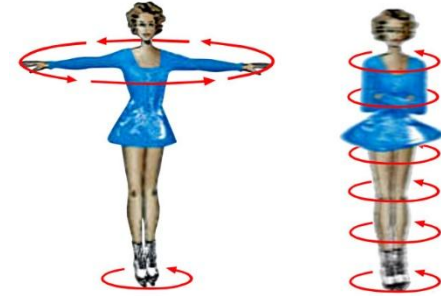


As $\vec{h}$ is constant, $\vec{r}$ and $\vec{v}$ must remain in the same plane, called the **orbital plane**

Angular momentum examples



- Example of the skater: h constant: change $r \rightarrow$ change v
 - Figure skaters use angular momentum conservation to control their jumps: They bring their arms closer to their bodies to increase the rotation speed when they are in the air, then they extend their arms on the landing to decrease the rotation speed.
- Example of riding a bicycle: **gyroscopic stability**
 - Bicycle @ 20 km/h $\rightarrow h = 7 \text{ km}^2/\text{s}^2$
 - Satellite @ 500 km, $v = \sqrt{\mu/r} = 7.6 \text{ km/s} \rightarrow h = 52\,360 \text{ km}^2/\text{s}^2$
 $\rightarrow$ difficulty in changing the orientation
- Example of the gyroscope in zero gravity
<https://youtu.be/xQb-N486mA4?si=SnhlIiUj39vL1tXe>



Angular momentum calculation: Equivalent formulas

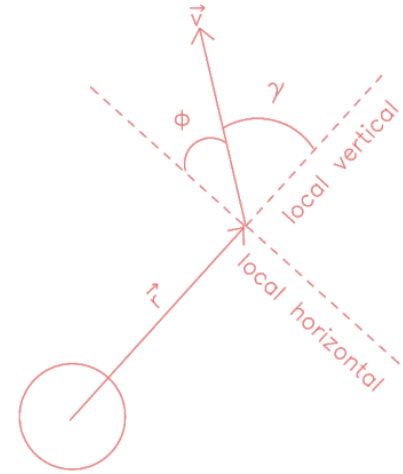


The magnitude of vector $\vec{h}$ is given by:

```
Orbit.h = sqrt(Orbit.mu * Orbit.p); % [km²/s]
        = sqrt(Orbit.a * Orbit.mu * (1 - Orbit.e^2));
        = sqrt(Orbit.r_P * Orbit.mu * (1 + Orbit.e * cos(θ)));
```

It can be geometrically derived from vectors $\vec{r}$ and $\vec{v}$ using $\vec{h} = \vec{r} \times \vec{v}$:

```
k = randi(length(Orbit.T)) % for any point within the orbit
Orbit.h = norm(cross(Orbit.ECI.pos(:,k), Orbit.ECI.vel(:,k)))
           = norm(Orbit.ECI.pos(:,k)) * norm(Orbit.ECI.vel(:,k)) * cosd(Orbit.phi(k))
where  $\phi$  is the flight path angle
           = Orbit.r_P * Orbit.v_P
           = Orbit.r_A * Orbit.v_A
```



Quiz: Compute the following angular momentum



1) Small sat

```
r = Earth.r1_km + 500; % [km]
v = sqrt(Earth.mu / r); % [km/s]
h = r * v % = 5.236e4 km²/s²
```

2) Moon-Earth

```
r = Moon.a; % [km]
mu = G/1000^3 * (Earth.mass_kg + Moon.mass_kg); % [km³/s²] ← here we don't neglect Moon's mass
v = sqrt(mu / r); % [km/s]
h = r * v % = 3.9e5 km²/s²
```

→ The Moon angular momentum is just 7 times that of a small artificial satellite?

Yes: 55 × farther, but 7 × slower in its orbit → 7.35 × larger overall (independent of mass)

3) Earth-Sun

```
r = Earth.a; % [km]
v = sqrt(Sun.mu / r); % [km/s]
h = r * v % = 4.4e9 km²/s²
```

Conservation of angular momentum



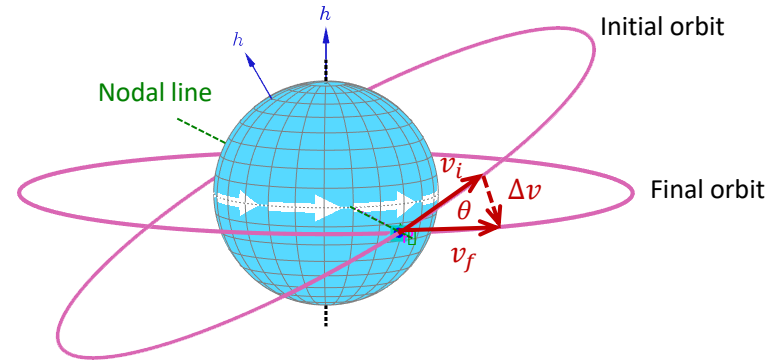
This property has huge implications for satellites in orbit

1. The orbital plane of a satellite remains fixed, unless an external pair is applied
2. It takes a lot of energy to change an orbital plane
e.g.: It takes as much fuel as the mass of a 500 kg satellite to change its 15° inclination
(see Change of inclination in the [Orbital Maneuvers](#) section)

Orbital inclination change:

To transfer the launch inclination (28.5° Cape Canaveral) to the desired final inclination (0° equatorial, for GEO):

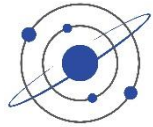
The change in inclination can be done by pushing up or down, perpendicular to the existing orbital plane, at one of the nodes (ascending or descending)



Equation of relative motion



Equation of relative motion



Newton was then able to combine his second law of motion, $F = ma = m \frac{d^2 r}{dt^2}$, with its law of universal gravitation, to obtain his [2-body differential equation of relative motion](#), which approximates very well the motion of artificial satellites and planets:

$$\frac{d^2 r}{dt^2} = -\frac{GM}{r^3} \vec{r}$$

$$\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r} = 0$$

← Although the denominator is r^3 , the equation is still an *inverse-square law* because $\vec{r}/r$ is a unit vector

where $\mu \equiv G \cdot M$ is the measure of the gravitational force of the main body → [Standard Gravitational Parameter](#)

→ For orbits around the Earth: $\mu_{\oplus} = G \cdot m_{\oplus} = 398600 \text{ [km}^3/\text{s}^2\text{]} = \textit{gravitational geocentric constant}$

Accurate calculations of orbits must take into account the [effect of the Earth's non-spherical symmetry](#), [perturbations due to other bodies](#), and [non-gravitational forces](#) (e.g. drag), but nearly all the foundations of orbit theory are direct extrapolations of the work foreseen by Newton himself

Equation of relative motion: Derivation



Applying **Newton's law of gravitation**

$$\vec{F}_g = -\frac{G m_1 m_2}{r^2} \hat{R} \qquad \hat{R} = \frac{\vec{r}}{r}; \vec{F}_g = m \vec{a} = m \ddot{\vec{r}}$$

to the Earth-satellite system:

$$\begin{aligned} M \ddot{\vec{r}} &= -\frac{G M}{r^3} \vec{r} \\ m \ddot{\vec{r}} &= -\frac{G m}{r^3} \vec{r} \end{aligned}$$

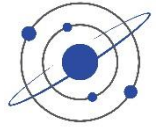
Subtracting one from another defines **the relative movement of one body relative to the other**:

$$\ddot{\vec{r}}_m - \ddot{\vec{r}}_M = \ddot{\vec{r}} = -\frac{G (M + m)}{r^3} \vec{r}$$

Using $\mu = G \cdot M$ where $m \ll M$ we finally find:

$$\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r} = 0$$

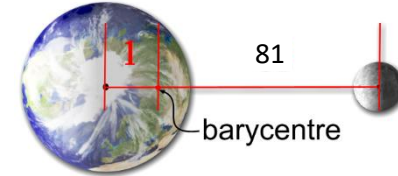
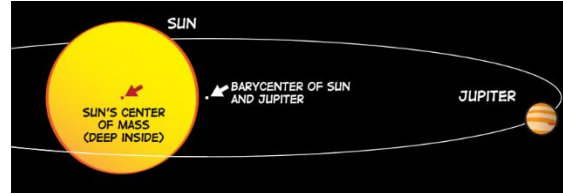
Center of mass



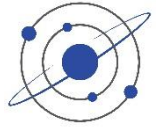
- The **barycenter** is the center of mass of two or more bodies orbiting each other and the point around which the bodies orbit
 - A barycenter is a dynamic point and not a physical object

$$\Delta x = \frac{m_1(x_1 = 0) + m_2 x_2}{m_1 + m_2}$$

Strictly speaking, it is not only the Earth that revolves around the Sun. The two orbit each other around their respective centers of mass.

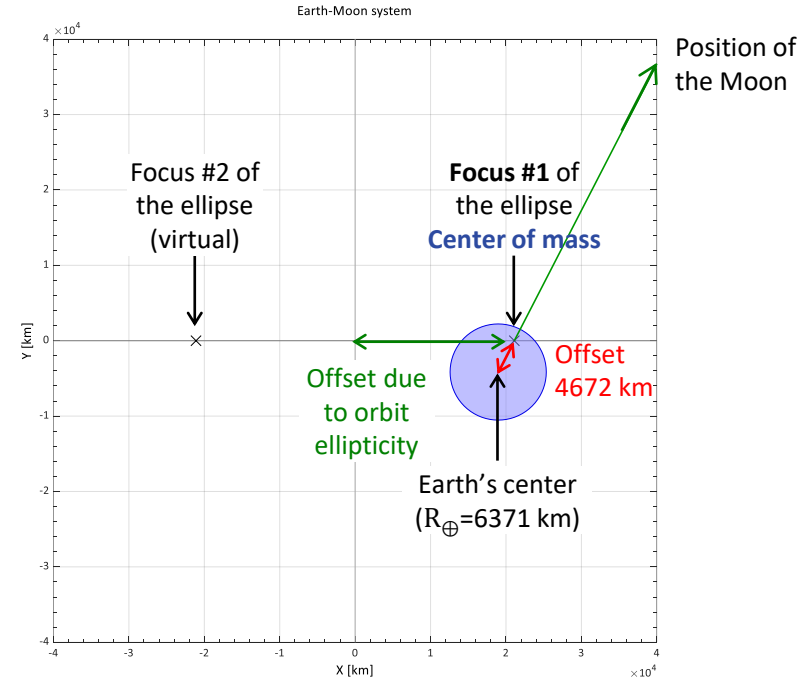
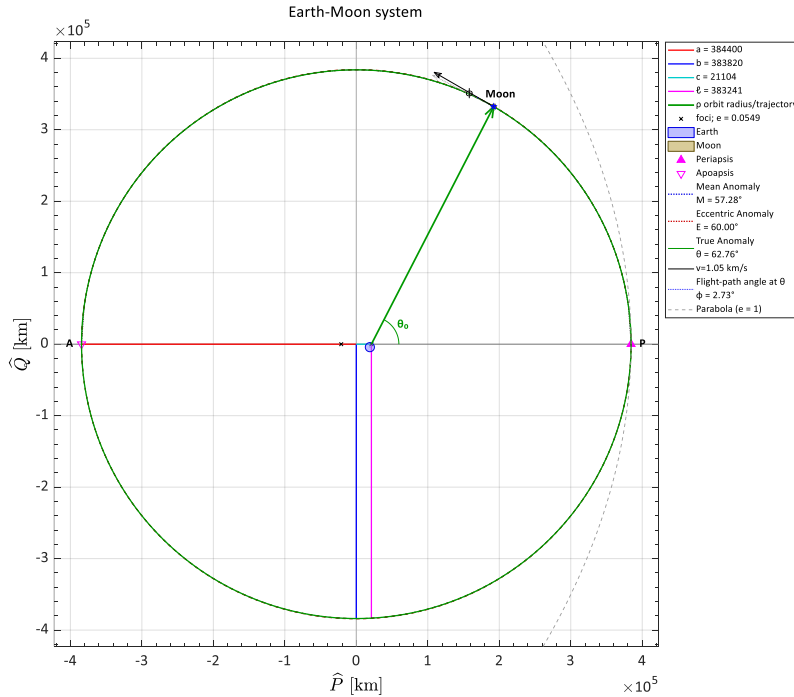


	m_1 [kg]	m_2 [kg]	d [km]	R_1 [km]	Δx [km]	$\Delta x/R_1$	
Sun – Jupiter	1.99E+30	1.90E+27	7.50E+08	695 700	715 242	102.8%	Just above Sun's surface
Earth – Moon	5.97E+24	7.35E+22	384400	6 371	4 672	73.3%	$\approx \frac{3}{4} R_{\oplus}$
Sun – Earth	1.99E+30	5.97E+24	1.496E+8	695 700	450	0.06%	≈ 500 km near Sun's center
Earth – ISS	5.97E+24	400 000	430	6 371	4.5E-21	< 1 billionth of a hydrogen atom	



Earth-Moon barycenter (to scale)

Here is displayed the elliptical orbit shape of the Moon and the offset barycenter of the Earth–Moon system



Orbital motion

Trajectory equation

- Restricted Two-body problem (2BK)
- Trajectory equation and Conics
- Geometry of the Ellipse
- Parabolic and Hyperbolic trajectories
- Reaching orbit
- Orbit propagation using Newton's model

Further reading:

Sellers 2004 [Understanding Space](#), Chapter 4

Orbital Motion

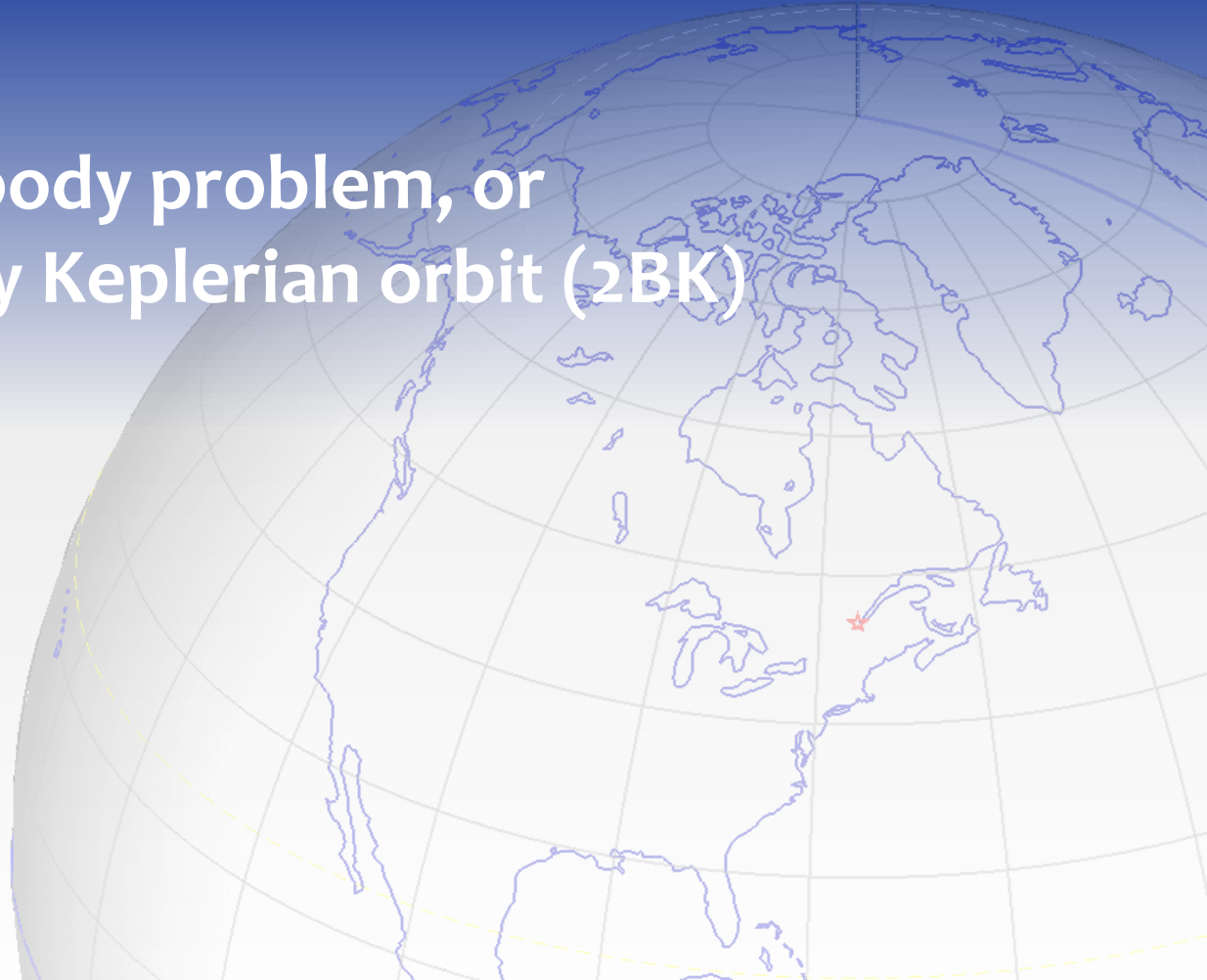
Newton's laws

Laws of Conservation

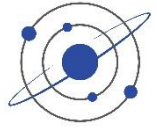
The Restricted Two-body Problem

Constants of Orbital Motion

Restricted Two-body problem, or Simplified 2-Body Keplerian orbit (2BK)



Simplified 2-Body Keplerian orbit (2BK)



- [Newton's universal law of gravitation](#) for n bodies is insoluble in a “closed form” solution
 - i.e. in an analytic form that can be expressed by a finite combination of elementary operations from certain numbers or reference functions
 - Solutions by *approximate numerical methods* can always be computed
 - However, the equation [can be solved exactly for \$n = two\$ bodies](#) (see [Appendix 3-Body problem](#))
 - where external influences (e.g. aerodynamic drag) and interaction with other masses (other satellites and planets) are neglected in the first order;
 - even the Moon and the Sun have little influence (because of their distance)
 - [will be treated as “perturbations”](#) (see Section [Orbital Perturbations](#))
- The orbit of each body is a [conical orbit](#), [one of whose foci coincides with the system's common center of mass](#), taken as the origin of the reference frame

Assumptions for deriving the Equation of motion



To treat orbital motion as a **2-Body problem**, we have to make the following important assumptions:

1. There are only two bodies involved: the mass of the satellite (m_2) is **negligible** compared to that of the attracting primary body (m_1), and **the masses are fixed and do not change**

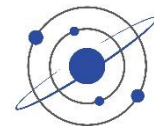
$$m_{\text{sat}} \ll m_{\oplus} (\approx 0) ; m_{\oplus} \ll m_{\odot} (0.0003\%) ; m_{\text{c}} \ll m_{\oplus} (1.23\%)$$

2. The **coordinate system chosen** for a particular problem is **inertial**
 - (this assumption removes derivatives of the coordinate system itself when differentiating vectors)
 - Earth-orbiting satellites use the *geocentric equatorial* coordinate system
 - Interplanetary probes use the *heliocentric system* (see Section: [Reference Frames & Coordinate Systems](#))
3. The bodies of the satellite and attracting primary body are **spherically symmetrical with uniform density**. This allows us to treat each as a **point mass** (Newton [Shell Theorem](#))
4. **No other forces act on the system**, except for gravitational forces that act along a line joining the centers of the two bodies (*non-perturbed* system, e.g. no aerodynamic drag)
 - This also means the masses are constant (i.e., no change in mass)

Trajectory equation & Conics

- Trajectory equation
- Conics
- Circular orbits
- Orbital velocities
- Escape velocity
- Orbit types



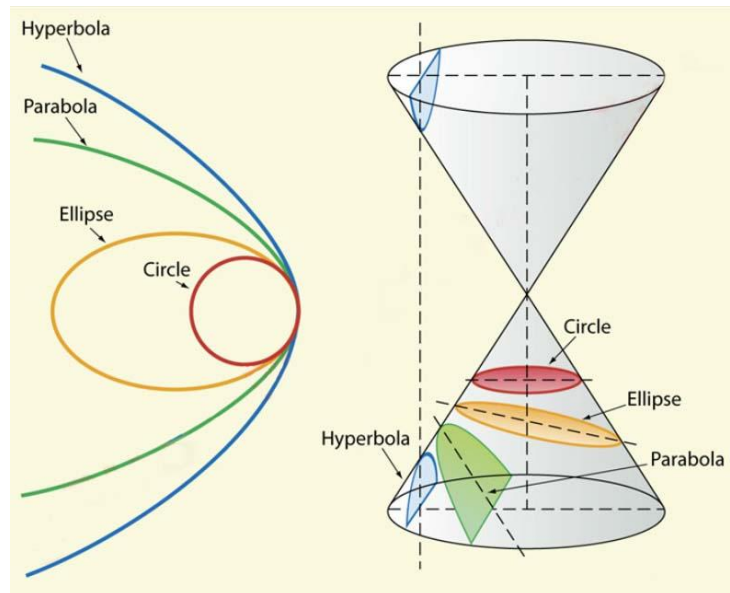


Trajectory equation

- The solution of **Newton's equation of motion** ($\ddot{\vec{r}} + \frac{\mu}{r^3} \vec{r} = 0$) for a small mass orbiting a central main body is given by a **conical section**
 - Curves obtained by *cutting a cone with a plane*
 - Circles, ellipses (closed loops),
Parabolas and hyperbolas (open conics)
- The solution *giving the shape and dimension* of the orbit is **the polar equation of the conic**:
 - (requires the algebraic differentiation of the differential equation second-order nonlinear vector $\rightarrow$ *rather tedious*)

$$r = \frac{p}{1 + e \cos(\nu)}$$

- p = Orbit size, *Semi parameter* $p \equiv a(1 - e^2) \equiv h^2 / \mu$ (also ℓ)
- e = eccentricity, determines the type of conic
- ν = angle between the ray r and the focus (**True anomaly**)

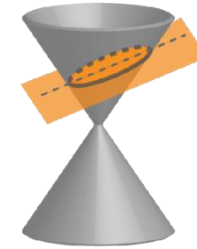


Conical sections (geometrical properties)

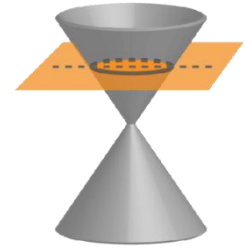


Conic sections are known since ancient times and some of its more interesting characteristics were widely known to the Greeks. Their name is a consequence of the fact that any of the basic 4 curves can be generated by the intersection of a plane (the orbital plane) and a right circular cone

- If the plane cuts only one of the sections of the cone, then the generated curve will be an **ellipse** which is the shape of all the orbits of the planets of the solar system. In fact, the **circle** is a special form of the ellipse (with eccentricity zero) and occurs when the intersection is parallel to the base of the cone. One of the characteristics of these curves is that **they are closed** and that is why all bodies which follow elliptic orbits are **periodic** and therefore remain orbiting the central body which governs their motion
- If the plane is parallel to the surface of the cone but only cuts one section, then the curve is a **parabola**. This curve usually represents the orbits of many Comets with the characteristic that being open such comets only come once through the solar system. Although such comets can be captured by the Sun or giant planets and become periodic as a consequence of perturbational gravitational forces
- If the plane cuts both sections of the cone, then the curve is a **hyperbola** with two branches. Although much less common, some comets are known to follow this type of orbits which are also of open shape → see [Hyperbolic Flight](#)



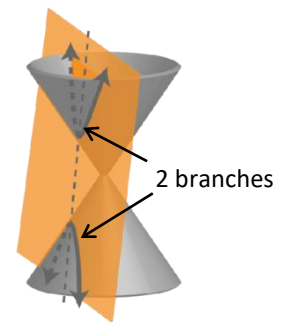
Ellipse



Circle



Parabola



Hyperbola

2 branches

Integration of the Equation of motion (advanced)



- See: Orbital Trajectory: Deep dive, To derive the orbit equation
www.studysmarter.co.uk/explanations/physics/rotational-dynamics/orbital-trajectory/
- See: Bate 1971, Section 1.5.1 Integration of the Equation of motion
- See: Sidi 2000, p.13–14
- See: Lissauer 2023, Fundamental Planetary Science Physics, Chemistry and Habitability.pdf → p.28
- See: Fitzpatrick 2012 Chap 1 & 2 up to Eq.(3.30) the equation of a conic section that is confocal with the origin (i.e. with the Sun)
- See Montebruck 2000, p.18
- See Valencia 1999 p.13–14
- See: Gleisner 2013, “Three solutions to the two-body problem”
www.diva-portal.org/smash/get/diva2:630427/FULLTEXT01.pdf

$$r = \frac{p}{1 + e \cos(\theta)}$$

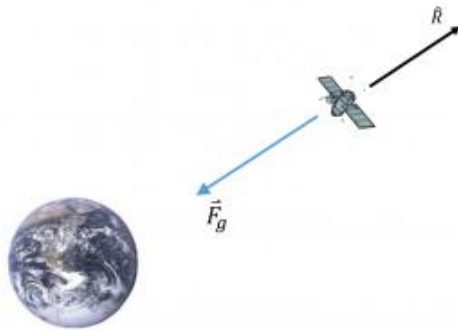
for the analytical derivation leading to the ellipse, with 3 approaches:

- | | | |
|-----------------------|-----------------------------------|---------------|
| 1) Newton’s solution | Uses K1, → proves K2, $1/r^2$ | see Eq.(5.20) |
| 2) Feynman’s solution | Uses K3, → proves K1, K2, $1/r^2$ | |
| 3) Modern solution | Uses $1/r^2$ → proves K1, K2 | |

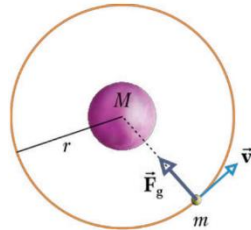
Circular orbits



- Mathematically, the orbits are all elliptical: however, most of the planets' orbits are *very close to circles*:
 - e.g., Sun-Venus: $e = 0.007$, Sun-Earth: $e = 0.017$, Earth-Moon: $e = 0.055$
- In this case, we can make the approximation that the motion of the planets or satellites is a **uniform circular motion** and that the centripetal force is equal to the gravitational force exerted by the central mass on the satellite:



$$\sum F_c = \frac{G M m}{r^2} = \frac{m v^2}{r}$$



A satellite of mass m orbiting a central mass M in a circular orbit

- This equation gives the modulus of the orbital velocity (*next page*)

Orbital velocities



- **Orbital velocity:** the (minimum) velocity needed to maintain a **circular trajectory** at a radius $r \equiv a (= r_{\oplus} + H)$

$$v^2 = \frac{G M_{\oplus}}{r} \rightarrow v_c = \sqrt{\frac{\mu}{r}}$$

ISS @ 430 km: $r = R_{\oplus} + 430 \text{ km} \rightarrow v = 7.65 \text{ km/s}$ (27 500 km/h) = `sqrt(Earth.mu/(Earth.r1_km+430))`

- For closed orbits, the **orbital period** of the satellite is:

$$P = 2\pi \sqrt{\frac{r^3}{\mu}} \quad (\text{Kepler's 3rd law, general for } r = a)$$

ISS @ 430 km: $T = 93.2 \text{ min}$

- **Escape velocity:** the minimum velocity required to escape a planet's gravity
 - The speed at which a probe can travel an infinite distance without falling back

$$v_{\text{esc}} = \sqrt{\frac{2\mu}{r}} \quad \leftarrow \text{escape velocity at distance } r$$

$$v_{\text{esc}\oplus} = 11.18 \text{ km/s}$$



Elliptical motion: Orbital velocities



Orbital velocity vs radius

Escape velocity or Escape speed?



- This confusion comes up a lot in physics. Strictly speaking:
 - **Velocity** is a vector quantity (it has magnitude and direction); **Speed** is a scalar (just the magnitude of velocity)
- Now, *escape velocity* is indeed usually treated as a scalar value in practice – the minimum speed needed to escape a gravitational potential well, regardless of direction (assuming you launch radially outward)
- So why do we use the word velocity?
 - **Historical usage:** The term escape velocity dates back to the 19th century, when physicists often used “velocity” more loosely, sometimes meaning simply “rate of motion” (i.e., what we’d now call speed). The distinction between “speed” and “velocity” wasn’t enforced as rigorously.
 - **Direction is implicit:** To actually escape, it’s not just the magnitude of the speed that matters – one must have a velocity vector pointing outward (radially, away from the gravitating body). If you fired sideways with that speed, orbital mechanics would give you a bound orbit, not escape. So the concept really involves a velocity vector of a certain minimum magnitude in a certain direction.
 - **Consistency with orbital mechanics:** In celestial mechanics, terms like *orbital velocity*, *injection velocity*, etc., are standard. Even if we’re talking about magnitudes, the convention is to say *velocity*
- So, while “escape speed” would be more precise in modern terminology, “escape velocity” is the historically rooted and still universally used phrase, with the understanding that we really mean the minimum speed along the outward radial direction



Escape velocity

$$v_{esc} = \sqrt{2 \frac{\mu_p}{r_p}} \text{ with } r \text{ is the radius of the planet } p$$

Planet	v_e [km/s]
Sun	617.67
Mercury	4.25
Venus	10.36
Earth	11.19
Moon	2.38
Mars	5.03
Jupiter	59.53
Saturn	35.48
Uranus	21.29
Neptune	23.49



Note that these speeds are for escaping the main body gravity alone, reaching infinite distance with zero residual speed relative to the body in question. It does not mean escaping the Sun

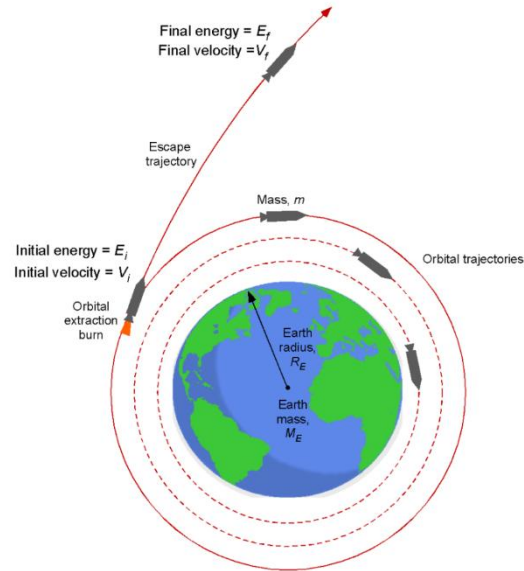
Rajouter le
mu en 2e
colonne

Escape velocity



- Introduction to Aerospace Flight Vehicles: Escape Velocity

<https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/chapter/astronautics/>

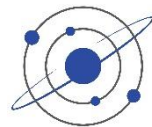


To “escape” the gravity of planet Earth and head out into deeper space, a spacecraft needs a velocity of over 11 km/s.

Application to the Escape of Atmospheres



- The kinetic theory of gases is that the volumes and pressures of gases are maintained by the mutual impacts of the individual molecules, which are, on the average, in a state of very rapid motion
- The rarity of the Earth's atmosphere and the fact that the pressure is ≈ 103 kPa, serve to give some idea of the high speed with which the molecules move and of the great frequency of their impacts
- The different molecules do not all move with the same speed in any given gas under fixed conditions; but the number of those which have a rate of motion different from the mean of the squares becomes less very rapidly as the differences increase. Theoretically, in all gases the range of the values of the velocities is from 0 to ∞ , although the extreme cases occur at infinitely rare intervals compared to the others. Under constant pressure the velocities are directly proportional to the square root of the absolute temperature, and inversely proportional to the square root of the molecular weight
- Since **in all gases all velocities exist** (kinetic theory of gases), some of the molecules of the gaseous envelopes of the heavenly bodies must be moving with velocities greater than the velocity from infinity
- If the molecules are near the upper limits of the atmosphere, and start away from the body to which they belong, they may miss collisions with other molecules and **escape never to return**



Escape to infinity

- The $v_{\text{esc}\oplus} = 11.2$ km/s escape velocity is the speed needed (ignoring atmosphere and rotation) to **escape Earth's gravity alone**, reaching infinite distance with zero residual speed relative to Earth
- It does not mean escaping the Sun. It simply means that, once free of Earth's gravitational influence, the object will still be in orbit around the Sun, moving roughly with Earth's orbital velocity:

```
Sun_mu = Planets.mu(1);
v_c_Earth = sqrt(Sun_mu / Earth.AU)  % = 29.78 km/s
v_esc_Earth = sqrt(2 * Earth.mu ./ Earth.r1_km)  % = 11.18 km/s
```

- To escape the Sun starting from Earth's surface, a higher energy is required. At Earth's orbital distance, the heliocentric escape speed is:
- ```
v_esc_Sun = sqrt(2 * Sun_mu / Earth.AU) % = 42.12 km/s
```
- Because Earth is already moving around the Sun at  $\approx 30$  km/s, a spacecraft leaving Earth can “add” speed to that. If it departs in the optimal prograde direction, it needs a hyperbolic excess speed relative to Earth of  $\approx 12.3$  km/s.
  - Combining that with escaping Earth's gravity gives a required Earth-surface launch speed of:

$$v_{\text{out}} \approx \sqrt{v_{\text{esc}\oplus}^2 + (v_{\text{esc}\odot} - v_{\text{c}\oplus})^2} \approx 16.6 \text{ km/s}$$

```
v_out = sqrt(v_esc_Earth^2 + (v_esc_Sun - v_c_Earth)^2) % = 16.65 km/s
```

# Launching into the Sun



- Launching something “directly into the Sun” is much harder than launching it out of the Solar System.
- Earth is already moving around the Sun at  $\approx 30$  km/s. To fall straight into the Sun, **an object must essentially cancel that orbital velocity**, so that its heliocentric speed becomes nearly zero and the Sun’s gravity can then pull it straight inward. This requirement dominates the problem; escaping Earth is the easy part.
- Ignoring atmosphere and practical losses:
  - Escape speed from Earth’s gravity (from the surface): 11.2 km/s
  - Required “hyperbolic excess” speed relative to Earth to cancel Earth’s orbital motion:  $\approx 30$  km/s
- These two combine in quadrature because one is relative to Earth and the other is relative to the Sun:

$$v_{\text{in}} \approx \sqrt{v_{\text{esc}\oplus}^2 + v_{c\oplus}^2} \approx 31.8 \text{ km/s}$$

$$v_{\text{in}} = \text{sqrt}(v_{\text{esc\_Earth}}^2 + v_{c\_Earth}^2) \% = 31.81 \text{ km/s}$$

- So,  $\Delta v = 32$  km/s is required for a “direct” launch that sends a spacecraft straight into the Sun, assuming an ideal retrograde burn opposite Earth’s orbital motion.
- In real life, rockets cannot do this directly; instead, missions use **gravity assists** (like [NASA Parker Solar Probe](#)) and long-duration propulsion to gradually remove orbital energy

# Escape speed vs $\Delta v$ budget



“Escape velocity from Earth = 9.2 km/s”, it is usually not referring to the pure gravitational escape speed

- The correct theoretical escape speed from the Earth’s surface in a *non-rotating*, atmosphere-free, isolated Earth model is:  $v_{\text{esc}} = \sqrt{2\mu_{\oplus}/R_{\oplus}} \approx 11.2 \text{ km/s}$
- Required  $\Delta v$  vs actual speed
  - Escape speed (instantaneous velocity) is 11.2 km/s, but a rocket does not have to accelerate from 0 directly to 11.2 km/s in vacuum. Due to gravity losses, staging, thrust profiles, and especially the *Oberth effect*, the  $\Delta v$  budget needed to reach a parabolic Earth-escape trajectory from the ground is typically quoted around:  $\Delta v \approx 9\text{--}10 \text{ km/s}$
  - So 9.2 km/s is often an engineering launch  $\Delta v$  figure, not the kinematic escape velocity.
- The Earth is not isolated – the Sun helps
  - Earth’s “escape velocity to infinity” assumes no other gravity. In reality, beyond Earth’s Hill sphere ( $\approx 1.5 \text{ million km}$ ), solar gravity dominates and actually helps pull the spacecraft away. Therefore, the speed needed to leave Earth’s gravitational control is slightly less than the ideal 11.2 km/s; more like  $\approx 10.9 \text{ km/s}$  from the equator, and even lower if efficiently trajectory-designed. Some popular sources simplify this down toward  $\approx 9\text{--}10 \text{ km/s}$
- Earth’s rotation reduces the requirement
  - Launching eastward at the equator gives you  $\approx 0.465 \text{ km/s}$  of “free” tangential speed, which reduces what the rocket must provide. That moves the number downward again, but only slightly (to  $\approx 10.7 \text{ km/s}$  speed equivalent)

# NASA Parker Probe



- **Parker Solar Probe** is a NASA heliophysics mission designed to perform **the first direct exploration of the Sun's outer atmosphere (the corona)** and to address long-standing questions about how the corona is heated and how the solar wind is accelerated. [https://en.wikipedia.org/wiki/Parker\\_Solar\\_Probe](https://en.wikipedia.org/wiki/Parker_Solar_Probe)
- Launched on **August 12, 2018**, it is the first spacecraft named after a living person, in honor of astrophysicist **Eugene Parker**, who predicted the existence of the solar wind decades earlier.
  - The mission is part of NASA's LIVING WITH A STAR program and is managed by the Goddard Space Flight Center, with design and operations led by the Johns Hopkins Applied Physics Laboratory
- Parker Solar Probe's scientific objectives focus on answering two fundamental problems: **why the Sun's corona is millions of degrees hotter than its surface**, and what mechanisms accelerate the high-speed stream of charged particles known as the solar wind
  - To achieve this, Parker carries four instrument suites that measure electric and magnetic fields, capture particles (electrons, protons, and ions), and image the structure and dynamics of the corona and solar wind. These instruments include *FIELDS* for electromagnetic fields, *SWEAP* for plasma particles, and *ISOIS* for energetic particles – allowing in-situ sampling of the Sun's environment far closer than any previous mission



# NASA Parker Probe (continued)

---

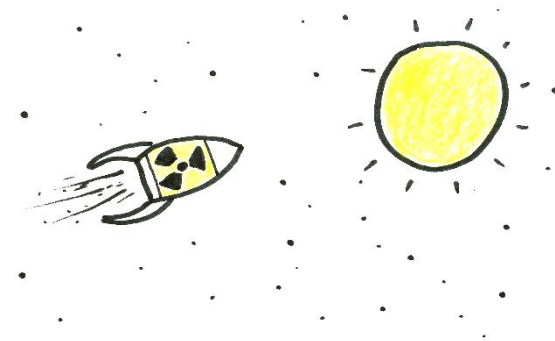


- One of the mission's defining technical achievements is its **Thermal Protection System (TPS)**, a 11.4 cm thick carbon-composite heat shield that allows the spacecraft to withstand temperatures exceeding 1370°C while keeping its instruments near room temperature
  - The spacecraft also uses an innovative solar array cooling system and autonomous pointing sensors to maintain safe orientation toward the Sun and protect instruments during periods of communication blackout when near perihelion
- Parker's trajectory uses **seven Venus gravity assists** over nearly seven years to gradually shrink its orbit around the Sun, ultimately reaching within  $\approx 6.2$  million km of the solar surface – closer than the orbit of Mercury
  - During each of its perihelion passes, Parker collects unprecedented measurements of the corona's plasma environment and magnetic structure. → See Section [Gravity Assist](#)
- The data returned by Parker Solar Probe are already transforming our understanding of the Sun and its influence on the solar system. Its detailed measurements of the solar wind and magnetic fields are improving models of **space weather** – phenomena that can affect satellites, astronauts, telecommunications, and power systems
- With the baseline mission successfully completed and continued operations planned through at least 2029, the probe's observations are expected to provide critical insight into stellar physics and the dynamic processes driving our closest star

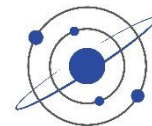


# What would it take to throw our garbage into the Sun?

- minutephysics: Hitting the Sun is HARD [3:12]  
[www.youtube.com/watch?v=LHvR1fRTW8g](http://www.youtube.com/watch?v=LHvR1fRTW8g)
- Astrum: Why It's Harder to Crash Into the Sun than to Leave The Solar System? [10.46]  
[www.youtube.com/watch?v=kC6CrUZcW-A](http://www.youtube.com/watch?v=kC6CrUZcW-A)
- What is the orbital speed of the Earth around the Sun?
  - $v_c = \sqrt{\mu/r} = \text{sqrt}(\text{Planets.mu}(1) ./ \text{Earth.AU}) = 29.78 \text{ km/s}$
- Comparative speeds:
  - Leaving the Earth's gravity influence: 11.2 km/s
  - Leave the solar system: 16.7 km/s
  - Crash into the Sun: **-31.8 km/s**

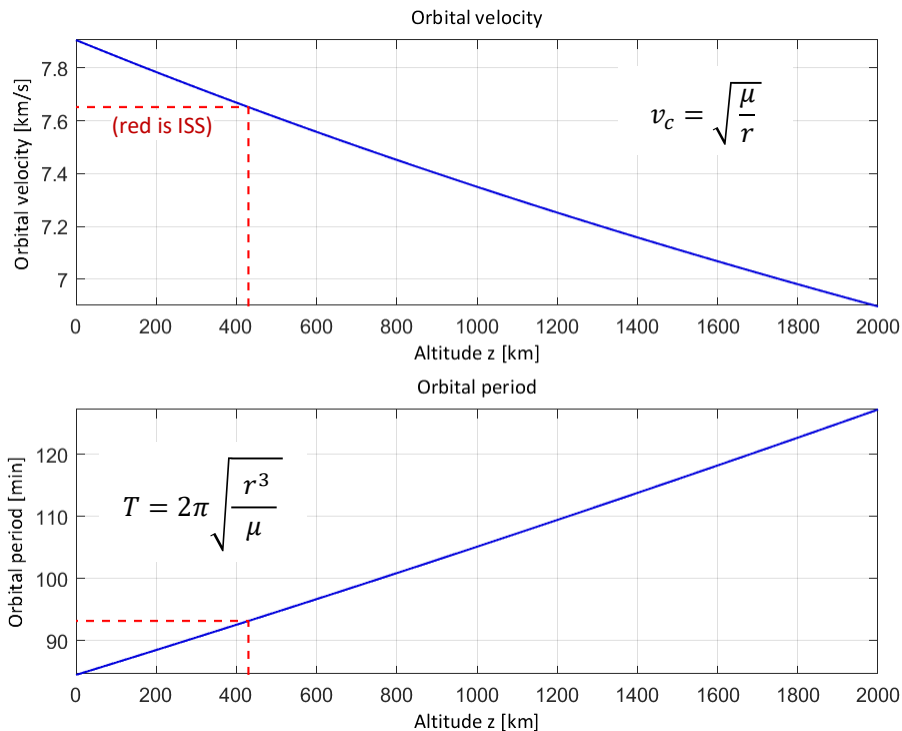


→ it takes 2× more energy to reach our own Sun than reaching other stars

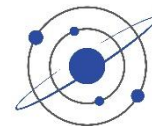


# Orbital velocity and period as a function of altitude

- Up to LEO Orbits:

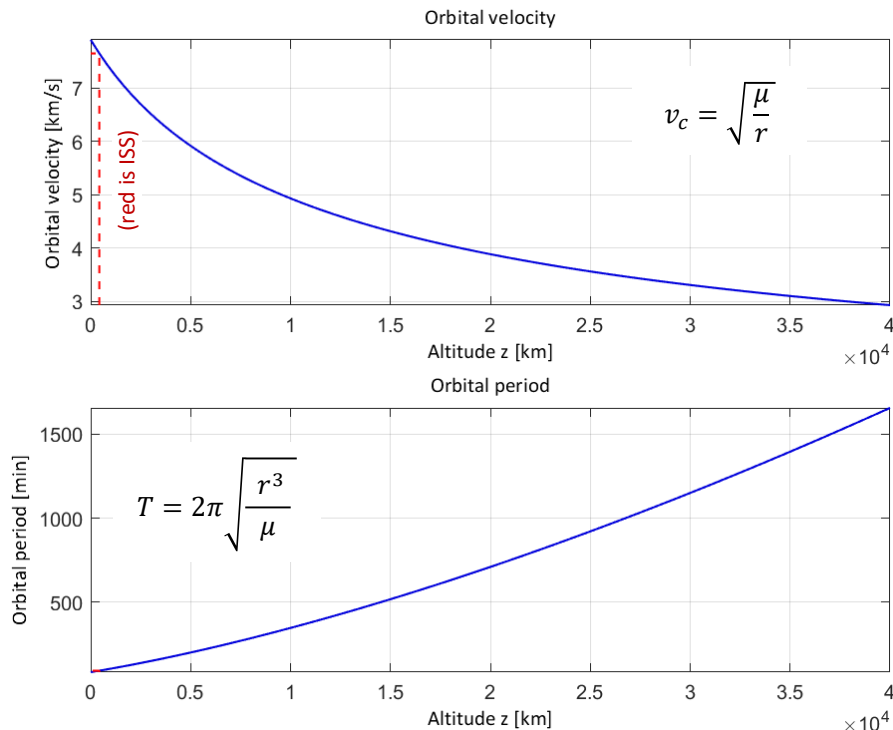




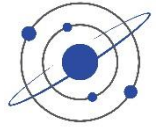


# Orbital velocity and period as a function of altitude

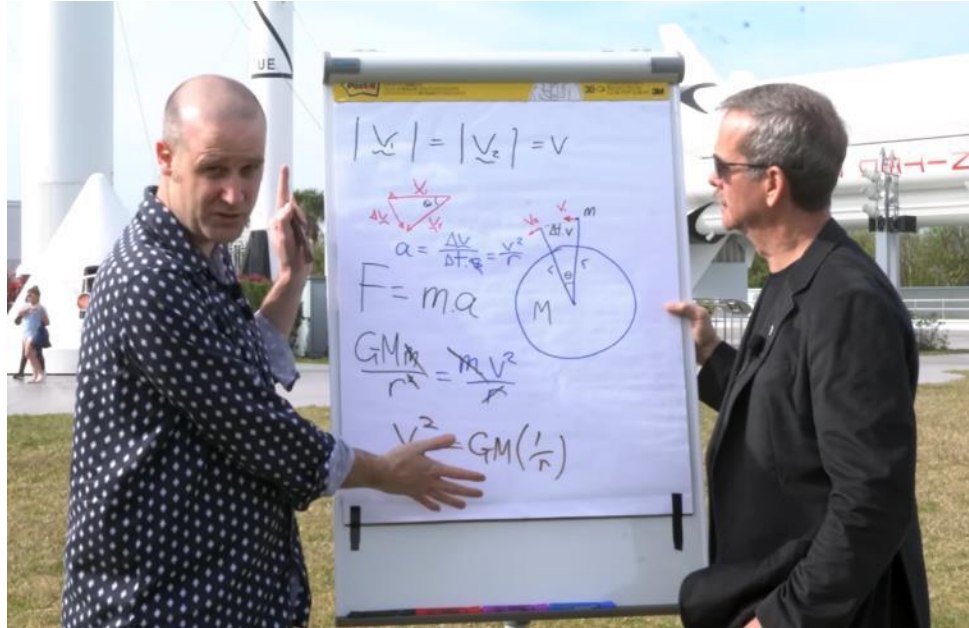
- Up to GEO Orbits:



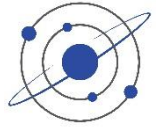
# Interesting video



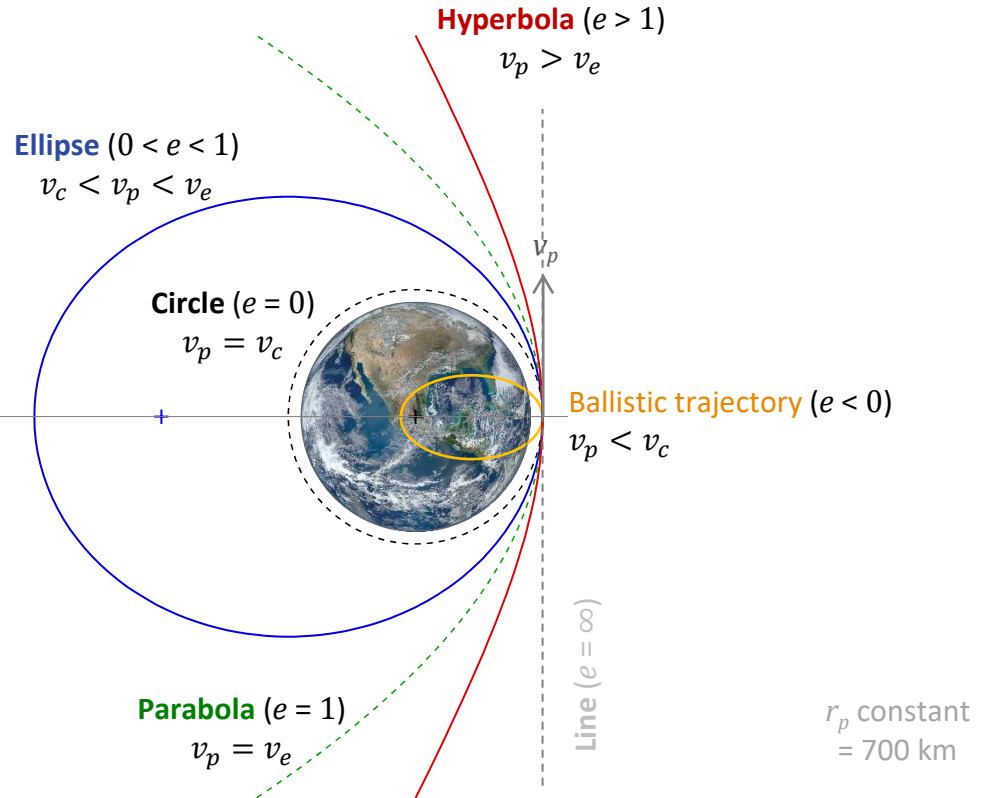
- Matt Parker & Chris Hadfield – Orbital Math [www.youtube.com/watch?v=PooFvQEN4n8](https://www.youtube.com/watch?v=PooFvQEN4n8)



# Orbit types



- Closed orbits → **Ellipses**
  - Periodic orbits
  - The *circle* is the limit case of an ellipse of zero eccentricity ( $e = 0$ ), where the two foci coincide
- Open trajectories → **Hyperbolas**
  - Escape trajectories
  - The *parabola* is the limit case between the ellipse and the hyperbola ( $e = 1$ )



Orbit types summary table

# Geometry of the ellipse

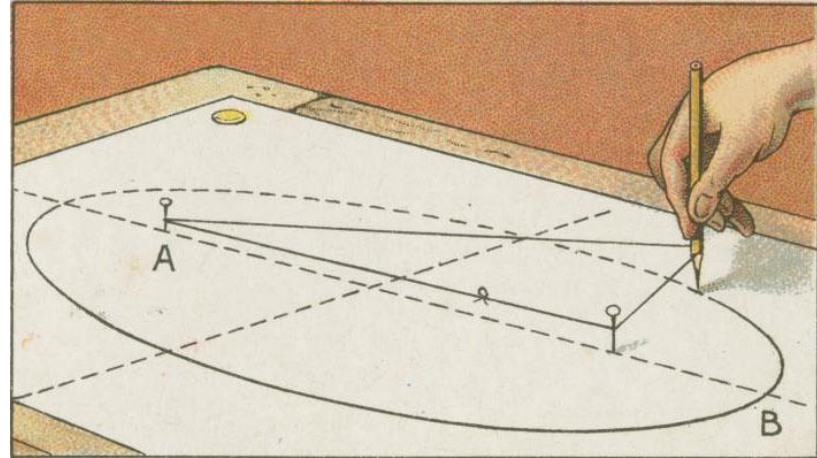
- See SOL Part I: Geocentric vs Geodetic



# Elliptical orbits: The most common



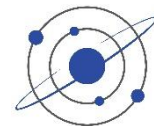
- Kepler's first law states that **planets' orbits are ellipses**
- These are **the most common orbits**, because one object is “captured” and orbits another larger object
- Not only do the planets and minor planets have elliptical orbits, most comets and binary stars also do
- The velocity in elliptical orbits is always less than that needed to escape from the central object's influence ( $v < v_E$ )
- The sum or difference of the distance from each focus to a point on an elliptical or hyperbolic orbit is a constant
  - This makes it possible to draw an ellipse with a string  $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$



# Conical sections



- Due to their symmetry, **conics have two foci**, one of which is called the **central focus** and contains the central body which governs the movement
- The **secondary focus** is empty and has no physical relevance in the study of the two body problem.
  - For the specific case of the parabola, which marks the transition between the closed and open conics, the curve has the secondary focus located at an infinite distance to the left of the central or main focus
- The chord that passes through the foci is called the **major axis** and is denoted by  $2a$ . In the circular orbits the major axis is nothing but the diameter of the orbit. In the elliptic orbit is customary to use half of this dimension and is then known as the semi-major axis of the orbit
  - In the parabolic orbits, the major axis is infinite; in the hyperbolic orbits is considered as negative
- Additionally, we have that the magnitude of the span of the curve passing through the focus, is twice the **parameter of the conic** and is represented by  $2p$  in the figure on next page
- The **distance between the two foci** is  $2c$ ,
  - It is zero for the circular orbits (for having the foci coincident in the center of the orbit)
  - For the parabolic and hyperbolic orbits, this distance is considered to be infinite



# Ellipse parameters

To understand elliptical orbits, we first need to familiarize ourselves with their geometry and characteristics:

$a$  = Semi-major axis ( $a > b$ )

$b$  = Semi-minor axis

$e$  = Eccentricity  $0 \leq e < 1$

$c$  = Half distance between foci

$p$  = Ellipse parameter (chord parallel to the directrix and passing through main focus) ( $\ell$ )

$\nu$  = **True Anomaly** (Greek letter “nu”  $\neq$  “v” for velocity)

$F$  = primary (occupied) and  $F'$  vacant foci of the ellipse

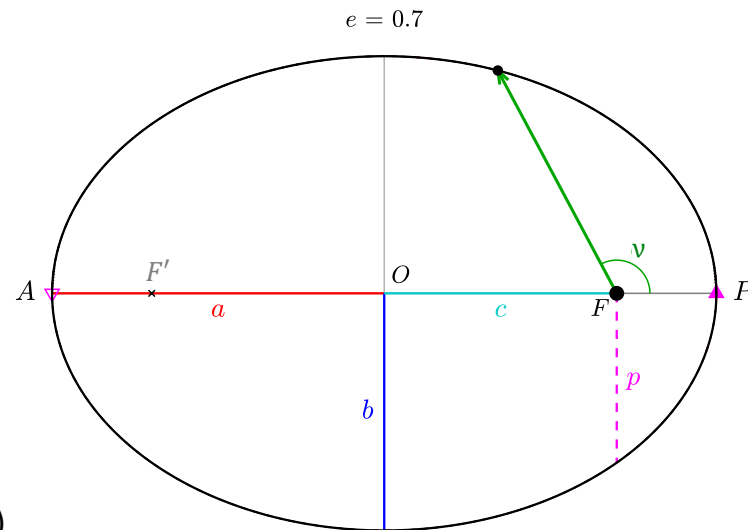
$P$  = Perihelion (nearest point)

$A$  = Aphelion (farthest point)

$r_p$  = Radius at perigee ( $\equiv a - c$ ),  $r_A$  = radius at the apogee ( $\equiv a + c$ )

$\vec{r}$  = Satellite position vector, measured from the center of the Earth

$\vec{v}$  = Satellite velocity vector, tangent to the orbital path



Note: To convert the distance measured from the surface of a planet to the distance measured from the center of gravity, the radius of the planet must be added.

$$a = c + R_{\oplus} + H_p$$

# *e* – eccentricity



- Quantity measuring the **degree of flattening of an ellipse** (also called *ellipticity*)
  - Defines the shape of the ellipse by comparing the ratio of the distance between the two foci and the length of the semi-major axis  $a$

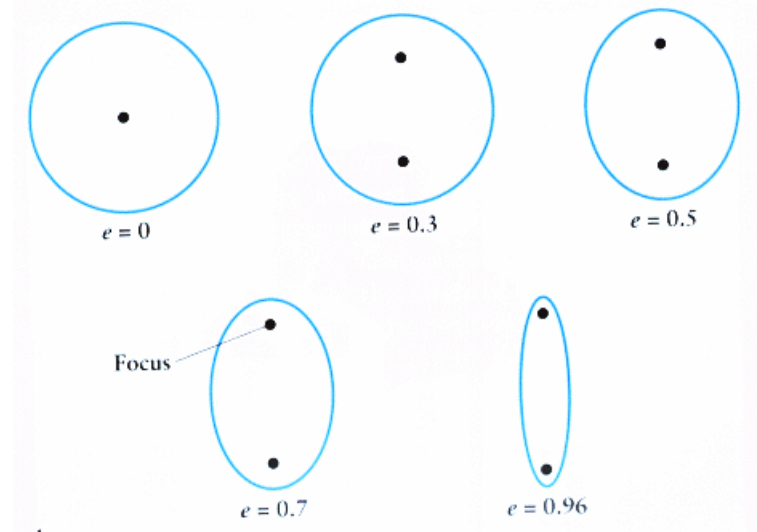
$$e = \frac{c}{a} = \sqrt{1 - \left(\frac{b}{a}\right)^2} = \frac{r_A - r_P}{r_A + r_P} \quad [\text{n. u.}]$$

$$e = \sqrt{1 + 2\varepsilon \left(\frac{h}{\mu}\right)^2}$$

$$e = \sqrt{1 + \left(\frac{h}{\mu}\right)^2 \left(v^2 - 2\frac{\mu}{r}\right)} \quad (\forall v \text{ \& } r)$$

→ [slider change](#) the shape of the orbit through  $e$

For ellipses:  $0 \leq e < 1$





# Ellipse geometry: basic relations



- If the ellipse is defined by its radii  $a$  and  $b$ :

$$c = \sqrt{a^2 - b^2}, \quad e = \frac{c}{a}, \quad p = \frac{b^2}{a},$$

- When the semi-major axis  $a$  and eccentricity  $e$  are known:

$$b = a\sqrt{1 - e^2}, \quad c = ae, \quad p = a(1 - e^2)$$

- Finally, in the bifocal definition of the ellipse where the lengths are known  $a$  and  $c$ :

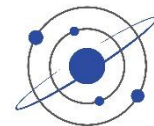
$$b = \sqrt{a^2 - c^2}, \quad e = \frac{c}{a}, \quad p = \frac{b^2}{a}$$

# Ellipse geometry: main parameters



|                               |                                                                                                                                     |                              |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Specific mechanical energy    | $\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} \quad \left[ \frac{\text{km}^2}{\text{s}^2} \right]$                 |                              |
| Specific angular momentum     | $h = \sqrt{a\mu(1-e^2)} = \sqrt{\mu p} = \sqrt{r_P \mu(1+e)} = r_P v_P = r_A v_A \quad \left[ \frac{\text{km}^2}{\text{s}} \right]$ |                              |
| Semi-parameter                | $p = a(1-e^2) = \frac{h^2}{\mu} = r_P(1+e) = r_A(1-e) \quad [\text{km}]$                                                            | « ellipse parameter $\ell$ » |
| Semi-major axis               | $a = -\frac{\mu}{2\varepsilon} = \frac{r_A + r_P}{2} = \frac{p}{1-e^2} \quad [\text{km}]$                                           | « ellipse dimension »        |
| Distance between the two foci | $c = ea = \sqrt{a^2 - b^2} = \frac{r_A - r_P}{2} \quad [\text{km}]$                                                                 | « linear eccentricity »      |
| Semi-minor axis               | $b = a\sqrt{1-e^2} = \sqrt{a^2 - c^2} = \frac{p}{\sqrt{1-e^2}} = \sqrt{r_A r_P} = \sqrt{p a} \quad [\text{km}]$                     |                              |
| Radius at perigee (min)       | $r_P = \frac{p}{1+e} = a(1-e) = a - c \quad [\text{km}]$                                                                            | } Apsides                    |
| Radius at the apogee (max)    | $r_A = \frac{p}{1-e} = a(1+e) = a + c \quad [\text{km}]$                                                                            |                              |
| Orbital period                | $P = 2\pi \sqrt{\frac{a^3}{\mu}} \quad [\text{s}]$                                                                                  |                              |

# Elliptical motion: basic relations



Orbital velocity:  $v(r) = \sqrt{\mu \left( \frac{2}{r} - \frac{1}{a} \right)} = \sqrt{2 \left( \frac{\mu}{r} + \varepsilon \right)}$  (*Vis-viva*)  $r = f(v)$

$v(\mathbf{v}) = \frac{\mu}{h} \sqrt{1 + 2 e \cos(\mathbf{v}) + e^2}$  → see [Other relationships based on Kepler anomalies](#)

Velocity at perigee ( $\mathbf{v} = 0$ ):  $v_P = \frac{\mu}{h} (1 + e) = \sqrt{\frac{\mu}{a} \left( \frac{1+e}{1-e} \right)} = \sqrt{\mu \left( \frac{2}{r_P} - \frac{1}{a} \right)}$

Velocity at apogee ( $\mathbf{v} = \pi$ ):  $v_A = \frac{\mu}{h} (1 - e) = \sqrt{\frac{\mu}{a} \left( \frac{1+e}{1-e} \right)} = \sqrt{\mu \left( \frac{2}{r_A} - \frac{1}{a} \right)}$

Radial velocity:  $v_r = \frac{\mu}{h} e \sin(\mathbf{v})$

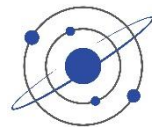
Normal velocity:  $v_n = \frac{\mu}{h} (1 + e \cos(\mathbf{v}))$  (perpendicular)  $\sqrt{\frac{\mu}{p}} \equiv \frac{\mu}{h}$



Orbital velocities



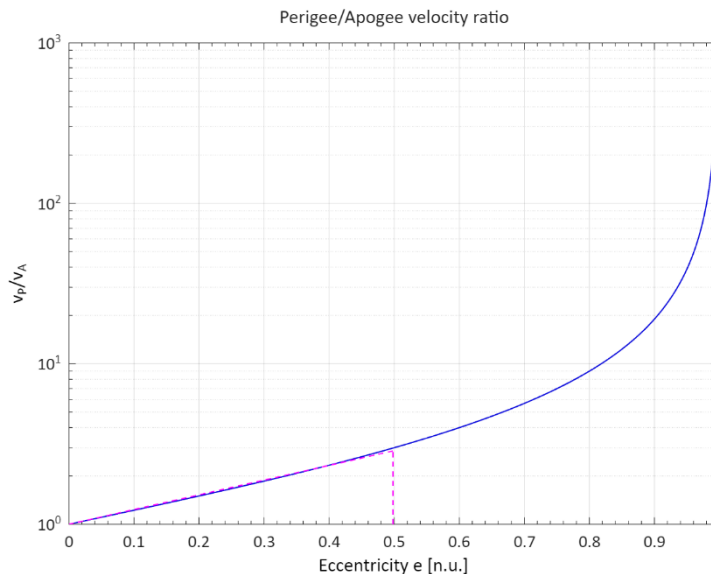
Other relationships based on Kepler anomalies

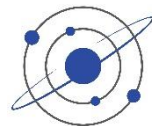


# Perigee/Apogee velocity ratio

- The ratio of theoretical maximum (**Perigee**) and minimum (**Apogee**) speeds depend only on the orbit eccentricity
- This ratio displays a rather linear behavior up to an eccentricity of  $e = 1/2$ , followed by a quite exponential growth from this point until right before  $e = 1$ ; from this point onwards the orbit will become open and cease to maintain a periodic movement

$$\frac{v_P}{v_A} = \frac{1+e}{1-e}$$





# “Average” distance – Mean orbit radius

- It is often said that the semi-major axis is the “average” distance between the primary focus of the ellipse and the orbiting body

→ This statement is not entirely accurate, as it **depends on what we define as the “mean” value**

1) Average distance over Eccentric Anomaly ( $E$ ) gives the semi-major axis  $a$

$$\langle r \rangle = a = (r_a + r_p)/2 \rightarrow \text{Arithmetic mean}$$

2) Average distance over Mean Anomaly ( $M$ ) gives the semi-minor axis  $b$

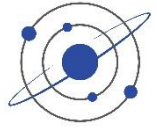
$$\langle r \rangle = b = \sqrt{r_a r_p} = a\sqrt{1 - e^2} \rightarrow \text{Geometrical mean}$$

3) Average distance *in time* over True Anomaly ( $\nu$ ) gives:

$$\langle r \rangle = a(1 + e^2/2)$$

Also:

- The radius of the circle with the same area is:  $\langle r \rangle = \sqrt{ab} = a\sqrt{1 - e^2}$
- The time-averaged value of the inverse of the radius is:  $\left\langle \frac{1}{r} \right\rangle = \frac{1}{a}$



# Ellipse radius equivalence for small eccentricities

% Ellipse radius

```
r = (1 - e.^2) ./ (1 + e .* cos(M));
```

% Sine approximation of the ellipse radius

```
r_ = e * sin(M - pi/2) + 1;
```

% Series expansion

```
r_ = b^2/a * (1 - e * cos(M) + (e * cos(M)).^2 - (e * cos(M)).^3 + 0);
```

```
n = 3;
```

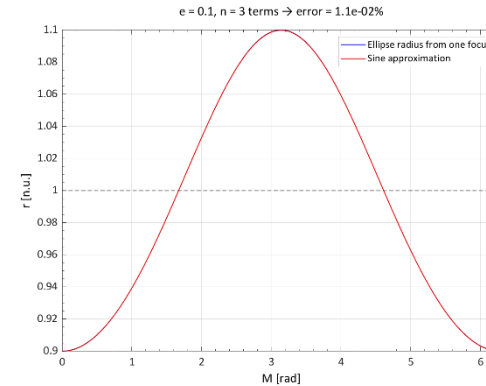
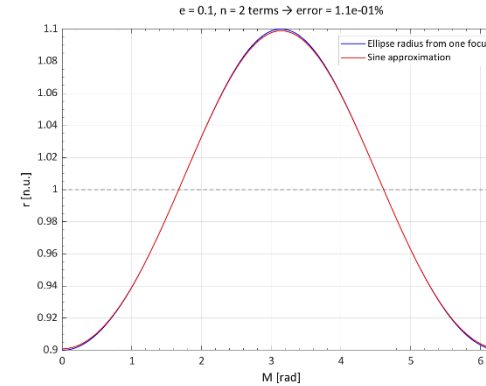
```
r_ = 0 * M;
```

```
for i = 0:n
```

```
 r_ = r_ + (-1)^i * (e * cos(M)).^i;
```

```
end
```

```
r_ = r_ * a * (1 - e^2);
```



# Summary of orbit characteristics



- Orbits are mainly classified by  $e$  or  $a$  which are the two independent parameters
- They can also be described by  $\varepsilon$  et  $v_p$  which are dependent parameters

$$\varepsilon = -\frac{\mu}{a} \quad v_p = \sqrt{\frac{\mu}{a} \left( \frac{1+e}{1-e} \right)}$$

| Conical section | Eccentricity | Semi-major axis | Energy                            | Velocity   | Earth     |
|-----------------|--------------|-----------------|-----------------------------------|------------|-----------|
| Ballistic       | $e < 0$      | $a < R$         | $\varepsilon < 0$                 | $v_p < vc$ |           |
| Circle          | $e = 0$      | $a = R$         | $\varepsilon_0 = \varepsilon < 0$ | $v_p = vc$ | 7.6 km/s* |
| Ellipse         | $0 < e < 1$  | $a > 0$         | $\varepsilon_0 < \varepsilon < 0$ | $v_p < ve$ |           |
| Parabola        | $e = 1$      | $a = \infty$    | $\varepsilon = 0$                 | $v_p = ve$ | 11.2 km/s |
| Hyperbola       | $e > 1$      | $a < 0$         | $\varepsilon > 0$                 | $v_p > ve$ |           |

$$\varepsilon_0 = -\frac{1}{2} \left( \frac{\mu}{h} \right)^2$$

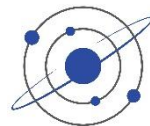
\*@ H = 430 km



Orbital energy



Orbit types



# Flight Path Angle $\phi$

The **flight path angle**  $\phi$  (or trajectory angle) is the angle between the orbiting body's velocity vector  $\vec{v}$  (equal to the vector tangent to the instantaneous orbit) **and the local horizontal**

$$h = r v \cos(\phi)$$

- $\phi$  = flight path angle between  $\vec{v}$  and local horizontal

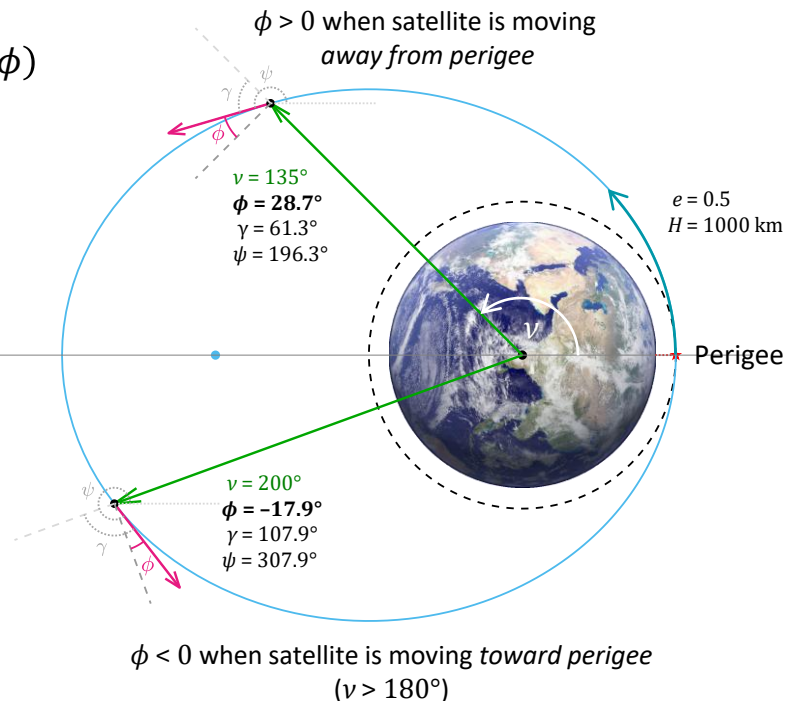
$$\phi = \text{atan}\left(\frac{e \sin(\nu)}{1 + e \cos(\nu)}\right) = \text{asin}\left(\frac{e \sin(E)}{\sqrt{1 - e^2 \cos(E)^2}}\right)$$

- $\gamma$  = zenith angle between  $\vec{v}$  and local vertical

$$\gamma = 90^\circ - \phi$$

- $\psi$  = angle between  $\vec{v}$  and the semi-major axis

$$\psi = \nu + 90^\circ - \phi + \omega$$







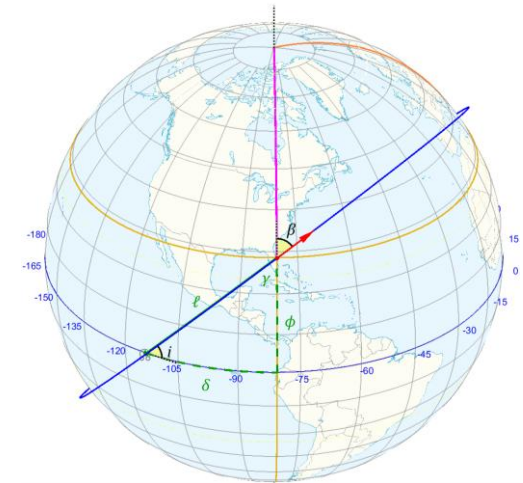
# Flight Path Azimuth angle $\beta$

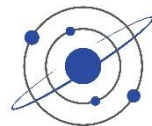
- The satellite **Azimuth angle  $\beta$**  is the angle between the satellite's velocity vector and the local north direction in inertial frame ECI → see [Launch Azimuth Angle](#)
  - ≠ Not with respect to Earth surface, as for see [Heading angle  \$\psi\$](#)  in ECEF
  - ≠ (unrelated to the [Sun Beta Angle](#))

$$\beta = \arcsin\left(\frac{\cos(i)}{\cos(\phi)}\right)$$

ECI

- Used to evaluate the *ground track velocity* (in ECEF, by definition of ground track)
  - see [Ground track speed](#)





# Heading angle $\psi$

- The **heading angle**  $\psi$  of a satellite is the angle between the satellite's velocity vector and the local north direction, as projected vertically onto the Earth's surface (the ground track)
- It gives the direction in which the satellite is moving over the Earth's surface at a specific point
- $\psi$  is the instantaneous azimuth of the satellite's ground track relative to true north at the subsatellite point
- Assuming a Keplerian orbit (no perturbations), the formula is:

$$\psi = \text{atan}\left(\frac{\dot{\lambda} \cos(\phi)}{\dot{\phi}}\right) \quad \text{ECEF}$$

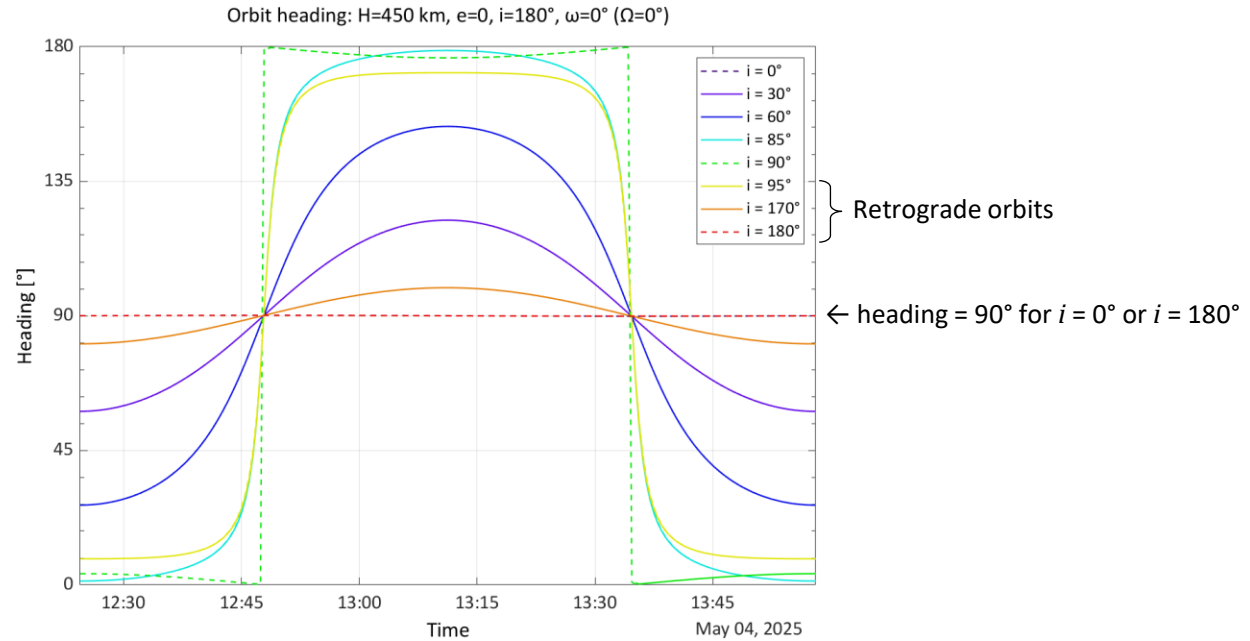
- Can be computed from the ECEF velocity vector:

```
% 1) Computes vector pointing to North: v_N = RotZ(λ) * RotY(-φ) * [0;0;1];
v_N = [cosd(Orbit.Lon) .* sind(-Orbit.Lat); sind(Orbit.Lon) .* sind(-Orbit.Lat); cosd(-Orbit.Lat)];

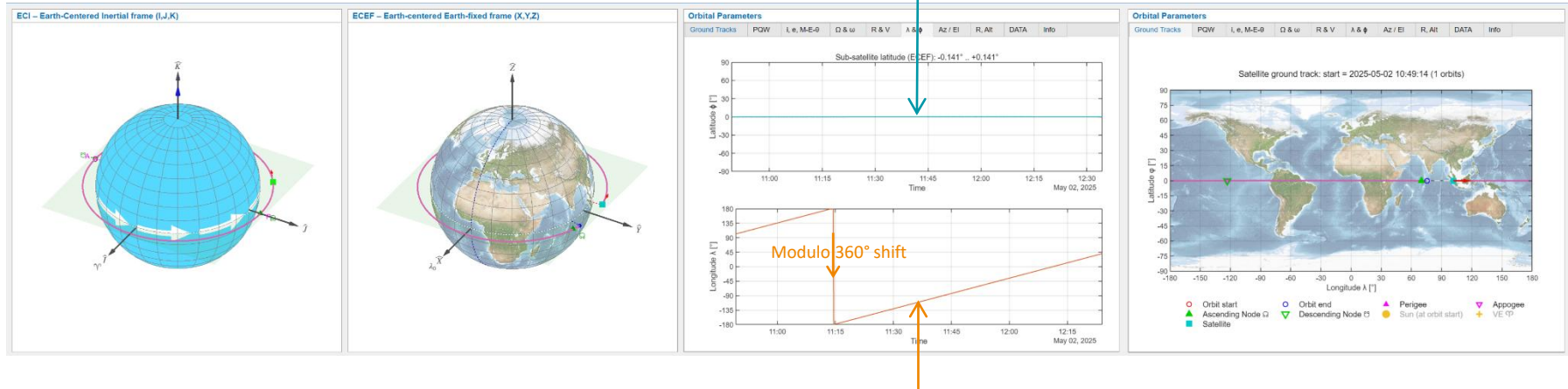
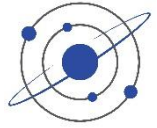
% 2) Computes angle between this vector and the ECEF velocity
Heading_ECEF = Angle(v_N, Orbit.ECEF.vel);
```



# Orbit Heading



$$i = 0, +180^\circ$$



Longitude  $\lambda$  linearly increases at a rate of  $360^\circ/P - \omega_\oplus$  [°/s],  
 slowed by Earth's rotation in prograde orbit (negative sign)  
 (use positive sign for  $i = +180^\circ$  where Earth's rotation is *added* to satellite speed)

# Mathematical study



```
A = 360 / Orbit.P_s; % == DEG(Orbit.n_)
B = Earth.omega_deg_s;
t = Orbit.time_s;
Lon = atan2d(sind(A * t) .* cosd(B * t) - cosd(A * t) .* sind(B * t), ...
 sind(A * t) .* sind(B * t) + cosd(A * t) .* cosd(B * t));
Lon = DEG(unwrap(RAD(Lon)));

Lon1 = (A - B) * t; % ← exact!
```

```
Taylor Lon=ArcTan.nb * - Wolfram Mathematica 13.0
File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

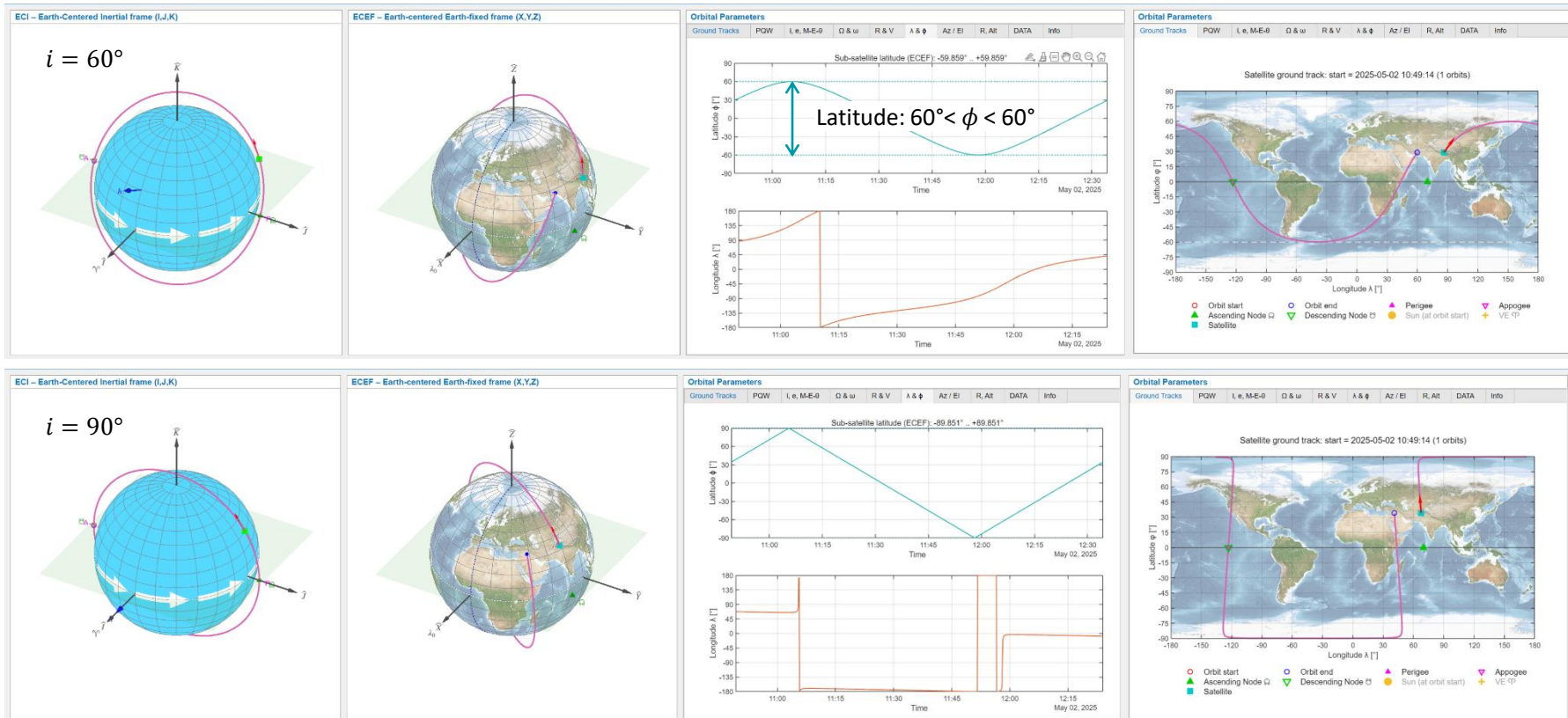
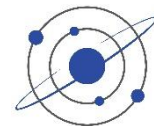
In[2]:= F[t] =
 ArcTan[(Sin[A t] Cos[B t] - Cos[A t] Sin[B t]) /
 (Sin[A t] Sin[B t] + Cos[A t] Cos[B t])]

Out[2]= ArcTan[Cos[B t] Sin[A t] - Cos[A t] Sin[B t] /
 Cos[A t] Cos[B t] + Sin[A t] Sin[B t]]

In[3]:= Series[F[t], {t, 0, 10}]

Out[3]= (A - B) t + O[t]^11
```

# Examples: $i = 60^\circ$ and $i = 90^\circ$ (with $\Omega = 90^\circ$ )





# Exercise

→ Find the  $a$ ,  $e$ ,  $P$ ,  $v_P$ , and  $v_A$  of the [Explorer VII](#) satellite on 2025-01-24 from its apogee and perigee distances

```
%--- Extract input data from real satellite
Orbit.SatName = 'Explorer 7';
Orbit.Epoch.Time = datenum(2025,01,24, 12,00,00);
Orbit = GetSatInfo(Orbit); % ← calls OrbitParameters() to compute
 associated parameters

Orbit.Launch_year % 1959 (info)
H_P = Orbit.r_P - Earth.r1_km % 453.4 km
H_A = Orbit.r_A - Earth.r1_km % 591.7 km

%--- Calculations: (done in
a = (Orbit.r_A + Orbit.r_P) / 2 % → 6900.7 km
e = (Orbit.r_A - Orbit.r_P) / (2 * a) % → 0.0100
P_s = 2*pi * sqrt(Orbit.a^3 / Orbit.mu) % = 5704.9 s → 1.585 hours
v_P = sqrt(Earth.mu / a * (1 + e) / (1 - e)) % → 7.677 km/s
v_A = sqrt(Earth.mu / a * (1 - e) / (1 + e)) % → 7.524 km/s
100 * (v_P/v_A - 1) % → 2% << 1 due to the small eccentricity
```



[Explorer VII](#) is a NASA satellite launched on 13 October 1959 from Cape Canaveral Air Force Station (CCAFS)

→ Compute the mass of the Earth from this data

```
M = (2*pi)^2 * (a*1000)^3 / (G * P_s^2) % → 5.972e+24 kg ≈ Earth.mass_kg
```

# Supplement: Ellipse circumference



If the area of an ellipse is  $A = \pi a b$ , what about the **circumference**?

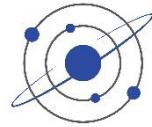
- There is **no simple expression** of the exact end result: the calculation is a complicated problem
- The solution is given as a function of the **elliptic integral of the second kind ( $E$ )** or of Gauss' **hypergeometric function ( ${}_2F_1$ )**
  - Cannot be solved in terms of elementary functions or expressed in combination of elementary integrals

$$C = 4a \int_0^{\pi/2} \sqrt{1 - e^2 \sin^2(t)} dt = 4a E(e) \quad (\text{here } E \neq \text{Eccentric Anomaly})$$

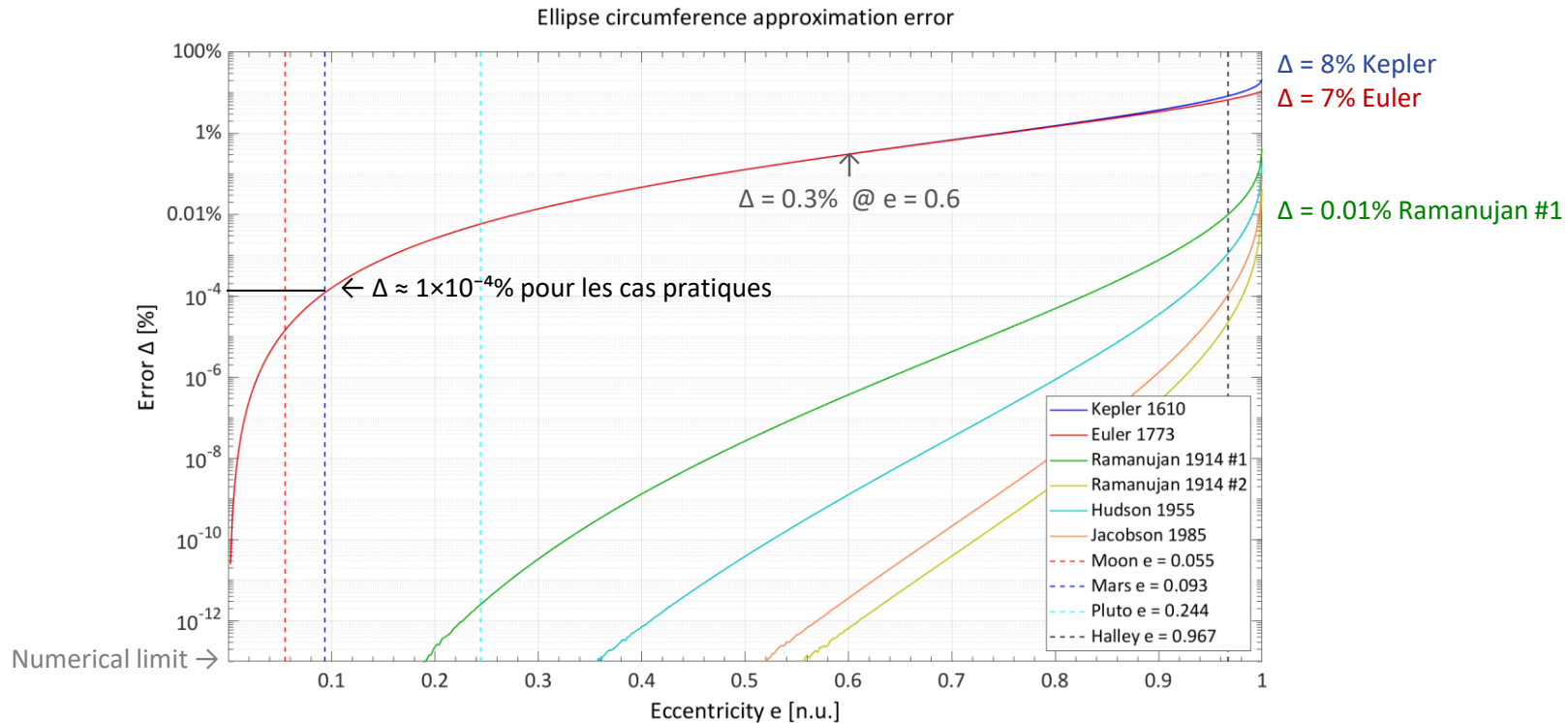
Approximations:

- Kepler 1610:  $C \approx \pi (a + b)$ , and Euler in 1773:  $C \approx \pi \sqrt{2(a^2 + b^2)}$ 
  - Giving errors of  $\Delta = 2 \times 10^{-4}\%$  for  $e=0.1$  et  $\Delta=0.3\%$  pour  $e=0.6$
- Ramanujan 1914:  $C \approx \pi (a + b) \left( 1 + \frac{3h}{10 + \sqrt{4 - 3h}} \right)$ , with  $h = \left( \frac{a-b}{a+b} \right)^2$ , error  $O(h^5)$ 
  - Giving errors of  $e = 0.1$ :  $\Delta = 4 \times 10^{-14}\%$   
 $e = 0.6$ :  $\Delta = 7 \times 10^{-13} \leftarrow \text{fantastic!}$





# Circumference approximations



# Parabolic and Hyperbolic trajectories



# Parabolic orbits



<https://oer.pressbooks.pub/lynnanegeorge/chapter/chapter-2-copy/>

Parabolic orbits (Figure 10) are rarely found in use, but they are important and interesting because they are the borderline case between the elliptical (closed) orbit and the hyperbolic (open) orbit.

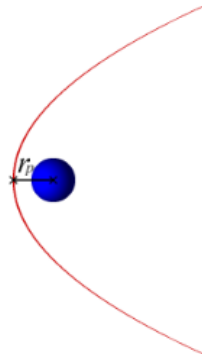


Figure 10: Parabolic Orbit Visualization  
(Source: Brandir, 2006).

The semi-major axis of a parabolic orbit is infinitely large, so an object traveling in a parabolic orbit is on a one-way trip to infinity and will never retrace the same path again. A unique property of a parabolic orbit is that the two arms of the parabola become more and more nearly parallel as they are extended further and further to the right of the focus.

Also, since the eccentricity of a parabolic orbit is 1, the radius to perigee equation used for elliptical orbits,  $R_p = a(1 - e)$ , cannot be used so the more generic form will be used:

$$R_p = \frac{p}{1 + e \cos \omega} \quad (15)$$

# Hyperbolic orbits



- <https://oer.pressbooks.pub/lynnanegeorge/chapter/chapter-2-copy/>

Hyperbolic orbits, as seen in Figure 12, are very useful, especially for interplanetary travel. An interplanetary probe must have some speed left over after it escapes the earth's gravitational field.

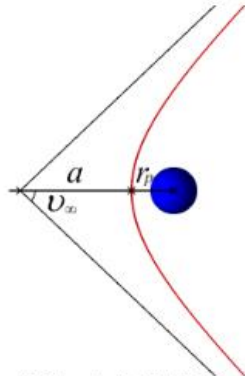


Figure 12: Hyperbolic Orbit Visualization  
(Source: Brandir, 2006).

The semi-major axis of a hyperbolic orbit is negative. This seems like a strange phenomenon, but it makes sense if you consider the specific mechanical energy of a hyperbolic orbit:

$$\epsilon = -\frac{\mu}{2a} \quad (19)$$

If the semi-major axis,  $a$ , is negative, this will make the specific mechanical energy positive. Thus in the energy equation,

$$\epsilon = \frac{V^2}{2} - \frac{\mu}{R} \quad (20)$$

# Reaching orbit

- Newton's Cannon
- Free fall
- Microgravity vs “Zero-g”
- Earth gravitational acceleration:  $g$  vs  $g_0$

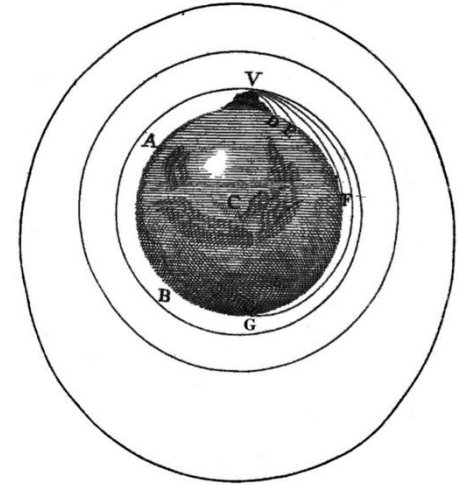




# How does a satellite get into orbit?

## Newton's Cannon thought experiment

- First, Imagine you are **at the top of a mountain and start firing a cannon from it**
- We can think of the cannonball as a projectile, an object that, when it is set in motion, continues in its path by its own inertia and is influenced only by the downward force of gravity. The horizontal motion of the projectile is the result of the tendency of any object in motion to remain in motion at a constant velocity. The vertical motion of the projective, or cannonball, is influenced by the gravitational acceleration,  $g$ , pulling downward
- The horizontal and vertical motions are independent, hence, the faster you fire the cannonball, the farther it will go
- Since the Earth is approximately spherically shaped (not entirely true), its surface drops  $\approx$  **5 m for every 8 km** the cannonball travels horizontally (or **8 inches/mile**)
- So, if we fire the cannonball faster and faster (assuming no air resistance), its path would eventually match the rate of curvature of the Earth



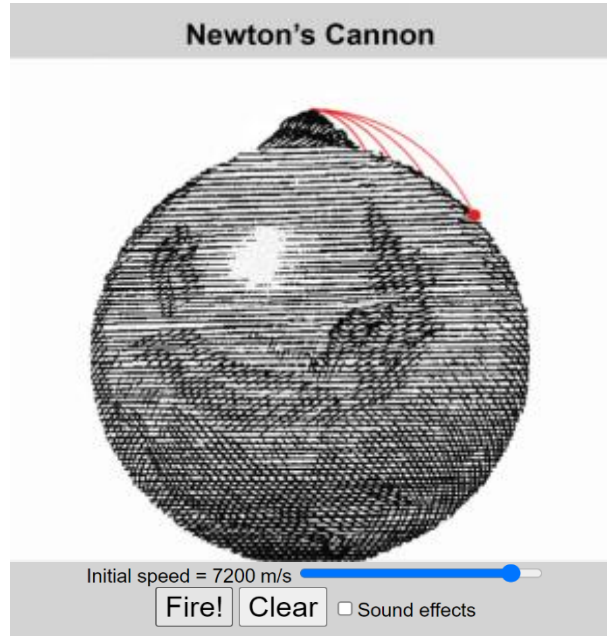
% "8 inches per mile" rule

```
R = 6378.137; % [km] Earth radius
dx = 1.6093; % [km] ≡ 1 mile
dy = sqrt(R^2 + dx^2) - R; % [km]
inches_per_m = 39.3701; % ["/m]
dy * 1000 * inches_per_m % → 7.99"
```

# Newton's Cannon



Try it yourself: <https://physics.weber.edu/schroeder/software/NewtonsCannon.html>



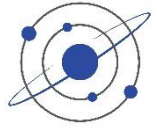
The fastest the cannon is fired horizontally, the farther before hitting the Earth, until orbital speed is reached:

e.g.  $7.9 \text{ km/s}$  for  $H = 8 \text{ km}$   
(Everest height, assuming no atmosphere),  
or  $7.7 \text{ km/s}$  for  $H = 400 \text{ km}$  (ISS altitude)

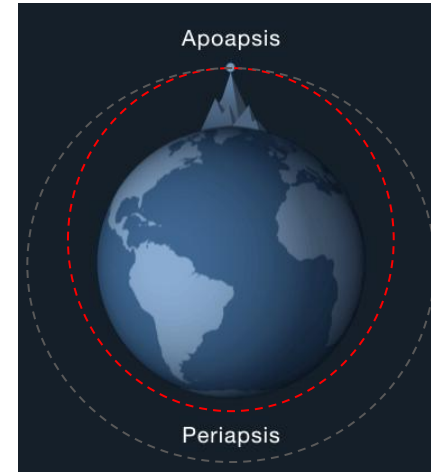
and the projectile remains in orbit

Faster launches lead to elliptical orbits, up to a point where escape velocity is attained and the projectile never returns to Earth

# Orbital speed



- Once fired, a cannonball falls to Earth
- The greater the speed, the farther it will travel before landing
- If fired with the proper speed, the cannonball would achieve a state of continuous free-fall around Earth, which we call *orbit*



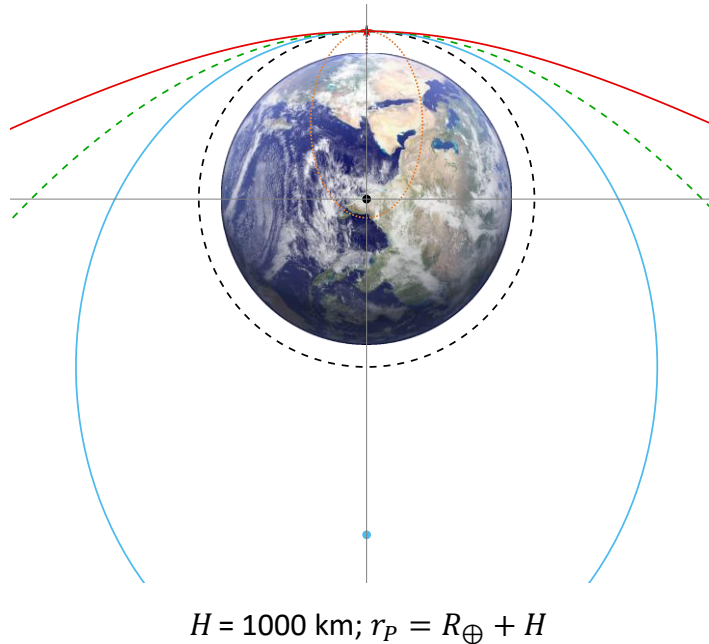
← when  $v_0 < v_c$ ,  
launch point  
corresponds to  
orbit's apogee



# Newton's Cannon

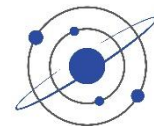


- A cannon located at the top of a very high mountain fires a projectile **horizontally**



- 0) Projectiles launched at low velocities (suborbital) fall back to the Earth in **quasi-parabolic ballistic trajectories** (ellipses internal to the Earth)
- 1) From a critical speed ( $v_c$ ), they enter **circular orbits**, corresponding to the simplest form avoiding falling back to Earth
- 2) As the launch speed increases, the orbit becomes more and more eccentric → **ellipses**
- 3) As the horizontal velocity increases, the orbit becomes **parabolic**, then **hyperbolic**, where the projectile “escapes” from Earth gravity and leave it without ever returning

# Newton's Cannon (continued)



- From  $p = h^2/\mu$ , we can conclude that the parameter of the conic  $p$  (or  $a = p/(1 - e^2)$ ) depends solely on the angular momentum  $h$ , and thus an increment in the angular momentum must have the consequence of increasing the size of the orbit
- From  $\vec{H} = \vec{R} \times \vec{V}$ , and  $h = r_p v_p$ , we can conclude that in a two-body motion, **the size of the orbit ( $a$ ) is determined by the speed of the satellite** ( $a = \mu/v_c^2$ )
- From  $\varepsilon = -\mu/(2a)$ , valid for all conics, we can conclude that semi-major axis  $a$  of an orbit depends only on the energy of the orbiting body, which in turn depends on the velocity and position of the body at a given point of the orbit

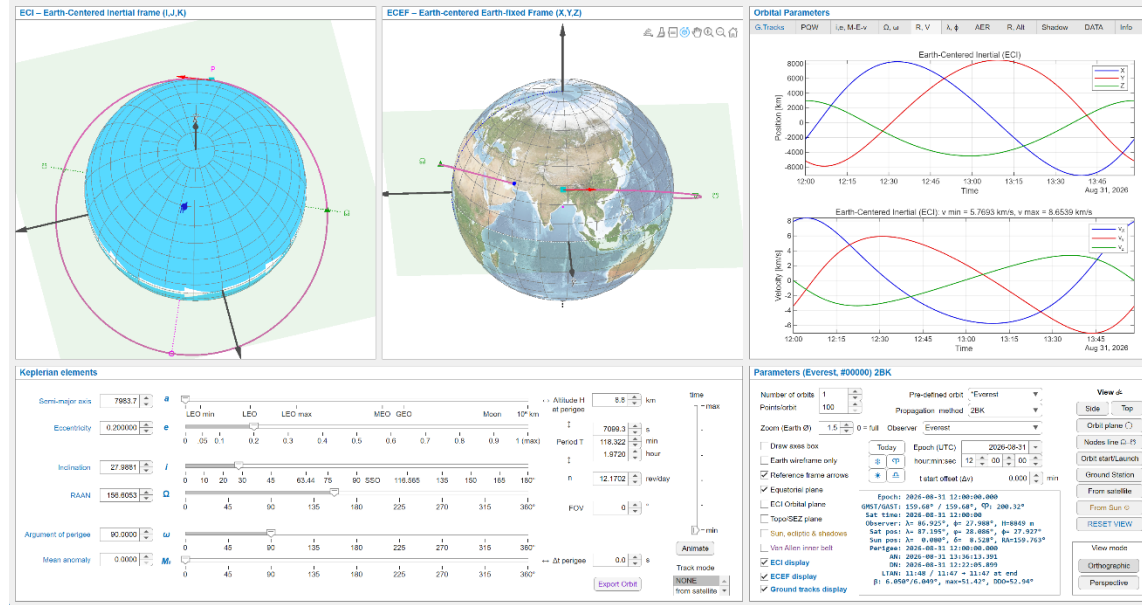


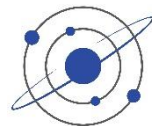
Vis-viva equation

# Re-create in Orbits101



- Neglecting atmosphere, re-create the Newton's cannon experiment: **Launch from Mount Everest**
- Adjust Altitude at Perigee, Eccentricity
- Evaluate velocity at launch and compare with theory





# Newton's cannon on Mount Everest

```
case '*Everest' % Newton's cannonball experiment from top of Mount Everest
```

```
Orbit.SatName = 'Everest';
```

```
Orbit.Observer = DefineObserverLocation('Everest'); % $\lambda \equiv 0^\circ$, $\varphi = 51.4934^\circ$
```

```
Orbit.Epoch.Time = datenum(2026,08,31, 12,00,00);
```

```
Orbit.Epoch.GMST = ComputeGMST(Orbit.Epoch.Time); % $[\circ] = 159.6803^\circ$
```

```
Orbit.e = 0.2; % Arbitrary
```

```
H_P = Orbit.Observer.Alt_m / 1000; % 8.84886 km @ Perigee
```

```
Orbit.a = (Earth.r1_km + H_P) / (1 - Orbit.e); % $\neq R_\oplus + H$!
```

```
Orbit.i = Orbit.Observer.Lat_geod; % to adjust orbit inclination equal to mountain geodetic latitude
```

```
Orbit.w = 90; % $[\circ]$ to adjust orbit perigee at launch point
```

```
Orbit.RAAN = Orbit.Epoch.GMST + Orbit.Observer.Lon - 90; % $[\circ]$ to adjust ECEF position on top of mountain at epoch
```

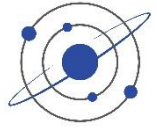
# Free fall



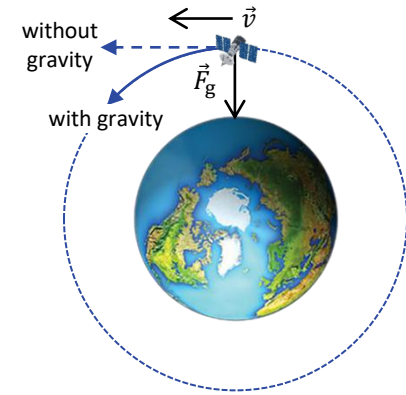
- If you ride along in an orbiting spacecraft, you feel as if you are falling, *as in fact you are*
- The condition is called **free fall**, where an object is *falling under the influence of gravity, free from any other forces*
  - Astronauts float around in their spacecraft as they experience no contact forces with Earth's surface
  - They are falling at the same rate as the spacecraft, which appears to be floating there around them
- It's what skydivers feel before reaching terminal velocity and before their parachutes open
- If you drop an apple on Earth, it falls at **1 g**
- If an astronaut on the space station drops an apple, it falls too with the same acceleration; it just doesn't look like it's falling
  - *That's because they're all falling together: the apple, the astronaut and the station*
  - But they're not falling *towards* Earth, they're **falling around it**



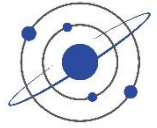
# Immense speed forward



- As soon as you **drop something** (like an apple) it is in a state of free fall
- The same is true if you **throw something**; it immediately starts falling towards Earth
- As seen with Newton's Cannon, if the horizontal speed is large enough, at some point *the continuous fall keeps missing the Earth*, and orbital trajectory is attained
  - It is the **satellite's immense forward speed** that appears to make Earth's surface curve away as it falls toward it
- While objects inside the ISS appear to be floating and motionless, they are actually traveling at the same orbital speed as their spacecraft: **27 700 km/h!** (7.7 km/s)
- Notice that an **orbiting satellite has not escaped Earth's gravity**, which is very much present – it is giving the mass the centripetal acceleration it needs to stay in circular orbit
  - Without the gravitational force, the satellite would move in a straight line
- It just happens to be **balanced out by the speed** that the rocket provided when it placed the spacecraft in orbit

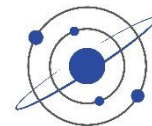


# Microgravity



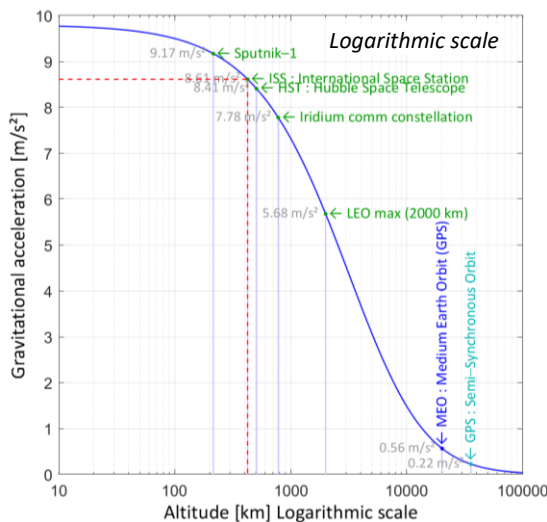
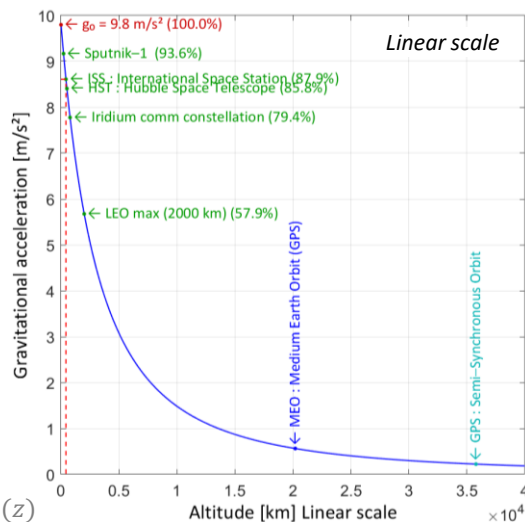
- Because they're all falling at the same rate, objects inside of the station appear to float in a state we call “weightlessness”, or more accurately **microgravity**
  - **Weightlessness** is a condition in which the effects of gravity are *not felt*
    - Your weight varies depending on whether you are on Earth, the Moon; it becomes zero when you are in orbit
    - In free fall you don't *feel* the force of gravity *even though gravity is present*
    - Your mass remains the same, but it would register “0” on a scale
  - “Zero-gravity” (0 g) is **technically incorrect and misleading**, as **the absence of gravity can never be achieved**
    - As we have seen, even if it lessens further away from Earth, it is never zero
    - Even in intergalactic space there's still the gravity of the distant galaxies
- So terms like “microgravity” and “weightless” and have to be taken with a grain of salt... gravity is still dominant, but some of its familiar effects are not apparent on orbit





# Gravity always present

- Gravity is a force that governs motion throughout the universe
  - It holds us to the ground, and it keeps the Moon in orbit around Earth and Earth in orbit around the Sun
- Many people mistakenly think that gravity does not exist in space
  - Gravity lessens further away from Earth, decreasing in proportion to the square of the distance
  - It is just weaker in LEO orbit, simply because we're farther from Earth's center, *but it's mostly there*



$$g(z) = g_0 \left( \frac{R_{\oplus}}{R_{\oplus} + z} \right)^2$$

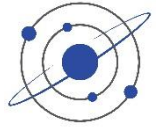
Typical orbital altitudes for human spaceflight vary between 200 – 580 km above Earth's surface

The gravitational field is still quite strong in these regions, since this is only about 1.8% the distance to the Moon

Earth's gravitational field at ~400 km above the surface (ISS altitude) is **88.8%** of its strength at the surface

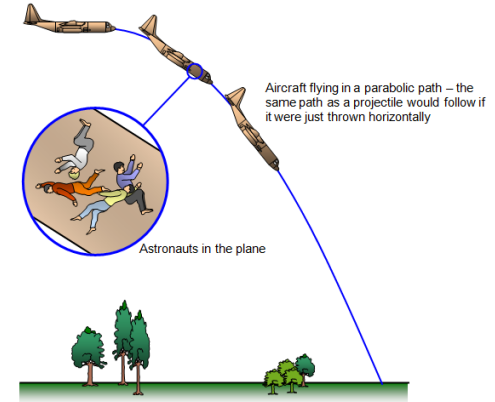


# Creating microgravity



## NASA parabolic flights

- One way to create microgravity environments is with **aircraft flying in parabolic arcs** for tests and simulations that last 20–25 seconds ( $\Delta H = t^2 g / 2 \rightarrow 2\text{--}3\text{ km}$  altitude drop)
  - NASA's operates a C-9 Flight Research aircraft also known as the “Vomit Comet”
  - The KC-135 aircraft was used to shoot weightless scenes in the movie *Apollo 13*



## NASA drop towers

- The Zero Gravity Research Facility is a large shaft some 150 m deep that allows test packages to free fall in vacuum for just over 5 s ( $\Delta t = \sqrt{2 H / g}$ )

## Rollercoasters

- You may have experienced weightlessness yourself without realizing it. Many amusement park rides create brief periods of feeling **free fall**
- Most **roller coasters** have a set of **parabolic (rolling) hills** that also create brief periods of weightlessness

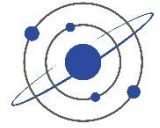


# Gravity and Gravitation



- Physical geodesy uses distinguishing definitions for concepts such as **gravitation** and **gravity**  
Those terms are closely related terms that are often used interchangeably, but they have subtle distinctions:
- Gravitation** refers to the fundamental force of attraction between any two objects with mass. It's the universal phenomenon described by **Newton's law of universal gravitation** and Einstein's general relativity  
Gravitation exists everywhere in the universe between all massive objects, regardless of their size
  - Gravitation emphasizes the universal nature of the attractive force  
e.g.: **the Moon's orbit around Earth is due to gravitation** (the universal attractive force) → **Astrodynamics** (Earth.g)
- Gravity** typically refers to the specific gravitational force experienced by objects near a massive body, especially planets. When we talk about "Earth's gravity" or objects "falling due to gravity," we're referring to the *local gravitational field effect* that we experience
  - Gravity emphasizes the local effects we observe, like objects falling toward Earth  
e.g.: **an apple falling from a tree is due to gravity** (the local gravitational field effect) (Earth.g0)
- In Einstein's general relativity, both concepts are understood as the curvature of spacetime caused by mass and energy, but the terminology distinction between the universal phenomenon (gravitation) and local effects (gravity) can still be useful for clarity

See: [https://en.wikipedia.org/wiki/Standard\\_gravity](https://en.wikipedia.org/wiki/Standard_gravity)  
[https://en.wikipedia.org/wiki/Gravity\\_of\\_Earth](https://en.wikipedia.org/wiki/Gravity_of_Earth)



# $g_0$ Earth gravitational acceleration

The value of gravitational acceleration **changes with latitude, altitude, and even local geology**.

That's because Earth isn't a perfect sphere – it's an oblate spheroid – and its mass isn't uniformly distributed.

Earth.g0 = **9.80665** [m/s<sup>2</sup>] → **Standard gravity** used for general physics, atmospheric models, and engineering.

No need for regional variations. *Conférence Générale des Poids et Mesures, CGPM 1901*

Earth.g = **9.79764** [m/s<sup>2</sup>] → Mean value of **Theoretical gravity** at the surface of the Earth; *for Orbital Mechanics*  $a = \mu/r^2$

For detailed simulations involving launch or re-entry trajectories; accounting for geodetic variations

Because the Earth isn't spherical, we may approximate the force of gravity by recognizing differences based on an observer's location,  $\phi_{gd}$ . The adopted value of theoretical gravity,  $g_{th}$ , at the Earth's surface is a function of the reference ellipsoid and the latitude of the observer. From NIMA (2000:4-1), the closed-gravity formula is:

$$g \equiv G \frac{M_{\oplus}}{R_{\oplus}^2} = \left( \frac{\mu}{r^2} \right)$$

$$g_{th} = g_{eq} \frac{1 + k_g \sin(\phi_{gd})^2}{\sqrt{1 - e_{\oplus}^2 \sin(\phi_{gd})^2}}$$

$$\text{with } k_g = \frac{b_{\oplus} g_{pole}}{a_{\oplus} g_{eq}} - 1$$

$g_{eq} = \text{Earth.g}_{eq}; \quad \% 9.7803253359 \text{ m/s}^2$   
gravity is weaker at equator due to centrifugal effects from Earth's rotation

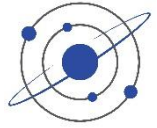
$g_{pole} = \text{Earth.g}_{pole}; \quad \% 9.8321849378 \text{ m/s}^2$   
Near the poles, gravity is stronger because the closer distance to Earth's center.

$a_{\oplus}$  and  $b_{\oplus} = \text{Earth.r1} \ \& \ \text{r2}$

$e_{\oplus} = \text{Earth.ecc}; \quad \% \text{ spheroid eccentricity}$

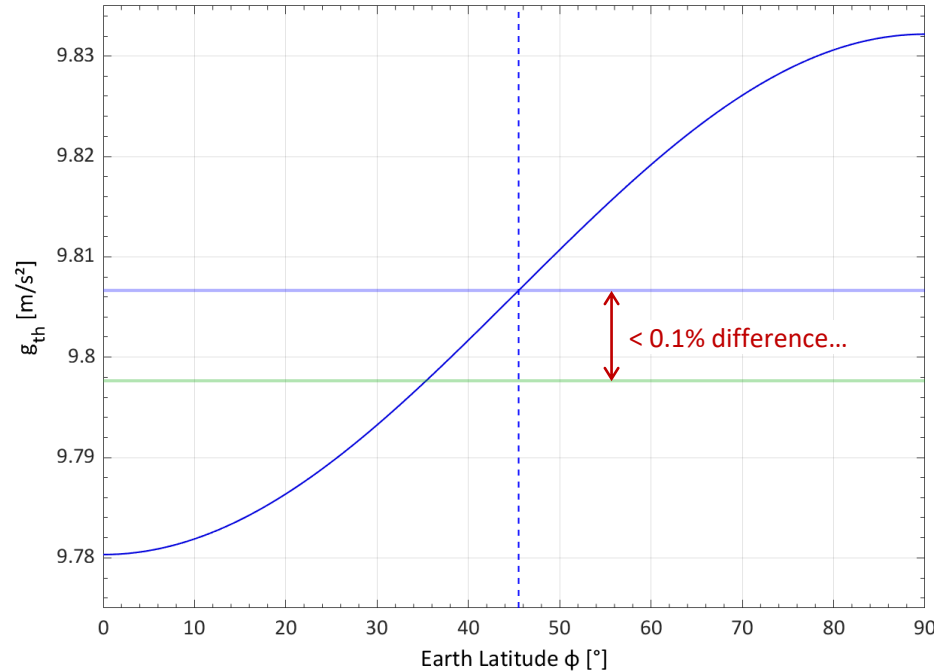
$g_{th} \approx \mathbf{9.79829} = \text{Earth.mu} / \text{Earth.r1\_km}^2 * 1000 \quad \rightarrow \text{Earth.g}$   
 $= G * \text{Earth.mass\_kg} / (\text{Earth.r1\_km} * 1000)^2 \quad \text{because Earth's mass is defined from } \mu$

$g_{th} = \mathbf{9.80665} \ @ \ \phi_{gd} = \mathbf{45.5^\circ} \ \text{kind of average over latitudes} \quad \rightarrow \text{Earth.g0} \equiv \text{EarthGravity}(0,45.5)$



# Theoretical gravity at Earth's surface

Theoretical gravity at Earth's surface as a function the latitude of the observer

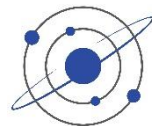


← 9.832 m/s<sup>2</sup>  $EarthGravity(0,90)$

← 9.80665 m/s<sup>2</sup>  $EarthGravity(0,45.5)$   
 $Earth.g_0$

← 9.79764 m/s<sup>2</sup>  $g = \mu/R_{\oplus}^2$   
 $Earth.g$

← 9.780 m/s<sup>2</sup>  $EarthGravity(0,0)$



# Notes on $g_0$

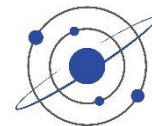
- The Standard Gravity  $g_0$  9.80665 m/s<sup>2</sup> is an **arbitrarily defined** unit that derives from gravitation;  
→ it is not defined as a property of the Earth;  
Unfortunately, some science and engineering texts inattentively list standard gravity under “properties of the Earth” or “physical constants”
- The Standard Gravity shall not be used as the mean value over the ellipsoid
- For general calculations, the mean value of theoretical gravity at the surface of the Earth shall be used [Vallado 2013, from NIMA (2000:4-2 to 4-8)]:

$$\text{Earth.g} = 9.797643222; \quad \% \text{ [m/s}^2\text{]}$$

- For real mission planning or high-fidelity simulation, engineers often use Earth gravity models like:
  - EGM96 / EGM2008 (Earth Gravitational Models): Includes thousands of spherical harmonics to model gravity variation
  - **WGS84 (World Geodetic System)**: Often used for GPS and aerospace applications
- These models help calculate perturbations like:
  - $J_2$  effect (Earth’s oblateness) → causes orbital precession
  - Atmospheric drag → important for reentry and low orbit decay
  - Third-body effects → e.g. Moon’s gravity impacting satellites

## ✓ Simple rule of thumb

- 9.80665 → *definition of “1 g”*
- 9.79–9.83 → *real gravity of Earth*



# Gravity as function of altitude

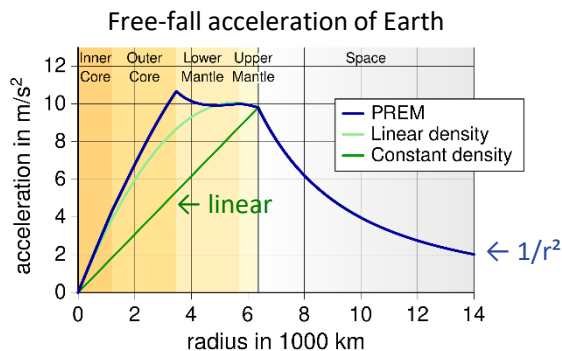
- The gravitational acceleration decreases as altitude increases, since the larger distance from Earth's center.

$$g(z) = g_0 \left( \frac{R_{\oplus}}{R_{\oplus} + z} \right)^2$$

```
R_gd = GeocentricRadius(phi_gd); % [km] Approx: R_gd = Earth.radius_km; % [km] "Mean Radius"
g = g_th * (R_gd ./ (R_gd + H_km)).^2; % [m/s^2] effect of altitude
```

- The formula treats the Earth as a perfect sphere with a radially symmetric distribution of mass

See [https://en.wikipedia.org/wiki/Gravity\\_of\\_Earth](https://en.wikipedia.org/wiki/Gravity_of_Earth) for more details



Newton's Shell Theorem



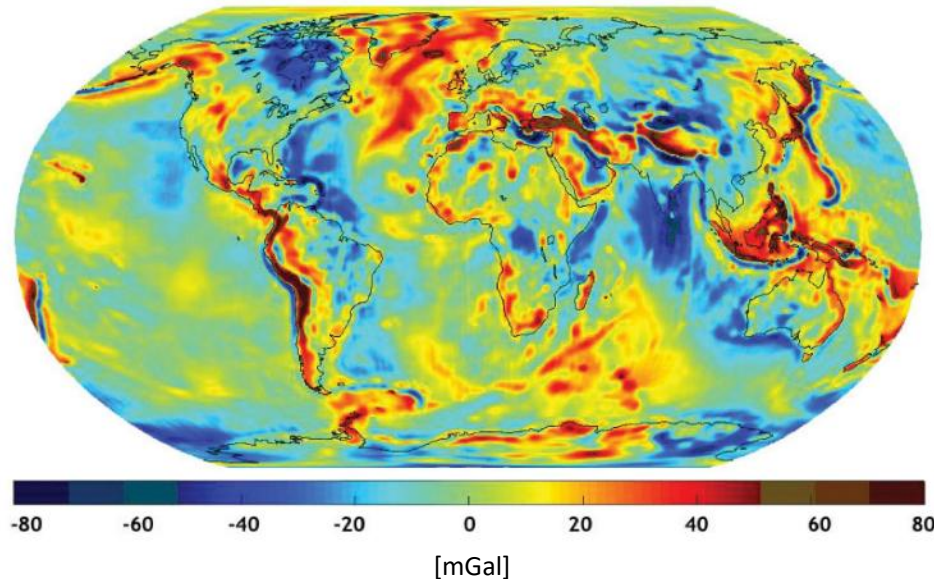
Gravity in function of altitude

# Gravity anomalies



- Gravity anomalies from ten years (2003–2013) of GRACE data (GGM05S)

<https://www2.csr.utexas.edu/grace/gravity/>



A milligal is a convenient unit for describing variations in gravity over the surface of the Earth.

1 milligal (or mGal) =  $0.00001 \text{ m/s}^2$ , which can be compared to the total gravity on the Earth's surface of  $\approx 9.8 \text{ m/s}^2$ .

Thus, a milligal is about 1 millionth of the standard acceleration on the Earth's surface.

## Earth extremes

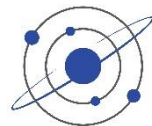
Standard models predict:

$9.7803 \text{ m/s}^2$  at equator

$9.8322 \text{ m/s}^2$  at the poles

Lowest:  **$9.7639 \text{ m/s}^2$**  Mount Nevado Huascarán, Peru

Highest:  **$9.8337 \text{ m/s}^2$**  Surface of the Arctic Ocean



# Earth ellipsoid properties

- The ellipsoid WGS-84, widely used for mapping and satellite navigation

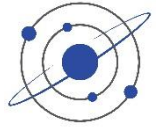
```
E = wgs84Ellipsoid('kilometer'); % << MATLAB Mapping Toolbox

 Code: 7030
 Name: 'WGS 84'
 LengthUnit: 'meter'
 SemimajorAxis: 6378.137
 SemiminorAxis: 6356.75231424518
 InverseFlattening: 298.257223563
 Eccentricity: 0.0818191908426215
 Flattening: 0.00335281066474748
 ThirdFlattening: 0.0016792203863837 (a-b)/(a+b)
 MeanRadius: 6371.00877141506 SurfaceArea: 510065621.724089 ≈ 4*pi * E.MeanRadius^2
 Volume: 1083207319801.41 ≈ 4*pi/3 * E.MeanRadius^3

Earth.r1_km = 6378.137; % [km] Earth equatorial radius ; Radius (at sea level) at equator
% = GeocentricRadius(0)
Earth.r2_km = 6356.75231424518; % [km] Earth polar radius (at sea level)
% = GeocentricRadius(90)
Earth.radius_km = 6371.00877141506; % [km] Earth average radius
% = (2 * Earth.r1_km + Earth.r2_km) / 3 as defined in the 1984 World Geodetic System revision
% ≈ (Earth.r1_km^2 * Earth.r2_km)^(1/3) Wertz 2001 9.1.5 Earth Oblateness
%
% IMP: r1 is used by most of all astrodynamics calculations
%
Earth.circ_km = 2*pi * Earth.r1_km; % [km] 40075.0
```



# “Why Gravity Is a Lie, explained in Zero G”



“GRAVITY IS FAKE\*” → *typical clickbait title !*

Cleo Abram, HUGE\* if true: <https://youtu.be/5zJbE7J3X8I?si=w3n1OnedAliBZLW0> [22:24]

→ Gravity do exist...



# Orbit propagation using Newton's methods

- Orbital State Vectors
- Numerical solution

→ Transferred to [SOL Part III – Orbit Propagators](#)



# Summary: Universal law of gravitation

---



- It is this gravitational force that continuously accelerates an orbiting body (a satellite) towards the elliptical focus of its orbital trajectory (e.g., the Earth) and prevents it from continuing in a straight line
- Using this law and his equations of motion, Newton was able to show that two point masses attracting each other would each follow a perfectly elliptical orbit, as Kepler had previously stated
  - Conical solution for the case of closed orbit (the circular orbit is a special case of an elliptical orbit)
  - If  $m_1$  is much larger than  $m_2$ , the largest mass will appear to be stationary at the focus of the elliptical orbit of the lightest mass
- This model can be applied approximately to the solar system
  - The mass of the Sun being much greater than that of the planets, the force acting on each of them is mainly due to the latter; The gravity of the planets on each other can be overlooked in 1<sup>st</sup> approximation

The combination of Kepler's and Newton's laws forms the basis  
of the modern understanding of orbits and astrodynamics

# Summary

---



## A: Universal gravitation law

- Law in  $1/r^2$

$$\vec{F}_g = -\frac{GMm}{r^3}\vec{r}$$

## B: Equation of motion

- $\mu = G m_{\oplus}$

$$\ddot{\vec{r}} + \frac{\mu}{r^3}\vec{r} = 0$$

## C: Trajectory equation

- Solution through conics

$$r = \frac{p}{1 + e \cos(\nu)}$$

# First law of orbital motion

---



Newton's formulation of Kepler's first law consists of two parts:

- 1) The shapes of the orbits are **conic sections**, representing the only possible trajectories for an orbiting object in the **2-Body problem**
    - Curves found by cutting a cone with a plane
    - Circles, Ellipses, Parabolas, and Hyperbolas
  - 2) The **center of mass** is located at one of the foci
    - Strictly speaking, the Earth is not the only one to rotate around the Sun: The Earth and the Sun orbit each other around their mutual center of mass
- The orbital motion takes place **in a plane** fixed in inertial space
  - The **specific angular momentum**  $\vec{h}$  of a satellite remains constant

# Kepler problem

- Kepler problem
- Anomalies
- Kepler's orbit calculation
- Numerical solution by Newton's method

Further reading:

Sellers 2004 [Understanding Space](#), Chapter 8  
Predicting an Orbit (Kepler's Problem)  
Orbital Perturbations  
Predicting Orbits in the Real World

# Kepler problem

---



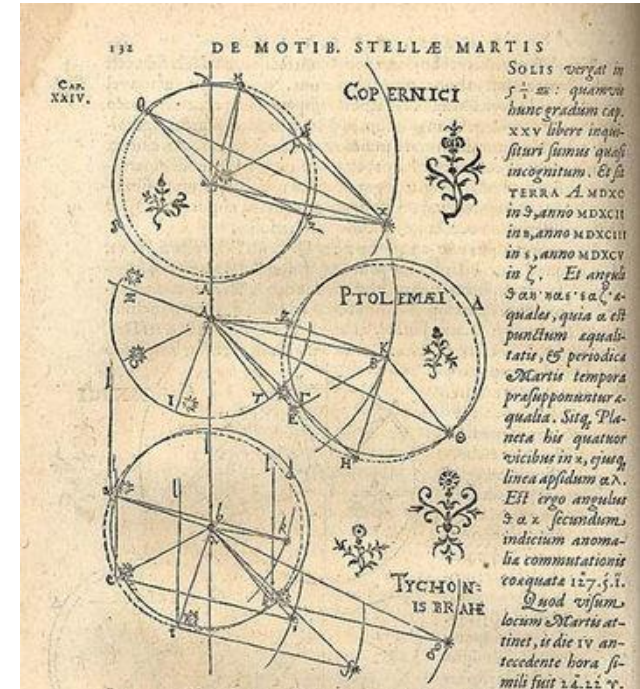
- We use “**Kepler problem**” instead of “Kepler’s **P** Problem”
- In modern technical writing (astrodynamics, celestial mechanics, physics), we typically use **attributive nouns rather than possessive forms**. So:
  - **Kepler problem** → the class of motion governed by inverse-square gravity
  - Analogous to:
    - Laplace equation
    - Navier–Stokes equations
    - Schrödinger equation
- This is consistent with standard terminology in astrodynamics and celestial mechanics
- “Kepler’s problem” is less common in technical literature and sounds more historical or descriptive
  - Occasionally used in textbooks or pedagogical contexts, in general prose or historical discussion

# Calculating orbits – Kepler prediction problem



- How to **know when and where a satellite will be over time**, or how long it takes a satellite to get to a particular location. These two scenarios make up **Kepler prediction problem**, or **Kepler problem**  
→ Both of these are crucial in astrodynamics
- Kepler used his first two laws to calculate the position of a planet as a function of time. His method involves solving a transcendent equation called “**Kepler’s equation**”
  - This equation was derived in 1609 in chapter 60 of his *Astronomia nova*, and in Book V of his *Epitome of Copernican Astronomy* (1621)
- The equation has played an important role in the history of both physics and mathematics, particularly classical celestial mechanics

Note: The following study focuses only on **closed orbits** (circular and elliptical orbits, as opposed to parabolic and hyperbolic trajectories)



*Epitome of Astronomiae Copernicanae* (1618)



# The Equation of Kepler

---

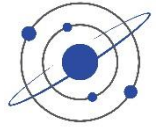


- Of all of the transcendental equations found in celestial mechanics and mathematics in general, [certainly few have generated so much study and interest as the Equation of Kepler](#)

*“A great number of analytical and graphical solutions have been discovered and almost all of the greatest mathematicians since Newton until the mid of last century have paid some of attention to the subject”* – F.R. Moulton 1914

- There is a recent book dedicated exclusively to the description and solution by a large number of methods titled “Solving Kepler’s equation over three centuries” Peter Colwell 1993
- [Friedrich W. Bessel](#) even has developed his famous functions to solve [Kepler equation](#) (see [Series expansion of Kepler equation](#))

# Mean motion $n$



- The **Mean motion**  $n$  is the mean number of orbits per day the object completes (TLE definition)

$$n = \frac{24 \cdot 60^2}{P} \left[ \frac{\text{rev}}{\text{day}} \right] \quad \leftarrow \text{The orbital period } P \text{ [s] refers to the solar (not sidereal) period (given in [TLE data](#))}$$

- $n'$  is the average angular velocity:

$$n' = \frac{2\pi}{P} = \sqrt{\frac{\mu}{a^3}} \left[ \frac{\text{rad}}{\text{s}} \right] \quad \text{Note: } n \text{ is in rev/day, and } n' \text{ is in rad/s}$$

- The angle of the satellite's ray-vector **Mean Anomaly**  $M$  at a given time  $t$  [s] is then simply:

$$M = n'(t - t_0) \text{ [rad]} \quad \leftarrow t_0 \text{ is the moment of passage at perigee}$$

- In relation to Kepler 3<sup>rd</sup> law

$$P = \frac{2\pi}{n'} = 2\pi \sqrt{\frac{a^3}{\mu}} \text{ [s]}$$



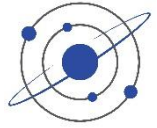
Kepler problem Summary

# Anomalies

- Mean Anomaly  $M$
- Eccentric Anomaly  $E$
- True Anomaly  $\nu$

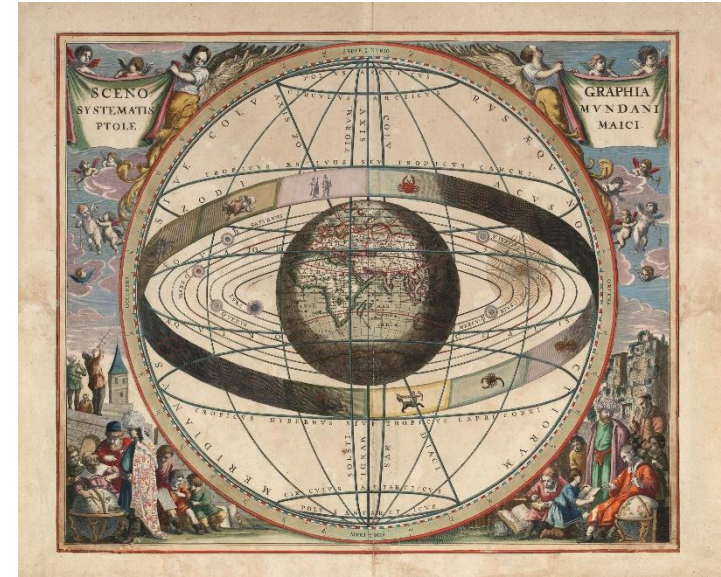


# “Anomaly”



- Before we carry out the derivation of Kepler equation, we shall introduce the concept of *orbital anomalies*
  - These anomalies are simply *auxiliary angles* that allow us to locate a planet on a given time considering its movement along an ellipse with a determined eccentricity
  - The term “anomaly” has its historical origin in the *ancient geocentric system* in which the ancients observed an *anomaly* of the orbit *in relation to the ideal circular orbit*
- There are *three angles* called anomalies allowing to determine the position of a planet around the Sun at a given instant (or a satellite around its main body)

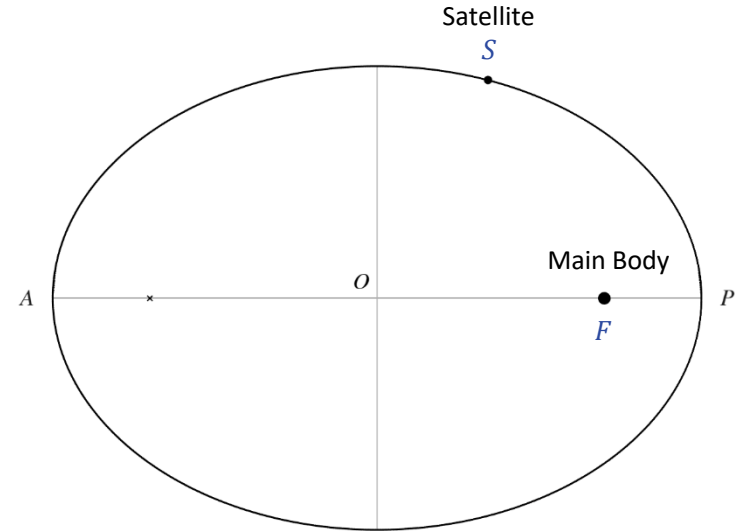
Figure showing the signs of the zodiac and the solar system with the Earth at its center, according to the Ptolemy system.  
*Harmonia Macroscopica*  
Andreas Cellarius, 1661



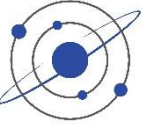


# Elliptical orbits calculations

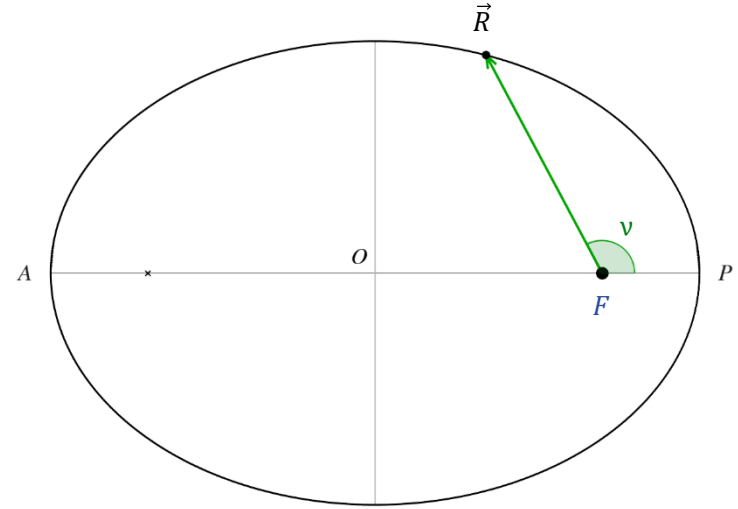
- For elliptical orbits, the speed and the associated true anomaly  $\nu$  of the object is *not changing uniformly* throughout the orbit
  - Because of this, **Kepler defined “anomalies” (angles)** used to define a planet’s orbit motion around the Sun
  - The three anomalies are:
    - **True Anomaly**  $\nu$
    - **Eccentric Anomaly**  $E$
    - **Mean Anomaly**  $M$
  - In an elliptical orbit, the Main Body (e.g. Sun or Earth) lies at one of the foci ( $F$ )
- The Earth is *not* at the center of the ellipse, which is one of the reasons why  $\nu$  and  $E$  do not change uniformly



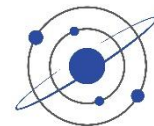
# $\nu$ – True Anomaly



- $\nu$  is the angle in polar coordinates that defines the current position of an object in its orbit
  - Calculated from perigee  $P$  to the position-vector  $\vec{R}$  of the satellite relating to the main focus  $F$
- True anomaly corresponds, as its name suggests, to an angle that actually exists in the orbit of a celestial body
- $\nu$  varies *non-linearly* with time
  - $\nu = 0$  radians ( $0^\circ$ )  $\equiv$  Perigee argument
  - $\nu = \pi$  ( $180^\circ$ )  $\equiv$  Apogee argument



$\nu$  – True Anomaly



# True Anomaly: $\theta$ vs $\nu$

- Both symbols  $\theta$  (theta) and  $\nu$  (nu) are used in orbital mechanics to represent the true anomaly, but  $\nu$  is more standard than the other depending on context

## Preferred Convention

- $\nu$  (Greek letter nu) is the preferred and most widely accepted symbol for the true anomaly in modern celestial mechanics and astrodynamics

## Alternative Notation

- $\theta$  is sometimes used, but it's *less precise* because it can also denote:
  - A polar angle in spherical coordinates, or
  - A generic angular position not specifically tied to orbital geometry
- Historically, older literature or simplified treatments (e.g., in basic physics or geometry of conic sections) sometimes used  $\theta$  for the true anomaly because it represents the angle from the periapsis in polar form:

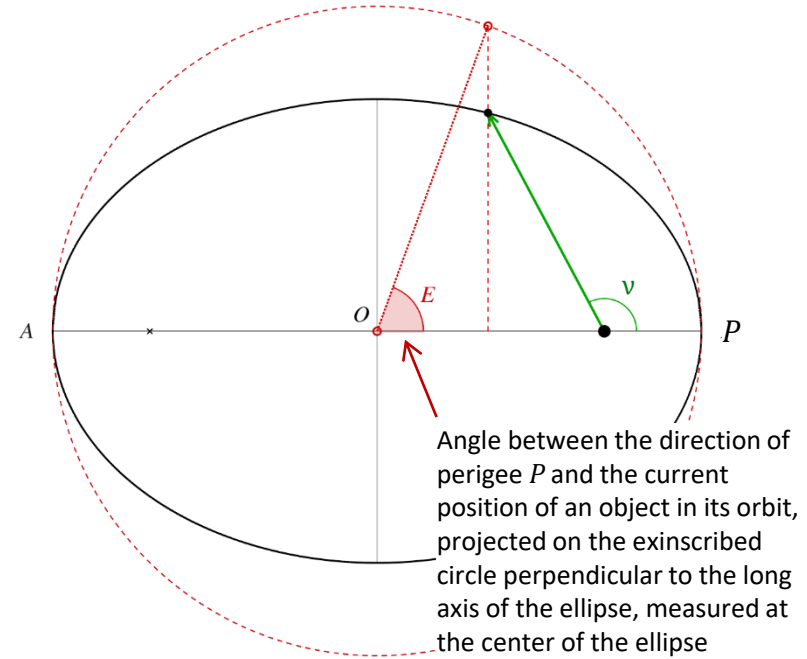
$$r = \frac{p}{1 + e \cos(\theta)}$$

→ But in orbital dynamics,  $\nu$  replaced  $\theta$  to avoid ambiguity

# $E$ – Eccentric Anomaly



- $E$  is an *intermediate* angular parameter that defines the position of a body moving along an elliptic Kepler orbit, using an auxiliary circle
  - This creates a direct link between the true anomaly  $\nu$  and a way to measure angles that are circular.
- Although it has no physical reality (we do not measure this angle, but only  $\nu$ ),  $E$  is of real interest, as it allows us to establish a relatively simple relationship between the distance  $r$  of the object to the focus  $F$  of the trajectory and the time  $t$ , in the form of a parametric equation
  - we do not have the exact relationship  $r(t)$ , but of a double relationship between  $r$  and  $E$ , and between  $t$  and  $E$
- $E$  still does not change uniformly
  - $E = \nu$  at perigee and apogee only for elliptical orbits (all the time for circular orbits)





# Relations between $E$ and $\nu$



$$\cos(E) = \frac{x}{a} = \frac{e + \cos(\nu)}{1 + e \cos(\nu)}$$

$$\sin(E) = \frac{y}{b} = \frac{\sqrt{1 - e^2} \sin(\nu)}{1 + e \cos(\nu)}$$

$$\tan\left(\frac{\nu}{2}\right) = \sqrt{\frac{1 + e}{1 - e}} \tan\left(\frac{E}{2}\right)$$

$$\nu = 2 \operatorname{atan}\left[\sqrt{\frac{1 + e}{1 - e}} \tan\left(\frac{E}{2}\right)\right]$$

$$\cos(\nu) = \frac{\cos(E) - e}{1 - e \cos(E)}$$

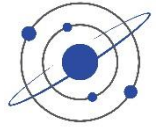
$$\sin(\nu) = \frac{\sqrt{1 - e^2} \sin(E)}{1 - e \cos(E)}$$

$$\tan\left(\frac{E}{2}\right) = \sqrt{\frac{1 - e}{1 + e}} \tan\left(\frac{\nu}{2}\right)$$

$$E = 2 \operatorname{atan}\left[\sqrt{\frac{1 - e}{1 + e}} \tan\left(\frac{\nu}{2}\right)\right]$$



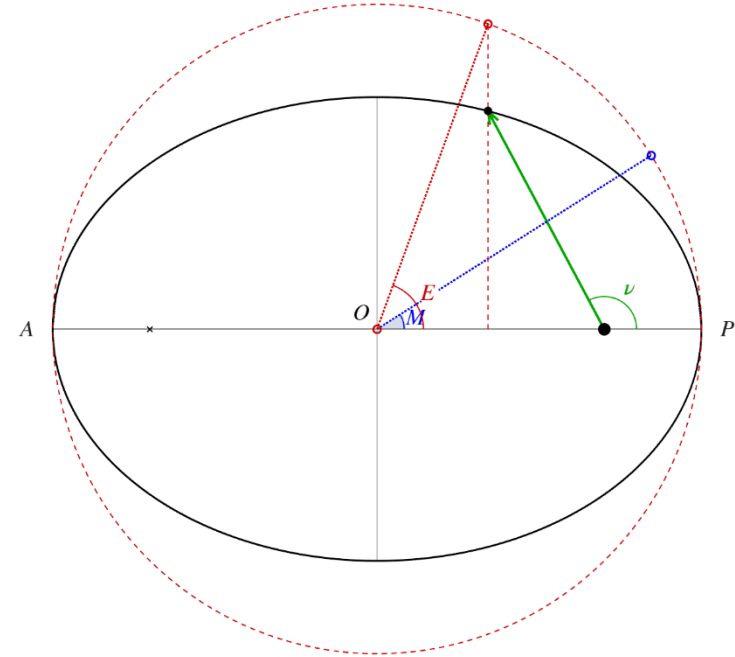
# M – Mean Anomaly



- $M$  is the angle between the perigee and the position of a fictitious body traveling in a circular orbit synchronous with the real body
- Rotates from  $0^\circ$  to  $360^\circ$  *linearly* with time:

$$M = n'(t - t_0)$$

- It is not as easy to relate  $E$  and  $M$  geometrically, as they are changing at different rates as opposed to  $E$  et  $\nu$
- For an elliptical orbit, they are connected by [Kepler equation](#) (see later):  $M = E - e \sin(E)$
- All three anomalies pass both perigee ( $\nu$ ,  $E$ , and  $M = 0$ ) and apogee ( $= \pi$ ) at the same time
  - This is why  $M$  is called the *mean* anomaly; it takes the average of the velocities of using eccentric anomaly and keeps the same period

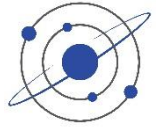


$M$  is the angle that the satellite would have moved since perigee if it were going at a constant speed (mean motion,  $n$ ) in an imaginary circular orbit with the same period as the actual elliptical orbit



Kepler problem Summary

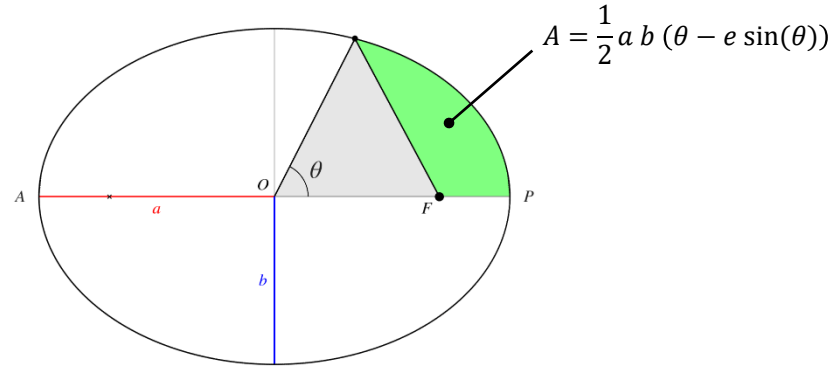
# Kepler equation derivation



- By **Kepler's second law** (law of areas), referring to a complete orbit:

$$\frac{M}{2\pi} = \frac{\text{sector area}}{\text{total ellipse area}}$$

- From geometry, the area of a sector in an ellipse defined from a focus is:



- Then simply:

$$\frac{M}{2\pi} = \frac{\frac{1}{2} a b (E - e \sin(E))}{\pi a b} \rightarrow \boxed{M = E - e \sin(E)}$$



# Verification of Kepler's second law of equal areas

Check sectors areas:

1) Polygon surface calculation vs

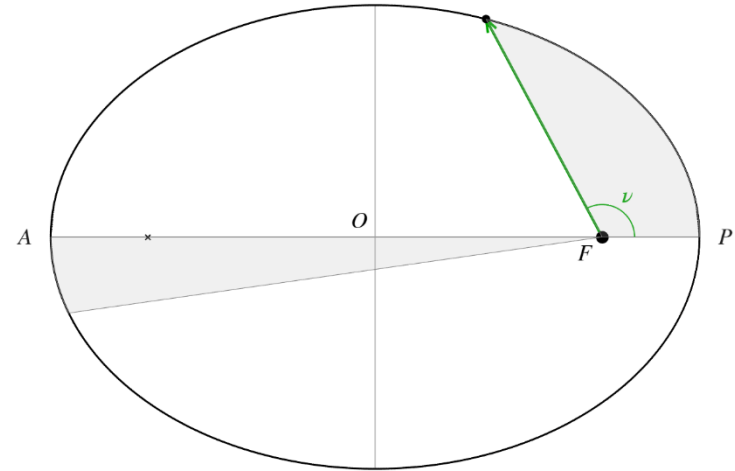
$A \approx \text{ComputeArea}(x, y)$  & polyshape

2) Analytical formulas

$A = \text{EllipseSectorArea}(a, b, e, \text{delta}) - a*b*e/2 * \sin(E)$

$$A = \frac{ab}{2} [(E_2 - e \sin(E_2)) - (E_1 - e \sin(E_1))]$$

$$A = \frac{ab}{2} (M_2 - M_1)$$



# Kepler equation



# Orbit calculation using Kepler equation

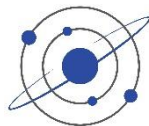


- For a given value of the eccentricity  $e$ , the initial problem consists in finding the true anomaly  $\nu$  from the mean anomaly  $M$ . The key to this problem is based in finding a solution for the **Kepler equation**
- One approach to calculating orbits is to use this Kepler equation, linking the two anomalies  $E$  and  $M$ :

$$M = E - e \sin(E)$$

**IMP:** All calculations shall always be done in *radians*

- Calculating  $M$  for a given value of  $E$  is straightforward; however, solving for  $E$  when  $M$  is given is more challenging
  - It is a *transcendent* equation in  $E$ , and there is no closed-form solution, i.e., **it cannot be solved for  $E$  analytically**
- But **it can be solved by inversion using iterative numerical methods** (see next pages)
- Even as it looks rather simple, its solution has always been a subject of interest and careful study
  - For *linear* temporal variation  $M(t)$  during orbit (related to the orbital period  $P$  or  $n$ ), we solve Kepler equation for  $E$ , and then **we infer the True Anomaly**,  $\nu(E(t))$  angle ▶
  - Knowing  $\nu(t)$ , one can finally **calculate the orbital state vectors** (radius and velocity) ▶



# Direct and inverse transformations

- Finding  $M_0$  from orbit starting at a given  $v_0$ :

$$E_0 = 2 \operatorname{atan} \left[ \sqrt{(1-e)/(1+e)} \tan(v_0/2) \right] \quad \underline{E \leftrightarrow v}$$

$$M_0 = E_0 - e \sin(E_0)$$

```
Orbit.v0 = 90; % [°]
E0 = 2 * atan(sqrt((1 - Orbit.e)/(1 + Orbit.e)) * tan(RAD(Orbit.v0) / 2)); % [rad]
M0 = E0 - Orbit.e * sin(E0); % [rad]
Orbit.M0 = DEG(M0); % M0(v0)
```

- Finding  $v_0$  from orbit starting at a given  $M_0$ :

$$E_0 = \mathbf{KeplerSolver}(M_0, e)$$

$$v_0 = 2 \operatorname{atan} \left[ \sqrt{(1+e)/(1-e)} \tan(E_0/2) \right] \quad \underline{E \leftrightarrow v}$$

```
Orbit.M0 = 90; % [°]
[~, v0] = KeplerSolver(RAD(Orbit.M0), Orbit.e); % [rad]
Orbit.v0 = DEG(v0); % v0(M0)
```

# The impossible equation at the heart of Kepler's laws



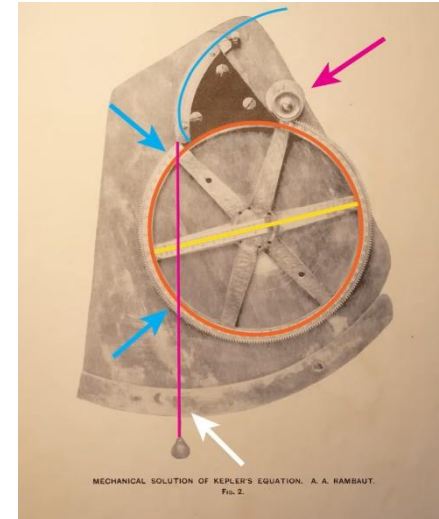
Welch Labs: [www.youtube.com/watch?v=hBkmyJ3TE0g](https://www.youtube.com/watch?v=hBkmyJ3TE0g) [22:41]

- Mathematicians used transcendental equations like Kepler equation to push the boundaries of mathematics, as happens so often with mathematically driven pursuits
- The search for a solution to Kepler equation led to the creation of entirely new mathematical tools that have broad applications outside of the initial problem statement

$$M = E - e \sin E$$

$$e = 0.21$$

| $\hat{E}$ | $\hat{M} = \hat{E} - e \sin \hat{E}$ | $M - \hat{M}$ | $\hat{E} + (M - \hat{M})$ |
|-----------|--------------------------------------|---------------|---------------------------|
| 145.0000° | 138.0987°                            | 6.9013°       | 151.9013°                 |
| 151.9013° | 146.2343°                            | -1.2343°      | 150.6670°                 |
| 150.6670° | 144.7727°                            | 0.2273°       | 150.8943°                 |
| 150.8943° | 145.0417°                            | -0.0417°      | 150.8527°                 |
| 150.8527° | 144.9924°                            | 0.0076°       | 150.8603°                 |





# Numerical solution of Kepler Equation by Newton's method

- Numerical iterative solution
- Convergence maps
- True Anomaly calculation
- Orbital state vectors

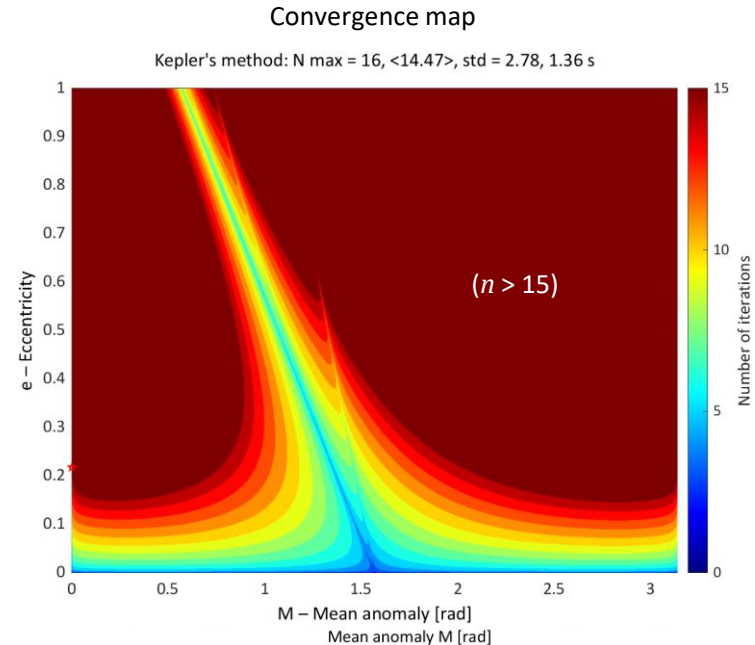


# Numerical solution by Kepler iterative method



- Intuitive iterative solution developed by Kepler in 1621
- The number of iterations  $n$  depends on the value of  $e$   
→ *Impractical for large eccentricities ( $e > 0.05$ )*

```
E = M; % initial guess $\hat{E} = M$
while 1
 M_ = E - e * sin(E); $\hat{M} = \hat{E} - e \sin(\hat{E})$
 err = M - M_;
 if abs(err) < eps
 break;
 end
 E = E + err; $\hat{E} = \hat{E} + (M - \hat{M})$
end
```



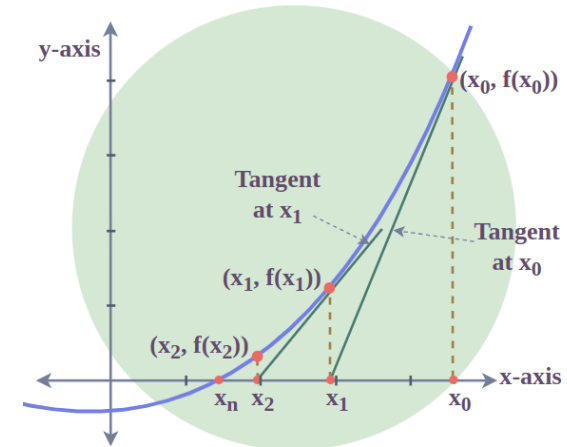
# Numerical solution by Newton's method

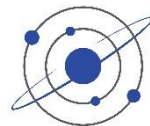


- **Newton's method** is a numerical method for iteratively solving a transcendent equation  $y(x) = 0$  whose algebraic derivative is known  $y'(x)$  or  $\left(\frac{dy}{dx}\right)$ . The value of the independent variable  $x$  quickly converges to the solution by recalculating the following relation a few times:

$$x_{n+1} = x_n - \frac{y(x_n)}{y'(x_n)}$$

- The iterative method consists of introducing a value for  $x_n$  on the right-hand side of the equation, to obtain a value  $x_{n+1}$ , and then use this value to introduce it again into the right-hand side of the equation
- This operation is repeated until the result converges to the solution, becoming similar to the previous result within a required numerical precision limit
  - In general, Newton's method allows for relatively rapid convergence





# Iterative solution of Kepler equation

- We can now use this method in order to calculate the final eccentric anomaly  $E$ , the unknown independent variable, as mean anomaly  $M$  and eccentricity  $e$  that are known
- The iterative formula of Newtons' method is:

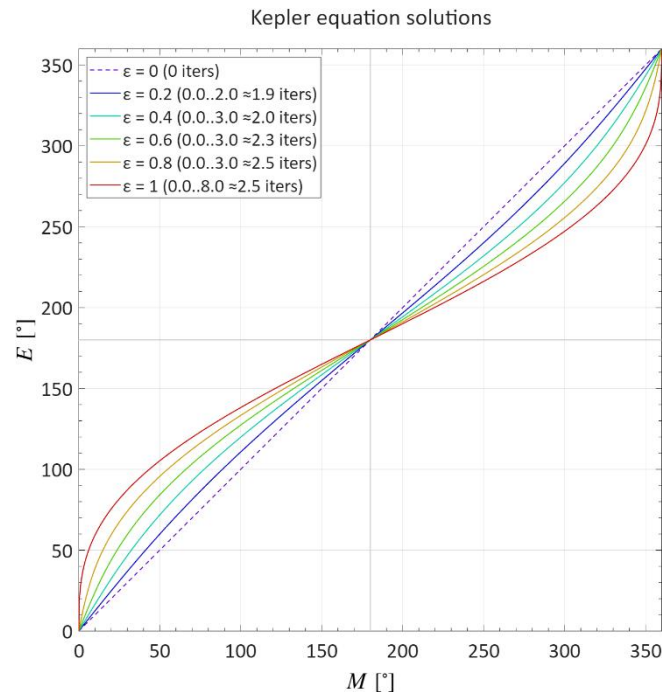
$$f(E) = M + e \sin(E) - E$$

$$f'(E) = e \cos(E) - 1$$

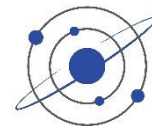
$$E_{n+1} = E_n - \frac{M + e \sin(E_n) - E_n}{e \cos(E_n) - 1}$$

- The first value  $E_0$  used is typically  $M$

→ The general map of solutions  $E(M)$  for different eccentricities  $e$  varying between 0 and 1 is given at right:



Kepler problem Summary



# Numerical solution

**%--- Solves Kepler equation  $M = E - e \cdot \sin(E)$  using iterative Newton Solution**

**% Input: Mean Anomaly  $M$  [rad] and Eccentricity  $e$**

**% Output Eccentric Anomaly  $E$  [rad] and True Anomaly  $v$  [rad]**

**eps = 1e-10; % Convergence tolerance ( $\geq 5e-14$ )**

**$E_n = M$ ; % [rad]  $\leftarrow$  (make sure  $M$  is in radians)**

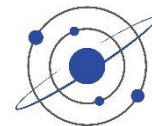
```
while 1
 E = E_n - (E_n - e * sin(E_n) - M) / (1 - e * cos(E_n));
 if abs(E - E_n) < eps
 break;
 end
 E_n = E; % $E_{n+1} = E_n$
end
```

**% Corresponding True Anomaly  $v(E)$**

**$v = \text{atan2}(\sin(E) \cdot \sqrt{1 - e.^2}, \cos(E) - e)$ ; %  $-\pi < v < \pi$**

**%  $\rightarrow$  Or using SOL toolbox subroutine:**

**$[E, v] = \text{KeplerSolver}(M, e)$ ;**



# Numerical solution

```
function [E, T] = KeplerSolver(M, e)
% Function solves Kepler equation: $M = E - e \sin(E)$ (M & E in radians)

 eps = 5e-14; % minimal convergence tolerance
 e = min(e, 0.999999); % otherwise will not converge...

 % Starting value / initial guess
 B = cos(e) - (pi/2 - e) .* sin(e); % Chobotov 2002 Eq.(4.30)
 E_n = M + e .* sin(M) ./ (B + min(M, 2*pi - M) .* sin(e)); % Adjusted for $M > \pi$

 % Vectorial implementation
 E = 0 * M;
 for i = 1:length(M)
 while 1
 % Newton-Raphson's method of successive approximation
 f = E_n(i) - e * sin(E_n(i)) - M(i); % $f(E_i)$
 f_ = 1 - e * cos(E_n(i)); % $f'(E_i)$
 dE = -f / f_; % $\Delta E_i = f(E_i) / f'(E_i)$
 E(i) = E_n(i) + dE; % $E_{i+1} = E_i + \Delta E$

 if abs(E(i) - E_n(i)) < eps
 break;
 end
 E_n(i) = E(i);
 end
 end

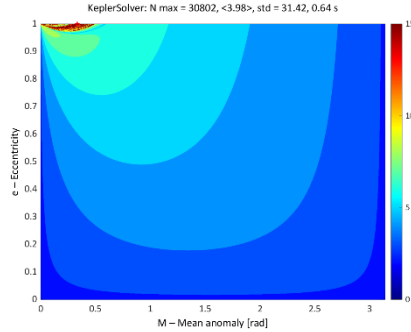
 %--- Computes the True Anomaly v from E by solving the equation: $(1 - \epsilon) \tan^2(v/2) = (1 + \epsilon) \tan(E/2)^2$
 T = atan2(sin(E) .* sqrt(1 - e.^2), cos(E) - e); % $-\pi < v < \pi$
 T = mod(T, 2*pi); % [rad]

end % function
```

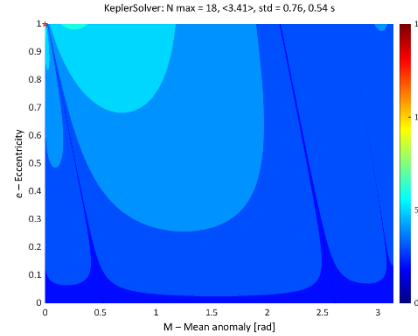


# Convergence maps

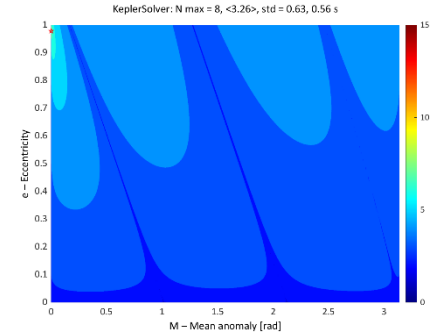
## 0) Baseline $E_n = M$



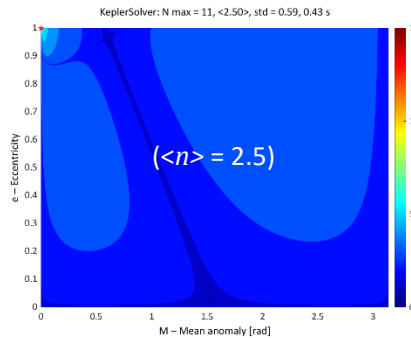
## 1) Vallado $E_n = M + \frac{1}{2} e \text{ sign}(\pi - M)$



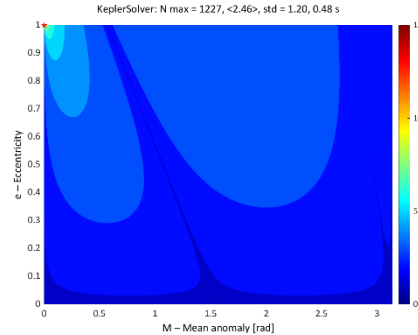
## 2) Danby $E_n = M + 0.85 e \text{ sign}(\sin(M))$



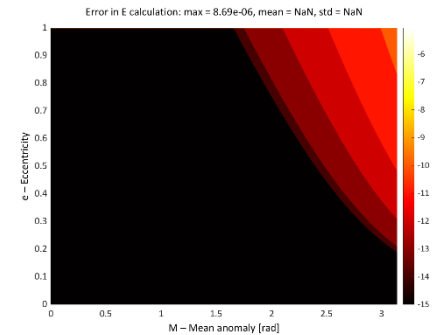
## 3) Chobotov (Best)

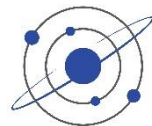


## 4) Battin



(without transcendental functions)





# Series expansion of Kepler equation

Series solution in Bessel functions of the first kind. This series converges for all  $e < 1$

$$E = M + \sum_{k=1}^{\infty} \frac{2}{k} J_k(k e) \sin(k M)$$

→ Friedrich Wilhelm Bessel (1784–1846) *invented Bessel functions to solve Kepler's equation*  
(but he didn't call them Bessel functions of course)

$$E = M + e^1 / (\text{factorial}(1) * 2^0) * 1^0 * \sin(1 * M) + \dots$$

$$e^2 / (\text{factorial}(2) * 2^1) * 2^1 * \sin(2 * M) - 2 * 0^1 * \sin(0 * M) + \dots$$

$$e^3 / (\text{factorial}(3) * 2^2) * (3^2 * \sin(3 * M) - 3 * 1^2 * \sin(1 * M)) + \dots$$

$$e^4 / (\text{factorial}(4) * 2^3) * (4^3 * \sin(4 * M) - 4 * 2^3 * \sin(2 * M)) + \dots$$

$$e^5 / (\text{factorial}(5) * 2^4) * (5^4 * \sin(5 * M) - 5 * 3^4 * \sin(3 * M) + 10 * \sin(M));$$

$$E = M + (e - 1/8 * e^3 + 1/192 * e^5 - 1/9216 * e^7) * \sin(M) \dots$$

$$+ (1/2 * e^2 - 1/6 * e^4 + 1/48 * e^6) * \sin(2 * M) \dots$$

$$+ (3/8 * e^3 - 27/128 * e^5 + 243/5120 * e^7) * \sin(3 * M) \dots$$

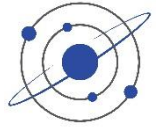
$$+ (1/3 * e^4 - 4/15 * e^6) * \sin(4 * M) + (125/384 * e^5 - 3125/9216 * e^7) * \sin(5 * M) \dots$$

$$+ 27/80 * e^6 * \sin(6 * M) + 16807/46080 * e^7 * \sin(7 * M);$$

$$E_- = 1/(1-e) * M - e/(1-e)^4 * M^3/\text{factorial}(3) + (9 * e^2 + e)/(1-e)^7 * M^5/\text{factorial}(5) - \dots$$

$$(225 * e^3 + 54 * e^2 + e)/(1-e)^{10} * M^7/\text{factorial}(7) + (11025 * e^4 + 4131 * e^3 + 243 * e^2 + e)/(1-e)^{13} * M^9/\text{factorial}(9)$$





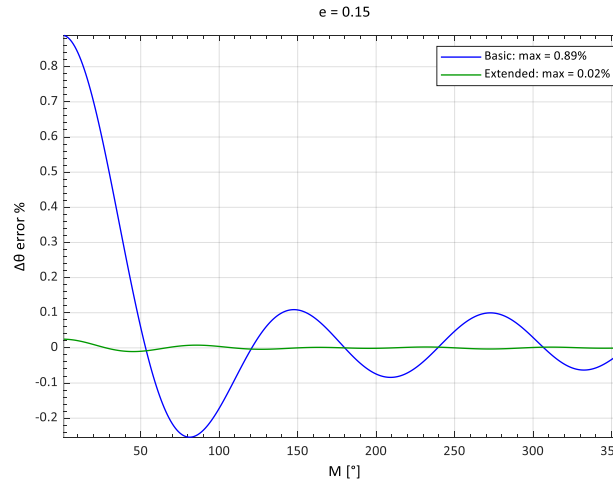
# Non-iterative approximations

Method #1:  $e \leq 0.15$  (max err < 1%)

$$\theta \approx M + 2e \sin(M) + \frac{5}{4}e^2 \sin(2M)$$

Method #2:  $e \leq 0.32$  (max err < 1%)

$$\theta \approx M + \left(2e - \frac{e^3}{4}\right) \sin(M) + \left(\frac{5}{4}e^2 - \frac{11}{24}e^4\right) \sin(2M) + \frac{13}{12}e^3 \sin(3M) + \frac{103}{96}e^4 \sin(4M)$$





# True anomaly calculation $\nu(E)$

$$\nu(E(t)) = 2 \operatorname{atan} \left( \sqrt{\frac{1+e}{1-e}} \tan \left( \frac{E}{2} \right) \right) \quad \leftarrow \text{preferred}$$

Equivalent forms:

$$\nu(E(t)) = \operatorname{atan} \left( \frac{\sin(E) \sqrt{1-e^2}}{\cos(E) - e} \right) \quad (\text{use atan2})$$

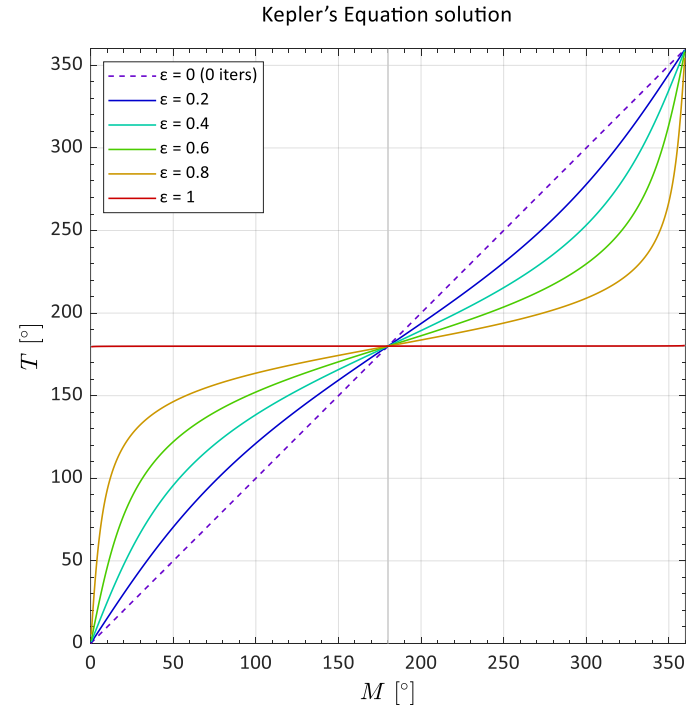
Avoids numerical issues when arguments are near  $\pm\pi$

$$\nu(E(t)) = E + 2 \operatorname{atan} \left( \frac{\sin(E)}{\beta - \cos(E)} \right), \beta = (1 + \sqrt{1-e^2})/e$$

$$\nu(E(t)) = \operatorname{acos} \left( \frac{\cos(E) - e}{1 - e \cos(E)} \right) \quad (\text{needs phase correction})$$

Optional:

$$\nu = \operatorname{unwrap}(\nu + 2\pi)$$



Orbit calculation using Kepler equation

# Radius and velocity calculation



- Knowing  $v(t)$ , We can then calculate the orbital radius by one the polar relation:

$$r(v(t)) = \frac{a(1 - e^2)}{1 + e \cos(v)} \quad \text{or } p \equiv a(1 - e^2)$$

$$r(E(t)) = a(1 - e \cos(E)) \quad \leftarrow \text{from } E \text{ only, if computation } v \text{ is not required}$$

- The modulus of the associated orbital velocity can be calculated in several ways:

Velocity  $v$  from True anomaly  $v$  :

$$v(v(t)) = \frac{\mu}{h} \sqrt{1 + 2 e \cos(v) + e^2}$$

$$v(E(t)) = \sqrt{\frac{\mu}{a} \left( \frac{1 + e \cos(E)}{1 - e \cos(E)} \right)} \quad \leftarrow \text{from } E \text{ only, if computation } v \text{ is not required}$$

$$v(r(t)) = \sqrt{\mu \left( \frac{2}{r} - \frac{1}{a} \right)}$$



Elliptical motion



Kepler problem Summary

# Orbital state vectors



- The temporal **perifocal position** (2D) of the satellite in the orbital plane from the polar coordinates:

$$P(t) = r(t) \cos(\mathbf{v}(t))$$

$$Q(t) = r(t) \sin(\mathbf{v}(t))$$

- The **perifocal velocity** (2D) is given by:

$$v_P = \dot{r} \cos(\mathbf{v}) - r \dot{\mathbf{v}} \sin(\mathbf{v}) = -\frac{\mu}{h} \sin(\mathbf{v})$$

$$v_Q = \dot{r} \sin(\mathbf{v}) + r \dot{\mathbf{v}} \cos(\mathbf{v}) = \frac{\mu}{h} (\cos(\mathbf{v}) + e)$$

$$v = \frac{\mu}{h} \sqrt{1 + 2e \cos(\mathbf{v}) + e^2}$$



Orbit calculation using Kepler equation



Kepler problem summary

# Orbital state vectors (3D)



- And finally to the **3D orbital state vectors** according to orbit COEs:

$$\vec{R}_{\text{ECI}} = R_Z(\Omega) \times R_X(i) \times R_Z(\omega) \times \begin{bmatrix} P(t) \\ Q(t) \\ 0 \end{bmatrix} \leftarrow \text{no z-component as in perifocal plane}$$

$$\vec{V}_{\text{ECI}} = \underbrace{R_Z(\Omega) \times R_X(i) \times R_Z(\omega)}_{\text{RotPeri2ECI}} \times \begin{bmatrix} v_P(t) \\ v_Q(t) \\ 0 \end{bmatrix}$$

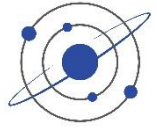


Orbit calculation using Kepler equation

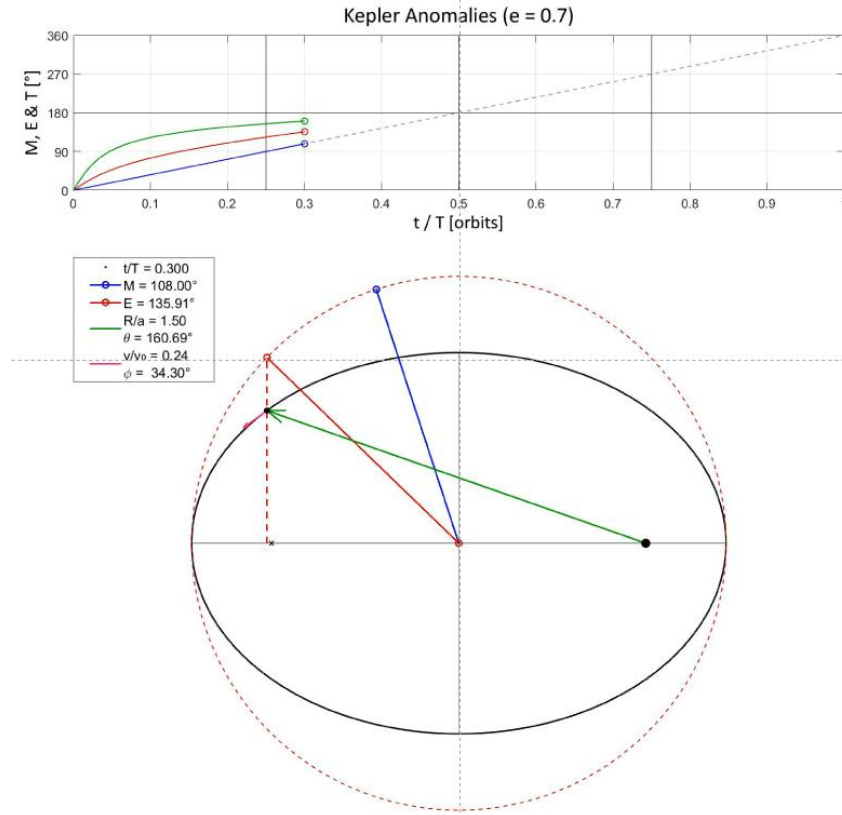


Kepler problem summary

# Animation



Video: [Kepler Anomalies.avi](#)



$$r(v) = \frac{p}{1 + e \cos(v)}$$

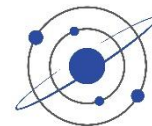
$$v(r) = \sqrt{\mu \left( \frac{2}{r} - \frac{1}{a} \right)}$$

$$M = E - e \sin(E)$$

$$r = a(1 - e \cos(E))$$

$$@ E = \frac{\pi}{2} \rightarrow \frac{r}{a} = 1$$

$$@ E = \pi \rightarrow \frac{r}{a} = 1 + e$$



# Numerical solution

%% 5) 2-Body Keplerian calculation (2BK)

%

% The theory of the Keplerian motion of a celestial body is the simplest analytical model for the propagation of an orbit.

% It perfectly returns to its initial orbit starting point after N orbits, when no perturbation are taken into account.

%

% IMP: all calculations are done in radians

%--- Defines angle linearly varying with time

M = linspace(0, 2\*pi, N); % [rad] with N = length(Orbit.time\_s) defined in Newton Solution

%== Solves the Kepler equation:  $M = E - e \cdot \sin(E)$

[E, v] = **KeplerSolver**(M, Orbit.e); % [rad] E = Eccentric Anomaly, T/v  $\equiv$  True Anomaly

%  $\rightarrow$  uniform in M, but not uniform in v to reflect varying speed along the orbit

%--- Calculation of satellite orbit radius magnitude (perifocal plane)

rho = Orbit.a \* (1 - Orbit.e^2) ./ (1 + Orbit.e .\* cos(v)); % Radius from v

vel = sqrt(Earth.mu .\* (2 ./ rho - 1 / Orbit.a)); % Velocity from v and R

%--- PQW perifocal

% Orbit ellipse: polar  $\rightarrow$  cartesian

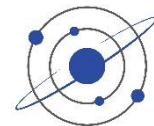
P = rho .\* cos(v);

Q = rho .\* sin(v);

W = 0 \* r; % (null in perifocal)

*Voila!* (same as [Newton's solution](#))

# Solution to Kepler problem (summary steps)

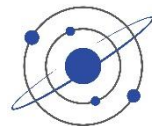


Compute new state vector (position & velocity) after **time of flight**  $\Delta t = t - t_0$  (epoch  $t$ , time at perigee  $t_0$ )

- |                                    |                                                    |
|------------------------------------|----------------------------------------------------|
| 0) $n' = f(a)$                     | <u>Mean motion</u> (orbit characteristic)          |
| $M = n' \Delta t$                  | <u>Mean Anomaly</u> (time since perigee)           |
| 1) $E = \text{KeplerSolver}(M)$    | <u>Eccentric anomaly</u> (intermediate relation)   |
| $v = f(E)$                         | <u><math>E \leftrightarrow v</math></u>            |
| 2) $r \ \& \ v = f(v)$             | <u>Orbit radius and velocity</u> (derived outputs) |
| 3) $P \ \& \ Q = f(v, r)$          | <u>Perifocal position</u> (2D), and                |
| $\vec{R} \ \& \ \vec{V} = f(P, Q)$ | <u>Orbital state vectors</u> (3D)                  |

Note:  $\mu$  is known from the Main Body the satellite is orbiting,





# Numerical Example

% Defines the wanted time of flight  $\Delta t$

```
t0 = Orbit.time_s(1);
```

```
dt = 60^2; % = 1 h
```

% → To get  $M$ ,  $n'$  is required, which is function of  $a$

% If  $a$  is not given, it can be derived from  $\epsilon$ ,  $h$  &  $e$ ,  
 %  $p$  &  $e$ ,  $r_P$  &  $e$ ,  $r_A$  &  $e$ , or  $b$  &  $c$

```
e = Orbit.e;
```

```
r_P = Orbit.r_P;
```

%%% Compute it step-by-step

```
mu = Orbit.mu; % Artificial satellite orbiting the Earth
```

```
h = sqrt(mu * r_P * (1 + e))
```

```
a = h^2 / (mu * (1 - e^2))
```

% Mean motion  $n'$

```
n_ = sqrt(mu / a^3)
```

```
M = n_ * (t0 + dt);
```

% Eccentric Anomaly  $E(e, M)$

```
E = KeplerSolver(M, e);
```

% True Anomaly

```
v = 2 * atan(sqrt((1 + e) / (1 - e)) * tan(E / 2)); % [rad]
```

% Orbit radius & velocity

```
rho = a * (1 - e^2) / (1 + e * cos(v)); % $\rho(v)$
```

```
vel = mu / h * sqrt(1 + 2 * e * cos(v) + e^2); % $vel(v)$
```

% Perifocal position & velocity

```
P = rho * cos(v); % Orbit.peri_P(k)
```

```
Q = rho * sin(v); % Orbit.peri_Q(k)
```

```
v_P = - mu / h * sin(v);
```

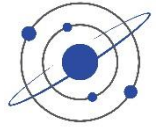
```
v_Q = mu / h * (cos(v) + e);
```

% rotation by  $w^\circ$  in perifocal plane, followed by  $i^\circ$  over  $x$   
 % axis, followed by  $\Omega^\circ$  over  $z$  axis

```
RotPeri2ECI = RotZ(RAAN) * RotX(i) * RotZ(w);
```

```
R = RotPeri2ECI * [P; Q; 0];
```

```
V = RotPeri2ECI * [v_P; v_Q; 0];
```

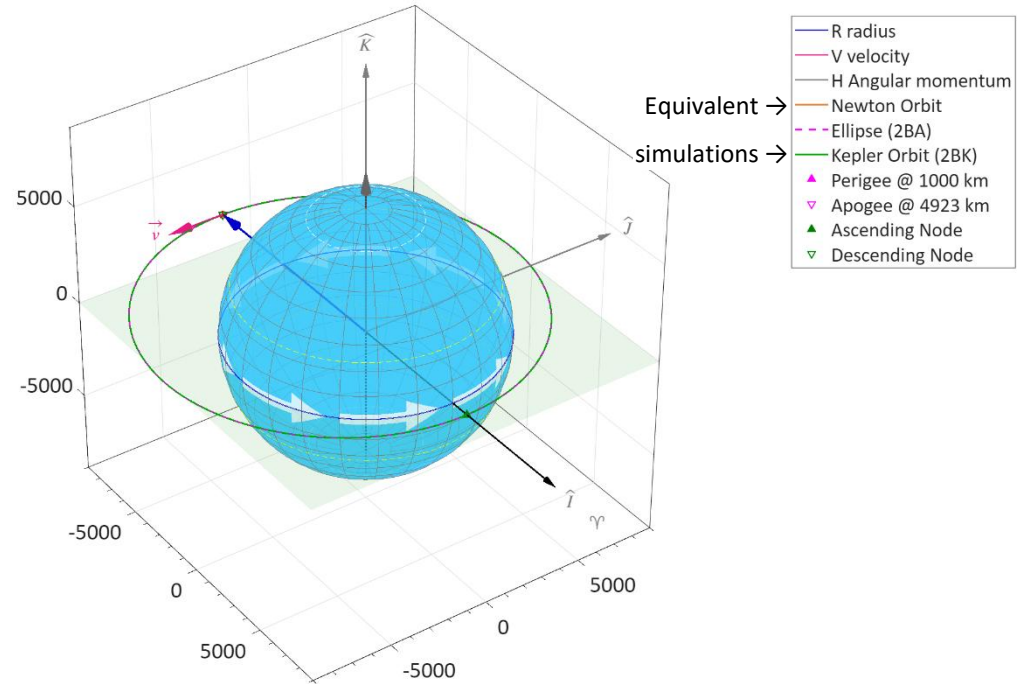
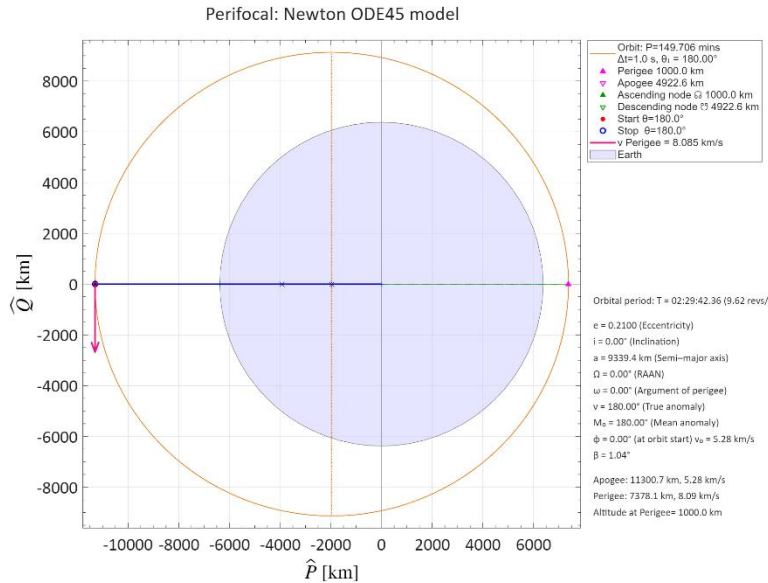


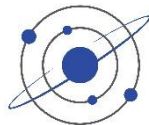
# Elliptical orbit

## Earth-Centered Inertial frame ( $I, J, K$ )

$H = 1000$  km (min at perigee)

Newton's orbit:  $H_{\min} = 1000$  km,  $e = 0.210$ ,  $i = 0^\circ$ ,  $\Omega = 0^\circ$ ,  $\omega = 0^\circ$ ,  $v = 180^\circ$





# Solution to Kepler problem (summary steps)

Compute the **time of flight**  $\Delta t = t - t_0$  for a change in true anomaly  $\Delta \nu$

- |    |                              |                                              |
|----|------------------------------|----------------------------------------------|
| 0) | $h = \sqrt{\mu r_p (1 + e)}$ | Given $e$ and $\vec{R}$ at perigee ( $r_p$ ) |
|    | $a = h^2 / \mu (1 - e^2)$    |                                              |
|    | $n' = \sqrt{\mu / a^3}$      |                                              |
| 1) | $E = f(\Delta \nu)$          | <u><math>E \leftrightarrow \nu</math></u>    |
| 2) | $M = E - e \sin(E)$          | <u>Kepler equation</u>                       |
| 3) | $\Delta t = M / n'$          | <u>Mean Anomaly</u> (time since perigee)     |

Note:  $\mu$  is known from the Main Body the satellite is orbiting,



# Numerical Example

```
% Defines targeted Δv change
v = RAD(45); % [rad]

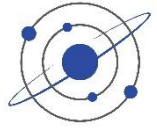
% retrieves True Anomaly from previous simulation
T = atan2(Orbit.peri_Q, Orbit.peri_P);
% finds index of closest point
k = find(T >= v, 1);
 v = T(k); % resets Δv value to actual one in simulation
 DEG(v)

% Computes Eccentric Anomaly
E = 2 * atan(sqrt((1 - e) / (1 + e)) * tan(v / 2)); % [rad]

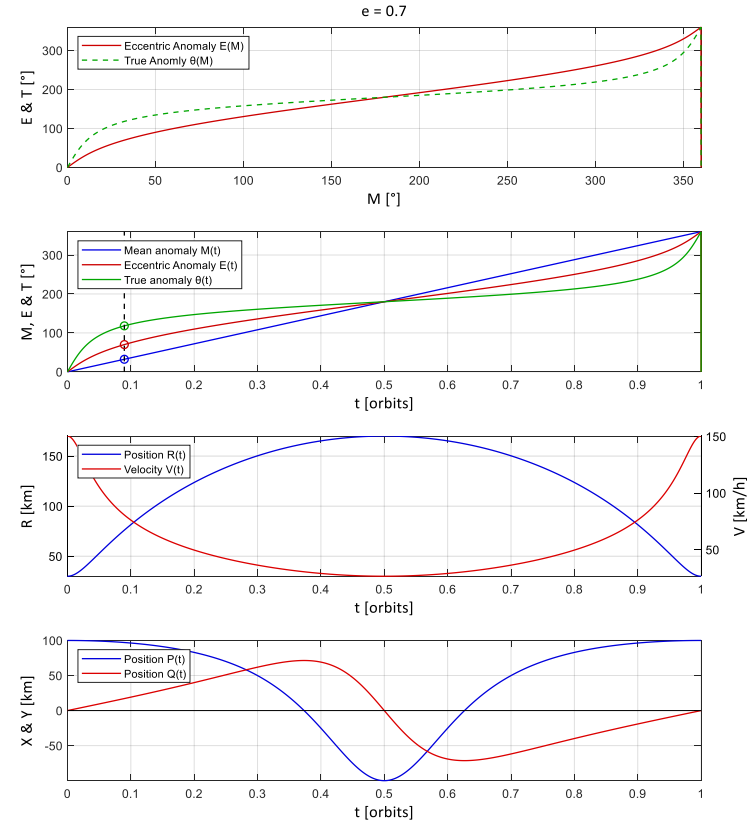
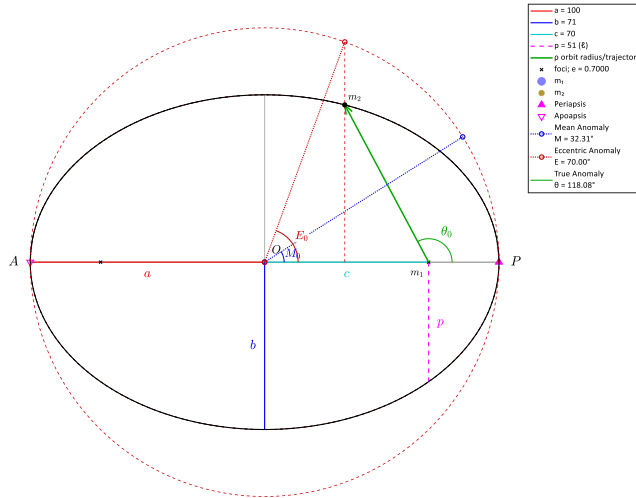
% Computes Mean Anomaly
M = E - e * sin(E); % [rad]

% Computes time of flight Δt
dt = M / n_ - t0; % [s]
[dt , Orbit.time_s(k) - t0]
```

# Summary



0. Define  $M(t) = [0 .. 2\pi]$  (for one orbit)
1. Solve numerically  $E(M)$
2. Compute  $v(E)$
3. Compute  $r(v)$  &  $v(r) \leftarrow$  Polar Solution
4. Compute  $P(t)$  &  $Q(t) \leftarrow$  PQW referential



# Simulation programming



## Orbit temporal characteristics

```
% Mean Motion: Number of revolutions per second; The orbital period refers to the solar (not sidereal) period
Orbit.n_ = sqrt(Orbit.mu / Orbit.a^3); % n' [rad/s] definition for an elliptical orbit

% Keplerian period
Orbit.P_s = 2*pi / Orbit.n_;
Orbit.n = (24*60^2) / Orbit.P_s; % n [rev/day] "day" here is solar day

% Simulation input (e.g. defined by TLE) with all other COEs
Orbit.Epoch.Time % at angular position M0 & v0 wrt. to Perigee
Orbit.M0 % Angular position of object at Epoch wrt to Perigee

% Corresponding v0
[~, v0] = KeplerSolver(RAD(Orbit.M0), Orbit.e); % [rad]
Orbit.v0 = DEG(v0); % [°]

% M0 defines Perigee & Apogee
Orbit.Perigee.Time = Orbit.Epoch.Time + sec_to_day(RAD(0 - Orbit.M0) / Orbit.n_); % [days]
 == Orbit.Epoch.Time + sec_to_day((0 - Orbit.M0)/360 * Orbit.P_s); % (alternate version using P_s)
Orbit.Apogee.Time = Orbit.Epoch.Time + sec_to_day(RAD(180 - Orbit.M0) / Orbit.n_); % [days]
 == Orbit.Perigee.Time + sec_to_day(Orbit.P_s / 2); % (alternate version using P_s)
```



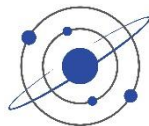
## Ascending Node calculation

```
% Angular position defined by Argument of Perigee ω
theta = -Orbit.w;

% → must use Kepler for proper time calculation (irrelevant for $v = 0$ & $\pi/180^\circ$ where $M \equiv E \equiv v$)
E = 2 * atan(tan(RAD(theta)/2) * sqrt((1 - Orbit.e) / (1 + Orbit.e)));
M = E - Orbit.e * sin(E);
M = mod(M, 2*pi);
AN.Time = Orbit.Perigee.Time + sec_to_day(DEG(M)/360 * Orbit.P_s); % [UTC]

AN.r = Orbit.a * (1 - Orbit.e^2) ./ (1 + Orbit.e * cosd(theta)); % [km]
u = theta + Orbit.w; % $u \equiv v + \omega$ Argument of latitude
AN.ECI.pos = AN.r * RotZ(Orbit.RAAN) * RotX(Orbit.i) * RotZ(u) * [1; 0; 0];
AN.ECI.Lon = atan2d(AN.ECI.pos(2), AN.ECI.pos(1));
AN.ECI.Lat = 0;

% → Descending node computed the same way with $\theta = 180 - \text{Orbit.w}$
```



# Time vector vs Anomalies

## Time

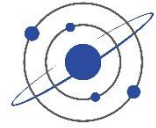
```
%--- Simulation time in datenum format [days] → Linear scale
% e.g. simulation over a given time-of-flight and time increment
Orbit.Time = Orbit.Epoch.Time + sec_to_day([0:dt_sec:TOF_min*60]); % [days]
% e.g. just between Apogee and Perigee
Orbit.Time = Orbit.Perigee.Time + sec_to_day(linspace(0, Orbit.P_s / 2, N));

%--- Corresponding time vector in seconds since perigee
Orbit.time_s = day_to_sec(Orbit.Time - Orbit.Epoch.Time) + RAD(Orbit.M0) / Orbit.n_; % [s] ← basic definition
 == day_to_sec(Orbit.Time - Orbit.Epoch.Time) + Orbit.M0/360 * Orbit.P_s; % [s] (equivalent)
 == (Orbit.Time - Orbit.Perigee.Time) * (24*60^2); % [s] (equivalent, but based on derived quantity)
Orbit.time_s(1) % == 0 s for orbit starting at Perigee
```

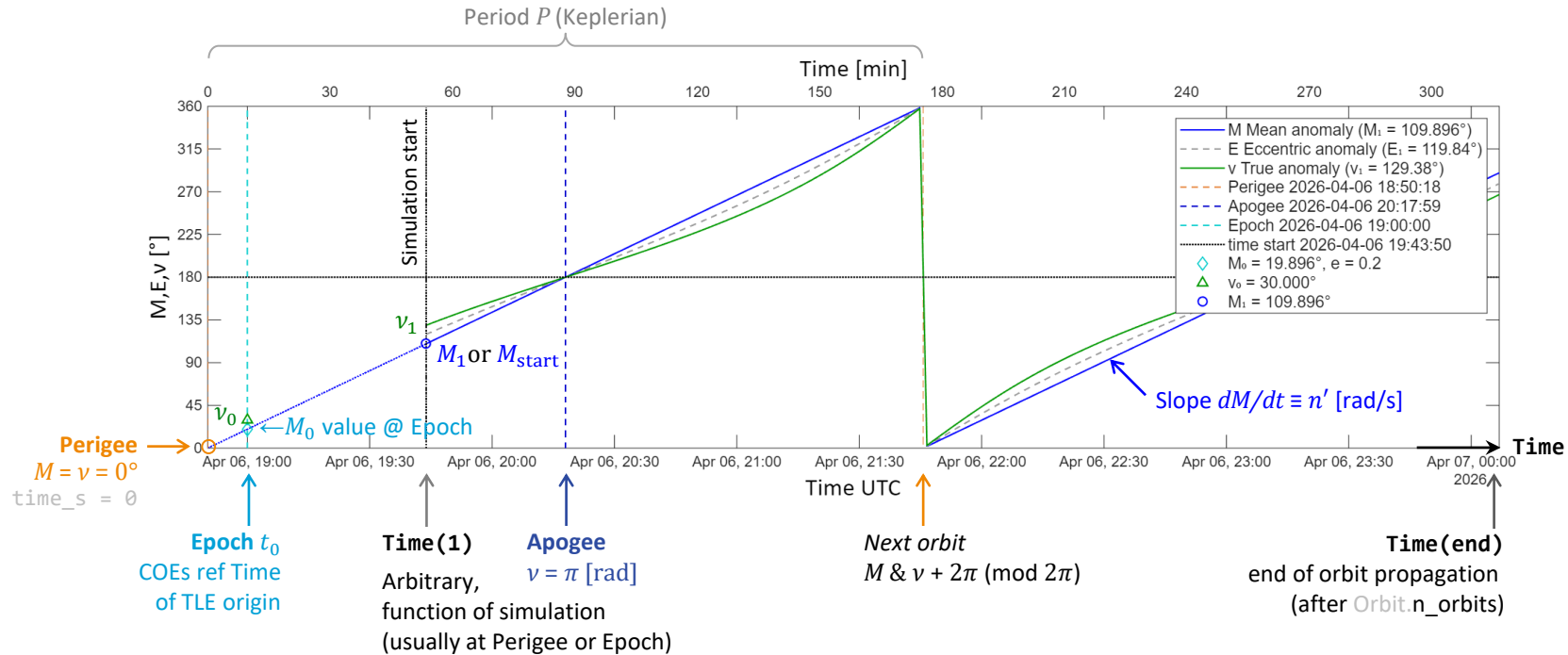
## Anomalies

```
%--- Mean Anomaly from time vector
Orbit.M = Orbit.n_ * Orbit.time_s; % [rad] Mean Anomaly ← here time_s contains offset wrt to M0
% Corresponding Eccentric and True Anomalies
[Orbit.E, Orbit.T] = KeplerSolver(Orbit.M, Orbit.e);
```





# Time vector vs Anomalies



$M_0$  is the Mean Anomaly angle value, measured from perigee at Epoch  $t_0$ ;  $v_0$  is the equivalent True Anomaly value  
 $\text{DEG}(M(1)) \neq M_0 \leftarrow$  only equal if orbit starts at Epoch

# Reference Frames & Coordinate Systems

- Reference Frames and Coordinate Systems
- [ECL](#) – Heliocentric–Ecliptic / J2000
- [PQW](#) – Perifocal
- [ECI](#) – Earth-Centered Inertial
- [ECEF](#) – Earth-Centered, Earth-Fixed *for Ground Tracks ( $\lambda, \varphi$ )*
- [Topo](#) – Topocentric (SEZ, ENU, NED) *for satellite tracking*
- [Frame instances](#) – ICRS, GCRS, TETE, TEME, ITRS, etc.
- [Transformation chains](#)

Further reading:

Sellers 2004 [Understanding Space](#), Chapter 4

**The restricted Two-Body Problem**

→ Coord Systems

<https://rhodesmill.org/skyfield/coordinates.html>

<https://rhodesmill.org/skyfield/positions.html>

Reference Frames — Orbital Mechanics & Astrodynamics

<https://orbital-mechanics.space/intro/reference-frames.html>

# Introduction

---

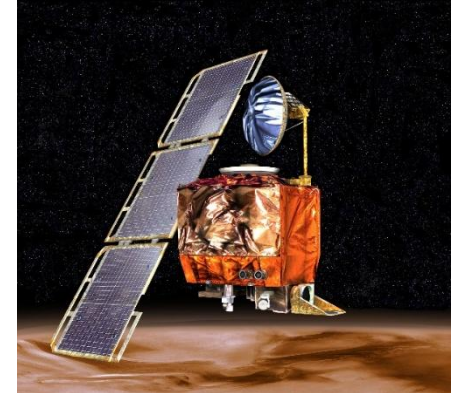


- Imagine you are an engineer responsible for tracking a spacecraft on its journey to Mars. For months, you have carefully analyzed its position and velocity, and the long-awaited day of arrival finally comes. As the spacecraft begins its descent, you expect a controlled approach to the Martian surface – but instead, it impacts the planet at high speed and the signal is lost
- The calculations appeared correct, so what went wrong? The problem was not the mathematics – it was the reference frame. A state vector or velocity expressed in the wrong frame can lead to a completely incorrect interpretation of the spacecraft's motion and, ultimately, to a catastrophic navigation error
- A multi-million-dollar mission can be lost because of a simple frame mismatch
  - *Don't be that engineer. Always check your reference frames*
- While this example may be somewhat exaggerated, the underlying risk is very real. There have been several occasions in which seemingly small engineering inconsistencies have contributed to major mission failures
- The **MARS CLIMATE ORBITER** is a well-known example: a mismatch in units ultimately led to the loss of a spacecraft worth hundreds of millions of dollars. In the same way, using an incorrect reference frame can produce serious errors in position, velocity, guidance, and navigation
- The lesson is simple: frame consistency is just as critical as unit consistency

# Mars Climate Orbiter disaster

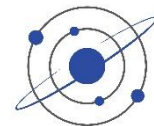


- The Mars Climate Orbiter was destroyed on Sept. 23, 1999, because ground software provided by Lockheed Martin calculated thruster data in US customary units (**pound-force seconds**), while NASA's Jet Propulsion Laboratory (JPL) navigation software expected metric SI units (**newton-seconds**)
  - This elementary unit mismatch caused a  $4.45\times$  calculation discrepancy, which steered the spacecraft drastically off course
    - Instead of stabilizing at a safe altitude of 150 km, the probe plunged to just 57 km above the surface, disintegrating in the thick Martian atmosphere
  - The cost of the mission was \$327.6 M (**\$586 M** in 2024) total for the orbiter and lander, comprising \$193.1 M (**\$346 M** in 2024) for spacecraft development, \$91.7 million (**\$164 M** in 2024) for launching it, and \$42.8 million (**\$77 M** in 2024) for mission operations
- See [https://en.wikipedia.org/wiki/Mars\\_Climate\\_Orbiter](https://en.wikipedia.org/wiki/Mars_Climate_Orbiter)
- NASA does not place the responsibility on Lockheed for the mission loss; instead, various officials at NASA have stated that NASA itself was at fault for **failing to make the appropriate checks and tests** that would have caught the discrepancy



# Mars Climate Orbiter disaster (continued)

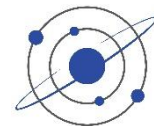
---



## The Mathematical Mismatch

- The core failure originated in a ground-based file named SM\_FORCES (small forces). The software tracked Angular Momentum Desaturation (AMD) maneuvers, which fired small thrusters to counteract solar wind pressure.
- The formula for impulse is the product of force and time:  $I = F \times t$ 
  - The Subcontractor's Output: Lockheed Martin coded the ground software to output impulse data using the imperial unit **pound-force seconds** ( $lbf \cdot s$ )
  - NASA's Expected Input: NASA's JPL navigation models expected the metric unit **newton-seconds** ( $N \cdot s$ ) as specified in the mission contract.
- Because  $1 \text{ lbf} \approx 4.45 \text{ N}$ , every calculated thruster correction was factored in at less than a quarter of its actual strength on the trajectory models

# Mars Climate Orbiter disaster (continued)



## Trajectory Deviation

- Over the nine-month, 742 million km journey to Mars, these tiny, uncorrected modeling errors compounded

| Parameter                | Planned Trajectory | Actual Trajectory | Result                                                  |
|--------------------------|--------------------|-------------------|---------------------------------------------------------|
| Orbit Insertion Altitude | 140 to 150 km      | 57 km             | Spacecraft entered the friction-dense upper atmosphere. |
| Minimum Safe Limit       | 85 km              | 57 km             | Atmospheric stresses ripped the probe apart.            |

## Systemic Failures Beyond the Math

- The Mars Climate Orbiter Mishap Investigation Board revealed that *the crash was ultimately a software and systems engineering failure rather than just a basic math error*:
  - Flawed End-to-End Testing**: The interface software between the two engineering teams was never fully audited or tested end-to-end with live data
  - Ignored Human Discrepancies**: During the cruise phase, navigators noticed minor trajectory anomalies but their concerns were dismissed *due to strict bureaucratic channels and budget constraints*
  - Unchecked Interface Assumptions**: Both teams assumed the other had integrated the correct standard unit guidelines across platform handoffs
- The disaster remains a classic cautionary tale in engineering schools globally regarding software interface control, rigorous verification protocols, and unit standardization.

# Reference frames and Coordinate systems



- **Reference frames** are essential tools in the field of astrodynamics and satellite navigation, and observational astronomy, providing a standardized framework for describing the motion of objects in space and on the Earth's surface, enabling precise calculations and measurements
  - A **reference frame** is paired with a specific **coordinate system** for practical use
- The key to most spherical geometry problems is to **define the reference frame correctly**  
Given the correct **coordinate system**, most problems can be solved algebraically very quickly
- A **reference frame** defines the physical or conventional basis from which positions and velocities are measured
  - Tells where the axes are (origin + axis orientation) and how they move (fixed inertial or rotating with Earth surface)
  - Examples: ICRF, GCRS/GCRF, ITRS/ITRF, TETE/TOD, TEME, etc. → **ECI & ECEF are general classes of reference frames**
- A **coordinate system** defines how numerical coordinates are assigned inside that frame
  - Mathematical coordinate description used within that frame, telling how to label positions within those axes
  - How a position is numerically represented within that frame (e.g., Cartesian  $x, y, z$ , spherical  $RA/Dec + \text{range } \rho$ )
- For example, one talks about cartesian coordinates, a state vector, expressed in a given reference frame
  - Analogy: **the reference frame is the stage**, and **cartesian coordinates is the grid painted** on the stage

In astrodynamics, the reference frame determines the physics of the vector,  
while the coordinate system determines its numerical representation

# Reference frames

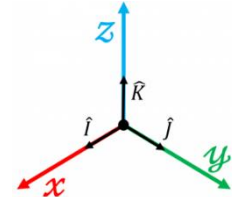




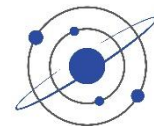
# Reference frames



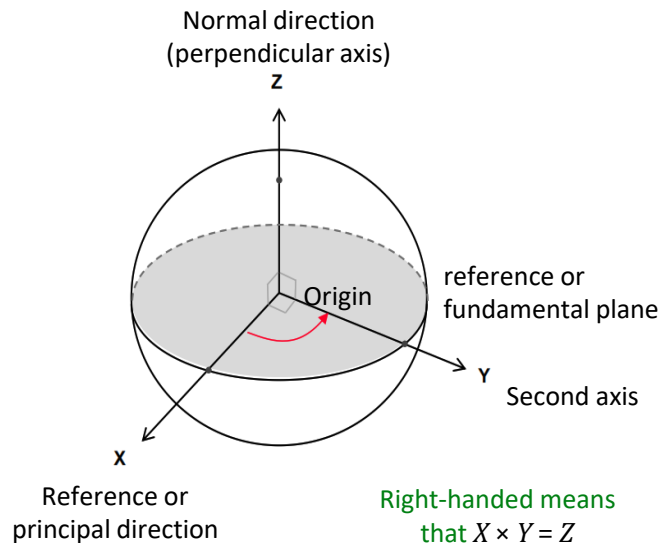
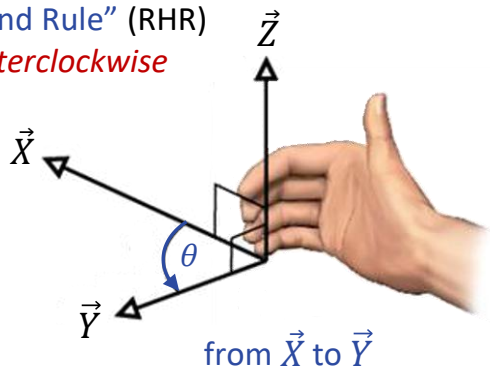
- In orbital mechanics, the motion of satellites and planets are tracked in a reference frame which consists in **three mutually perpendicular axes** (at right angle) describing the 3D coordinates of all points in space
- Each particular frame is defined by three geometric concepts:
  1. An **origin**, defining the center from which the axes originate.  
The main options for origins are: **barycentric (SSB)**, **heliocentric**, **geocentric**, and **topocentric**
  2. A **reference direction**, and
  3. A **fundamental plane**
    - The  $x$  and  $y$  axes define a reference plane from which right ascension or longitude is measured
    - The  $x$ -direction defines the origin from which **right ascension or longitude** is measured
    - The  $z$ -direction is at right angles to that  $x - y$  plane and defines the “pole”
- **Spatial coordinates** and **timing conventions** are adopted to consistently identify locations and motions of the trajectory of a spacecraft or the orbit of a satellite with respect to an observer
  - Relativity is assumed to be negligible, so a single universal clock is sufficient to specify the time for all reference frames
- Reference frames are essential because they provide the context needed to define an object's **position**, **velocity**, and **orientation**
- They specify the geometric basis with respect to which these quantities are measured



# Right-Hand Rule (RHR)



“Right Hand Rule” (RHR)  
 $\equiv$  *counterclockwise*



- The third perpendicular axis completes the **right-hand system** defined by the other two vectors:

Perifocal:  $\hat{P} \times \hat{Q} = \hat{W}$

ECI:  $\hat{I} \times \hat{J} = \hat{K}$

ECEF:  $\hat{X} \times \hat{Y} = \hat{Z}$

SEZ:  $\hat{S} \times \hat{E} = \hat{Z}$



Rotation matrices

# Coordinate systems



# Coordinates

---



- Coordinates by themselves have no physical meaning unless the associated reference frame is specified
  - A set of values such as  $(x, y, z)$ , or any other coordinate representation, is incomplete without knowing the frame in which those coordinates are expressed
  - It is the reference frame that gives coordinates their meaning
- Coordinates are the numbers we use to name astronomical positions. For any given position, one can calculate any of at least a **half-dozen different sets of coordinates** naming that position
- Remember that **coordinates are merely names for a position**

# Coordinate systems



- “Coordinates” are the numbers used to name astronomical positions
- Examples, all potentially describing a given vector in 3D space
  - Cartesian  $x, y, z$
  - Spherical range  $r$ , right ascension  $\alpha$ , declination  $\delta$  or  $r, \lambda, \phi$
  - Geodetic longitude  $\lambda$ , latitude  $\varphi$ , height  $h$
  - Cylindrical  $\rho, \theta, z$
  - Local horizon azimuth, elevation, range
  - Orbital elements  $a, e, i, \Omega, \omega, \nu$
- Coordinates are merely names for a position. Asking for a different coordinate doesn’t change the position itself  
`[Az, El, rho] = Cart2Sph([500; 1000; 7000]); xyz = Sph2Cart(Az, El, rho)`

# Clarification



- Changing the coordinate system does not change the physical position of the object; it only changes how that same direction is expressed numerically
- For a single physical direction in the sky, different coordinate representations – such as equatorial, ecliptic, or horizontal coordinates – [refer to the same line of sight](#)
  - They are simply different mathematical descriptions of the same point

Same physical direction → different coordinate values

- In practical terms, once a particular position has been defined, changing coordinates does not change where the telescope should point—it only changes the numbers used to describe that direction
- By contrast, [astrometric](#) and [apparent](#) positions are not merely different coordinate descriptions of the same direction. They represent different physical directions [because different corrections have been applied](#)
  - An [astrometric](#) position typically includes light-time effects, whereas an [apparent](#) position additionally includes effects such as aberration and gravitational light deflection (see [SOL Part III – 1.2 Reference positions of celestial objects](#))

Astrometric position  $\neq$  Apparent position

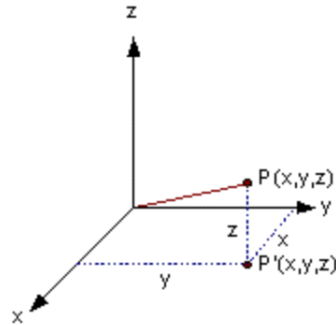
- Astrometric and apparent positions generally correspond to slightly different pointing directions in the sky

# Coordinate systems

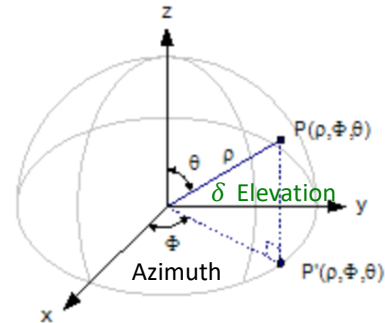


- A **coordinate system** specifies the method used to locate a point *within* a particular **reference frame**
- Once the underlying reference frame has been defined, most work is done in **either rectangular ( $x, y, z$ ) or spherical ( $\phi, \theta$ )** coordinates or some combination of the two (also cylindrical, generally not used)
  - A given problem may be easier in one or the other, but so long as the computations are done correctly, the accuracy of the solution will be the same in either set of coordinates
- Positions or 3D spatial coordinates in these systems may be expressed either as:

**Rectangular** or **Cartesian** coordinates ( $I, J, K$  or  $X, Y, Z$ )



**Spherical** coordinates ( $\phi, \theta$ , or  $\delta, \alpha$ )



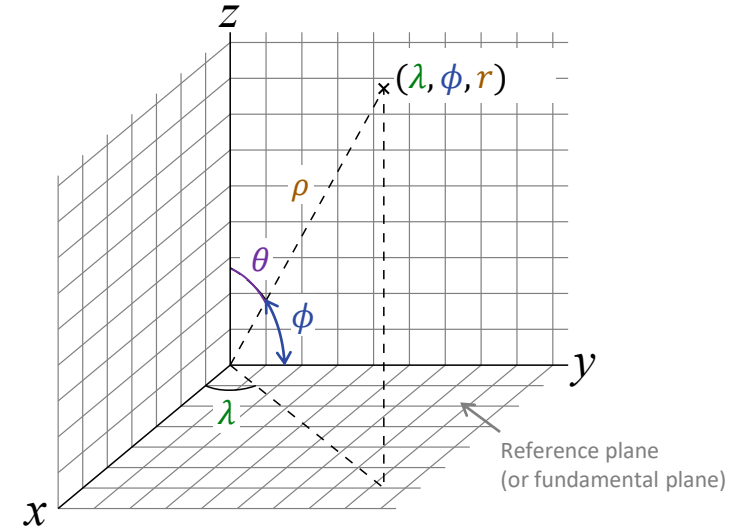
- Transforming between them is both conceptually and numerically straightforward  Spherical ↔ Cartesian

# Spherical coordinates



- In mathematics, a **spherical coordinate system** is a coordinate system for 3D space where the position of a given point in space is specified by three real numbers:
  - the **radial distance**  $r$  along the radial line connecting the point to the fixed point of origin;
  - the **polar angle**  $\theta$  between the radial line and a given polar axis;
  - and the **azimuthal angle**  $\phi$  as the angle of rotation of the radial line around the polar axis.
- Once the radius is fixed, the three coordinates  $(\lambda, \phi, \rho)$ , provide a coordinate system on a sphere, typically called the **spherical polar coordinates**

$\lambda$  = Longitude / Right Ascension (RA) / Azimuth  $\alpha$   
 $\phi$  = Latitude (or  $\varphi$ ) / Declination (Dec) / Elevation  $\delta$   
 $\theta$  = **Colatitude** or **polar angle** (used in some coordinate systems)  
=  $90 - \phi$   
 $\rho$  = Radial distance (or  $r$ )



Spherical  $\leftrightarrow$  Cartesian





# Spherical $\leftrightarrow$ Cartesian

- The mapping from **spherical coordinates** to **3D cartesian coordinates** is given by:

$$x = \rho \cos(\phi) \cos(\lambda)$$

$$y = \rho \cos(\phi) \sin(\lambda)$$

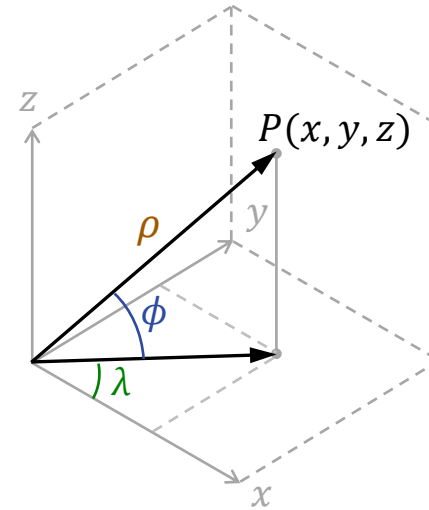
$$z = \rho \sin(\phi)$$

$$h = \text{hypot}(x, y)$$

$$\phi = \text{atan2}(z, h)$$

$$\lambda = \text{atan2}(y, x)$$

$$\rho = \text{hypot}(h, z)$$

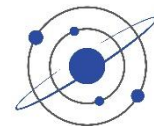


$$[x, y, z] = \text{Sph2Cart}(az, el, r)$$

$$[az, el, r] = \text{Cart2Sph}(x, y, z)$$



See Reference frames & Coordinate Systems



# Spherical $\leftrightarrow$ Cartesian transformations

```
function [x, y, z] = Sph2Cart(az, el, r)
% Transforms spherical to Cartesian coordinates [°]
%
% az is the counterclockwise angle (α) in the xy plane
% measured from the positive x axis.
% el is the elevation angle (δ) from the xy plane.

if nargin < 3
 r = 1;
end

az = az(:)'; % → line vector
el = el(:)'; % → line vector

z = r .* sind(el);
rcoselev = r .* cosd(el);
x = rcoselev .* cosd(az);
y = rcoselev .* sind(az);

% Allows the assignation of 3D points matrix directly
if narginout == 1
 x = [x; y; z]; % → output a 3 × n matrix
end

end % function
```

```
function [az, el, r] = Cart2Sph(x, y, z)
% Transforms Cartesian to spherical coordinates [°]

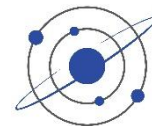
if nargin == 1
 z = x(3,:);
 y = x(2,:);
 x = x(1,:);
end

h = hypot(x, y);

r = hypot(h, z);
el = atan2d(z, h);
az = atan2d(y, x);

end % function
```

Note: those functions are defined in degrees [°]



# Spherical $\leftrightarrow$ Cartesian transformations

```
def Sph2Cart(az, el, r = 1.0):
 # Transform spherical to Cartesian coordinates, angles in °
```

```
 az = np.ravel(np.asarray(az, dtype=float))
 el = np.ravel(np.asarray(el, dtype=float))
 r = np.asarray(r, dtype=float)
```

```
 rcoselev = r * cosd(el)
 x = rcoselev * cosd(az)
 y = rcoselev * sind(az)
 z = r * sind(el)
```

```
 return np.vstack((x, y, z)) # 3xn array
 # vs return x, y, z (3-element list)
```

```
end of Sph2Cart()
```

```
def Cart2Sph(xyz):
 # Converts Cartesian coordinates to spherical lon/lat
```

```
 x = xyz[0, :]
 y = xyz[1, :]
 z = xyz[2, :]
```

```
 r = np.linalg.norm(xyz, axis=0)
```

```
 lon = atan2d(y, x) % 360.0
 lat = asind(z / r)
```

```
 return lon, lat, r
```

```
end of Cart2Sph()
```

# Types of Reference Frames

---



- The two types of reference frames are:
  - Inertial frames
  - Non-Inertial frames

# Inertial vs non-inertial

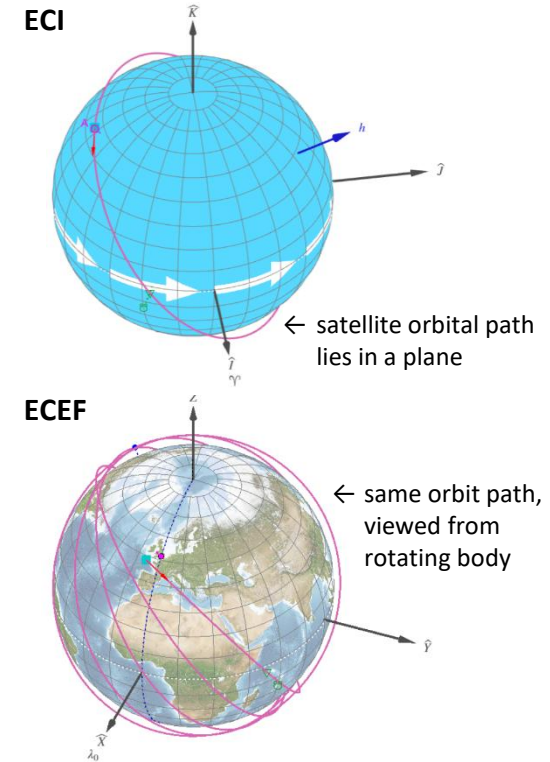


## ■ Inertial reference frame

- Non-rotating system with respect to stars
  - Non-accelerating relative to a certain reference point or origin
  - Velocity is typically non-zero, but acceleration is negligible (e.g. ECI, ECL)
- Suitable inertial frames are required for orbit description  
(Newton's second law is to be expressed in an inertial frame)

## ■ Non-Inertial reference frame

- Accelerating, mainly by rotation
- Examples:
  1. Body-fixed (ECEF): Associated with a natural body (e.g. planets, satellites)
  2. Topocentric (SEZ): Associated with an observer /object on (or near) the surface of a natural body
  3. Spacecraft: Associated with the main spacecraft structure (the “bus”)
  4. Instrument: One or more frames are usually associated with each instrument

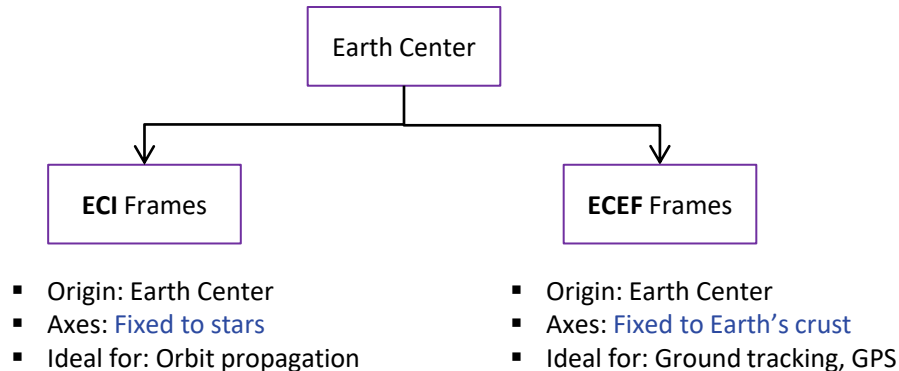


# Earth centered frames: ECI & ECEF



- In astrodynamics and satellite navigation, selecting the right reference frame is critical
- A mistake of just a few " or meters can cause a satellite to miss its target orbit entirely or degrade GPS accuracy
- At the highest level, celestial mechanics divides these systems into **two foundational categories**:
  - **ECI** (Earth-Centered Inertial) and **ECEF** (Earth-Centered, Earth-Fixed) frames

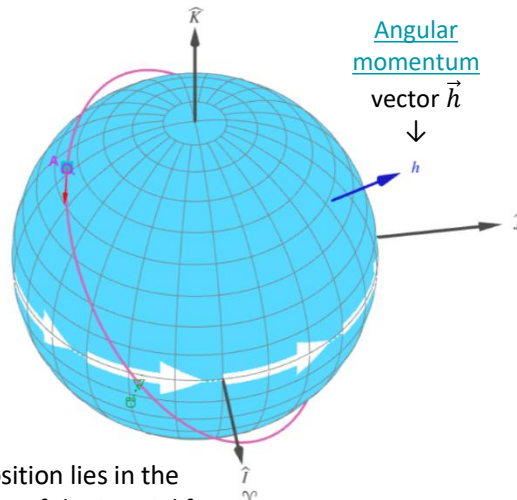
The Core Categories: ECI vs. ECEF



# Orbits101 main 3D displays: ECI & ECEF

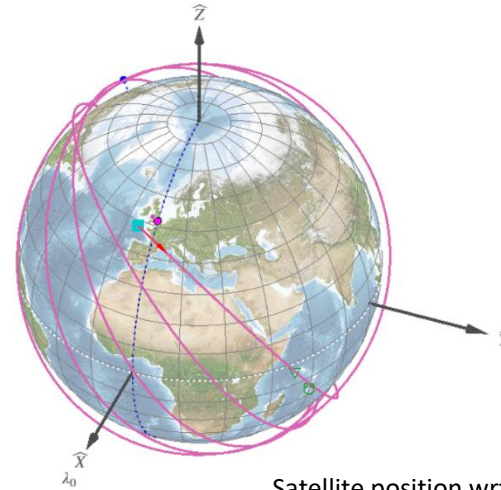


ECI – Earth-Centered Inertial frame (I,J,K)



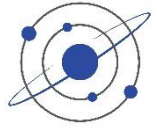
Satellite position lies in the orbital plane of the inertial frame  
(*independent of Earth's rotation*)

ECEF – Earth-centered Earth-fixed Frame (X,Y,Z)

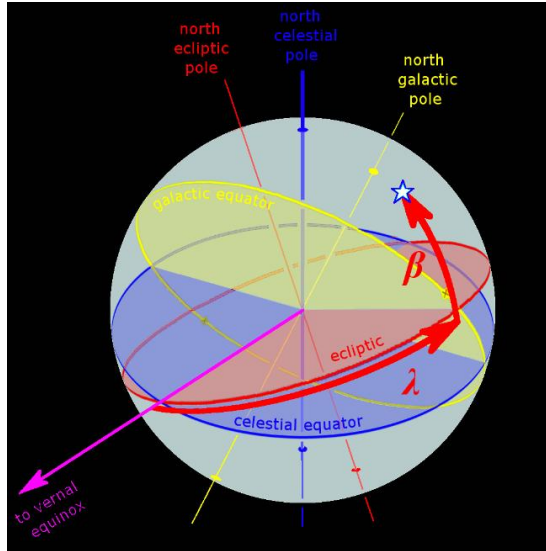


Satellite position wrt to Earth surface *as a function of time*

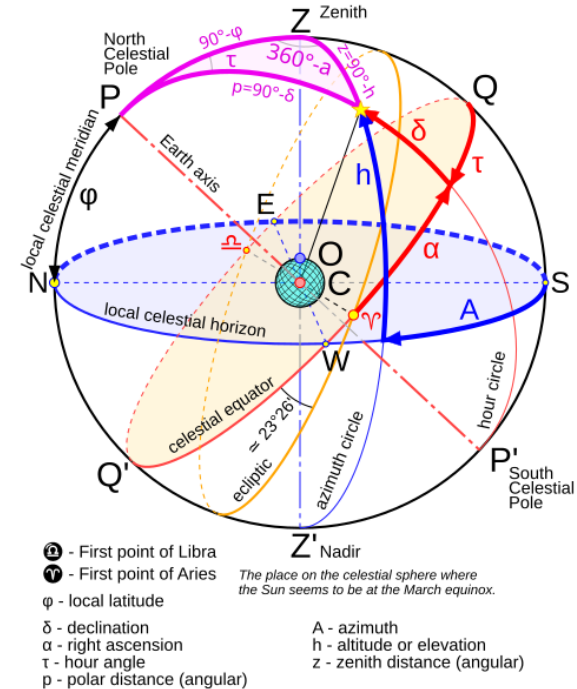
# Astronomical coordinate systems



- Animation: [https://en.wikipedia.org/wiki/Astronomical\\_coordinate\\_systems](https://en.wikipedia.org/wiki/Astronomical_coordinate_systems)



A star's ■ galactic, ■ ecliptic, and ■ equatorial coordinates, as projected on the celestial sphere. Ecliptic and equatorial coordinates share the ■ March equinox as the primary direction, and galactic coordinates are referred to the ■ galactic center

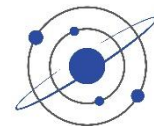




# ECL – Heliocentric-Ecliptic reference frame

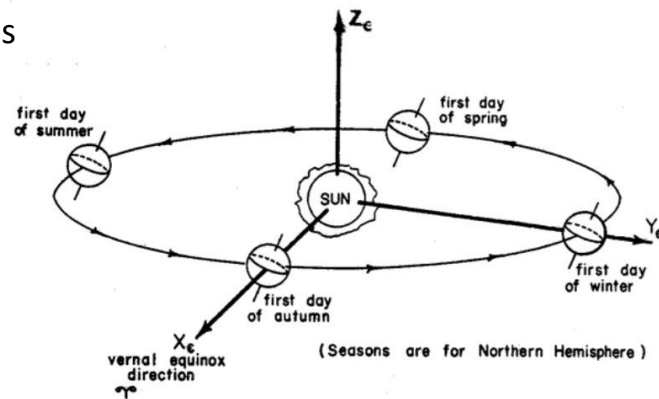


# Heliocentric-Ecliptic (ECL) reference frame

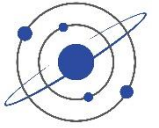


- The **Heliocentric Ecliptic (ECL)** reference frame is a 3D inertial system in astronomy to map the solar system with the **Sun at the origin**, using the ecliptic plane (Earth's orbit plane) as the fundamental reference plane
    - It defines the  $x$ -axis (principal direction) toward the First Point of Aries, or vernal equinox  $\Upsilon$ ,
    - the  $z$ -axis normal to the ecliptic, and
    - the  $y$ -axis to complete a right-handed system
  - The frame has an inertial nature and is generally fixed relative to the stars, though it can rotate slightly over long periods due to the precession of the equinoxes
- It is commonly used for analyzing and propagating the trajectories of planets, comets, asteroids, and **interplanetary spacecraft**

It is particularly convenient because most major planet orbits lie close to the ecliptic plane, so their inclinations are relatively small and their motion is easier to visualize and describe in this reference frame

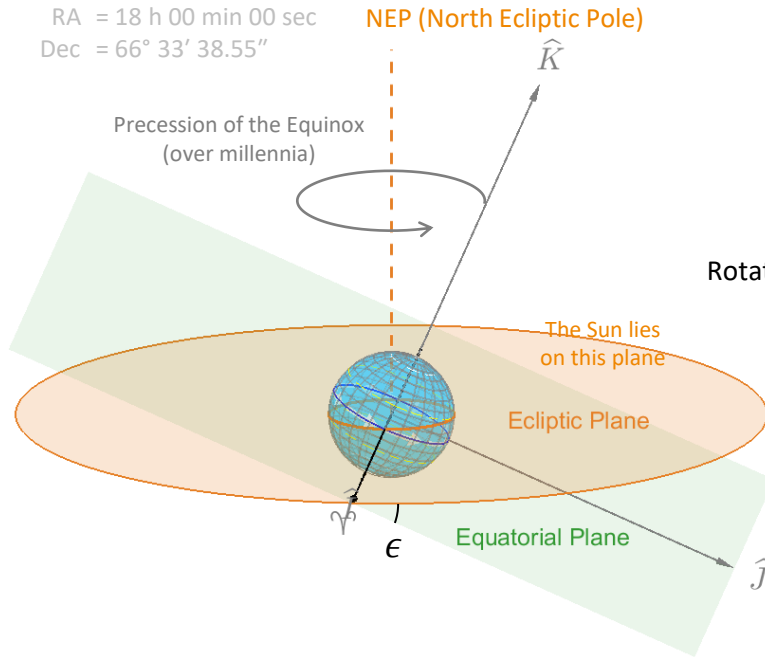


See [SOL Part I – Seasons](#)



# ECL vs ECI

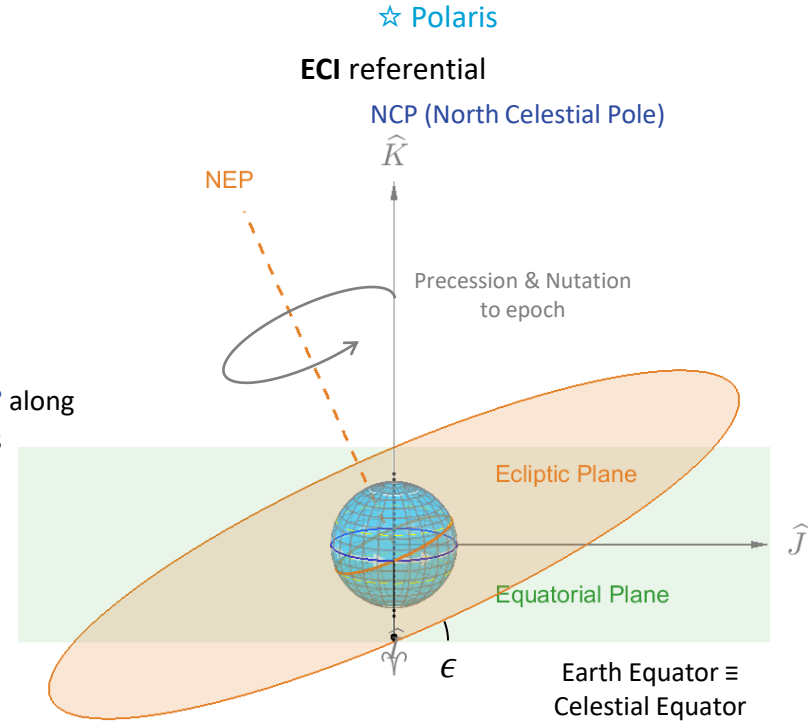
**ECL referential**

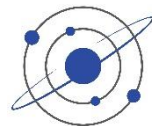


Rotation by  $+\epsilon^\circ$  along  
the  $\hat{I}$  axis



**ECI referential**





# ECL calculation from ECI

## ▪ QueryHorizons

```
k = strfind(data{1}, 'IAU76 obliquity of ');
if isempty(k)
 % EarthObliquity(datenum(2000,01,01, 12,00,00), 1)
 SV.Rot_obl = Earth.e_J2000; % = 23.4392911111111°
else
 SV.Rot_obl = str2num(data{1}(k+19 + [0:9])) / 60^2; % [°] = 23.4392911111111°
end
SV.ECL.pos = RotX(-SV.Rot_obl) * SV.ECI.pos; % - sign here
SV.ECL.vel = RotX(-SV.Rot_obl) * SV.ECI.vel; %
```

## ▪ JPL\_DE\_Evaluate

```
% JPL_DE positions are given in ECI geocentric
% Computes Ecliptic position (ECL) from ECI (not used often, as we're mainly interested in ECI positions)
Rot_e = RotX(-Earth.e_J2000); % e = 23.4392911111111°
DE.(object_name).geoc.ECL.pos(:, d) = Rot_e * DE.(object_name).geoc.ECI.pos(:, d);
DE.(object_name).geoc.ECL.vel(:, d) = Rot_e * DE.(object_name).geoc.ECI.vel(:, d);
```

## ▪ SOL\_Planets

→ All study is plotted in ECL



# Galactic coordinates

- Galactic coordinates are measured against the disc of our own Milky Way galaxy, as measured from our vantage point here inside the Orion Arm
- **Galactic Coordinates**
  - x-y plane: Galactic plane
  - x-axis: Galactic center
  - z-axis: North galactic pole
  - $\updownarrow$  Latitude  $\pm 90^\circ$  above galactic plane
  - $\leftrightarrow$  Longitude  $0^\circ$ – $360^\circ$  east from galactic center
- Astronomers have generated a series of more and more precise estimates of our galaxy's orientation over the past hundred years
- The IAU 1958 Galactic System II is believed to be accurate to within  $\pm 0.1^\circ$

[www.astro.rug.nl/software/kapteyn/celestialbackground.html](http://www.astro.rug.nl/software/kapteyn/celestialbackground.html)

The composed rotation matrix for FK5 to Galactic coordinates from celestial is:

$$J_{2000} = \begin{bmatrix} -0.054876, & -0.873437, & -0.483835, \\ 0.494109, & -0.444830, & 0.746982, \\ -0.867666, & -0.198076, & 0.455984 \end{bmatrix};$$

which is consistent with the transpose of the matrix in eq. 33 of Murray (1989)

# PQW – Perifocal reference frame

- $P, Q, W$

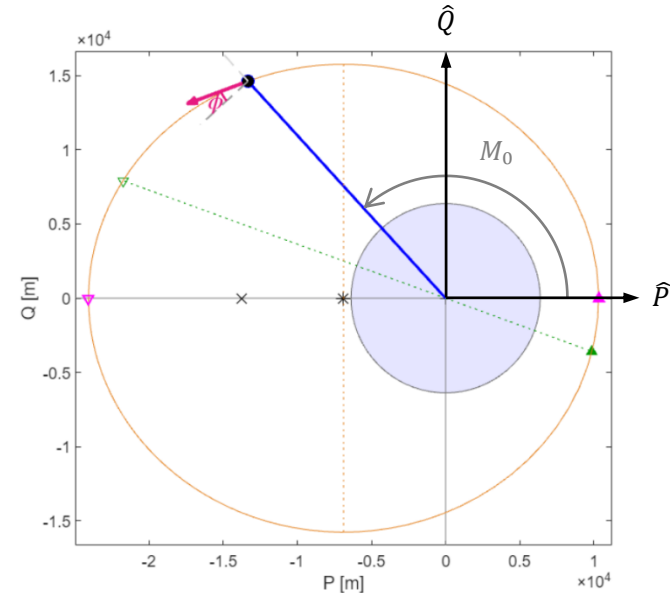


# Perifocal reference frame ( $PQW$ )



- A convenient system for processing satellite observations is the **perifocal coordinate system**
  - Used to describe **unperturbed orbits**, where the motion of the satellite is **entirely confined to a plane of motion**
  - The perifocal system **reduces orbital motion from 3 to 2 dimensions**, greatly **simplifying computations**
- The coordinates of the perifocal system are expressed as  $PQW$ 
  - the **fundamental plane** is the satellite orbit, and the **origin is at the center of the Earth**
  - the  $\hat{P}$  axis **points towards perigee**, and the  $\hat{Q}$  axis is perpendicular in the direction of satellite motion
  - the  $\hat{W}$  axis is normal to the orbit, and **point in the direction of angular momentum  $\hat{h}$**
- The position and velocity vectors exist entirely in the orbital plane and can be described exclusively in terms of  $\hat{P}$  and  $\hat{Q}$

$$\mathbf{r} = r(v) \cos(v) \hat{P} + r(v) \sin(v) \hat{Q}$$



▲ Perigee    ▼ Apogee    ▲ AN  $\Omega$     ▼ DN  $\Omega$     ● Start  $\theta_0 = 132.3^\circ$     ○ Stop  $\theta_0 = 132.3^\circ$

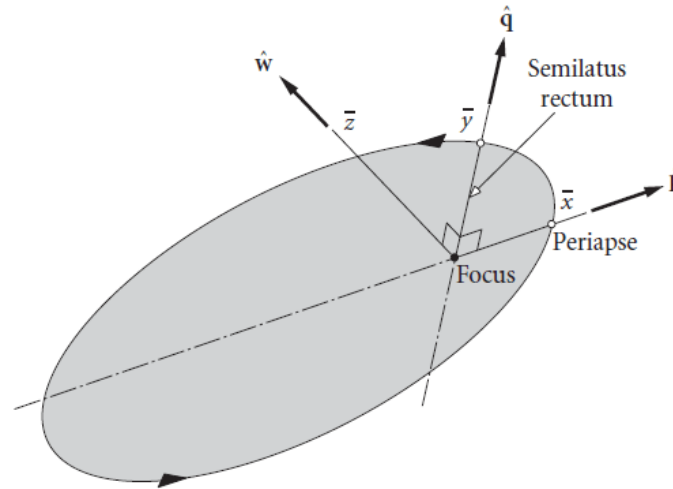


Rotations

# Perifocal reference frame



- This is popularly known as the “natural frame” for an orbit
- This frame is centered at the center of the Earth, the orbital plane is the fundamental plane ( $X - Y$  plane) itself,  $X$  axis is directed to the eccentricity vector,  $Z$  axis is in the direction of the satellite’s angular momentum which lies perpendicular to the orbital plane, and the  $Y$  axis completes the [right hand](#) set of co-ordinate axis





# ECI – Earth-Centered Inertial reference frame

- J2000
- ICRS/ICRF
- $I, J, K$

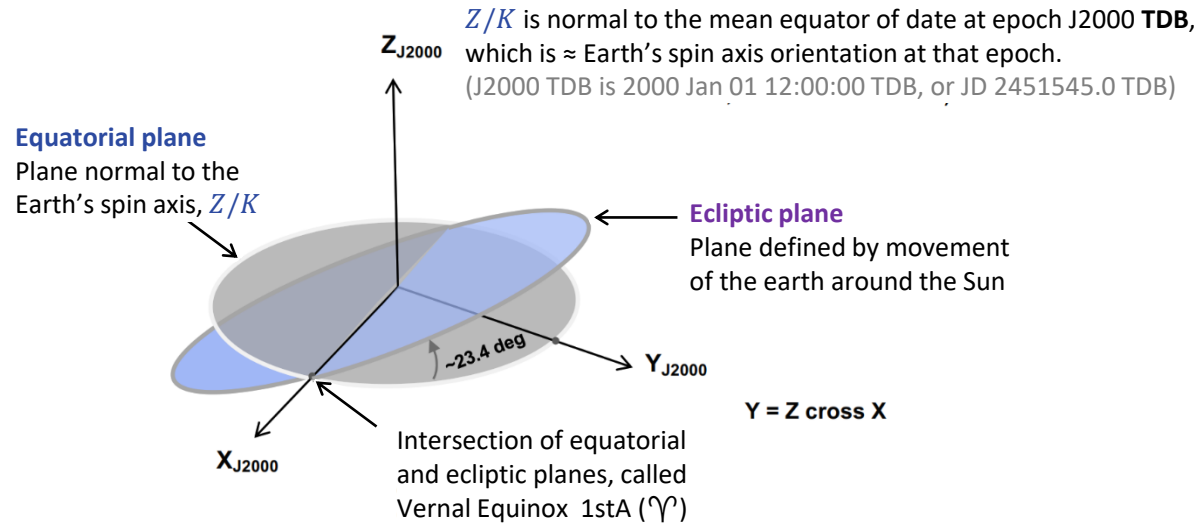
See: [Earth Centered Inertial Frames](#) (Equatorial = ECI and Ecliptic = ECL)

Alfonso Gonzalez: Orbital Mechanics with Python 39: [www.youtube.com/watch?v=nOthGWnO2Ns&t=295s](https://www.youtube.com/watch?v=nOthGWnO2Ns&t=295s) [06:57]



# The J2000 Inertial Frame

- The J2000 (aka EME2000) frame definition is based on the Earth's equator and equinox, determined from observations of planetary motions, plus other data (see NASA: [Reference Frames Coordinate Systems](#))
  - Caution: The name "J2000" is also used to refer to the zero epoch of the ephemeris time system (ET, also known as TDB)



J2000/EME2000 frame

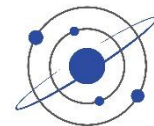


Transformation between reference frames

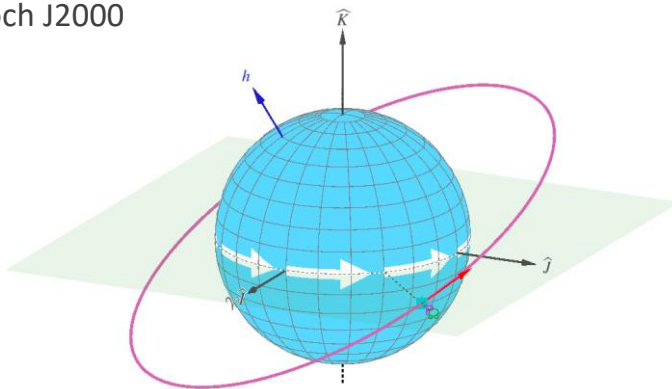
% Earth obliquity at J2000.0

e\_J2000 = EarthObliquity(datenum(2000,01,01, 12,00,00)); %  $\equiv 23.43928^\circ$

# Earth-Centered Inertial (ECI) frame



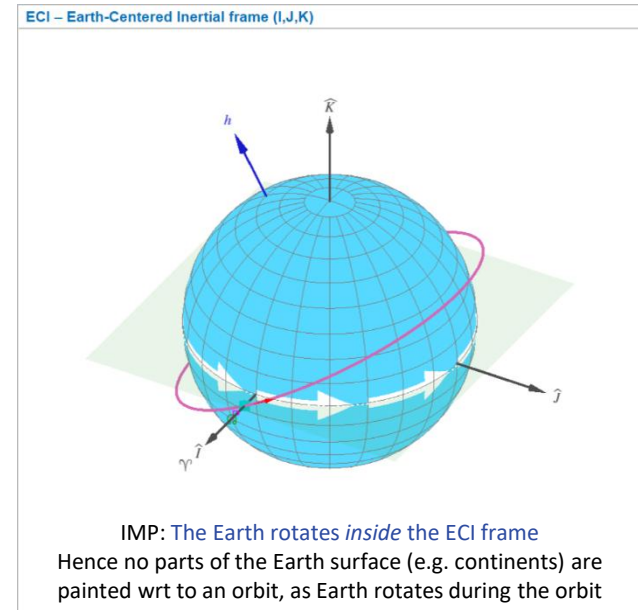
- As the name implies, this frame is inertial, meaning it has **negligible acceleration** (it can have a constant velocity though) and is **non-rotating**
- **Newton's laws are valid in inertial reference frames, making these frames convenient for orbital analysis**
- One of the most common inertial frames used for Earth-orbiting satellites is the J2000 frame
  - Origin: Fixed at the center of the Earth
  - X-axis: Points towards the vernal equinox (where the equatorial and ecliptic plane intersect) at epoch J2000
  - Y-axis: Perpendicular to X and Z axes (Z crossed with X)
  - Z-axis: Earth's rotation axis at epoch J2000

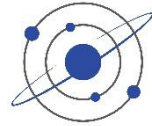


# ECI – Earth-Centered Inertial ( $IJK$ )



- In the ECI **geocentric-equatorial frame**, Newton's laws of motion apply directly, making it ideal for describing the motion of objects in space over long periods of time. Satellites in orbit around the Earth are often described using ECI coordinates, facilitating calculations of orbital elements, trajectory predictions, and space mission planning
  - It is an *inertial frame* fixed with respect to the celestial sphere
  - This frame uses capital letters  $\vec{I}, \vec{J}, \vec{K}$  to represent reference axes
    - Origin is the center of the Earth-mass (geocenter)
    - Axes fixed relative to distant stars (non-rotating)
    - The fundamental plane  $\vec{I}-\vec{J}$  is in the equatorial plane
    - The spin axis  $\vec{K}$  axis points towards the North pole
    - The **principal direction**  $\vec{I}$  axis points towards the 1stA ( $\Upsilon$ )
- IMP: Axes **are fixed** and **do not rotate** with the Earth
- Common realizations include:
    - J2000 frame
    - MOD (Mean of Date)
    - TOD (True of Date)





# Transformations between reference frames

- Using a spacecraft/satellite/planet [Orbital Elements](#), transformation between reference frames is computed with a 3D rotation matrix combining the orbit orientation parameters  $i$ ,  $\theta$ , and  $\omega$  as follows:
- Perifocal PQW  $\rightarrow$  ECI : when COEs are in ECI (e.g. for artificial satellite orbiting the Earth)

```
% 3x3 Coordinate transformation matrix (constant over all time points)
Orbit.RotPeri2ECI = RotZ(Orbit.RAAN) * RotX(Orbit.i) * RotZ(Orbit.w); % ← combined rotation matrix
Orbit.peri_P = Orbit.Rho .* cos(Orbit.T); % T is the True Anomaly ν [rad]
Orbit.peri_Q = Orbit.Rho .* sin(Orbit.T);
Orbit.ECI.pos = Orbit.RotPeri2ECI * [Orbit.peri_P; Orbit.peri_Q; 0 * Orbit.peri_P];
```

- Perifocal PQW  $\rightarrow$  ECL : when COEs are in ECL (e.g. for planets orbiting the Sun)

```
% Computes the ECL coordinates in the J2000.0 ecliptic plane, with the x-axis aligned toward the equinox:
Orbit.ECL.pos = zeros(size(Orbit.ECI.pos)); % (initializes matrix)
for i = 1:size(Orbit.ECI.pos, 2)
 RotPeri2ECL = RotZ(COE.Omega(i)) * RotX(COE.i(i)) * RotZ(COE.omega(i)); % ← COE values at each time point
 Orbit.ECL.pos(:, i) = RotPeri2ECL * Orbit.ECI.pos(:, i);
end
```

- ECL  $\rightarrow$  ECI (see [The J2000 Inertial Frame](#), [SQL Part I – Ecliptic plane & Obliquity](#))

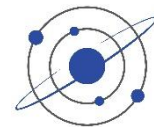
```
% Computes the equatorial coordinates in the "ICRF", or "J2000.0 frame":
RotECL2ECI = RotX(+Earth.e_J2000); % Earth's obliquity at J2000.0 = 23.4392794° → use negative sign for ECI→ECL
Orbit.ECI.pos = RotECL2ECI * Orbit.ECL.pos;
```

# ECEF – Earth-Centered, Earth-Fixed Reference frame

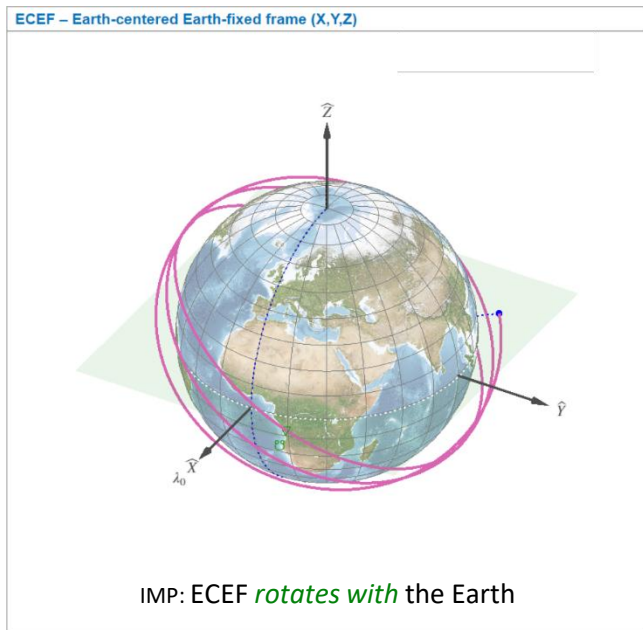
- $X, Y, Z$
- For Ground Tracks ( $\lambda / \varphi$ )



# ECEF – Earth-Centered, Earth-Fixed (XYZ)



- The Earth-centered, Earth-fixed (ECEF) frame is a *non-inertial frame*, with its origin at the center of the Earth
- The main difference from the ECI is that **the axes of the ECEF frame are fixed relative to Earth's surface and rotate with the planet at the same angular rate**
- This frame uses capital letters  $\vec{X}, \vec{Y}, \vec{Z}$  to represent reference axes
  - Origin is the center of the Earth-mass (geocenter)
  - The fundamental plane  $\vec{X}-\vec{Y}$  is in the equatorial plane
  - The spin axis  $\vec{Z}$  axis points towards the North pole
  - Since the ECEF rotates with the Earth, the  $\vec{X}$  axis always points the **Prime Meridian**
- Every 24 hours, the ECEF and ECI are aligned
  - Thus, the angular distance between  $\vec{I}$  and  $\vec{X}$  is  $\theta_G$ ,  
which increases at the rate  $\omega_{\oplus}$
- ECEF coordinates are commonly used in satellite navigation systems such as GPS (Global Positioning System) to provide accurate positioning information for ground-based receivers



# ECI $\leftrightarrow$ ECEF transformation



- The ECI to ECEF transformation converts a position (or state vector) from an inertial frame fixed in space to a frame that rotates with the Earth
  - This is fundamental in astrodynamics for linking orbital motion to ground locations
- The transformation is essentially a time-dependent rotation about Earth's spin axis
- In its simplest form, the transformation is a rotation about the z-axis:

$$r_{\text{ECEF}} = R_z(\theta(t)) r_{\text{ECI}}$$

- The angle  $\theta$  is the Earth rotation angle, derived from time
- Historically, this uses Greenwich Sidereal Time (GST):

$$\theta = \text{GMST}(t)$$

- This accounts for Earth's rotation relative to inertial space

→ The precise ECI to ECEF transformation also includes precession, nutation, and rotation effects, as well as pole wander and frame corrections (see later)



# Sidereal time & GMST



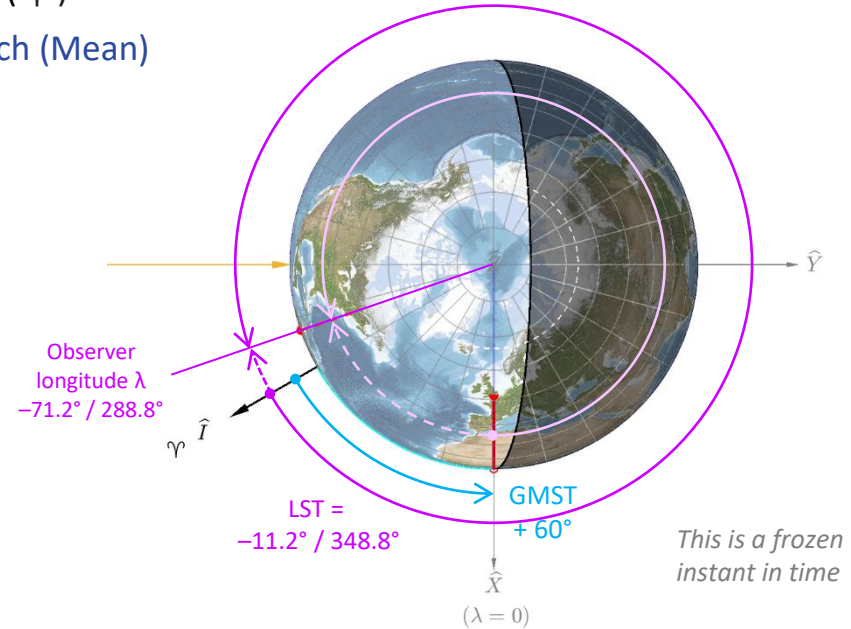
- **Sidereal time**, unlike solar time, is inertial and is measured as the angle between a given longitude line and the 1stA/vernal equinox ( $\Upsilon$ )
- The time associated with longitude  $0^\circ$  is defined as **Greenwich (Mean) Sidereal Time (GMST)**

GMST  $\equiv$  rotation from  $\Upsilon$  ( $\vec{I}$ ) to  $\lambda=0^\circ$  ( $\vec{X}$ ) at given Epoch

→ Relates ECI to ECEF reference frames (function of time)

- The sidereal time at any given longitude is defined as **Local Sidereal Time, LST**

$$LST = GMST + \lambda$$



ECI  $\leftrightarrow$  ECEF

# Sun longitude vs Sun Right Ascension



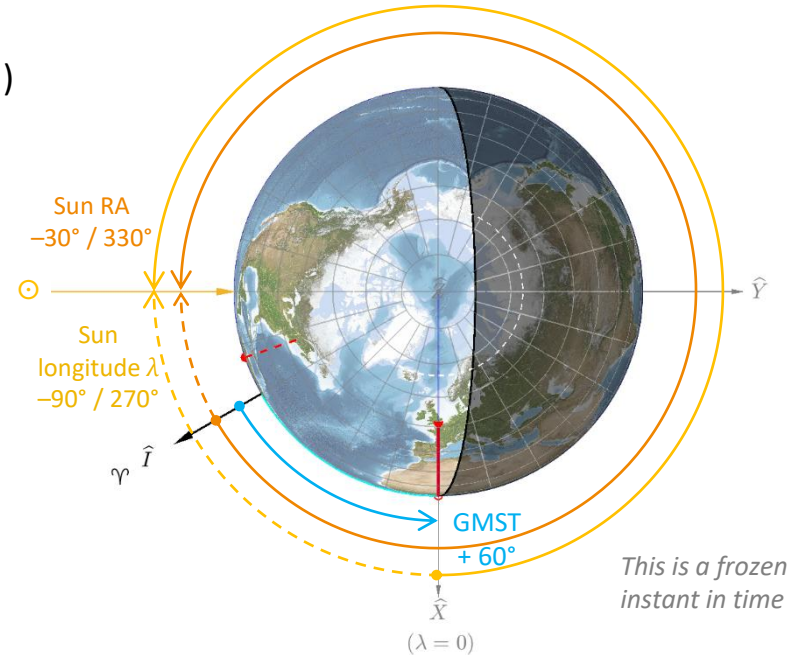
- **Sun longitude** ( $\lambda$ ) is angle in ECEF measured to Prime Meridian ( $\lambda = 0^\circ$ )  
→ Changes by  $\approx 360^\circ$  in 24 hours, following Earth rotation
- **Sun Right Ascension (RA)** is angle in ECI measured from 1stA ( $\gamma$ )  
→ Changes by  $\approx 360^\circ$  in 1 year, following Earth revolution

“Time of meridian transit”

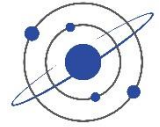
“(Apparent) Ecliptic Longitude” ( $\neq$  Astrometric)

$$\odot_{RA} = GMST + \odot_{\lambda}$$

Note: both are independent of observer's location



*This is a frozen instant in time*



# GMST computation

- Translating between mean solar time and sidereal time requires the knowledge of the vernal equinox at some previous date, and a propagation model must be used to compute this important value

```
function GMST = ComputeGMST(epoch)
% Computes Greenwich Mean Sidereal Time (GST) in degrees (iau-82)
% GMST ≡ rotation from First Point of Aries γ to λ=0° (Greenwich) at given Epoch

JD_UT1 = JulianDate(epoch); % Julian date of UT1 (days from 4713 BC)

% Julian centuries from jan 1, 2000 12 h epoch (UT1)
T_UT1 = (JD_UT1 - 2451545.0) / 36525.0;
T_UT1 = (JD_UT1 - JulianDate([2000,01,01, 12,00,00])) / 36525.0;

% Vallado 2013, Section 3.5.2, Eq. (3-47) [pp. 184-188] → from Astronomical Almanac 1984
GMST_s = 67310.54841 + (876600 * 3600 + 8640184.812866) * T_UT1 + 0.093104 * T_UT1.^2 - 6.2e-6 * T_UT1.^3; % [s]

% 360/86400 = 1/240, to deg, to rad
GMST = mod(360 * temp / (24*60^2), 360);

(→ See other formulas inside the code)

end % function
```

- Contributions of the quadratic and cubic terms are usually small but measurable for precise applications

# GMST vs GAST



- The choice between GMST and GAST depends entirely on how accurately you need to know the Sun's position, and whether your solar coordinates account for the nutation of Earth's axis
  - Use GAST if you are calculating the True/Apparent position of the Sun (high precision)
  - Use GMST if you are calculating the Mean position of the Sun (lower precision or simplified models)

| Feature            | GMST (Greenwich Mean)                                                           | GAST (Greenwich Apparent)                                                       |
|--------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| What it measures   | Uniform rotation relative to a smoothly moving (mean) equinox.                  | Real-time rotation relative to the true, shifting (apparent) equinox.           |
| Nutation included? | No. It ignores the tiny gravitational wobbles caused by the Moon and Sun.       | Yes. It adds the Equation of the Equinoxes to GMST to correct for nutation.     |
| Maximum Difference | Differs from GAST by up to $\pm 1.2$ seconds of time (or $\sim 18$ arcseconds). | Differs from GMST dynamically over an 18.6-year cycle.                          |
| Primary Use Case   | Broad structural conversions, almanacs, and simplified tracking algorithms.     | Precision astrometry, satellite tracking, and high-accuracy telescope pointing. |

# GMST vs GAST



## ■ When to Use GMST

- You should use GMST if you are working with Mean Solar Elements (such as low-precision analytical formulas or standard orbital elements like TLEs for satellites)
- If your formulas for the Sun's position provide the Mean Longitude or Mean Right Ascension (which also ignore nutation), using GAST would actually introduce an error. The two “mean” values belong together because their errors cancel each other out
- Examples: [Standard sunrise/sunset apps](#), solar panel tilts, or basic hobbyist astrophotography

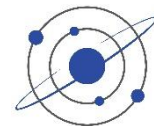
## ■ When to Use GAST

- You must use GAST when you are calculating the Apparent Right Ascension ( $\alpha$ ) and Declination ( $\delta$ ) of the Sun. High-precision ephemerides (like JPL's DE440 or the standard IAU 2000/2006 models) calculate where the Sun actually appears in the real sky, including all gravitational distortions and nutation
- To map those precise coordinates to your actual horizon (Altitude and Azimuth), your Earth rotation angle must also include those exact same nutation distortions
- Examples: Solar eclipse mapping, professional observatory tracking, and space mission navigation.

$$\text{Local Hour Angle (LHA)} = \text{GAST} + \text{Local Longitude } (\lambda) - \text{Sun's Apparent Right Ascension (RA)}$$

# The Bridge: Earth Orientation Parameters (EOP)

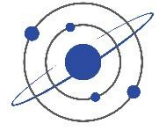
---



- To transform coordinates from an ECI frame to an ECEF frame, one must account for the complex, irregular motion of the Earth in space. This transformation is driven by four factors, collectively tracked as [Earth Orientation Parameters \(EOP\)](#):
  1. **Precession:** The long-term, slow cone-like sweeping of Earth's rotational axis ( $\approx 26,000$ -year cycle)
  2. **Nutation:** The short-term, periodic “wobble” superimposed on precession, mostly caused by the Moon's gravitational pull
  3. **Earth Rotation** (GMST/GAST or ERA): The spinning of the Earth on its axis. Because Earth's spin rate fluctuates due to tidal friction and core movements, this requires precise atomic time tracking ( $\Delta\text{UT1}$ )
  4. **Polar Motion:** The physical shifting of the Earth's crust relative to its rotational axis (the North Pole moves around in an irregular circle the size of a tennis court over the span of months)

See [SQL Part I: EOP – Earth Orientation Parameters](#)

- **Earth Orientation Parameters (EOP)**
- Earth rotational velocity variation
- Polar Motion, ComputePNE
- UTC – Coordinated Universal Time, Leap seconds



# Transformation from ECI to ECEF

%== Simplified transformation only taking into account Earth rotation to compute ECEF  
 % Precession, nutation, polar motion ignored  
 % → approximation to true ECEF, allowing initial good estimates of  $\lambda$  &  $\phi$   
 % e.g. ISS: Max  $\Delta R$  error computation = 34.9 m, 0.52%,  $\Delta\lambda$ :  $0.24 \pm 0.08^\circ$ ,  $\Delta\phi$ :  $-0.00 \pm 0.13^\circ$   
 % Given a satellite's position vector  $R$ , we find its RA/Dec relative to the geocentric equatorial XYZ frame, which is  
 % fixed in space. The Earth rotates at an angular velocity  $\omega_{\text{Earth}}$  relative to this system

```
for i = 1:length(Orbit.ECI.pos) % for each point in orbit
```

```
 if 0 % A) Approximation: GMST at origin, rotation with Earth.omega_deg_s * Δt
```

```
 ECEF_angle = Earth.omega_deg_s * (Orbit.time_s(i) - Orbit.Epoch.dtP_s); % [$^\circ/s \times s = ^\circ$] rotation in time
```

```
 ECEF_angle = -ECEF_angle; % → negative because must cancel ECEF Earth rotation during the orbit
```

```
 GMST_0 = ComputeGMST(Orbit.Epoch.Time); % [$^\circ$] Greenwich Sidereal Time at Epoch
```

```
 ECEF_angle = ECEF_angle - GMST_0; % removes offset wrt start time
```

```
 else % B) True angle: GMST at each point
```

```
 GMST = ComputeGMST(Orbit.Time(i));
```

```
 ECEF_angle = -GMST; % → negative because must cancel ECEF Earth rotation during the orbit
```

```
 end
```

```
 Earth_Rot = RotZ(ECEF_angle); % counter-clockwise rotation along Earth rotation axis
```

```
 Orbit.ECEF.pos(:, i) = Earth_Rot * Orbit.ECI.pos(:, i); % Position (3 x n)
```

```
 Orbit.ECEF.vel(:, i) = Earth_Rot * Orbit.ECI.vel(:, i); % Velocity
```

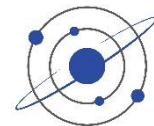


GMST



Time vector and Anomalies

# ECI $\leftrightarrow$ ECEF transformation (continued)



- In precise astrodynamics, the transformation is not just one rotation. It includes:

$$r_{\text{ECEF}} = W \cdot R \cdot N \cdot P \cdot r_{\text{ECI}}$$

where:  $P$  = **Precession**, long-term motion of Earth's axis

$N$  = **Nutation**, short-period oscillations

$R$  = **Earth rotation**, sidereal rotation (main effect)

$W$  = **Polar motion**, small wobble of Earth's crust

- For most applications (e.g., basic orbit propagation):  $r_{\text{ECEF}} = R_z(\theta) r_{\text{ECI}}$ 
  - This is accurate to meters to kilometers depending on time span

- The velocity transform is:

$$v_{\text{ECEF}} = R_z(\theta) v_{\text{ECI}} - \omega_{\oplus} \times r_{\text{ECEF}}$$

where  $\omega_{\oplus}$  is the Earth rotation vector. This extra term accounts for the rotating frame.





# Transformation from ECI to ECEF (precise)

```
% PNE is structure containing: Precession, Nutation, Equinoxes (Polar Motion-Wandering)
% for the precise computation of ECEF from ECI
% → Here the E refers to Equinoxes correction, not the Earth's rotation for ECEF calculation like used in ECI_to_ECEF
Orbit.PNE = ComputePNE(Orbit.Epoch.Time);

 LOD = Excess Length Of Day
 Pi = Polar motion matrix
 NPB = bias-precession-nutation matrix

% Earth rotation matrix
GMST_0 = ComputeGMST(Orbit.time(i)); % [°]
Theta = RotZ(-GMST_0);

Omega = RAD(Earth.omega_deg_s) - 0.843994809 * 1e-9 * Orbit.PNE.LOD; % IERS

S = zeros(3);
S(1,2) = 1;
S(2,1) = -1; % Derivative of Earth rotation
dTheta = Omega * S * Theta; % matrix [1/s]

U = Orbit.PNE.Pi * Theta * Orbit.PNE.NPB; % ICRS to ITRS transformation
dU = Orbit.PNE.Pi * dTheta * Orbit.PNE.NPB; % Derivative [1/s]

% Transformation from ECI to ECEF
Orbit.ECEF.pos(:, i) = U * Orbit.ECI.pos(:, i);
Orbit.ECEF.vel(:, i) = U * Orbit.ECI.vel(:, i) + dU * Orbit.ECI.pos(:, i);
```

# Exercises

---



- → see file [Exercises ECI, ECEF.docx](#)

# Topocentric-horizon Reference frame

- **SEZ**, ENU, NED
- For astronomy/satellite tracking, geodesy, and aircraft / navigation / guidance



# Topocentric/Topographical Frame



The most common choice depends on the community:

Astronomy / satellite tracking: SEZ or ENU ← Vallado only uses SEZ

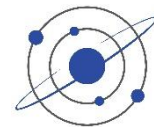
Geodesy / GNSS / mapping: ENU

Aircraft / navigation / guidance: NED

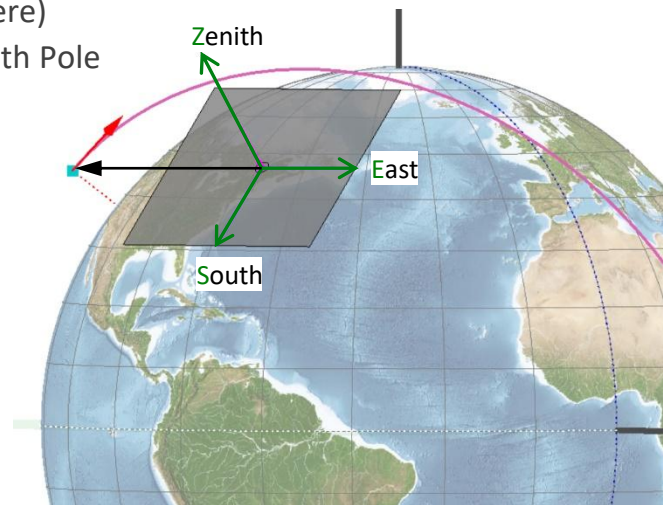
For **ground-based topocentric referencing of satellites**, I would say:

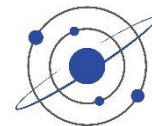
ENU is now the most broadly used practical convention, but SEZ remains very common in classical astrodynamics textbooks.

# Topocentric-Horizon (SEZ) on the Earth surface



- The **topocentric-horizon system**, or South-East-Zenith (SEZ), is a non-inertial local reference frame with its **origin at a specific point on the Earth's surface**, such as a **ground-based observer's position**
  - The surface at the observer position forms the fundamental plane, that is tangent to the surface of the Earth
  - The origin of this coordinate system is fixed to the observer; The SEZ system rotates with the site
  - At a given instant, the position of this coordinate system can be determined relative to the ECEF frame by specifying the **longitude angle  $\lambda$**  and the **geodetic latitude angle  $\varphi$**
  - The  **$\hat{S}$**  axis points due south from the site (even in the Southern Hemisphere)
  - The  **$\hat{E}$**  axis points east from the site and is undefined for the North or South Pole
  - The  **$\hat{Z}$**  axis (zenith) points radially outward from the site, along the site's local vertical; is **normal to the reference spheroid** (surface of the Earth), parallel to the gravity gradient\*, and does not necessarily pass through the center of the Earth
- Topocentric coordinates are often used in astronomy and terrestrial navigation for observations of stars, planets, and other celestial bodies from specific observation points on the Earth's surface





# Local topocentric frames: SEZ, ENU, NED + AER

- Once the observer position in ECEF/ITRF is known, one can transform satellite or celestial vectors into a **local tangent frame** (topocentric)
- The common local frames are
  - **SEZ**: South, East, Zenith      Common in classical astrodynamics texts
  - **ENU**: East, North, Up,      Common in geodesy and GPS
  - **NED**: North, East, Down      Common in aerospace navigation and aircraft dynamics
  - **AER**: Azimuth, Elevation, Range      Spherical pointing angles (see Cart2Sph)
- For a topocentric vector:

$$\rho_{ECEF} = r_{\text{object}, ECEF} - r_{\text{site}, ECEF}$$

- Then you rotate it into SEZ or ENU, then compute:

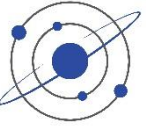
Azimuth, Elevation(/Altitude), Range

→ Note that these are not global reference frames; they are **observer-local frames**

```
[Azim, Elev, slantRange] = ECEF_to_AER(R_ECEF, Observer);
```

# Skip the rest for the moment...

---



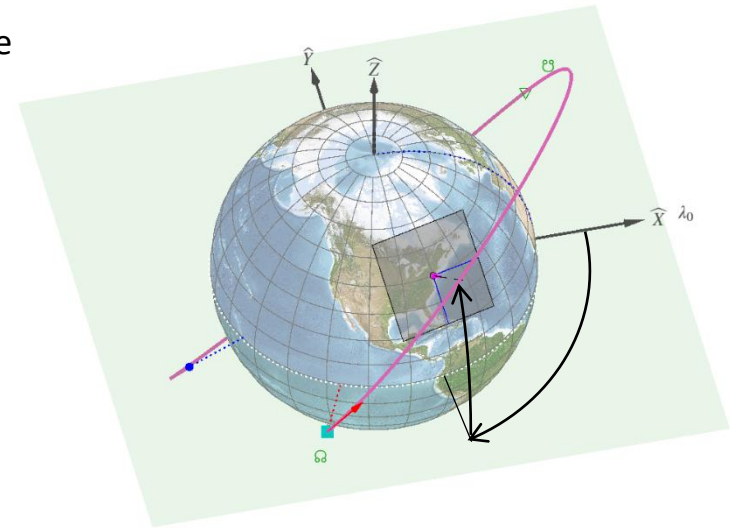
- Frame instances
- Transformation chains

→ [Orbital Elements](#)

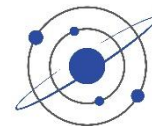
# Topographical Frame



- The topographical frame is defined much different than the other four reference frames
- The frame's origin is at the surface of the planet
  - The SEZ – South, East, Z (up) – is used in astronomy and satellite tracking
  - The ENU – East-North-Up frame, is a typical topographical frames
- It is a *non-inertial frame* that rotates on the surface of the plane
  - Origin: Fixed at or near the Earth's surface
  - S-axis: Points South towards the axis of rotation
  - E-axis: Points East, perpendicular to S and Z axes ( $S \times Z$ )
  - Z-axis: Points up, or parallel to the gravity gradient
- Applications:
  - Rocket launch tracking,
  - Telecommunications stations,
  - Landers or rovers







# SEZ derivation

For a ray vector expressed in ECEF, the clean two-rotation transformation is usually made into the local SEZ frame:

$$\boxed{\text{ECEF} \rightarrow \text{SEZ}}$$

where:

$$\text{SEZ} = \begin{bmatrix} S \\ E \\ Z \end{bmatrix} = \begin{bmatrix} \text{South} \\ \text{East} \\ \text{Zenith} \end{bmatrix}$$

Let the observer geodetic latitude and longitude be:

$$\phi = \text{observer latitude}$$

$$\lambda = \text{observer longitude, positive east}$$

and let the ECEF line-of-sight vector be:

$$\rho_{\text{ECEF}} = \mathbf{r}_{\text{target,ECEF}} - \mathbf{r}_{\text{observer,ECEF}}$$

Using the common astrodynamics passive-rotation convention,

$$R_z(\lambda) = \begin{bmatrix} \cos \lambda & \sin \lambda & 0 \\ -\sin \lambda & \cos \lambda & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$R_y\left(\frac{\pi}{2} - \phi\right) = \begin{bmatrix} \sin \phi & 0 & -\cos \phi \\ 0 & 1 & 0 \\ \cos \phi & 0 & \sin \phi \end{bmatrix}$$

Then:

$$\boxed{\rho_{\text{SEZ}} = R_y\left(\frac{\pi}{2} - \phi\right) R_z(\lambda) \rho_{\text{ECEF}}}$$

which gives:

$$\begin{bmatrix} \rho_S \\ \rho_E \\ \rho_Z \end{bmatrix} = \begin{bmatrix} \sin \phi \cos \lambda & \sin \phi \sin \lambda & -\cos \phi \\ -\sin \lambda & \cos \lambda & 0 \\ \cos \phi \cos \lambda & \cos \phi \sin \lambda & \sin \phi \end{bmatrix} \begin{bmatrix} \rho_x \\ \rho_y \\ \rho_z \end{bmatrix}$$

Then the topocentric range, azimuth, and elevation are:

$$\rho = \sqrt{\rho_S^2 + \rho_E^2 + \rho_Z^2}$$

$$\text{El} = \sin^{-1}\left(\frac{\rho_Z}{\rho}\right)$$



or, numerically better,

$$\text{El} = \text{atan2}\left(\rho_Z, \sqrt{\rho_S^2 + \rho_E^2}\right)$$

For azimuth measured from **North toward East**:

$$\text{Az} = \text{atan2}(\rho_E, -\rho_S)$$

and reduce to:

$$\boxed{0^\circ \leq \text{Az} < 360^\circ}$$

because in the SEZ frame:

$$N = -S$$

so:

$$\text{Az} = \text{atan2}(E, N) = \text{atan2}(E, -S)$$

# Frame instances

- ICRS, GCRS, TETE, TEME, ITRS, etc.
- These various acronyms represent fundamental, interconnected reference frames used in space mission design, satellite tracking, and geodesy



# Summary of the main frames categories



Here are the main frames of references; there are numerous others not described here (see references for list)

— With few exceptions, most reference frames are simply variations of the ones presented here

| Frame of reference<br>general class           | Inertial | Origin                                | Fundamental<br>Plane | Reference Direction                            | Horizontal Angle                             | Vertical Angle                               |
|-----------------------------------------------|----------|---------------------------------------|----------------------|------------------------------------------------|----------------------------------------------|----------------------------------------------|
| ECL – Ecliptic                                | Yes      | Barycentric ( $SSB$ )<br>Heliocentric | Ecliptic             | 1stA/Vernal Equinox                            | Ecliptic longitude $\lambda$                 | Ecliptic Latitude $\beta$                    |
| Perifocal ( $PQW$ )                           | Yes      | Central body                          | Orbital plane        | Periapsis $P$<br>eccentricity vector $\vec{e}$ | True anomaly $\nu$                           | N/A or $Q$                                   |
| ECI – Earth-centered<br>inertial ( $IJK$ )    | Yes      | Geocentric                            | Equatorial plane     | 1stA/Vernal Equinox                            | Right Ascension $\alpha$                     | Declination $\delta$                         |
| ECEF Earth-centered,<br>Earth-Fixed ( $XYZ$ ) | No       | Geocentric                            | Equator              | Prime Meridian                                 | Geodetic Longitude $\lambda$ /<br>hour angle | Geodetic Latitude $\varphi$ /<br>declination |
| Topocentric Horizon<br>( $SEZ$ )              | No       | Topocentric                           | Local Horizon        | North/South point $S$                          | Azimuth $Z$                                  | Elevation $E$                                |

Note: The present study mainly consists in the analysis of artificial satellites orbiting the Earth, described in the last 3 frames

# Summary ECI/ECEF, and continuation



- ECI and ECEF are not single frames: they are large *categories*:

**ECI** = Earth-centered, non-Earth-rotating celestial frames

**ECEF** = Earth-centered, Earth-fixed terrestrial frames

- Within these categories, there are **several practical frame instances** for practical applications:
  - **ICRF, GCRF, TETE/TEME, ITRF**; and intermediate or older frames like TOD, MOD, PEF, CIRS, TIRS, etc.
- First distinction: reference *system* versus reference *frame*
  - A **reference system** is the theoretical definition: origin, metric, axes, time scale, relativistic conventions, etc.
  - A **reference frame** is the practical realization of that system through catalogs, stations, ephemerides, or numerical models

| Reference System | Practical Frame (realization)                         |
|------------------|-------------------------------------------------------|
| ICRS             | ICRF                                                  |
| GCRS             | GCRF (often treated as ICRF-aligned geocentric frame) |
| ITRS             | ITRF                                                  |

(note: Astropy generally refers to the System and not the Frame)

- The IAU/SOFA astrometry chain explicitly distinguishes the transformations among ICRS, GCRS, and ITRS, **which is the modern rigorous path from inertial celestial coordinates to terrestrial/observed coordinates.**

# The broad ECI family



- An ECI frame has its origin at or near the Earth's center but does not rotate with the Earth
- It is used for: Orbit propagation, inertial dynamics, satellite state vectors, ephemerides, and celestial pointing
- But “ECI” alone is ambiguous; it can mean different things depending on the software or standard
- The ECI frame has common possibilities, and the precise transformation depends on the source frame
- Important ECI-like frames include:

| Frame                  | Meaning                                             | Main use                                                         |
|------------------------|-----------------------------------------------------|------------------------------------------------------------------|
| <b>GCRS</b> / GCRF     | Geocentric Celestial Reference System / Frame       | Modern high-precision geocentric inertial frame                  |
| <b>J2000</b> / EME2000 | Mean equator and equinox of J2000 ( $\approx$ GCRS) | Legacy astrodynamics, SPICE/STK-style work                       |
| MOD                    | Mean of Date                                        | Mean equator and mean equinox of date                            |
| TOD<br><b>TETE</b>     | True of Date<br>True Equator, True Equinox          | True equator and true equinox of date<br>Apparent RA/Dec of date |
| CIRS                   | Celestial Intermediate Reference System             | Modern CIO-based apparent celestial frame                        |
| <b>TEME</b>            | True Equator, Mean Equinox                          | SGP4/TLE output frame                                            |

# Specific frames and real-world instances

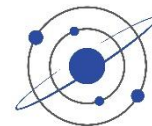


- Different specialized reference frames are used depending on how strictly one needs to account for precession, nutation, and rotation

## Modern Standards (IAU 2000/2006 Resolutions)

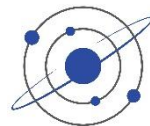
- Modern astrodynamics has largely shifted away from frames defined by shifting equinoxes to kinematically rigid, mathematically precise frames fixed to distant quasars
- ⊙ **ICRS/ICRF** (International Celestial Reference System/Frame)
  - The gold standard of **ECI** frames. Its origin is actually at the **Solar System Barycenter** (center of mass of the solar system), and its axes are defined by the measured positions of hundreds of distant, active galactic nuclei (quasars)
  - It is highly stable and does not change over time
- ⊕ **GCRS/GCRF** (Geocentric Celestial Reference System/Frame)
  - This is the **ECI** counterpart to the ICRF, shifted to be **centered on the Earth's center of mass**. It is the modern replacement for older inertial frames like FK5 or J2000. Does not rotate with Earth, nor moves with precession or nutation
- 🌐 **ITRS/ITRF** (International Terrestrial Reference System/Frame)
  - The ultimate **ECF** frame. It is maintained by a global network of GPS, SLR, and VLBI stations that track continental drift
  - It accounts for all tectonic movements, meaning coordinates on Earth's surface are accurate down to the mm level

# Reference frames list



XXXS/XXXF → Reference System / Reference Frame

| Frame name                                    | Category               | Category / Origin    | Rotates with $\oplus$ ? | Standard Reference Axis                            | Main meaning ; Primary use                                                                                                        |
|-----------------------------------------------|------------------------|----------------------|-------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>ICRS/ICRF</b>                              | ECI Barycentric        | <b>Celestial SSB</b> | No                      | Quasars / Extra-galactic sources                   | Fundamental quasi-inertial celestial system;<br>High-precision deep space navigation & astronomy<br>→ Horizons/DE SSB coordinates |
| BCRS                                          | Celestial relativistic | SSB                  | No                      |                                                    | Barycentric relativistic coordinate system                                                                                        |
| HCRS                                          | Celestial              | Sun                  | No                      |                                                    | Heliocentric ICRS-aligned system                                                                                                  |
| <b>GCRS/GCRF</b><br>Geocentric Celestial      | <b>ECI Geocentric</b>  | <b>Earth center</b>  | No                      | Aligned with ICRF →                                | Modern geocentric celestial system for satellite orbit propagation (LEO to GEO)                                                   |
| <b>J2000 / EME2000</b>                        | ECI Geocentric         | Earth center         | No                      | Mean Equinox at Epoch J2000                        | Mean equator/equinox of J2000; Legacy mission planning (FK5), basic orbital mechanics                                             |
| MOD / TOD<br><b>TETE</b>                      | ECI Geocentric         | Earth center         | No                      | True Equator, Mean/True Equinox of the target date | Mid-transformation steps in classical astronomical math<br>→ for Ephemeris calculations                                           |
| CIRS                                          | ECI-like apparent      | Earth center         | No                      | Celestial Intermediate                             | CIO-based apparent celestial system                                                                                               |
| TIRS                                          | ECEF terrestrial       | Earth center         | Rotating                | Terrestrial Intermediate                           | Earth rotation applied, before polar motion, <i>intermediate</i>                                                                  |
| PEF                                           | ECEF terrestrial       | Earth center         | Rotating                | Greenwich Meridian (No Polar Motion)               | Older pseudo-Earth-fixed frame; <i>Intermediate</i> step between space vectors and ground coordinates                             |
| <b>TEME</b>                                   | ECI Geocentric         | Earth center         | No                      | NORAD defined Equinox/Equator                      | <i>Mandatory</i> baseline frame for SGP4 / TLE processing                                                                         |
| <b>ITRS/ITRF</b><br>International Terrestrial | <b>ECEF Geocentric</b> | Earth center         | Yes                     | Prime Meridian & Earth Crust                       | Theoretical terrestrial system; Precise GPS, ground tracking, geophysics, and map mapping                                         |
| WGS84 ECEF                                    | ECEF/geodetic          | Earth center         | Yes                     |                                                    | GPS/geodetic engineering frame                                                                                                    |
| <b>SEZ / ENU / NED</b>                        | Local topocentric      | Observer             | Yes                     | Origin at observer / various axes                  | Local tangent frames                                                                                                              |



# ICRS/ICRF – International Celestial Reference

- The ICRS, [International Celestial Reference System](#), is the fundamental quasi-inertial celestial system  
The ICRF is the practical realization of the ICRS
- Its axes are anchored by extremely distant, fixed extragalactic radio sources (like quasars) rather than the Earth, so it is effectively non-rotating for Solar System and Earth-orbit work
- Used for deep-space navigation, astronomical catalogs, and as the baseline for computing celestial mechanics
- For Solar System barycentric work, [JPL DE ephemerides](#) and [Horizons vectors](#) are normally expressed in an [ICRF-aligned barycentric frame](#)  
→ From which other view angles (RA/Dec, Azim/Elev) are computed with astrometric or apparent frame transformations

## JPL\_Ephemerides:

```
%-- Earth's barycentric & heliocentric state vectors at time of interest (geometric: no correction)
Earth_SSB = [DE.Earth.SSB.pos(:,d); DE.Earth.SSB.vel(:, d)];
Earth_helio = [DE.Earth.helio.ECI.pos(:, d); DE.Earth.helio.ECI.vel(:, d)];
```

## Astropy\_Ephemerics:

```
body_state = coord.get_body_barycentric_posvel(body, target_date, pyargs('ephemeris', ephemeris_name));
```

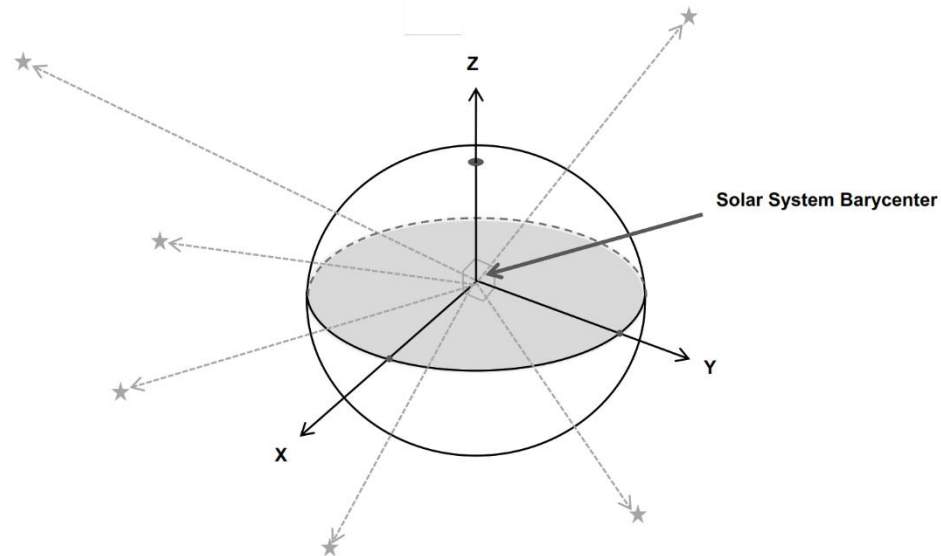
- For Earth-orbit work, the origin is usually shifted from the Solar System barycenter (SSB) to Earth's center, giving the [GCRS/GCRF](#) family



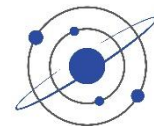
# The ICRF Inertial Frame



- The **International Celestial Reference Frame (ICRF)** frame is defined by the adopted locations of **295 extragalactic radio sources**
    - ICRF coincides almost exactly with the J2000 frame (The difference is very small: a rotation of less than  $0.1''$  ( $0.0003^\circ$ ))
- NASA/JPL Horizons reference frame



# J2000 / EME2000



- J2000 is often used informally to mean an ECI frame with axes tied to the mean equator and mean equinox of J2000.0

- In astrodynamics software, especially older tools, this is often called:

EME2000 = Earth Mean Equator and Equinox of J2000

- It is close to GCRF/ICRF but not exactly identical. The small difference is the [Frame Bias](#) between the older dynamical mean equator/equinox system and the ICRS/GCRS system

→ For many engineering applications, people [treat J2000, EME2000, and GCRF as practically equivalent](#)

- For sub-arcsecond astrometry, occultations, eclipses, VLBI, or precise orbit determination, they should *not* be blindly interchanged



The J2000 inertial frame



- Frame bias is the small, fixed rotation between the modern ICRS/GCRS axes and the older mean equator and mean equinox of J2000.0 system. In astrodynamics language:

Frame bias is the small rotation needed to go between GCRF/GCRS and dynamical J2000/EME2000

- It is not precession, not nutation, and not Earth rotation. It is a constant offset at J2000.0.

## Why frame bias exists

- Historically, the “J2000” celestial frame was based on the mean equator and mean equinox of J2000.0, tied to planetary dynamics and FK5-type astronomy. The USNO describes the earlier astronomical reference system as based on the equator and equinox of J2000.0 determined from planetary-motion observations.
  - The modern frame, however, the ICRS is realized by the ICRF, using distant extragalactic radio sources. The GCRS/GCRF axes are aligned with the ICRS/ICRF axes.
  - The two were designed to be very close, but improved observations showed that they are not exactly identical
- The small residual rotation between them is called *frame bias*



# GCRS/GCRF – Geocentric Celestial Reference

- The GCRS is the [modern rigorous ECI frame](#), the preferred modern ECI frame for high-precision astrodynamics
- It is an Earth-centered relativistic coordinate system whose axes are kinematically non-rotating and aligned with the ICRS (the distant stars)
- In engineering language, the practical realization is often called GCRF

GCRF  $\approx$  Earth-centered ICRF-aligned inertial frame

- Use GCRF/GCRS for:
  - [High-precision orbit propagation](#)
  - Numerical integration of satellite orbits
  - [Comparison with precise ephemerides](#) ( $\rightarrow$  STUDY\_Eclipse2024)
  - Transformations to/from ITRF
  - Modern astrometry and [apparent-place calculations](#).
- Earth's modern, standard inertial frame
- Does not change with the date. It is origin-centered on Earth, but its axes are fixed to the distant stars (aligned with ICRS)
- This is the default starting point for most geocentric orbital calculations before factoring in Earth's daily orientation.

A GCRS/GCRF state vector is generally:  $R_{GCRS} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$  with origin at Earth's center and axes not rotating with Earth

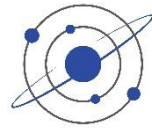
# GCRS/GCRF (continued)

---



- The standard inertial coordinate system for the Earth and it is the geocentric counterpart of the ICRF
  - It has been the adopted celestial reference frame for the IERS since January 1, 1997
- The axes of the GCRF are closely aligned to the frame of IAU-76/FK5 (J2000 in Sec 3.7) to provide continuity with former IAU systems
  - Because there was no official nutation theory compatible with the ICRF in 1997, the IERS maintains tabulated corrections for the IAU-76/FK5 theory to relate it to the GCRF. Applying these corrections effectively defines a more accurate (but different) conventional theory than the IAU-76/FK5
- The IAU-2000 Resolutions reference the GCRF directly

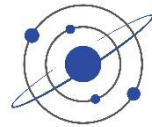
# GCRS/GCRF (I-J-K) (continued)



- To describe a point in space, one needs a frame of reference that does not rotate with respect to the stars
- The GCRF, with the origin at the Earth's center and orthogonal vectors  $\hat{I}$ ,  $\hat{J}$ , and  $\hat{K}$ , is used as frame of reference
- The fundamental plane is the  $\hat{I}$ ,  $\hat{J}$  plane, which is closely aligned with the equator with a small offset, and  $\hat{K}$  aligns closely with the north pole
- To describe the location of a satellite using a position vector and a velocity vector in the geocentric-equatorial coordinate system
- GCRS is the **Earth-centered counterpart of the International Celestial Reference Frame (ICRF) (IAU)** having its origin is at the barycenter of the Solar System, with axes that are intended to show no global rotation with respect to a set of distant extragalactic objects
  - Defined in a way that the transformation between the ICRS and GCRS contains no rotation.
  - This means its orientation is the same as the orientation of the ICRS. But due to its position at Earth's centre of mass the transformation has a changing translational component
- Created in 2000 by the International Astronomical Union (IAU)
- Standard coordinate system used to specify the location and motions of near-Earth objects, such as satellites

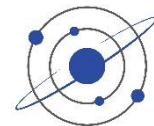
# GCRS/GCRF (I-J-K) (continued)

---



- Earth-centered inertial coordinate frame, where the origin is placed at the center of mass of Earth and the coordinate frame is fixed with respect to the stars (and therefore not fixed with respect to the Earth surface in its rotation)
    - X-axis is aligned with the mean equinox of Earth of J2000.0 (at Terrestrial Time 2000-01-01 12:00:00) and does not move with precession and nutation
    - Z-axis is aligned with the Earth's rotation axis
  - It is almost equivalent to the J2000 frame of reference (also called EME2000)
  - GCRF is an Earth-centered inertial coordinate frame, where the origin is placed at the center of mass of Earth and the coordinate frame is fixed with respect to the stars (and therefore not fixed with respect to the Earth surface in its rotation)
  - GCRF is the modern, relativistic replacement for the older J2000 inertial frame
    - The Origin: Center of mass of the Earth
    - The Anchors: It is mathematically fixed against the International Celestial Reference System (ICRS), which is defined by the coordinates of hundreds of stable, deep-space quasars
    - The Advantage: Because it is truly inertial (unaccelerated and non-rotating), Newton's laws of motion apply directly
- Satellite tracking software propagates orbits in GCRF to avoid dealing with complex coordinate accelerations

# Difference between “Mean” and “True”



- In this astrometry / Earth-orientation context, *Mean* and *True* refer mainly to whether the short-period nutation of Earth's rotation axis is included

**Mean** = precession included, nutation averaged out

**True** = precession included, nutation included

→ So “true” does not mean “more physically real” in an absolute sense. It means “the instantaneous, nutating equator/equinox of date”

- [https://en.wikipedia.org/wiki/Equatorial\\_coordinate\\_system](https://en.wikipedia.org/wiki/Equatorial_coordinate_system)
- “true” is when all variations are accounted for and “mean” is when oscillating variations are ignored (e.g. “small periodic oscillations of nutation”)
- “true Equator” means the plane perpendicular to the CEP at a given specified time. The mean Equator is the plane perpendicular to an axis computed from the CEP from which the nutation terms are excluded
- ALSO in [https://ai-solutions.com/help\\_files/orbit\\_reference\\_frames.htm#achr\\_todearthequator](https://ai-solutions.com/help_files/orbit_reference_frames.htm#achr_todearthequator)
  - True of Date Earth Equator
  - True Equator Mean Equinox
- → See and extract detailed info in doc: [Reference frames used in astrodynamics + ERA.docx](#)

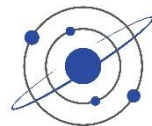


# TETE – True Equator, True Equinox

---



- **TETE** (True Equinox, True Equator): A fully rigorously calculated version of TEME that applies complete nutation corrections. While rarely used today, it represents the mathematically rigorous astronomical equivalent of what TEME approximated
- **TETE** is a highly specific, time-dependent celestial coordinate system used in astronomy
  - It matches the true physical equator and the true equinox of the target date. It calculates and applies both precession (long-term drift) and nutation (short-term wobble)
  - Use this when comparing your data directly to JPL Horizons outputs, or when you need to calculate an accurate local hour angle for a telescope
- **Primary Use Case**
  - **JPL Horizons matching & ground pointing**
- **ECEF** is the standard framework for anything happening on or near the surface of the Earth.
  - The Origin (0,0,0): Located at the Earth's center of mass.
  - The Real-World Standard: ITRF and WGS84 (used by GPS) are specific, highly accurate realizations of ECEF
  - Movement: If you sit still in a chair, your ECEF coordinates do not change, even though the Earth is moving through space



# TETE/TOD – True of Date

- TOD, or True of Date, usually means: True Equator and True Equinox of Date
- It **includes both: precession + nutation**
- So: J2000 → precession → MOD → nutation → TOD/TETE
- In practical astronomy, TETE is the frame used for traditional apparent right ascension and declination of date
- This frame is important for:
  - Apparent RA/Dec of stars & planets (→ STUDY\_Eclipse2024)
  - Telescope pointing in equatorial coordinates
  - Comparison with older almanac-style coordinates

→ Use TETE/TOD when you want traditional apparent RA/Dec of date measured from the equinox

TOD = (mean equinox) True (equator) Of Date

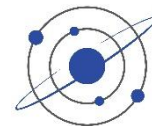
True of Date (TOD) is the most accurate system of celestial coordinates used to define a body's position relative to the center of Earth.

→ This coordinate system incorporates Earth's rotation, UT corrections, precession, nutation, and polar wandering.

→ Required to compute the RA & Dec from JPL DE440: The Sun's position must be in the TOD coordinate system to compute right ascension.

see [https://ai-solutions.com/help/Files/orbit\\_reference\\_frames.htm](https://ai-solutions.com/help/Files/orbit_reference_frames.htm)

# TETE vs GCRF (both ECI)



- TETE is an instantaneous, moving celestial frame, while GCRF is a highly precise, fixed inertial frame used as the modern standard for space navigation
- The primary difference lies in how they handle Earth's axial movements
  - GCRF is completely frozen in time and does not move with Earth's wobbles, making it perfect for calculating orbital trajectories
  - TETE constantly shifts to align with the Earth's instantaneous equator and orientation, making it a useful intermediate step when transitioning from space coordinates to ground tracking.

| Feature               | GCRF (Geocentric Celestial Reference Frame)          | TETE (True Equator, True Equinox)         |
|-----------------------|------------------------------------------------------|-------------------------------------------|
| Frame Type            | Inertial / Fixed Celestial                           | Apparent / Moving Celestial               |
| Time Dependency       | Fixed (Tied to distant quasars)                      | Ephemeral (Changes continuously)          |
| Precession & Nutation | Excluded (Axes remain perfectly stationary)          | Included (Axes move with Earth's wobbles) |
| Z-Axis                | Earth's rotation axis at J2000.0 epoch (approximate) | Instantaneous rotation axis of Earth      |
| X-Axis                | <a href="#">ICRS</a> Celestial Intermediate Origin   | Instantaneous true vernal equinox         |
| Primary Use           | Orbit propagation, deep space navigation             | Intermediate step to ECEF/ITRF            |

# TETE – True Equator, True Equinox of date



- TETE is a “now” frame that mirrors the physical orientation of the Earth in space at a specific split second.
  - The Geometry: It continuously adapts to both precession (the 26,000-year slow cone drift of the axis) and nutation (the short-term, 18.6-year cyclic nodding of the axis).
  - The Advantage: Because its Z-axis matches the actual spin axis *right now*, converting from TETE to a spinning Earth frame (like ECEF or ITRF) only requires a simple, single-axis rotation around the Earth’s current Greenwich Apparent Sidereal Time (GAST)
- To transform a satellite vector from the fixed GCRF to the moving TETE, you must apply a transformation matrix that strips away the Earth’s frame motions for that exact epoch:

$$R_{TETE} = B \cdot P \cdot N \cdot R_{GCRF}$$

$B$  = Frame Bias matrix (aligning GCRF to the mean frame)

$P$  = Precession matrix (accounting for long-term drift)

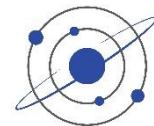
$N$  = Nutation matrix (accounting for short-term wobble)

- Once you are in TETE, you are only one rotation away from the Earth’s surface.
- If you are writing software for this, tell me which library you plan to use (such as astropy, skyfield, or orekit). I can provide the exact code snippets to handle this GCRF-to-TETE transformation.
- ➔ Link this directly to the calculation in ComputeViewAngles➔**LOS\_directions**

# TETE

TETE = true equator and true equinox of date; This is the traditional apparent RA/Dec-of-date frame.

TETE: incl precession & nutation → used for apparent



- TETE is an intermediate “apparent” frame used to bridge the gap between deep space and Earth observation.
  - The Equator: It uses the true equator, meaning it accounts for both precession (long-term wobbling) and nutation (short-term periodic nodding) of the Earth’s axis
  - The Equinox: The X-axis points to the true equinox, which is the intersection of the true equator and the ecliptic plane at that exact moment
  - Movement: A satellite or star will have relatively stable coordinates in a celestial frame like TETE, but a ground station will sweep through 360° of longitude every day
- How They Connect
- To transform coordinates from TETE to an ECEF frame (like ITRF), you primarily need to apply a single rotation matrix based on Greenwich Apparent Sidereal Time (GAST). This angle represents the precise rotation of the Greenwich Meridian relative to the true vernal equinox at a specific Julian Date

$$R_{ECEF} = Rot_Z(\theta_{GAST}) \cdot R_{TETE}$$

- **TOD**
  - Required for proper RA & Declination in correct frame (see exchanges with Mahooti; SunEphemeris & ComputePropagation.m & STUDY\_Orbit.m) Conversion from Astrometric (ECI, no PNE) to Apparent (TOD, with PNE)

What it is: Also commonly referred to as True of Date (TOD). It is an intermediate, rotating reference frame defined by the actual position of the Earth’s equator and the true equinox at a specific, given moment in time.

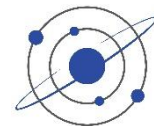
Use Case: It factors in real-time wobble (precession and nutation) caused by gravitational pulls on Earth, making it a critical stepping stone when transforming coordinates between the inertial ICRF and the rotating ECEF frames.

# TEME – True Equator, Mean Equinox – SGP4/TLE specific

---



- When working with Two-Line Element sets (TLEs) from NORAD/Space Command, we encounter unique frames designed to optimize computing time in the 1960s and 70s.
  - TEME: A unique, hybrid ECI frame used exclusively by the SGP4 satellite propagation model. It accounts for precession and the long-term components of nutation, but intentionally strips out specific mathematical terms to speed up calculations (during the 1970's) Crucial Rule: You must ingest TLE data using TEME; treating TLE data as if it were in J2000 or GCRF introduces kilometers of positioning error
  - TEME is in the class of ECI (non-rotating) reference systems
    - with the prime direction (X) pointing along the Vernal Equinox and the Z direction is the CEP (Celestial Ephemeris Pole)
    - But this Z direction moves wrt to the stars (Celestial Sphere); Earth orbital trajectories have a nutation and a precession
    - A very specific variation of an equatorial frame.
    - It uses Earth's true equator but aligns its Right Ascension with a mean (averaged) equinox.
- This is the exact frame required if you are working with satellite Two-Line Elements (TLEs) and the SGP4 orbital propagation model



# TEME – True Equator, Mean Equinox of date

- TEME is the *legacy operational* output frame of the classical **NORAD/TLE/SGP4 propagator**
- The name means:
  - True equator: includes nutation in the equator
  - Mean equinox: → does not use the true equinox in the same way TOD/TETE does
- TEME is not part of the modern IAU transformation chain in the same clean way as GCRS, CIRS, TIRS, and ITRS
- The workflow is:
  - call SGP4 to obtain position and velocity vectors in TEME, then
  - convert from TEME to ECEF for latitude/longitude or look-angle calculations
- Important warning: Do not treat TEME as GCRF or J2000
- For TLE work, the correct chain is usually:

TLE → SGP4 → **TEME** → **ITRF/ECEF** → topocentric **SEZ/ENU**

# TEME – True Equator Mean Equinox of date

---



- TEME – True Equator Mean Equinox of Date (used for TLEs !!2)
- Earth-centered inertial, the ECI frame used for the NORAD two-line elements is sometimes called true equator, mean equinox (TEME) although it does not use the conventional mean equinox.
  - <https://astronomy.stackexchange.com/questions/44140/can-someone-explain-to-me-the-teme-reference-frame-in-defining-orbital-coordinat>
  - <https://search.r-project.org/CRAN/refmans/asteRisk/html/TEMEtoGCRF.html>
- (Vallado, 2007:228-229):
  - The actual SGP4 model has little need for any specific coordinate or time system (e.g., the near-Earth part is rotationally symmetric about the pole), but when used for propagating TLE generated by DoD it becomes important to use the same coordinate system as the DoD orbit determination routines use. The commonly accepted output coordinate system is that of the “true equator, mean equinox” (TEME) (Herrick, 1971). An exact operational definition of TEME is very difficult to find in the literature, but conceptually its primary direction is related to the “uniform equinox”
  - → The intent was to provide an efficient, if approximate, coordinate system for use with the AFSPC analytical theories.
  - Technically, the direction of the uniform equinox resides along the true equator “between” the origin of the intermediate Pseudo Earth Fixed (PEF) and True of Date (TOD) frames



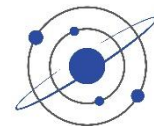
# TEME vs GCRF

---



- The TEME (True Equator, Mean Equinox) and GCRF (Geocentric Celestial Reference Frame) are both ECI frames of reference, i.e., Earth-centered inertial coordinate frames, where the origin is placed at the center of mass of Earth and the coordinate frame is fixed with respect to the stars (and therefore not fixed with respect to the Earth surface in its rotation)
- The difference between the two resides in the fact that in the GCRF frame, the X-axis and Z-axis are aligned respectively with the mean equinox and rotation axis of Earth at 12:00 Terrestrial Time on the 1st of January, 2000, while in the TEME frame they are aligned with the mean equinox and rotation axis at the time of the corresponding TLE
- Due to the change of the direction of the vernal equinox and the rotation axis over time, coordinates in the two frames differ slightly

# TEME vs GCRF (continued) → older info; ECI is a class



- While GCRF and ECI are related and can be used interchangeably for some purposes, they are different coordinate systems with distinct characteristics:
  - **GCRF** is a fixed reference frame centered at the center of mass of the Earth (geocenter). It is based on the International Celestial Reference Frame (ICRF), which provides the positions of distant extragalactic objects with high accuracy. GCRF is primarily used for specifying the positions of stars, quasars, and other celestial objects relative to the Earth's center, without considering the motion of the Earth itself. It is static over short time periods (days to years) but may experience long-term changes due to tectonic plate movements or other Earth-related phenomena
  - The **ECI** reference frame is a non-rotating, inertial reference frame centered at the Earth's center of mass. It takes into account the rotation of the Earth and its orbital motion around the Sun, allowing for more accurate representation of the positions and motions of objects as observed from space. The ECI frame is often used in satellite navigation and astrodynamics, as it simplifies the equations of motion for objects in space, especially in low Earth orbits. It is typically used for short-term predictions of satellite positions and is continually updated to account for the Earth's rotation and orbit
- In summary, GCRF is a fixed reference frame based on celestial objects' positions relative to the Earth's center, while ECI is an inertial frame that considers the motion of the Earth itself. ECI is more suitable for satellite navigation and orbit predictions, while GCRF is commonly used for specifying the positions of stars and other celestial objects relative to the Earth

INERTIAL (fixed wrt stars)

NON-INERTIAL (rotates with Earth)

GEODETIC

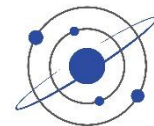
GCRF/ECI (X → vernal equinox)

ECEF (X → Prime Meridian)

LLA (Lat, Lon, Alt)

# ECEF/ITRF – Earth Centered, Earth Fixed

---



- An ECEF frame is fixed to and *rotates with the Earth*. Ground stations, geodetic latitude and longitude, topography, and map data are naturally expressed in ECEF or ECEF-like terrestrial frames

## ITRS (International Terrestrial Reference System frame)

- The rigorous modern terrestrial frame is ITRS, and its practical realizations are: ITRF2008, ITRF2014, [ITRF2020](#), etc.
- In many engineering contexts, “ECEF” means a WGS84-like or ITRF-like Earth-fixed Cartesian frame
  - But for precise geodesy, ITRF realization and epoch matter because stations move due to plate tectonics, loading, and other geophysical effects
- The ITRS/ITRF frame has:
  - Origin near Earth’s center of mass
  - z-axis close to the terrestrial pole
  - x-axis near the Greenwich meridian
  - *Rotates with Earth*
  - Includes station motion, tectonics, *polar motion*, and Earth orientation modeling at high precision

# ECEF/ITRF – Body-Fixed Coordinate System

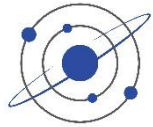


- A geocentric coordinate system **fixed to the rotating Earth**, results in the International Terrestrial Reference Frame (ITRF), coordinate system
  - Its origin is at the center of the Earth and the axes are realized by the adopted coordinates of defining stations on the Earth's surface
  - Because these stations are affected by plate tectonic motion ( $\approx \text{cm/yr}$ ), the ITRF is regularly re-estimated as a weighted solution combining several analysis center solutions so there is no net rotation with previous ITRF realizations
  - The ITRF solutions are usually identified with the year of their publication (ITRF-08).
  - Conversions between ITRF solutions involves translation, scale, and rotation. Typically, we use this system to process actual observations and acceleration calculations during orbit propagation
    - In some applications, a different meridian is chosen
- Confusion may exist because the ITRF system is frequently called the Earth-Centered, Earth-Fixed (ECEF) coordinate frame. The term “Earth-fixed” describes a terrestrial reference system whose net global orientation remains unchanged over time with respect to the crust of the Earth. However, imprecise use of this term invites confusion

What it is: A non-inertial reference frame that has its origin at the Earth's center of mass, but rotates along with the Earth. The Z-axis aligns with the Earth's rotation axis, while the X-axis passes through the prime meridian.

Use Case: Used for terrestrial GPS tracking, mapping, and determining exact satellite coverage footprints on the Earth's surface.

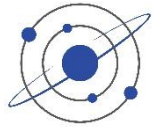
# ECEF/ITRF (X-Y-Z) (continued)



- ECEF is the “conventional terrestrial” coordinate system
- GPS uses ECEF as it’s primary coordinate system and derives all other coordinates from it
- The fixed-frame used is the International Terrestrial Reference Frame (ITRF)
- This reference frame is realized by the IAU2000/2006 reduction from the ICRF coordinate system.
- **Non-inertial system that rotates with the Earth.** Its origin is fixed at the center of the Earth
  - x-axis points outward along the intersection of the Earth’s equatorial plane and → Prime Meridian
  - y-axis points into the eastward quadrant, perpendicular to the x-z plane so as to satisfy the RH rule
  - z-axis points northward along the Earth’s rotation axis.
- System uses the Cartesian coordinates  $(X, Y, Z)$  to represent position relative to the center of the reference ellipsoid. The distance between the center of the ellipsoid and the center of the Earth depends on the reference ellipsoid (e.g. WGS84)
  - positive X-axis intersects the surface of the ellipsoid at  $0^\circ$  latitude and  $0^\circ$  longitude, where the equator meets the prime meridian
  - positive Y-axis intersects the surface of the ellipsoid at  $0^\circ$  latitude and  $90^\circ$  longitude
  - positive Z-axis intersects the surface of the ellipsoid at  $90^\circ$  latitude and  $0^\circ$  longitude, the North Pole.
- → Since the ECEF rotates with the earth, the axis always points through the equator and the prime meridian

# ECEF/ITRF (X-Y-Z) (continued)

---



- ITRF the origin is at the center of mass of Earth, and the frame rotates with respect to the stars to remain fixed with respect to the Earth surface as it rotates
- The Z-axis extends along the true North as defined by the IERS reference pole, and the X-axis extends towards the intersection between the equator and the Greenwich meridian at any time.
- ITRF is the terrestrial reference frame primarily used for geodetic and GPS applications on or near the Earth's surface; is updated to account for crustal movements and geophysical changes, while ECI is fixed relative to distant stars and does not consider Earth's rotation



- The International Terrestrial Reference System (ITRS) is a geocentric coordinate system defined by the IERS that provides a consistent framework for high-precision positioning, geodetic applications, and Earth orientation studies
- It serves as the theoretical foundation for Earth-fixed frames like the International Terrestrial Reference Frame (ITRF)

They are closely related but not strictly identical.

## 1. What is ITRS?

The **International Terrestrial Reference System (ITRS)** is a precisely defined reference frame:

- origin: Earth's center of mass
- axes: tied to Earth's crust
- includes:
  - Earth rotation
  - polar motion
  - tectonic plate motion
- defined by the **International Earth Rotation and Reference Systems Service (IERS)**

It is a **rigorous, time-dependent, geophysical** reference system.

## 2. What is ECEF?

**ECEF (Earth-Centered Earth-Fixed)** is a generic engineering term meaning:

Any frame that rotates with the Earth and is fixed to its surface

It usually implies:

- Cartesian coordinates  $(x, y, z)$
- origin at Earth's center
- rotating with Earth

But it does **not specify**:

- which realization
- which epoch
- how polar motion is handled

## 3. Relationship between ITRS and ECEF

- **ITRS** → theoretical definition
- **ITRF (realizations)** → actual coordinate frames (e.g., ITRF2014)
- **ECEF** → informal name for such frames in practice

So in most engineering contexts:

$ECEF \approx ITRF$  realization of ITRS

## 4. When the distinction matters

The difference becomes important in:

- high-precision GNSS
- geodesy
- Earth orientation modeling
- long-term coordinate accuracy

Because:

- ITRS includes **tectonic motion + Earth orientation corrections**
- "ECEF" may ignore or simplify these

## 5. Practical interpretation

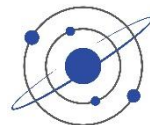
| Term | Meaning                         | Precision   |
|------|---------------------------------|-------------|
| ITRS | rigorous reference system       | highest     |
| ITRF | realized frame (e.g., ITRF2014) | operational |
| ECEF | generic Earth-fixed frame       | approximate |

## 6. Rule of thumb

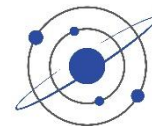
- In **astrodynamics / engineering** → "ECEF" is fine
- In **high-precision geodesy** → use ITRS / ITRF

## Final takeaway

ECEF is the practical, informal implementation of the ITRS

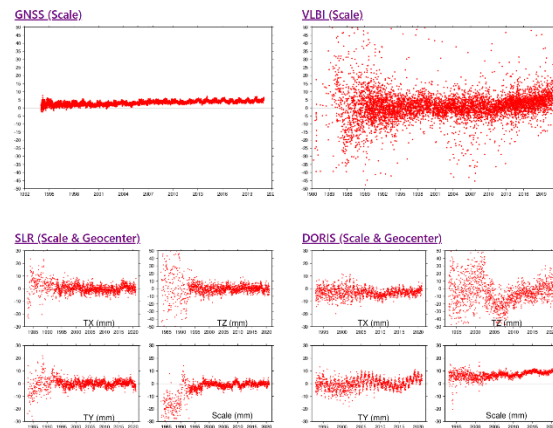






- **ITRF2020** is the new realization of the International Terrestrial Reference System (ITRS)
- Following the procedure already used for previous ITRF solutions, the ITRF2020 uses as input data time series of station positions and Earth Orientation Parameters (EOPs) provided by the Technique Centers of the four space geodetic techniques (VLBI, SLR, GNSS and DORIS), as well as local ties at colocation sites
- Based on completely reprocessed solutions of the four techniques, the ITRF2020 is expected to be an improved solution compared to ITRF2014
- A number of innovations were introduced in the new processing, including:
  - The time series of the four techniques were stacked all together, adding local ties and equating station velocities and seasonal signals at colocation sites;
  - Annual and semi-annual terms were estimated for stations of the 4 techniques with sufficient time spans; Post-Seismic Deformation (PSD) models for stations subject to major earthquakes were determined by fitting GNSS/IGS data
  - The PSD models were then applied to the 3 other technique time series at earthquake colocation sites
- Consult web site <https://itrf.ign.fr/en/solutions/ITRF2020> for more info on:
  - Computation strategy, Input data, Frame Definition: origin, scale, orientation

Scale and Geocenter time series  
Plots with respect to ITRF2020



## ■ ITRF

- [ComputeViewAngles.m](#)
- [ECI to ECEF.m](#)
- [teme2ecef.m](#)
- [Orbit Propagation\STUDY Orbit.m](#)
- [Orbit Propagation\SGP4 2.1.0 \(Mahooti, Oct 2022\)\SGP4\teme2ecef.m](#)
- [Orbit Propagation\SGP4 2.1.0 \(Mahooti, Oct 2022\)\SGP4 Vectorized\eci2ecef.m](#)
- [Orbit Propagation\SGP4 2.1.0 \(Mahooti, Oct 2022\)\SGP4 Vectorized\teme2ecef.m](#)
- [SPICE-MICE](#) => MANY FUNCTIONS
- [STUDY\\_PNE.m](#)
- [STUDY Sun ephemeris.m](#)
- [Geostationary Satellites Tracking\private\teme2ecef.m](#)
- [ISSTracker\computePasses.m](#)
- [ISSTracker\getPropagator.m](#)
- [ISSTracker\initializeEarthView.m](#)
- [ISSTracker\ISSTracker.m](#)
- [ISSTracker0\computePasses.m](#)
- [ISSTracker0\getPropagator.m](#)
- [ISSTracker0\initializeEarthView.m](#)
- [ISSTracker0\ISSTracker.m](#)

# Exercise



- Consider a Global Navigation Satellite System (GNSS) tracking station located on the Pacific Plate. According to [ITRF2020 tracking data](#), this station moves rapidly toward the northwest

- Given Parameters** (Epoch  $t_0 = 2020.0$ ):

Initial Position (X, Y, Z):  $X(2020.0) = -6,134,220.0000$  m,  $Y(2020.0) = -1,311,700.0000$  m,  $Z(2020.0) = +3,512,100.0000$  m

Tectonic Velocity ( $\dot{X}$ ,  $\dot{Y}$ ,  $\dot{Z}$ ):  $\dot{X} = -1.5$  cm/year,  $\dot{Y} = +3.2$  cm/year,  $\dot{Z} = -1.1$  cm/year

- Target Date:**

We want to find the coordinates for today, **July 12, 2026** ( $t = 2026.527$ ) → Time elapsed:  $t - t_0 = 2026.527 - 2020.0 = 6.527$  years

- Compute the New Coordinates**

Multiply the velocities by the elapsed time (6.527 years) and add them to the reference position:

$\Delta X = -0.0150 \times 6.527 = -0.0979$  m ;  $X(2026.527) = -6,134,220.0000 + (-0.0979) = -6,134,220.0979$  m

$\Delta Y = +0.0320 \times 6.527 = +0.2089$  m ;  $Y(2026.527) = -1,311,700.0000 + 0.2089 = -1,311,699.7911$  m

$\Delta Z = -0.0110 \times 6.527 = -0.0718$  m ;  $Z(2026.527) = 3,512,100.0000 + (-0.0718) = +3,512,099.9282$  m

- Total 3D Displacement**

- To see how far the station physically moved over those 6.5 years, use the 3D distance formula:

$$d = \sqrt{\Delta X^2 + \Delta Y^2 + \Delta Z^2} = \sqrt{(-0.0979)^2 + (0.2089)^2 + (-0.0718)^2} = 0.2415 \text{ m}$$

- Final Result**

- In the 6.527 years since the reference epoch, this tracking station drifted **24.15 cm** across space. Without updating the ITRF coordinates to match this movement, GPS tracking calculations would slowly lose centimeter-level accuracy.



# Classical & Legacy Frames (Equinox-Based)

---



- Historically, computers lacked the processing power to constantly compute modern IAU 2000 matrix transformations, or models were built around the “Mean Equinox of Date.”
- **TOD** (True of Date): An ECI frame that takes a fixed baseline and applies both Precession and Nutation up to the exact date/time requested. The X-axis points to the *True Equinox* of that day. In general astronomy, TOD (True of Date) means any system aligned with Earth’s real, physical equator on a specific day
- **MOD** (Mean of Date): An ECI frame similar to TOD, but it only applies Precession, completely ignoring the short-term, wavy effects of Nutation
- **J2000**: A specific snapshot of the Mean of Date frame taken exactly at noon on January 1, 2000. For decades, it was the default ECI frame for space missions and is very close (within milliarcseconds) to the modern GCRF
- **PEF** (Pseudo-Earth Fixed): An intermediate ECEF frame. It takes a TOD or MOD frame and applies the Earth’s daily rotation, pointing the X-axis at the Greenwich meridian. However, it *ignores* Polar Motion. It serves as a stepping stone between classical ECI and true ECEF

# MOD – Mean of Date

---

- MOD means: Mean Equator and Mean Equinox of Date
- It *includes precession from J2000 to the date*, but *not* nutation
- Conceptually: J2000/EME2000 → precession → MOD
- MOD is useful when you want coordinates referred to the slowly moving mean equator and equinox of the observation date. → Not used often, archaism du temps où ordinateurs plus lents??

# CIRS: the modern apparent celestial frame



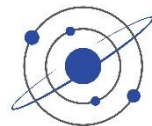
- CIRS means: Celestial Intermediate Reference System
- It is the modern IAU frame associated with the Celestial Intermediate Pole, CIP, and the Celestial Intermediate Origin, CIO
- CIRS replaces the older true-equator/true-equinox machinery with a cleaner CIO-based formulation. Instead of measuring right ascension from the moving equinox, CIRS uses the CIO
- The modern chain is approximately

GCRS → CIRS → TIRS → ITRS

(see same below)

- The IERS (The International Earth Rotation and Reference Systems) Conventions describe the transformation between the celestial and terrestrial systems using the CIP and Earth rotation, while SOFA summarizes the astrometric chain as ICRS/GCRS/CIRS to observed frames
- Use CIRS when you want a modern IAU-compliant apparent geocentric celestial frame

[>] [Part I – Earth Orientation Parameters \(EOP\)](#)



# PEF – Pseudo Earth Fixed

- PEF is an *intermediate* frame between inertial/celestial frames and fully Earth-fixed frames
- In the older equinox-based transformation chain, one commonly sees:

$$\text{TOD} \rightarrow \text{PEF} \rightarrow \text{ITRF}$$

- where:
  1. TOD to PEF applies Earth rotation, often through Greenwich apparent sidereal time
  2. PEF to ITRF applies polar motion
- Conceptually:

$$R_{PEF} = \text{Rot}_Z(\theta_{GAST})$$

then

$$R_{ECEF/ITRF} = W \times R_{PEF}$$

where  $W$  is the polar motion matrix.

- PEF is useful in older SGP4 and Vallado-style transformation chains
- In the modern IAU chain, the analogous intermediate frame is usually **TIRS**

# TIRS – Terrestrial Intermediate Reference System

---



- TIRS is the modern Earth-rotated frame after applying Earth rotation but *before* polar motion
- Modern chain:

CIRS  $\rightarrow$  TIRS  $\rightarrow$  ITRS

- where:
  - CIRS to TIRS uses the Earth Rotation Angle, ERA
  - TIRS to ITRS uses polar motion and the terrestrial intermediate origin, TIO
- This is the IAU/IERS replacement for older sidereal-time/equinox-based transformations
- The IERS conventions describe the GCRS-to-ITRS transformation through these intermediate pole/origin concepts.



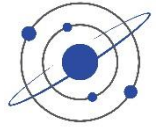
# TIRS – Terrestrial Intermediate Reference System

---

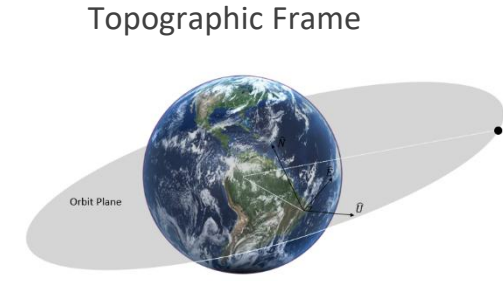
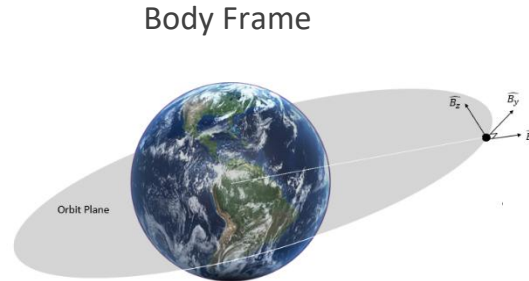
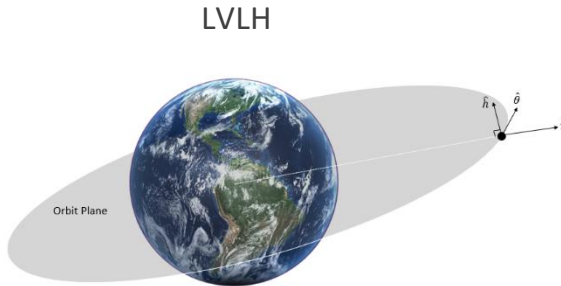
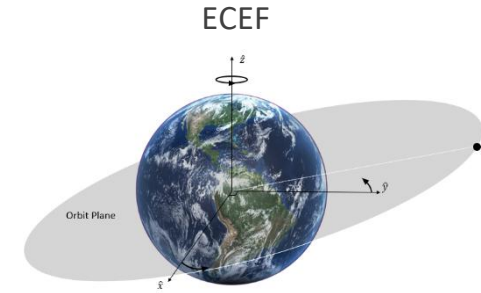
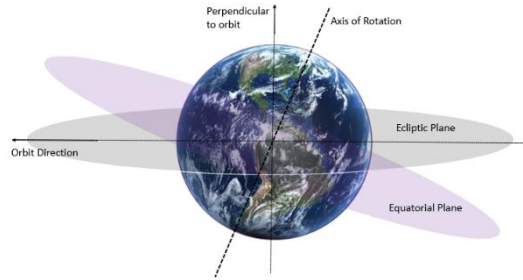
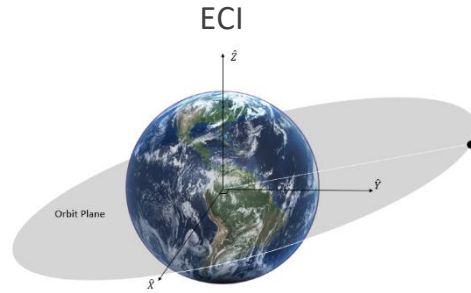


- The Terrestrial Intermediate Reference System (TIRS) is a geocentric coordinate system defined by the intermediate equator of the Celestial Intermediate Pole (CIP) and the Terrestrial Intermediate Origin (TIO). It acts as an intermediate frame linking the International Terrestrial Reference System (ITRS) to the Celestial Intermediate Reference System (CIRS), accounting for polar motion.
- Key Details About TIRS
  - **Definition:** TIRS is a geocentric system defined by the intermediate equator of the CIP and the TIO, providing a transient, rotating frame.
  - **Purpose:** It bridges the ITRS and CIRS, separated by the Earth Rotation Angle (ERA) around the CIP.
  - **Transformation:** TIRS is related to the ITRS by polar motion and to the CIRS by the Earth rotation angle, making it crucial for studying Earth's rotation.
  - **TIO:** The Terrestrial Intermediate Origin (TIO) acts as the longitude origin, moving slightly based on the TIO locator.
  - **Context:** TIRS is part of the IAU 2000/2006 conventions for Earth rotation and orientation, which separates the motion of the reference pole into terrestrial (polar motion) and celestial (precession-nutation) components.
- TIRS is used to model the Earth's non-rigid, complex rotational motion relative to the celestial framework

# SUMMARY: (TO SORT)



<https://medium.com/@zackfizell10/reference-frames-every-aerospace-engineer-should-know-db10638b6d7a>



# Orbit propagation in ECI, then position translated to ECEF

---



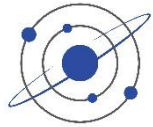
- Engineers must calculate satellite movement inside the ECI/GCRS/(CIS) because Newton's laws of motion only apply in an inertial frame
  - If you tried to calculate a spacecraft's orbit within the rotating Earth frame (ITRF), centrifugal and Coriolis forces would introduce massive complexities
- Instead, flight dynamics software tracks the satellite's smooth elliptical path in the GCRS, and then translates those coordinates into the ITRF only when they need to know exactly which city or ocean the satellite is flying over

# Transformation chains

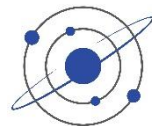
- Here in Part II are presented and described reference systems and frames
- → here are presented some transformations
- ➔ Dans Part III on les utilise pour le positional astronomy, dans les processing chains

# Transformation chains

---



- Coordinate transformations between two reference frames involve **rotation** and **translation**
- Transformations between reference frames can be performed efficiently using **matrix operations**  
(→ see Part I – Rotation Matrices)
  - A particularly useful property of rotation matrices is that **their inverse is simply their transpose**
- For each transformation, there is a transformation matrix  $R$  such that  $Q' = R Q$ , where  $Q$  is a vector in the first coordinate system (e.g. **state vector**  $\vec{R}$  or  $\vec{V}$ ), and  $Q'$  is the same vector in the second coordinate system
  - The transformation is thus totally described by the nine components of the matrix  $R$



# Basic transformations

## 0) Solar System Barycenter (SSB) referenced to (0,0,0): JPL DE ephemeris basic reference frame

- referenced to (0,0,0); the Sun is not at the center (see Part I – Solar System Barycenter)

```
DE.(object_name).SSB.pos(:, d) = r.(target)(1:3);
DE.(object_name).SSB.vel(:, d) = r.(target)(4:6);
```

## 1) Heliocentric, referenced to the Sun

- Sun heliocentric position  $\equiv [0;0;0]$  by definition

```
DE.(object_name).helio.ECI.pos(:, d) = r.(target)(1:3) - r.Sun_SSB(1:3);
DE.(object_name).helio.ECI.vel(:, d) = r.(target)(4:6) - r.Sun_SSB(4:6);
```

## 2) Geocentric, referenced to the Earth

```
DE.(object_name).geoc.ECI.pos(:, d) = r.(target)(1:3) - r.Earth_SSB(1:3);
DE.(object_name).geoc.ECI.vel(:, d) = r.(target)(4:6) - r.Earth_SSB(4:6);
```

→ Those reference frame transformations include only *translations*

## Ecliptic position from ECI

```
Rot_e = RotX(-Earth.e_J2000); % e = 23.439291111°
DE.(object_name).geoc.ECL.pos(:, d) = Rot_e * DE.(object_name).geoc.ECI.pos(:, d);
DE.(object_name).geoc.ECL.vel(:, d) = Rot_e * DE.(object_name).geoc.ECI.vel(:, d);
```

# ICRS/ICRF



This is from Part III

- **ICRS:** International Celestial Reference System

- The current standard celestial reference system adopted by the International Astronomical Union (IAU).
- Its origin is at the barycenter of the Solar System (SSB), with the goal to “show no global rotation with respect to a set of distant extragalactic objects”
- The coordinates are basically the declination  $\alpha$  and right ascension  $\delta$  based on the mean celestial equator and mean celestial equinox at the J2000 epoch

- **ICRF:** International Celestial Reference Frame

- The physical realization of the ICRS reference system, using reference celestial sources observed at radio wavelengths
- The RF is the set of numerical coordinates of the reference sources, derived using the procedures spelled out by the ICRS

→ JPL DE ephemerides are in a barycentric reference system whose axes are aligned with the [ICRF/ICRS](#)

- More precisely:

- Origin: Solar System barycenter, SSB
- Coordinate system: barycentric relativistic solar-system coordinates, often identified with the BCRS
- Axes/orientation: aligned to the International Celestial Reference Frame, ICRF
- For DE440/DE441: Horizons states that the JPL planetary ephemeris solution is closely aligned with the ICRF-3 realization, differing by at most about 0.0002”

# ECI → ECEF with EOP



- To bridge the ICRS and ITRS, scientists use Earth Orientation Parameters (EOP), which measure the daily irregularities in Earth's rotation and orientation
- Because the Earth is not a perfectly rigid sphere, its axis wobbles and its rotation speed fluctuates due to gravitational pulls from the Moon and Sun, atmospheric winds, and ocean currents.
- To transform a position vector from the ECI celestial frame ( $R_{ICRF}$ ) to the ECEF terrestrial frame ( $R_{ITRF}$ ), a series of four consecutive matrix rotations must be applied using the EOP data:

- The Transformation Formula:

$$R_{ITRF} = W \cdot R \cdot N \cdot P \cdot R_{ICRF}$$

- Matrix Roles
  - $W$  (Polar Motion Matrix): Corrects for the wobble of the crust relative to the rotation axis
  - $R$  (Earth Rotation Matrix): Accounts for the daily spin around the axis based on UT1
  - $N$  (Nutation Matrix): Adjusts for the short-term, periodic nodding of the axis in space
  - $P$  (Precession Matrix): Adjusts for the long-term, slow conical sweep of the axis over a 26,000-year cycle
- → Validate this equation with my code

→ Link to Part I – EOP

## The Five Earth Orientation Parameters

The connection between the celestial and terrestrial reference frames in space requires tracking several parameters:

- 1) Polar Motion (x, y): The displacement of the Earth's axis across the physical surface.
- 2) Universal Time Difference (UT-TCR): The discrepancy between the actual, irregular rotation and the uniform atomic time.
- 3) Celestial Pole Offset (CPO): Corrections to precession and nutation.



Here is the **full transformation chain** used in modern astrodynamics to go from a truly inertial frame (ICRS) down to the **Earth-fixed frame (ITRF/ECEF)**. This is essentially what high-precision orbit determination software implements following **International Earth Rotation and Reference Systems Service (IERS)** conventions.

1. Overall transformation chain

$$\mathbf{r}_{ITRF} = W \cdot R \cdot N \cdot P \cdot B \cdot \mathbf{r}_{ICRS}$$

Each matrix has a distinct physical meaning.

2. Step-by-step breakdown

(1) *B*: Frame bias (ICRS → mean J2000)

- Small constant rotation (~arcseconds)
- Aligns ICRS axes with the classical J2000 dynamical frame

$$\mathbf{r}_{J2000} = B \mathbf{r}_{ICRS}$$

(2) *P*: Precession

- Long-term motion of Earth’s rotation axis (~26,000-year cycle)

$$\mathbf{r}_{MOD} = P \mathbf{r}_{J2000}$$

(3) *N*: Nutation

- Short-period oscillations (mainly lunar/solar effects)

$$\mathbf{r}_{TOD} = N \mathbf{r}_{MOD}$$

(4) *R*: Earth rotation

- Rotation about Earth’s axis using **Earth Rotation Angle (ERA)**

$$\mathbf{r}_{PEF} = R \mathbf{r}_{TOD}$$

This is the dominant term (≈ 360° per sidereal day).

(5) *W*: Polar motion

- Small wobble of Earth’s crust relative to rotation axis

$$\mathbf{r}_{ITRF} = W \mathbf{r}_{PEF}$$

3. Physical interpretation

| Step     | Meaning                             |
|----------|-------------------------------------|
| ICRS     | inertial, barycentric               |
| <i>B</i> | align to Earth-based inertial frame |
| <i>P</i> | slow axis drift                     |
| <i>N</i> | periodic wobble                     |
| <i>R</i> | daily Earth rotation                |
| <i>W</i> | surface wobble                      |

4. Simplified astrodynamics version

In many engineering applications:

$$\mathbf{r}_{ECEF} \approx R_3(\theta) \mathbf{r}_{ECI}$$

i.e., only Earth rotation is used.

Accuracy:

- meters to kilometers depending on application

5. Velocity transformation (important)

When transforming velocity:

$$\mathbf{v}_{ITRF} = W R N P B \mathbf{v}_{ICRS} - \boldsymbol{\omega}_E \times \mathbf{r}_{ITRF}$$

The extra term arises because:

- ITRF is a **rotating frame**

6. Why this chain matters

You need the full chain when:

- GNSS positioning (cm-level accuracy)
- satellite laser ranging
- deep-space navigation
- Earth orientation studies

7. Key insight

All Earth-based frames are obtained by progressively “de-inertializing” ICRS

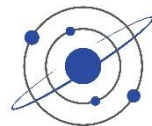
- Start with a perfect inertial frame
- Add real Earth physics step by step



8. Practical takeaway

- ICRS → true inertial reference
- ECI (J2000, TOD, etc.) → intermediate frames
- ITRF / ECEF → Earth-fixed

If you want, I can provide a **MATLAB implementation using real IERS formulas (ERA, precession-nutation series)** or a **numerical example converting a satellite position from ICRS to latitude/longitude**.



# (from code ComputeViewAngles, discussing **ECI\_to\_ECEF**)

The easiest rigorous solution is to use the IAU SOFA algorithms

UTC → TT → UT1

iauPn06a (precession/nutation)

iauEra00 (Earth rotation angle)

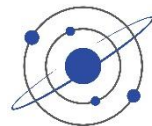
iauPom00 (polar motion)

$$R_{ECEF/ITRS} = W \cdot R \cdot Q \cdot R_{ECI/GCRS}$$

This reproduces the same transformations used by:

- Horizons ✓
- SPICE ✓
- Orekit (TODO)
- GMAT (TODO)
- STK (TODO)
- modern orbit determination systems

to the millimeter level.



# (from code ComputeViewAngles, discussing **ECI\_to\_ECEF**)

For astrodynamics practice

-----

The ECI coordinates come from:

- DE440 / Horizons vectors
- SPICE SPK kernels
- Astropy library

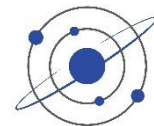
and you need:

- ground tracks,
- topocentric coordinates,
- satellite visibility,

then the recommended chain is:

ICRF/GCRS → **Q** → CIRS → **R** → TIRS → **W** → **ITRF/ECEF**

using IAU 2006 precession, IAU 2000A nutation, ERA-GAST, and IERS EOP corrections.  
This is the current international standard.



# (from code ComputeViewAngles, discussing ECI\_to\_ECEF)

% 1) UTC → UT1

% R) Earth Rotation Angle

R = RotZ(-GAST);

% Q) Precession-nutation matrix

% → The rigorous IAU 2006/2000A model uses 1365 nutation terms

Q = RotZ(-E) \* RotY(-d) \* RotZ(-(-E - s)); → Rot\_PN

% transforms GCRS → CIRS

% W) Rolar motion (tiny correction)

W = RotY(+DEG(x\_pole)) \* RotX(+DEG(y\_pole));

% rotates from TIRS → ITRF

% Final

R\_ECEF = W \* R \* Q \* R\_ECI;

%-- Velocity transformation (Velocity is more subtle)

Q = Rot\_PN;

r\_C = Q \* r\_ECI;

v\_C = Q \* v\_ECI;

% After Earth rotation

r\_T = R \* r\_C;

w\_E = Earth.omega; % [rad/s]

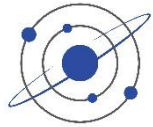
v\_T = R \* (v\_C - w\_E \* r\_C);

v\_ECEF = W \* v\_T;

Revisit & Clarify

Together with  
ECI\_to\_ECEF

# Practical transformation chains



## A. High-precision celestial-to-terrestrial chain

- Modern IAU/IERS chain:

$$\text{GCRS} \xrightarrow{\text{bias, precession, nutation}} \text{CIRS} \xrightarrow{\text{Earth rotation angle}} \text{TIRS} \xrightarrow{\text{polar motion}} \text{ITRS}$$

→ This is the rigorous modern way

## B. Older equinox-based chain

- Classical astrodynamics chain:

$$\text{J2000} \xrightarrow{\text{precession}} \text{MOD} \xrightarrow{\text{nutration}} \text{TOD} \xrightarrow{\text{GAST}} \text{PEF} \xrightarrow{\text{polar motion}} \text{ITRF}$$

→ This is still common in textbooks and legacy codes

## C. TLE / SGP4 chain

For TLEs:

$$\text{TLE} \rightarrow \text{SGP4} \rightarrow \text{TEME} \rightarrow \text{PEF/ITRF} \rightarrow \text{topocentric}$$

Do not skip the TEME-specific transformation. A frequent mistake is to use:  $\text{TEME} \equiv \text{J2000}$ , which is wrong



# Practical transformation chains



## D. Ground station pointing chain

- For a satellite or Solar System object:

$$r_{\text{object}, ITRF} - r_{\text{site}, ITRF} = \rho_{\text{topo}}$$

then:

$$\rho_{\text{topo}} \rightarrow ENU \text{ or } SEZ \rightarrow A, h, \rho$$

where:

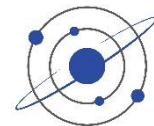
- $A$  = azimuth
- $h$  = elevation/altitude
- $\rho$  = range

# Transformations



- **GCRS → ICRS:** considered as identical in CelestLab ([www.connectbycnes.fr/en/celestlab](http://www.connectbycnes.fr/en/celestlab))
- **GCRS → EME2000:**
  - Constant rotation (IAU2006 model), at an angle is close to  $6.4 \times 10^{-6}^\circ$ , which corresponds to  $\approx 0.7\text{m}$  at 1 Earth radius.
- **GCRS → MOD:**
  - [3,1,3,1] rotation considering Fukushima-Williams precession angles (IAU2006 model).
- **MOD → EOD:**
  - Rotation about the X axis of the mean obliquity (IAU2006 model).
- **MOD → TOD:**
  - [1,3,1] rotation using (IAU2006/2000A model) nutation angles.
- **GCRS → CIRS:**
  - Can be done in several ways. The method implemented in CelestLab is the one using the classical angles (IAU2006/2000A Model, CIO based). For computation efficiency and for dates in the time period [2000, 2100], the results are obtained by interpolation of pre-computed data.
  - Another method (IAU2006/2000A Model, CIO based, using XY series) is also available (See remark below).
- **CIRS → TIRS:**
  - Rotation around the Z axis of Earth Rotation Angle (ERA) (IAU2000 model).
  - Ocean tidal and libration effects in UT1 are ignored in CelestLab. The amplitude of the effect is a few microseconds on UT1. See chapter 5.5.3 of IERS 2010 conventions (IERS Technical Note No. 36) for more details.

# Transformations (continued)



- **TIRS → PEF:**

- Rotation about the Z axis of the TIO locator (s) angle. The TIO locator is obtained from polar motion observations by numerical integration, and so is in essence unpredictable. However, it is dominated by a secular drift of 47 microarcseconds per century which is the approximation implemented (IAU2000 model).

- **PEF → ITRS:**

- The transformation from PEF to ITRS is a [2,1] rotation of polar motion angles (-xp,-yp). The polar motion angles are unpredictable and do not have any model associated with them (\*). Typical values range from 0.1 to 0.5 arcseconds.
- They can be retrieved in IERS bulletins at: [www.iers.org/IERSEN/Publications/Bulletins/bulletins.html](http://www.iers.org/IERSEN/Publications/Bulletins/bulletins.html) .

\*: Polar motion is also affected by two (small) effects for which correction models exist but they are ignored in CelestLab.

- Diurnal and semi-diurnal variations due to the oceanic tides, with amplitudes of a fraction of milliarcsecond.
- Libration in polar motion with an amplitude of about 10 microarcseconds.
- See chapter 5.5.1 of IERS 2010 conventions (IERS Technical Note No. 36) for more details.

- **Veis → PEF:**

- Rotation about the Z axis of the Veis sidereal time. (Reference: Georges Veis - the system reference, in Sao special report 200, 1st volume, 1966 Veis sidereal time).

- **TEME (of date) → PEF:**

- Rotation about the Z axis of the Greenwich mean sidereal time (IAU1982 model).



# Some relations



- Some relations

$$R_{TIRS} = Rot_Z(\theta_{ERA})R_{CIRS}$$

$$R_{TETE} = B \cdot P \cdot N \cdot R_{GCRF}$$

$$R_{ECEF} = Rot_Z(\theta_{GAST}) \cdot R_{TETE}$$

$$R_{ECEF/ITRS} = W \cdot R \cdot (N \cdot P \cdot B) \cdot R_{ECI/GCRS}$$



# Astropy

---

- Correct coordinate chain:

J2000 ICRS catalog position

↓ `apply_space_motion()`

ASTROMETRIC ICRS position at target epoch

↓ `transform_to(TETE(obstime=target_date))`

Geocentric APPARENT RA/Dec of date

↓ `transform_to(AltAz(..., pressure=0))`

Unrefracted TOPOCENTRIC Alt/Az

↓ `transform_to(AltAz(..., pressure>0))`

Refracted observed Alt/Az

# Practical recommendations

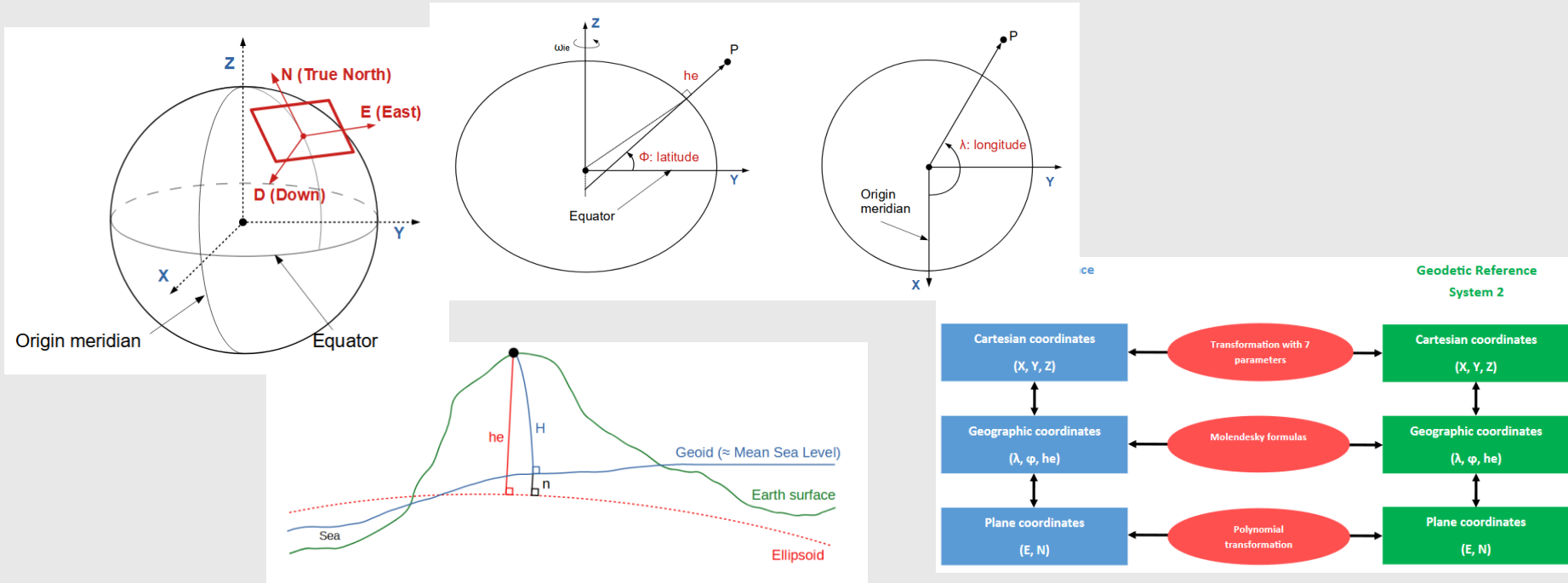


- For high-precision numerical orbit propagation: GCRF/GCRS
- For TLE propagation: TEME from SGP4, then transform properly
- For ground station coordinates: ITRF/ITRS or WGS84 ECEF
- For apparent RA/Dec of date: TETE or CIRS
- For azimuth/elevation: ITRF → topocentric ENU/SEZ
- For modern rigorous celestial-to-terrestrial transformations: GCRS → CIRS → TIRS → ITRS
- For classical textbook transformations, you will often see: J2000 → MOD → TOD → PEF → ITRF  
→ These two last chains are related, but the first is the modern IAU/IAUERS formulation, while the second is the older equinox/sidereal-time formulation.

# Reference coordinate frames

**HIDDEN SLIDE**

- <https://support.sbg-systems.com/sc/kb/latest/underlying-maths-conventions/reference-coordinate-frames>



# TO PRINT: Vallado

---

HIDDEN SLIDE

- P. 169, Algo 12 (p. 172) & 13 (p.173)

3.3 Coordinate Systems pp. 145–159

3.4 Coordinate Transformations: pp. 159–174

$\Sigma = 30$  pages

3.7 Transforming.. P. 203 .. ECEF to ECI

# Orbital Coordinate Systems (by Dr. T.S. Kelso)

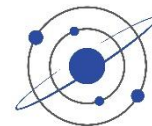
---

HIDDEN SLIDE

- Part I & II
- **TODO: Extract info** from the printed kit



# NASA/NAIF: Reference Frames



## Reference Frames

[Abstract](#)

[Purpose](#)

[Intended Audience](#)

## Using Frames

[Frame Functions in CSPICE](#)

[Frame Transformation Functions](#)

[Frame Information Functions](#)

[Frames Supported in SPICE](#)

[ICRF vs J2000](#)

[Kernels Needed For Computing Frame Transformations](#)

## Creating a Frame Kernel

[Frame Classes](#)

[Specifying a New Frame](#)

[Guidelines for Frame Specification](#)

[Selecting a Name](#)

[Selecting a Frame ID](#)

[Selecting the Class](#)

[Selecting the Center](#)

[Selecting a Class ID](#)

[Frame IDs Reserved for Public Use](#)

[Why have a Frame ID and a Class ID?](#)

[Putting the Pieces Together](#)

[Connecting an Object to its Body-fixed Frame](#)

[The rest of the frame information](#)

[https://naif.jpl.nasa.gov/pub/naif/toolkit\\_docs/C/req/frames.html#Reference%20Frames](https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/req/frames.html#Reference%20Frames)

## Inertial Frames

[PCK Frames](#)

[CK Frames](#)

[SCLK and SPK ID codes](#)

## TK Frames

[Defining a TK Frame Using a Matrix](#)

[Defining a TK Frame Using Euler Angles](#)

[Defining a TK Frame Using a SPICE-style Quaternion](#)

[Gaining Flexibility via TK Frames](#)

## Dynamic Frames

## Parameterized Dynamic Frame Families

[Notation](#)

[Required Keywords for Parameterized Dynamic Frames](#)

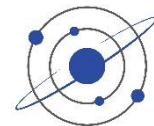
[Conditional Keywords for Parameterized Dynamic Frames](#)

[Rotation State](#)

[Freeze Epoch](#)



# NASA/NAIF : Reference Frames



## Two-Vector Frames

Defining a Two-Vector Frame in a Frame Kernel

Kernel Availability

Specifying the Base Frame

Specifying the Frame Family

Specifying the Rotation state or Freeze Epoch

Specifying the Angular Separation Tolerance

Frame Axis Labels

## Vector Specifications

Observer-Target Position Vectors

Target Near point Vectors

Observer-Target Velocity Vectors

Constant Vectors

## Mean Equator and Equinox of Date Frames

Defining a Mean Equator and Equinox of Date Frame in a Frame Kernel

Specifying the Base Frame

Specifying the Frame Family

Specifying the Precession Model

Specifying a Rotation State or Freeze Epoch

## True Equator and Equinox of Date Frames

Defining a True Equator and Equinox of Date Frame in a Frame Kernel

Specifying the Base Frame

Specifying the Frame Family

Specifying the Precession Model

Specifying the Nutation Model

Specifying a Rotation State or Freeze Epoch

## Mean Ecliptic and Equinox of Date Frames

Defining a Mean Ecliptic and Equinox of Date Frame in a Frame Kernel

Specifying the Base Frame

Specifying the Frame Family

Specifying the Precession Model

Specifying the Mean Obliquity Model

Specifying a Rotation State or Freeze Epoch

## Euler Frames

Defining an Euler Frame in a Frame Kernel

Specifying the Base Frame

Specifying the Frame Family

Specifying the Epoch

Specifying the Euler Angles

## Product Frames

Defining a Product Frame in a Frame Kernel

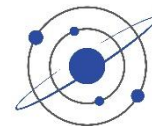
Specifying the Base Frame

Specifying the Frame Family

Specifying the Factors



# NASA/NAIF : Reference Frames



## [Dynamic Frame Implementation Considerations](#)

[Introduction](#)

[Simulated Recursion](#)

[The Need for Recursion in the CSPICE Frame Subsystem](#)

[Implementation of Limited Simulated Recursion](#)

[Limits on Recursion in Frame Definitions](#)

[Frame Derivative Accuracy](#)

[Degenerate Geometry](#)

[Efficiency Concerns](#)

## [Switch Frames](#)

[Specifying Switch Frames](#)

[The Base Frame List](#)

[Time Intervals Associated with Base Frames](#)

[Binary Search](#)

[Switch Frame Connections](#)

[Switch Frame Buffering](#)

## [Appendix. ``Built in'' Inertial Reference Frames](#)

[Complete List of ``Built in'' Inertial Reference Frames](#)

[Inertial Reference Frame References](#)

[Low Level Inertial Reference Frame Functions](#)

## [Appendix. ``Built in'' PCK-Based IAU Body-Fixed Reference Frames](#)

## [Appendix. High Precision Earth Fixed Frames](#)

## [Appendix. Frame Identifiers Reserved for Earth Fixed Frames](#)

## [Appendix. Frame Definition Examples](#)

[Inertial Frames](#)

[PCK Frames](#)

[CK Frames](#)

[TK frames](#)

[TK frame --- Alias](#)

[TK frame --- Topographic](#)

[TK frame --- Instrument](#)

## [Examples of Two-Vector Parameterized Dynamic Frames](#)

[Geocentric Solar Ecliptic \(GSE\) Frame](#)

[Geocentric Solar Magnetospheric \(GSM\) Frame](#)

[Mercury Solar Equatorial \(MSEQ\) Frame](#)

[Example: Nadir Frame for Mars Orbiting Spacecraft](#)

[Example: Roll-Celestial Spacecraft Frame](#)

## [Examples of Mean Equator and Equinox of Date Frames](#)

[Earth Mean Equator and Equinox of Date Frames](#)

## [Examples of True Equator and Equinox of Date Frames](#)

## [Example of a Mean Ecliptic and Equinox of Date Frame](#)

## [Example of an Euler Frame](#)

## [Examples of Product Frames](#)

[IAU\\_EARTH Frame, Augmented with Nutation Model](#)

[Dog-Leg Frame for Saturn](#)

## [Examples of Switch Frames](#)

[Switch Frame Using Reconstructed and Predict CKs](#)

[Switch Frame Using CK and Dynamic Frames](#)

[Predicted Attitude Profile for Observation Planning](#)

# IAU SOFA C function call from Matlab

- Compile C function from Matlab using wrapper
- Call the mex generated function

## iauEra00\_mex.c

```
#include "mex.h"
#include "T:\Work\Matlab\SPACE\IAU-SOFA\C\src\sofa.h"
/*
 iauEra00_mex.c
 MATLAB wrapper for SOFA iauEra00.
 Usage in MATLAB: era = iauEra00_mex(ut1a, ut1b);
 Inputs: ut1a, ut1b : two-part UT1 Julian Date
 Output: era : Earth Rotation Angle, radians

 Compilation instructions
 cd('T:\Work\Matlab\SPACE\IAU-SOFA\C\')
 mex iauEra00_mex.c ...
 "T:\Work\Matlab\SPACE\IAU-SOFA\C\src\era00.c" ...
 "T:\Work\Matlab\SPACE\IAU-SOFA\C\src\anp.c"

 → creates iauEra00_mex.mexw64
 → called from within ComputeERA.m
*/
```

```
void mexFunction(int nlhs, mxArray *plhs[],
 int nrhs, const mxArray *prhs[])
{
 double ut1a, ut1b;
 double era;

 /* (...arguments checking...) */

 ut1a = mxGetScalar(prhs[0]);
 ut1b = mxGetScalar(prhs[1]);

 era = iauEra00(ut1a, ut1b);

 plhs[0] = mxCreateDoubleScalar(era);

} /* end function */
```

TODO: explore further, including **iauGst06a** for example



# Orbital Elements

## Orbit Parameters (COEs & TLEs)

### THE SIX ORBITAL ELEMENTS:

$a$  = Semi-major axis (*size*)

$e$  = Eccentricity (*shape*)

$i$  = inclination (*tilt*)

$\omega$  = argument of perigee (*twist*)

$\Omega$  = longitude of the ascending node (*pin*)

$\nu$  = mean anomaly (*angle now*)

Further reading:

Sellers 2004 [Understanding Space](#), Chapter 5

Orbital Elements

Computing Orbital Elements

Spacecraft Ground Tracks

# Orbital State Vectors → 6 parameters



- An orbit follows the laws of Newton and Kepler around its central gravitational body (primary focus) and its motion is very predictable without major outside disturbances (e.g., drag) or internal disturbances (e.g., thrust)
- For any physical object, at least six variables **6 parameters are absolutely required to completely represent an elliptic orbit** and describe its location in 3D space.
  - Given the masses of the 2 bodies, they determine the full orbit in an inertial plane with respect to the to central body
- The state of an orbiting satellites can be described at any given time by the **3D cartesian coordinates vectors**:

$$\text{Position: } \vec{R} = r_x \hat{I} + r_y \hat{J} + r_z \hat{K}$$

$$\text{Velocity: } \vec{V} = v_x \hat{I} + v_y \hat{J} + v_z \hat{K}$$

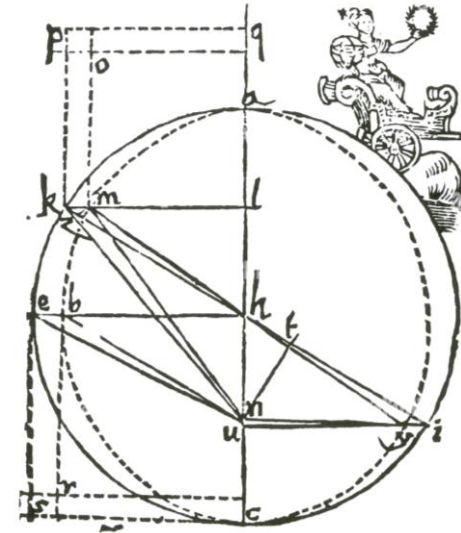
→ This set of **6 variables**, function of time  $t$ , are called the **Orbital State Vectors**

- The two most general cases with these 6 degrees of freedom are the **elliptic** and the **hyperbolic** orbit
  - Special cases with fewer degrees of freedom are the **circular** and **parabolic** orbit

# COE – Classical Orbital Elements



- However, the orbital state vectors  $\vec{R}$  and  $\vec{V}$  are constantly changing as a satellite moves around its orbit, and it is difficult to actually “visualize” the situation; it’s difficult to tell much about the orbit’s size or shape, nor where the satellite is located in it
  - $\vec{R}$  and  $\vec{V}$  do not directly yield much information about the orbit. One cannot even infer from them what type of conic the orbit represents!
  - Another set of six variables, which is much more descriptive of the orbit, is needed
- Kepler defined a set of **Classical Orbital Elements (COEs)**, sometimes called the *Keplerian elements*, that define the orbit and the position of the satellite in the orbit
  - Five of the COEs define the orbit, and the sixth defines the position of the satellite in the orbit at the given time
  - There are six *primary* COEs, and *alternatives* for degenerate cases



Johannes Kepler’s conceptualization of the orbit of Mars 1609

# Orbit element types or sets



- **Cartesian:**  $r_x, r_y, r_z, v_x, v_y, v_z$ 
  - Defines the position and velocity of a satellite with respect to the origin of an inertial rectangular coordinate system. Positions and velocities are expressed by their components along the three principal axes of the ICRF reference frame.
- **Keplerian:**  $a, e, i, \Omega, \omega, \nu$ 
  - Defines six parameters that describe the shape, size, and orientation of a satellite orbit, as well as its current location in the orbit (represented by the True Anomaly  $\nu$ ). Assumes perfect Keplerian orbital motion, so all gravitational forces are treated as point masses and *no perturbing forces are considered* when calculating these parameters.
- **Spherical:**  $|R|, RA, DEC, |V|$ , velocity azimuth angle,  $VFPA$  (Vertical Flight Path Angle)
  - Defines the position and velocity of a satellite with respect to the ICRF reference frame in spherical coordinates.
- **Spherical (Lat-Lon) :**  $R, \lambda, \varphi, |V|$ , velocity azimuth angle,  $HFPA$  (Horizontal Flight Path Angle)
  - Defines the position and velocity of a satellite with respect to the ECEF in spherical coordinates, using the Earth's latitude and longitude in place of right ascension and declination. This element set is intended for use only when the satellite's central body is Earth.
- **SGP4:**  $Epoch, TD_1, TD_2, B^*, i, \Omega, e, \omega, M_0, n$ 
  - The SGP4 derived elements are **associated with the Two-Line Element (TLE) definitions**. This element set is intended for use only when the satellite's central body is Earth, and *should be used with the SGP4 propagator*.

# Classical Orbital Elements (COE)

- Semi-major axis ( $a$ )
- Eccentricity ( $e$ )
- Inclination ( $i$ )
- Right Ascension of the Ascending Node ( $RAAN/\Omega$ )
- Latitude Argument of Perigee ( $\omega$ )
- Mean anomaly ( $M_0$ )



# Describing orbits

---



- Previously we looked at the restricted two-body problem and developed an equation of motion to describe in strictly mathematical terms, how satellites and spacecraft move through space
  - But many times it's not enough to generate a list of numbers that give a spacecraft's position and velocity in inertial space (e.g. state vectors)
- Often, *we want to visualize its orbit* with respect to points on Earth
- Two important tools to “see” satellite motion are:
    1. [Classical Orbital Elements](#) (COEs)
    2. [Satellite Ground Tracks](#)



# Classical Orbital Elements



An elliptical orbit is described by means of two planes (the **orbital plane** and the reference **equatorial plane**) and 6 parameters, allowing you to locate a satellite:

## Size and shape

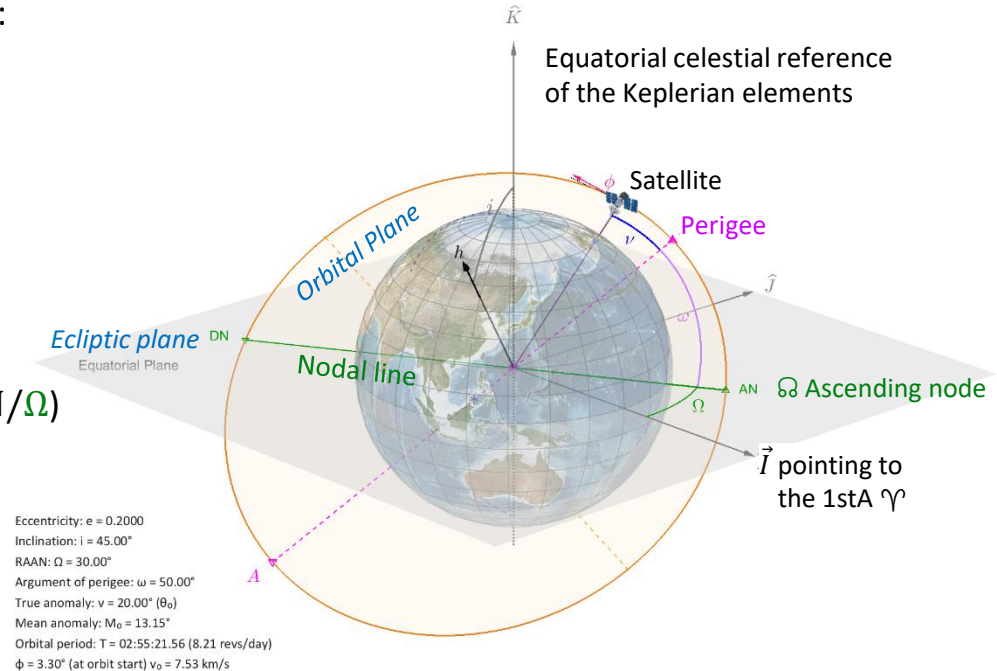
- Semi-major axis ( $a$ )
- Eccentricity ( $e$ )

## Orbital plane orientation

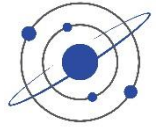
- Inclination ( $i$ )
- Right Ascension of the Ascending Node (RAAN/ $\Omega$ )
- Latitude Argument of Perigee ( $\omega$ )

## Temporal position

- Mean anomaly ( $M_0$ )



# $a$ & $e$ – Ellipse shape parameters

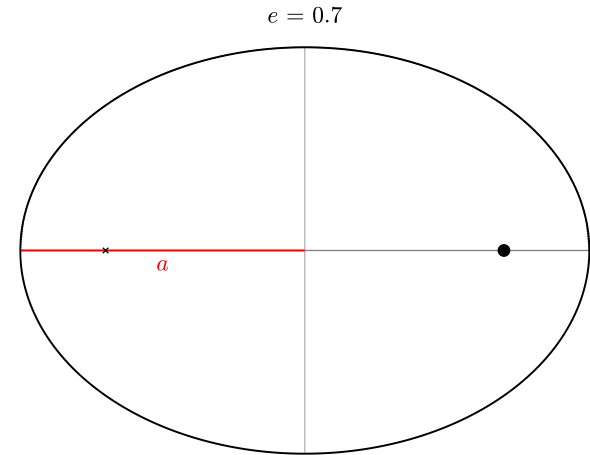


## $a$ – Semi-Major Axis

- The first orbital element describes [the orbit's size](#)
- $a$  measures half the distance across the major (long) axis of the orbit
- It is a *constant parameter* in the absence of forces other than gravity since the orbit's specific mechanical energy does not change

## $e$ – [Eccentricity](#)

- The second orbital element describes [the shape of the orbit](#)
  - It tells how “out of roundness” the conic section is
- Equal to the ratio of the distance between the two foci and the total length of the major axis:  $e = c/a$
- It is a unitless quantity [n.u.]



# Line of Nodes



- The **line of nodes** lies along the intersection of the orbital plane and a reference plane
  - In astronautics, the reference plane is either the **equatorial plane** of the planet around which the artificial satellite is orbiting, or the **ecliptic** (ECL) for an interplanetary probe placed in a heliocentric orbit
- **Nodes** are the points where an orbit crosses the Earth's equatorial plane
  - The **ascending node** ( $\oslash$ ) is the point where the satellite crosses the equatorial plane from the southern to northern hemisphere
  - While the **descending node** ( $\ominus$ ) is the point where the satellite crosses the equatorial plane in the southerly direction. The descending node is located  $180^\circ$  (half-way) further around the orbit from the ascending node

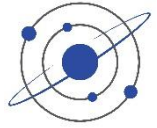
- $\vec{n}$ , the **ascending node vector**, is a vector on the line of nodes that points in the direction of the ascending node

$$\vec{n} = \vec{K} \times \vec{h}$$

- $\vec{e}$ , the **eccentricity vector**, is a vector on the apse line that points in the direction of the perigee of the ellipse

$$\vec{e} = \frac{\vec{v} \times \vec{h}}{\mu} - \frac{\vec{r}}{r}$$

# Key vectors



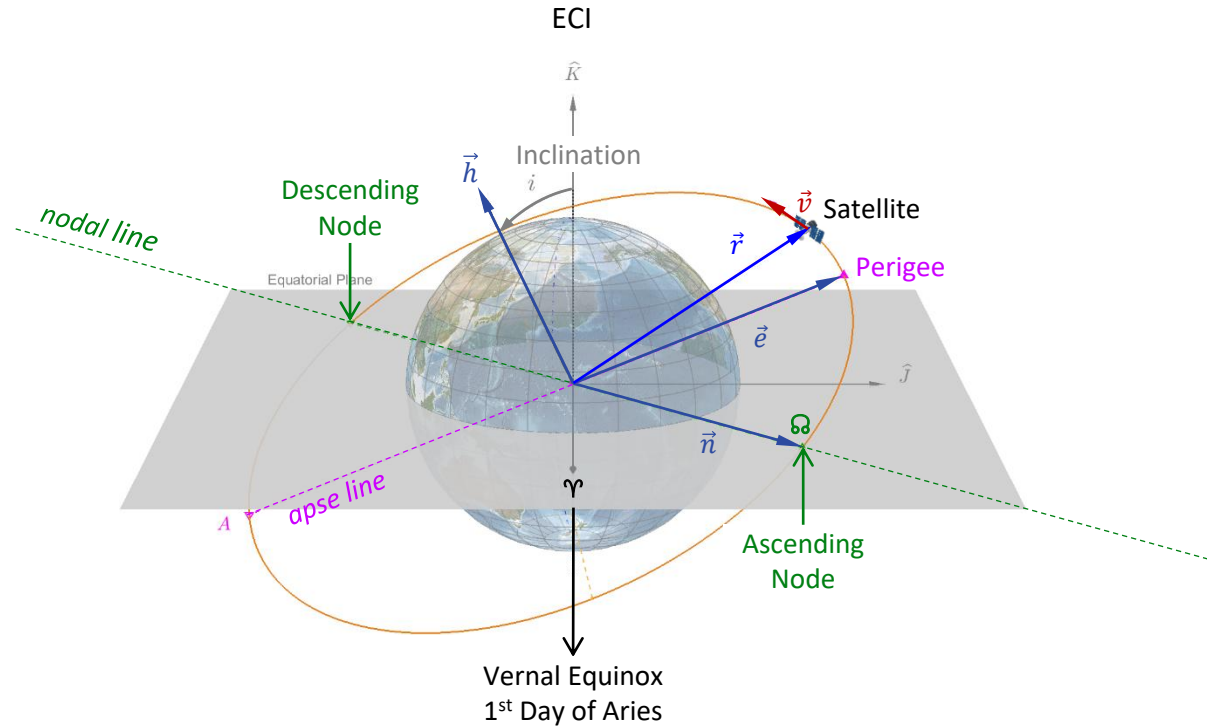
$\vec{r}$  = Satellite position vector

$\vec{v}$  = Satellite velocity vector

$\vec{h}$  = Angular Momentum vector  
perpendicular to orbital plane

$\vec{n}$  = Ascending Node vector  
in equatorial plane

$\vec{e}$  = Eccentricity vector  
in orbital plane

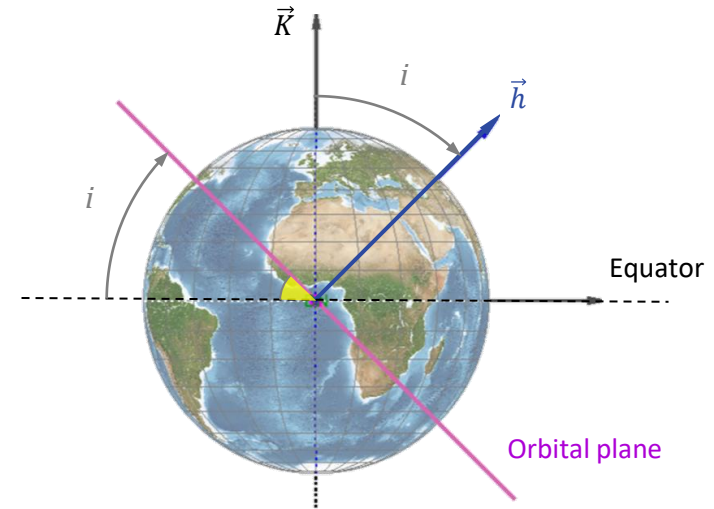


SV\_to\_COE

# $i$ – Inclination



- The angle between the **satellite's orbital plane** and the **equatorial plane** of the celestial body it orbits (e.g. Earth) **measured at the ascending node** (the location where the orbit passes upward through the reference plane).
  - Computed as the angle between Earth's spin axis ( $\vec{K}$  or  $\vec{Z}$ ) and the angular momentum vector ( $\vec{h}$ ) of the orbital plane
  - Pivots along the line of nodes
- Inclination can range from **0° to 180°**
  - $i = 0^\circ$  is called an **equatorial orbit** because satellites in these orbits travel above Earth's equator
  - $i = 63.4^\circ$  is called a **critical inclination**, because they have zero apogee drift (→ more on this later with perturbations)
  - $i = 90^\circ$  is called **polar orbit** as it travels over the north and south geographical poles; used for Earth Observation
  - $i \approx 98^\circ$  is called **Sun synchronous orbit**, in which the angle between the orbital plane and the direction of the Sun remains constant (→ more on this later with types of orbit)
  - $i = 180^\circ$  or  $i < 0^\circ$  is theoretically possible but not used in practice

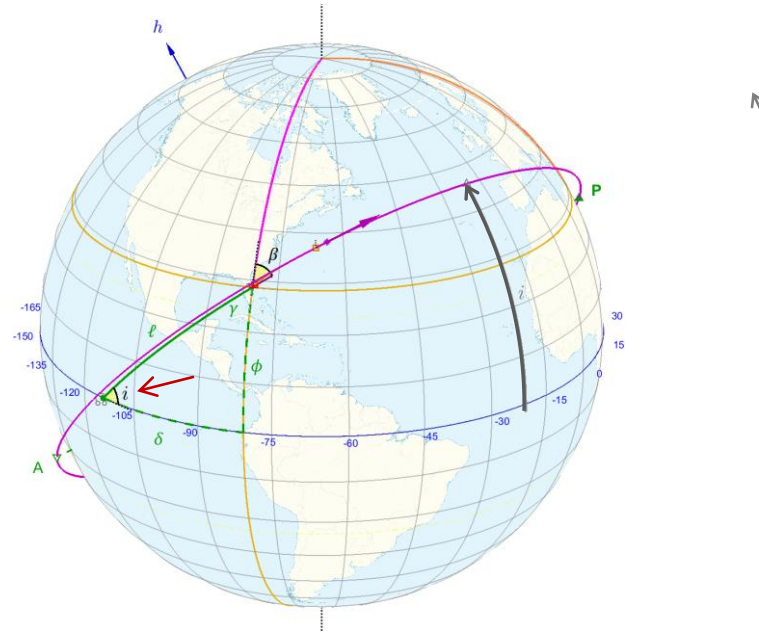


Orbit types



# $i$ – Inclination

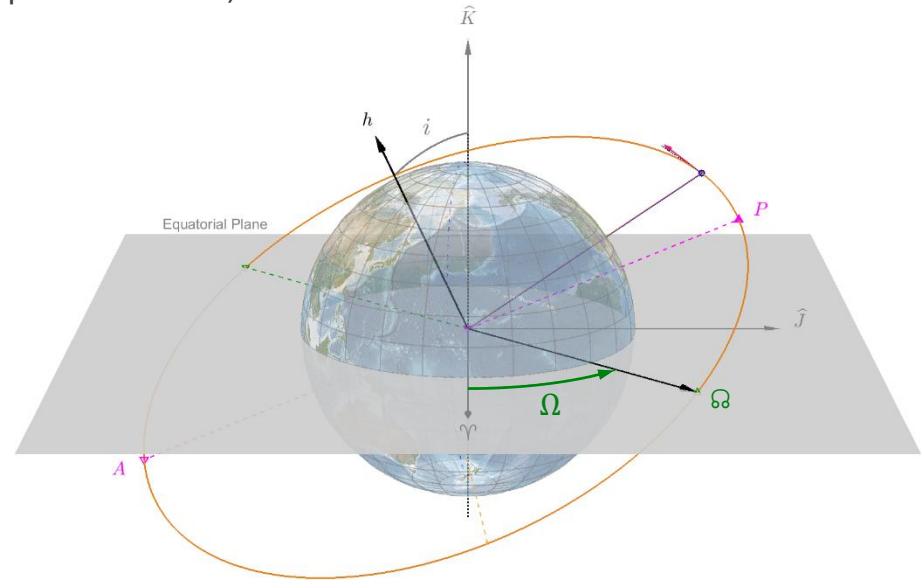
- Angle at equator intersection

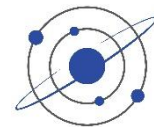


# $\Omega$ – Right Ascension of Ascending Node (RAAN)



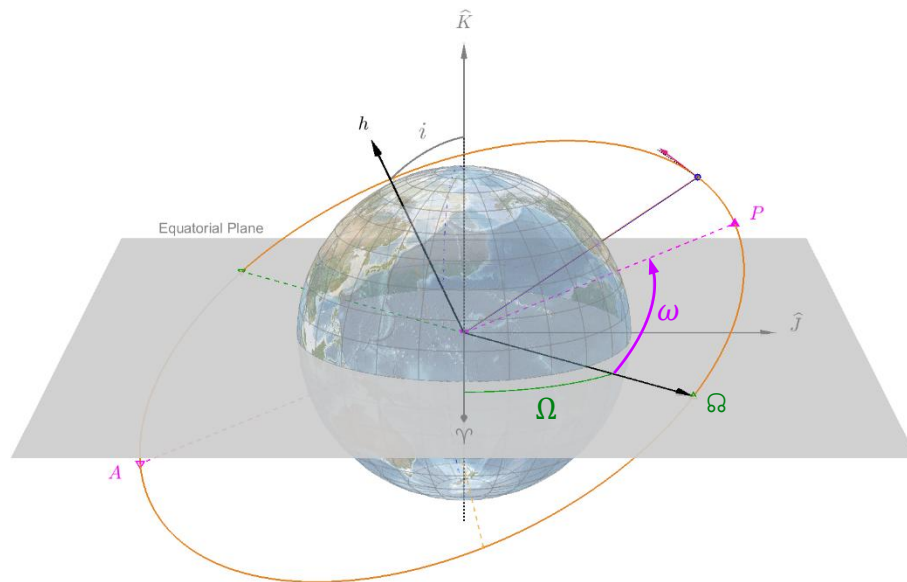
- $\Omega$ , or RAAN, indicates the **swivel** or orientation of an orbit within the equatorial plane
- $\Omega$  is the angle measured from the **vernal equinox** direction  $\Upsilon$ , or  $\hat{I}$  axis, to the **ascending node** vector  $\vec{n}$ 
  - If the ascending node vector points along the vernal equinox direction, then  $\Omega = 0^\circ$
- $\Omega$  can take any value between  $0^\circ$  and  $360^\circ$
- Note that  $\Omega$  is undefined for equatorial orbits ( $i = 0^\circ$ ), where there is no ascending node
  - Use of Longitude of the Ascending Node (see alternate COEs)





# $\omega$ – Argument of Perigee

- Defines the **orientation of the orbit** within the orbital plane, or where its perigee is located in the ellipse
- $\omega$  is the angle measured between the ascending node vector,  $\vec{n}$ , and the location of perigee, measured in the direction of satellite motion
- $\omega$  can take any value between  $0^\circ$  and  $360^\circ$
- If perigee is at the ascending node, then  $\omega = 0^\circ$
- Note that  $\omega$  is not defined for equatorial orbits ( $i = 0^\circ$ ), where there is no ascending node
- It is not defined for circular orbits ( $e = 0$ ) since there is no perigee in a circular orbit

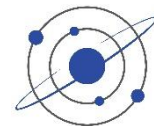




# $\varpi$ – Longitude of perihelion



- The **argument of perigee (or periapsis)  $\omega$**  and **longitude of perihelion (or periapsis)  $\varpi$**  are closely related orbital elements, but they refer to different reference frames and include different angular components.
  - $\omega$  is the angle measured **in the orbital plane from the ascending node to the periapsis** (perigee, perihelion, etc.)
  - It tells you where the closest point (periapsis) lies within the orbit, measured from the point where the orbit crosses the reference plane going northward (ascending node).  $\omega = \angle(\text{ascending node} \rightarrow \text{periapsis})$
- Mainly **used for artificial satellites** around the Earth
- $\varpi$  is the angle measured **in the reference plane** (not the orbital plane) from the reference direction (usually the **vernal equinox direction**) to the perihelion.
  - It combines two elements:  $\varpi = \Omega + \omega$ 
    - $\Omega$ : tells you **which way the plane is rotated** around a vertical axis (in space)
    - $i$ : tells you **how much the plane is tilted**.
    - $\omega$ : tells you **where on that plane the low point (periapsis)** is located.
    - $\varpi$ : tells you **the direction of that low point (periapsis) in space**, combining both the tilt and the rotation.
- Commonly **used in planetary astronomy**



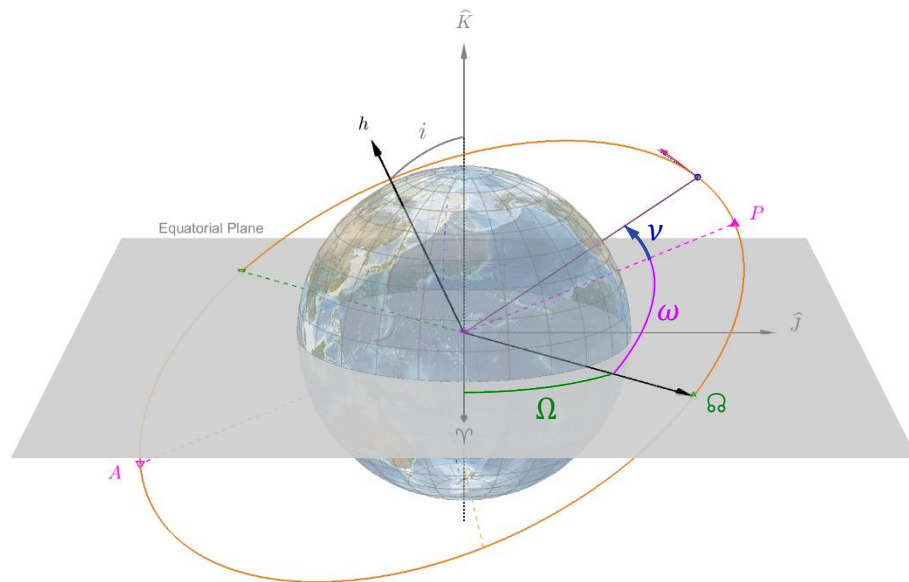
# $\nu$ – True Anomaly

- $\nu$  is the varying angle during the orbit, scanning 0 to 360°, in correlation with  $M$  and  $E$  anomalies
- $\nu$  is the angle measured from perigee to the position vector to the satellite,  $\vec{r}$
- $\nu$  can take any value between 0° and 360°
- $\nu$  is not defined for circular orbits ( $e = 0$ ) since there is no perigee in a circular orbit

## In Orbits101:

- $\nu$  is the starting point for the orbit propagation
- indicates location of the satellite in the orbit at  $t_{\text{start}}$

`Orbit.v [°] ≡ DEG(Orbit.T(1))`



$\nu$  – True Anomaly

# Summary of Classical Orbital Elements



| Element  | Name                                  | Description                                                                          | Range of Values                           | Undefined                                                                  |
|----------|---------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------|
| a        | Semimajor axis                        | Size                                                                                 | Depends on the conic section              | Never                                                                      |
| e        | Eccentricity                          | Shape                                                                                | $e = 0$ : circle<br>$0 < e < 1$ : ellipse | Never                                                                      |
| i        | Inclination                           | Tilt, angle from $\hat{K}$ unit vector to specific angular momentum vector $\hat{h}$ | $0 \leq i \leq 180^\circ$                 | Never                                                                      |
| $\Omega$ | Right ascension of the ascending node | Swivel, angle from vernal equinox to ascending node                                  | $0 \leq \Omega < 360^\circ$               | When $i = 0$ or $180^\circ$ (equatorial orbit)                             |
| $\omega$ | Argument of perigee                   | Angle from ascending node to perigee                                                 | $0 \leq \omega < 360^\circ$               | When $i = 0$ or $180^\circ$ (equatorial orbit) or $e = 0$ (circular orbit) |
| $\nu$    | True anomaly                          | Angle from perigee to the spacecraft's position                                      | $0 \leq \nu < 360^\circ$                  | When $e = 0$ (circular orbit)                                              |

# Orbital Elements for various missions

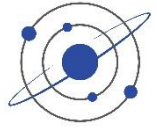


| Mission                                                                                                             | Orbital Type     | Semimajor Axis<br>(Altitude)                       | Period  | Inclination                | Other                             |
|---------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------|---------|----------------------------|-----------------------------------|
| <ul style="list-style-type: none"><li>• Communication</li><li>• Early warning</li><li>• Nuclear detection</li></ul> | Geostationary    | 42,158 km<br>(35,780 km)                           | ~24 hr  | ~0°                        | $e \cong 0$                       |
| <ul style="list-style-type: none"><li>• Remote sensing</li></ul>                                                    | Sun-synchronous  | ~6500 – 7300 km<br>(~150 – 900 km)                 | ~90 min | ~95°                       | $e \cong 0$                       |
| <ul style="list-style-type: none"><li>• Navigation<br/>– GPS</li></ul>                                              | Semi-synchronous | 26,610 km<br>(20,232 km)                           | 12 hr   | 55°                        | $e \cong 0$                       |
| <ul style="list-style-type: none"><li>• Space Shuttle</li></ul>                                                     | Low-Earth orbit  | ~6700 km (~300 km)                                 | ~90 min | 28.5°, 39°,<br>51°, or 57° | $e \cong 0$                       |
| <ul style="list-style-type: none"><li>• Communication/<br/>intelligence</li></ul>                                   | Molniya          | 26,571 km ( $R_p = 7971$ km;<br>$R_a = 45,170$ km) | 12 hr   | 63.4°                      | $\omega = 270^\circ$<br>$e = 0.7$ |

# Computing/retrieving COEs



# Computing Orbital Elements



- In real life, COEs can't be measured directly, but a spacecraft's inertial position and velocity,  $\vec{R}$  and  $\vec{V}$  state vectors, can be determined using ground-tracking sites
- Still, we need some way to convert the state vectors to COEs, so we can make sense of an orbit
- All of the orbital elements can be retrieved with just a position vector,  $\vec{R}$ , and a velocity vector,  $\vec{V}$ , at a single point in time, as six pieces of information are needed to define an orbit: the three components of  $\vec{R}$  and the three components of  $\vec{V}$
- From a state vector, six different quantities can be computed – the COEs – to better visualize the orbit

$$\begin{array}{ccc} & r_x & e \\ \vec{R} = & r_y & a \\ & r_z & i \\ & v_x & \Omega \\ \vec{V} = & v_y & \omega \\ & v_z & \nu \end{array} \quad \rightarrow$$

# Measuring satellite state vectors from Earth

---



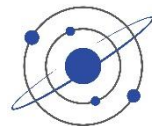
- Ground-based tracking of satellites relies on:
  - 1) [Acquisition of observational data](#), followed by
  - 2) [Orbit reconstruction](#) through estimation techniques
- Two primary sensor types are employed: radar and optical systems
  - **Radar** measurements provide range (distance), range rate (via Doppler), and angular information (azimuth and elevation), yielding direct geometric constraints on the satellite's position and partial velocity
  - **Optical systems**, by contrast, measure angular quantities such as right ascension and declination or azimuth and elevation, offering high directional accuracy but *lacking direct range information*. As a result, optical observations require multiple measurements over time to reconstruct the full three-dimensional position
- These observations are first expressed in a [local topocentric frame](#) tied to the ground station, then transformed into an Earth-Centered Inertial (ECI) frame by accounting for the station's position and Earth orientation
  - With a sequence of such measurements, an initial orbit is obtained using classical initial [Orbit Determination Methods](#) (e.g., Gauss or Laplace for angles-only data)
  - When range data are available, as in radar tracking, the reconstruction is more straightforward and typically more accurate at the initial stage

# Measuring satellite state vectors from Earth (continued)



- The preliminary orbit is subsequently refined through orbit determination techniques based on estimation theory, such as batch least-squares or Kalman filtering
    - These methods adjust the satellite's state vector – its position and velocity in an inertial frame – by minimizing the discrepancy between observed measurements and those predicted by a dynamical model
    - The model [incorporates relevant Orbital Perturbations](#), including Earth's non-spherical gravity field, atmospheric drag, third-body effects, and solar radiation pressure, thereby yielding a high-fidelity estimate of the satellite's trajectory
  - Finally, the estimated orbit is converted into a set of orbital elements (COEs) and fitted to the standardized [Two-Line Element \(TLE\)](#) format
  - This fitting process is not a simple parameter conversion; rather, it involves adjusting [mean orbital elements](#) so that their propagation under the [SGP4 model](#) reproduces the observed motion as closely as possible
  - [The entire process is iterative](#), with [new observations continuously updating the orbit solution and the associated TLE](#), ensuring accurate prediction of satellite positions over time
- [Ground stations do not directly measure orbits – they measure observables, then use physics and estimation theory to reconstruct the satellite's inertial state vector, which is finally compressed into a TLE for operational use](#)





# SV → COE

```
function COE = SV_to_COE(SV_R, SV_V, mu)
% Computes the classical orbital elements (COEs)
% from the state vector (R,V)
%{
 mu - gravitational parameter [km³/s²] μ

 R - Position vector [km]
 V - Velocity vector [km/s]
 in the geocentric equatorial frame

 Hv - Angular momentum vector [km²/s]
 Nv - Node line vector [km²/s]
 Ev - Eccentricity vector [n.u.]

 e - Eccentricity (magnitude of E)
 a - Semimajor axis [km]
 i - Inclination of the orbit [°]
 RAAN - Right Ascension of the Ascending Node [°] Ω
 w - argument of perigee [°] ω
 T - True anomaly [rad] v

 E - Eccentric Anomaly [rad]
 M - Mean Anomaly [rad]
 epsilon - Specific Mechanical Energy [km²/s²] ε
 h - Angular momentum [km²/s]

%}
```



Key vectors

```
 $\vec{r}$ R = SV_R(:, j); % (when looping on j)
 $\vec{v}$ V = SV_V(:, j);

%--- Input State Vectors
r = norm(R); % [km] magnitude of R
v = norm(V); % [km/s] magnitude of V
vr = dot(R,V) / r; % [km/s] radial velocity component

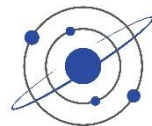
% Specific Mechanical Energy ε
ε epsilon = v^2/2 - mu/r; % [km²/s²]

%--- Key vectors
 $\hat{I}$ I = [1;0;0]; % [n.u.] (normalized units)
%J = [0;1;0]; % [n.u.] (unused)
 $\vec{K}$ K = [0;0;1]; % [n.u.]

% Specific Angular Momentum vector
 $\vec{h}$ Hv = cross(R, V); % H = R × V
h = norm(Hv); % [km²/s] magnitude of H

% Line of nodes vector
 $\vec{n}$ Nv = cross(K, Hv); % N = K × H

% Eccentricity vector: First integral of motion
 $\vec{e}$ Ev = 1/mu * ((v^2 - mu/r) * R - r * vr * V); % [n.u.]
```



# SV → COE (continued)

```
%--- Ellipse eccentricity (First integral of motion)
e = norm(Ev); % [n.u.]
```

```
%--- Semimajor axis a through Vis-viva $\epsilon = -\mu/(2a)$
a = -mu / (2 * epsilon); % [km]
```

```
% → the following are found by geometrical arguments
% Angle(A,B) = acosd(dot(A,B)/(norm(A)*norm(B)))
```

```
%--- Inclination i
i = Angle(K, Hv); % [°]
```

```
%--- RAAN Ω
if i ~= 0 % or n ~= 0
 RAAN = Angle(I, Nv); % [°]
 if Nv(2) < 0 RAAN = 360 - RAAN; end
else
 RAAN = 0; % [°] undefined for equatorial orbit (i = 0°)
end
```

```
%--- Argument of Perigee w
if i ~= 0
 if e > eps
 w = Angle(Nv, Ev); % [°]
 if Ev(3) < 0 w = 360 - w; end
 else
 w = 0; % [°] undefined for circular orbit (e = 0)
 end
else
 w = 0; % [°] undefined for equatorial orbit (i = 0°)
end
```

```
%--- True Anomaly v
if e > eps
 v = Angle(Ev, R); % [°]
 if vr < 0 v = 360 - v; end
else % Circular orbit
 v = Angle(Nv, R); % [°]
 cp = cross(Nv, R);
 if cp(3) < 0 v = 360 - v; end
end
```

# COE exercise



- Given the following COEs, which of the following apply to the satellite's orbit and position (check all that apply)  
 $a = 10000 \text{ km}$ ,  $e = 0 / 0.2$ ,  $i = 30^\circ$ ,  $\Omega = 180^\circ$ ,  $\omega = 90^\circ$ ,  $\nu = 90^\circ$
- |                                                                                    |                                                                                       |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <input type="checkbox"/> Orbit is circular                                         | → when $e = 0$                                                                        |
| <input type="checkbox"/> Orbit is elliptical                                       | → when $e > 0$                                                                        |
| <input type="checkbox"/> Orbit is equatorial                                       | → when $i = 0^\circ$                                                                  |
| <input type="checkbox"/> Orbit is polar                                            | → when $i = 90^\circ$                                                                 |
| <input type="checkbox"/> Orbit is prograde                                         | → when $0 < i \leq 90^\circ$ (ambiguous for $i = 90^\circ$ ; not defined)             |
| <input type="checkbox"/> Orbit is retrograde                                       | → when $i > 90^\circ$                                                                 |
| <input type="checkbox"/> Satellite is at Perigee                                   | → when $\nu = 0^\circ$                                                                |
| <input type="checkbox"/> Satellite is at Apogee                                    | → when $\nu = 180^\circ$                                                              |
| <input type="checkbox"/> Satellite is at Ascending Node $\oslash$                  | → when $\omega + \nu = 0^\circ$ with $e = 0$ (or numerical solution)                  |
| <input type="checkbox"/> Satellite is at Descending Node $\oslash$                 | → when $\omega + \nu = 180^\circ$ with $e = 0$                                        |
| <input type="checkbox"/> Satellite is along $\hat{I}$ axis in the + or – direction | → when $\Omega = 0^\circ/180^\circ$ , $\omega + \nu = 0^\circ/180^\circ$ with $e = 0$ |
| <input type="checkbox"/> Satellite is along $\hat{J}$ axis in the + or – direction | → when $\Omega = \pm 90^\circ$ , $\omega + \nu = 0^\circ/180^\circ$ with $e = 0$      |
| <input type="checkbox"/> Satellite is along $\hat{K}$ axis in the + or – direction | → when $e = 0$ , $i = 0^\circ$ , $\omega + \nu = \pm 90^\circ$                        |

# COE exercise (continued)

---

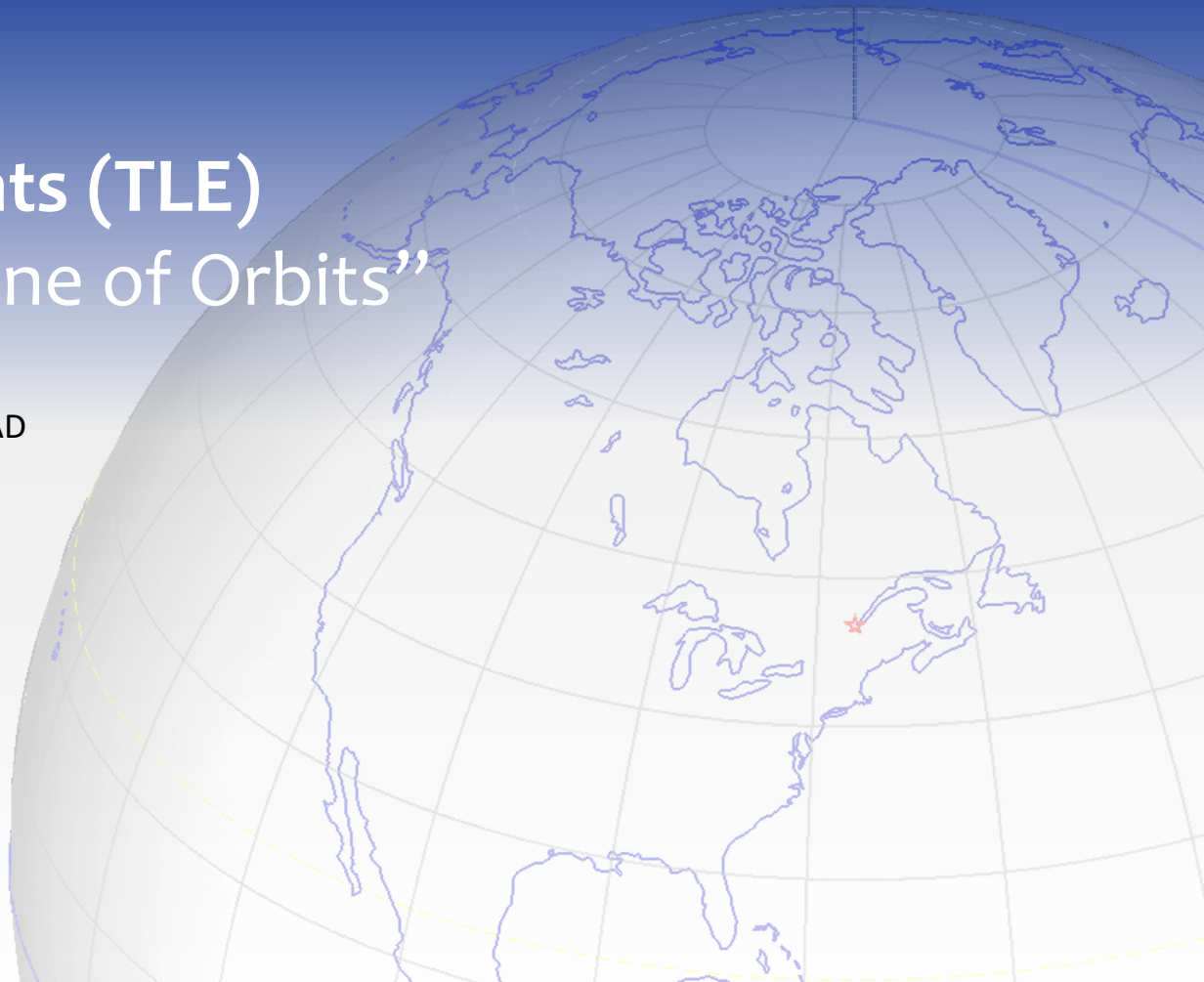


- Given the following COEs, which of the following apply to the satellite's orbit and position (check all that apply)  
 $a = 10000 \text{ km}$ ,  $e = 0 / 0.2$ ,  $i = 30^\circ$ ,  $\Omega = 180^\circ$ ,  $\omega = 90^\circ$ ,  $\nu = 90^\circ$ 
  - ☐ Satellite is over the Equator  $\rightarrow$  when  $e = 0$ ,  $\omega + \nu = 0^\circ$  or  $180^\circ$ ,  $\forall i$
  - ☐ Satellite is over North Pole  $\rightarrow$  when  $e = 0$ ,  $i = 90^\circ$ ,  $\omega + \nu = 90^\circ$ ,  $\forall \Omega$
  - ☐ Satellite is over South Pole  $\rightarrow$  when  $e = 0$ ,  $i = 90^\circ$ ,  $\omega + \nu = 270^\circ$  or  $-90^\circ$ ,  $\forall \Omega$
  - ☐ Satellite is gaining potential energy  $\rightarrow$  when satellite going towards perigee ( $e \neq 0$ )
  - ☐ Satellite is losing potential energy  $\rightarrow$  when satellite going towards apogee ( $e \neq 0$ )

# Two-Line Elements (TLE)

## “The Rosetta Stone of Orbits”

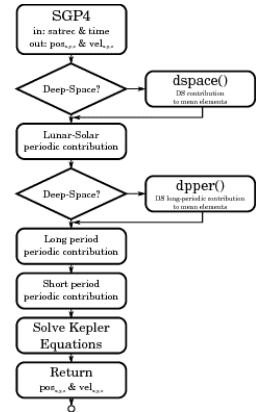
- Data sets for monitoring from NORAD



# TLE – NORAD Two-Line Element set



- A **two-line element set** (TLE) is a data format encoding a set of *mean* orbital elements describing the trajectory of an Earth-orbiting object **for a given point in time**, the *epoch*
  - Developed for the *North American Aerospace Defense Command* (NORAD) in the early 1970s
  - TLEs are generated from the Space Surveillance Network observations
  - De facto standard widely used for the distribution of satellite information
- TLE information is used as input for computing the state vector (position and velocity) at any point in the past or future for the orbital tracks of satellites
- TLE data representation is **specific to the Simplified Perturbations Models (SGP4, SDP4, SDP8)**
  - **SGP4 model** accounts for orbital perturbations caused by Earth's oblateness and atmospheric drag, and applicable to near-Earth satellites ( $P \leq 225$  min, i.e.  $H \leq 5876$  km) → *described later in details*
  - The SGP4 model has an **error  $\approx 1$  km at the epoch** of the element set and **grows at  $\approx 1\text{--}3$  km per day**
  - This data is **updated frequently** in NASA and NORAD sources due to this error



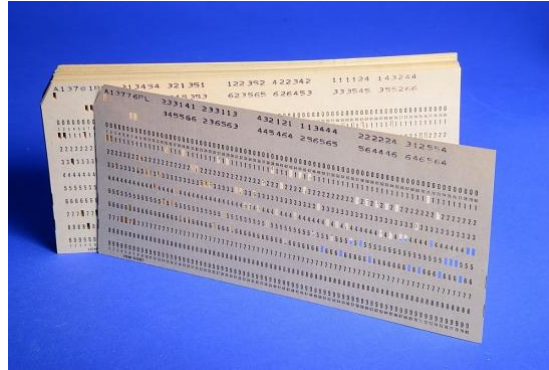
Note: The **ISS study** shows a **10 km error at epoch** and growing at **tens of km per day**, limiting high-accuracy predictions to **within a week (< 100 orbits)**

# TLE Format



- The format was **originally intended for punched cards**, encoding a set of elements on two standard 80-column cards. This format was eventually replaced by **text files** as punch card systems became obsolete, with each set of elements written to two 69-column ASCII lines preceded by a title line

[https://en.wikipedia.org/wiki/Two-line\\_element\\_set](https://en.wikipedia.org/wiki/Two-line_element_set)



- **3LE:** A TLE set may include a title line preceding the element data, so each listing may take up three lines in the file, in which case the TLE is referred to as a **three-line element set (3LE)**
    - The title is not strictly required, as each data line includes a unique object identifier code
- all times are measured **mean solar** rather than sidereal time units

# TLE data format parser

---



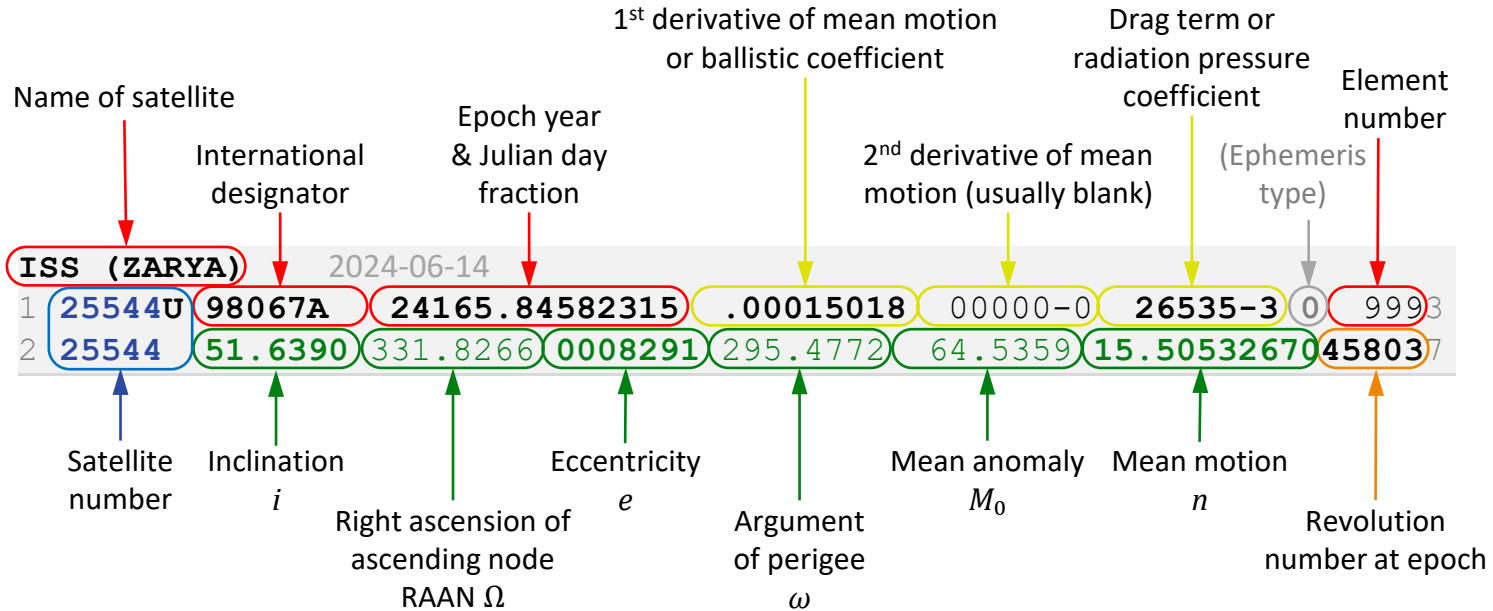
```
ISS (ZARYA) 2024-06-14
1 25544U 98067A 24165.84582315 .00015018 00000-0 26535-3 0 9993
2 25544 51.6390 331.8266 0008291 295.4772 64.5359 15.50532670458037
```

(adapted from: <https://ntrs.nasa.gov/api/citations/20180008672/downloads/20180008672.pdf>)





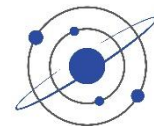
# TLE data format parser



(adapted from: <https://ntrs.nasa.gov/api/citations/20180008672/downloads/20180008672.pdf>)

# Traffic Coordination System for Space (TraCSS)

---



- <https://tracss.gov/>

- TraCSS delivers cutting-edge space situational awareness to protect your assets in an increasingly crowded orbital environment. Real-time collision warnings and comprehensive analysis empower operators to make critical decisions with confidence. The future of spaceflight safety -- powered by the Office of Space Commerce

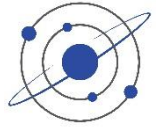
- <https://space.commerce.gov/traffic-coordination-system-for-space-tracss>



- NOAA's Office of Space Commerce (OSC) is developing the Traffic Coordination System for Space (TraCSS) to provide basic space situational awareness (SSA) data and services to civil and private space operators in support of spaceflight safety

- <https://www.parsons.com/products/tracss/>

# TLE production

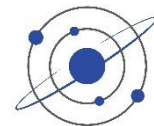


- The satellite Two-Line Elements (TLEs) are under the responsibility of the [Combined Force Space Component Command \(CFSCC\)](#), formerly the JFSCC, and operates under [U.S. Space Command \(USSPACECOM\)](#)
- These commands track all detectable objects in Earth orbit, creating a corresponding TLE for each object, and release and makes publicly available the satellite orbital parameters and historical satellite data of many of the space objects on websites like SPACE-TRACK and CELESTRAK
  - Some TLEs are kept unpublished for security purposes (military or classified objects)



The [U.S. Space Force](#) is a military branch that organizes, trains, and equips space professionals (“Guardians”). [U.S. Space Command](#) is a combatant command that utilizes and directs those forces to defend space interests. Space Force (Department of the Air Force) provides the personnel, while Space Command operates them

# Public data sources



**SPACE-TRACK:** [www.space-track.org/auth/login](http://www.space-track.org/auth/login)

**SPACE-TRACK.ORG**

- Website maintained by the 18<sup>th</sup> Space Defense Squadron that provides public, authorized, and commercial users with space situational awareness (SSA) data. It promotes space flight safety by sharing satellite tracking data, such as TLEs and reentry predictions, to prevent collisions in orbit

**CELESTRAK:** <https://celestrak.org/NORAD/elements>



- Celestrak is a prominent, non-profit resource providing free, public, and high-accuracy tracking data for Earth-orbiting satellites, including Starlink, commercial, and scientific satellites. It maintains a database of Two-Line Elements (TLEs) from the US Space Force to provide real-time satellite locations, conjunction assessments (collision monitoring) through SOCRATES, and advanced orbit visualization tools
- Download latest [Earth Orientation Parameters \(EOP\)](#) data (see LoadEOP)

**SatNOGS:** <https://db.satnogs.org>

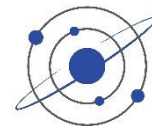
**SatNOGS DB**

- SatNOGS DB is an effort to create an holistic, unified, global database for all artificial objects in space (satellites and spacecrafts). Data viewer, and export data with API

**ISON:** International Scientific Optical Network: <http://spacedata.vimpel.ru>



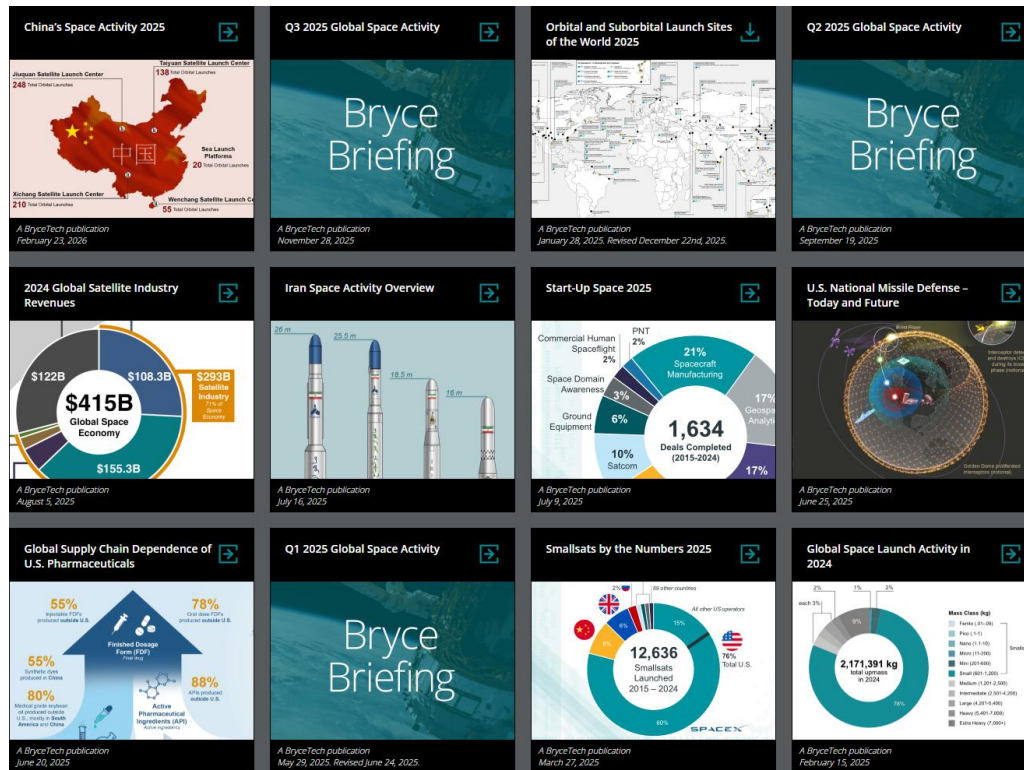
- Russian site providing a weekly update of a catalog listing manmade objects in Earth orbit



BryceTech is an analytics and engineering firm that provides data-driven strategic advisory, systems engineering, and management services to aerospace, defense, and government clients, including NASA and the DoD

<https://brycotech.com>

<https://brycotech.com/reports>



# Space-Track data download



## [SPACE-TRACK](#) : Login

> Element Set (ELSET) Data > Bulk Catalog Data Downloads > Full Catalog · 3LE (→ save link as...)

SPACE-TRACK.ORG

HOME ▾ HELP ▾

Welcome Box Score SATCAT Decay/Reentry Query Builder

**ELEMENT SET (ELSET) DATA**  
Retrieve ELSET Data by Satellite Catalog Number  
Bulk Catalog Data Downloads  
TLE Format Description  
CCSDS OMM Standard

**SATELLITE CATALOG (SATCAT) DATA**  
Satellite Box Score (API)  
Satellite Search  
Catalog Change Report - now parts 4 & 5 of the SSR  
Geosynchronous Report (API)  
Satellite Situation Report

**SATELLITE DECAY & REENTRY DATA**  
Satellite Decays, Predictions, and TIP Messages  
Search the SATCAT by Satellite Decay Date

| SPACE SCOREBOARD |                |
|------------------|----------------|
| OBJECT TYPE      | APPROX OBJECTS |
| Active Payloads  | 12200          |
| Analyst Objects  | 16700          |
| Debris           | 21200          |
| Total            | 50100          |

Note that de-orbited satellites are removed from latest lists, and prior lists might not contain newer satellites

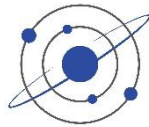
Example message:

SPACE-TRACK.ORG

LOGIN HELP ▾ Current Time (UTC)  
2024-06-29 16:48:37

The 18th Space Defense Squadron (18 SDS) has confirmed the breakup of RESURS P1 (#39186, 2013-030A), which likely occurred June 26, 2024, between 1512-1640 UTC. As of June 26, 18 SDS is tracking 100+ associated pieces at an estimated 350 km altitude, which are being incorporated into routine conjunction assessment to support spaceflight safety. Analysis is ongoing.

# CELESTRAK data download



**CELESTRAK** : [Current Data \(GP\)](#)

TLE/3LE (Three Line Elements) > Active Satellites (→ save link as...)

### NORAD GP Element Sets Current Data

Current as of 2024 Jun 14 12:05:34 UTC (Day 166)

**A New Way to Obtain GP Data (aka TLEs)**

TLE/3LE 2LE OMM XML OMM KVN JSON JSON PP CSV

**Supplemental GP Data**

|                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special-Interest Satellites</b>                                                                                                                                                                                                                                      |
| Last 30 Days' Launches                                                                              |
| Space Stations                                                                                        |
| 100 (or so) Brightest                                                                               |
|  Active Satellites   |
| Analyst Satellites                                                                                  |
| Russian ASAT Test Debris (COSMOS 1408)                                                            |
| Chinese ASAT Test Debris (FENGYUN 1C)                                                             |
| IRIDIUM 33 Debris                                                                                   |
| COSMOS 2251 Debris                                                                                  |

Example message:

**URGENT:** We will run out of 5-digit catalog numbers at 69999 *not* 99999, which is **estimated to occur around 2026-07-20** (we're currently at 68408). At that point, newly cataloged objects will have 6-digit catalog numbers of 100000+ and GP data will not be available for them using the TLE format. **CelesTrak developed new formats** that removed this limitation (and finally fixed the Y2K problem) in May 2020 and immediately began providing GP data in those formats for software developers. The same limitations apply to the **legacy fixed-field SATCAT**. Follow **@TSKelso on Bluesky** for the latest updates, tutorials, and changes as we approach this transition. Spread the word! - TS

# Satellite databases



Most recent element sets for every object in specific groups that has received an update

## SPACE-TRACK [www.space-track.org/auth/login](http://www.space-track.org/auth/login)

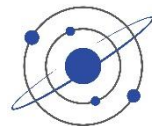
- Element Set (ELSET) Data > Bulk Catalog Data Downloads > Full Catalog · 3LE
- Current Catalog Files (2026-03-19):
  - Full = 30517 satellite entries
  - LEO = 15225 → Starlink = 10123
  - MEO = 264; GEO = 1045; HEO = 192

|                                                 |                        |
|-------------------------------------------------|------------------------|
| 'Space-Track Full Catalog 3LE (2023-03-27).txt' | % 24990 entries        |
| 'Space-Track Full Catalog 3LE (2024-03-04).txt' | % 25860 entries        |
| 'Space-Track Full Catalog 3LE (2026-03-27).txt' | % <b>30517</b> entries |

## CelesTrak <https://celestrak.org/NORAD/elements>

|                                                   |                        |
|---------------------------------------------------|------------------------|
| 'CelesTrak NORAD All Active 3LE (2023-04-13).txt' | % 7542 entries         |
| 'CelesTrak NORAD All Active 3LE (2026-02-18).txt' | % 14373 entries        |
| 'CelesTrak NORAD All Active 3LE (2026-03-29).txt' | % <b>14868</b> entries |

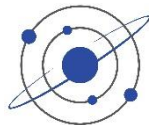




# Data structure: Line 1

|    |       |                                                                                      |                       |
|----|-------|--------------------------------------------------------------------------------------|-----------------------|
| 1  | 01    | Line number                                                                          | 1                     |
| 2  | 03-07 | <b>Satellite catalog number</b>                                                      | 25544 ( $\equiv$ ISS) |
| 3  | 08    | Classification (U: unclassified, C: classified, S: secret)                           | U                     |
| 4  | 10-11 | International Designator (last two digits of launch year)                            | 98                    |
| 5  | 12-14 | International Designator (launch number of the year)                                 | 067                   |
| 6  | 15-17 | International Designator (piece of the launch)                                       | A                     |
| 7  | 19-20 | <b>Epoch year</b> (last two digits of year)                                          | 08                    |
| 8  | 21-32 | <b>Epoch</b> (day of the year and fractional portion of the day)                     | 264.51782528          |
| 9  | 34-43 | First derivative of mean motion; the <b>ballistic coefficient</b>                    | -.00002182            |
| 10 | 45-52 | Second derivative of mean motion (decimal point assumed)                             | 00000-0               |
| 11 | 54-61 | B*, the <b>drag term</b> , or radiation pressure coefficient (decimal point assumed) | -11606-4              |
| 12 | 63-63 | Ephemeris type (always zero; only used in undistributed TLE data)                    | 0                     |
| 13 | 65-68 | Element set number. Incremented when a new TLE is generated for this object.         | 292                   |
| 14 | 69    | Checksum (modulo 10)                                                                 | 7                     |

- For historical reasons, some parameters are split in two (e.g. Epoch), and decimal point is dropped, and other idiosyncrasies are kept for compatibility.



# Epoch extraction

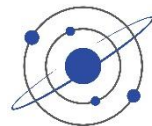
```
% Epoch year (last two digits of year)
Ephemeris.Epoch_year = str2num(Line1(19:20));

% Julian day fraction (day of the year and fractional portion of the day)
Ephemeris.Epoch_day = str2num(Line1(21:32));

if Ephemeris.Epoch_year < 57
 Year = Ephemeris.Epoch_year + 2000;
else
 Year = Ephemeris.Epoch_year + 1900;
End

[Month, Day, Hour, Minute, Sec] = days2mdh(Year, Ephemeris.Epoch_day);

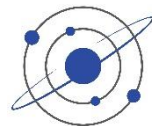
Ephemeris.Epoch = datenum(Year, Month, Day, Hour, Minute, Sec);
% Ephemeris.Epoch.TimeZone = 'UTC'; % (only if needed)
```



# Data structure: Line 2

|      |       |                                                                             |             |
|------|-------|-----------------------------------------------------------------------------|-------------|
| 1    | 01    | Line number                                                                 | 2           |
| 2    | 03-07 | Satellite Catalog number                                                    | 25544       |
| 3    | 09-16 | <b>Inclination</b> (degrees) $\rightarrow i$                                | 51.6416     |
| 4    | 18-25 | <b>Right ascension of the ascending node</b> (degrees) $\rightarrow \Omega$ | 247.4627    |
| 5    | 27-33 | <b>Eccentricity</b> (decimal point assumed) $\rightarrow e$                 | 0006703     |
| 6    | 35-42 | <b>Argument of perigee</b> (degrees) $\rightarrow \omega$                   | 130.5360    |
| 7    | 44-51 | <b>Mean anomaly</b> (degrees) $\rightarrow M_0$                             | 325.0288    |
| 8    | 53-63 | <b>Mean motion</b> (revolutions per day) $\rightarrow n$                    | 15.72125391 |
| 9    | 64-68 | Revolution number at epoch (revolutions)                                    | 56353       |
| 1069 |       | Checksum (modulo 10)                                                        | 7           |

- Six of these values correspond to Keplerian Elements:  $i, \Omega, e, \omega, M_0, n$
  - Notice that the semi-major axis  $a$  and the true anomaly  $\nu$  are *not* given, and must be inferred from the others parameters (next page)
- $\rightarrow$  Some of these parameters are fixed (e.g. Catalog number), while most COEs constantly change with time (although some faster than others, like  $e, a$ , and  $i$  which mostly stays the same for example)



# Computation of missing COEs

```

Ephemeris.n = str2num(Line2(53:63)); % [rev/day] Mean number of revolutions per day

%--- From n, one can compute series of other key parameters:
Ephemeris.mu = Earth.mu % ($\mu \equiv$ Earth.mu for TLE data describing Earth satellites)

Ephemeris.P_s = (24*60^2) / Ephemeris.n; % [s] Orbit period
Ephemeris.n_ = 2*pi / Ephemeris.P_s; % [rad/s] revolution angular speed

Ephemeris.a = Earth.mu^(1/3) / Ephemeris.n_^(2/3); % Semi-major axis [km]

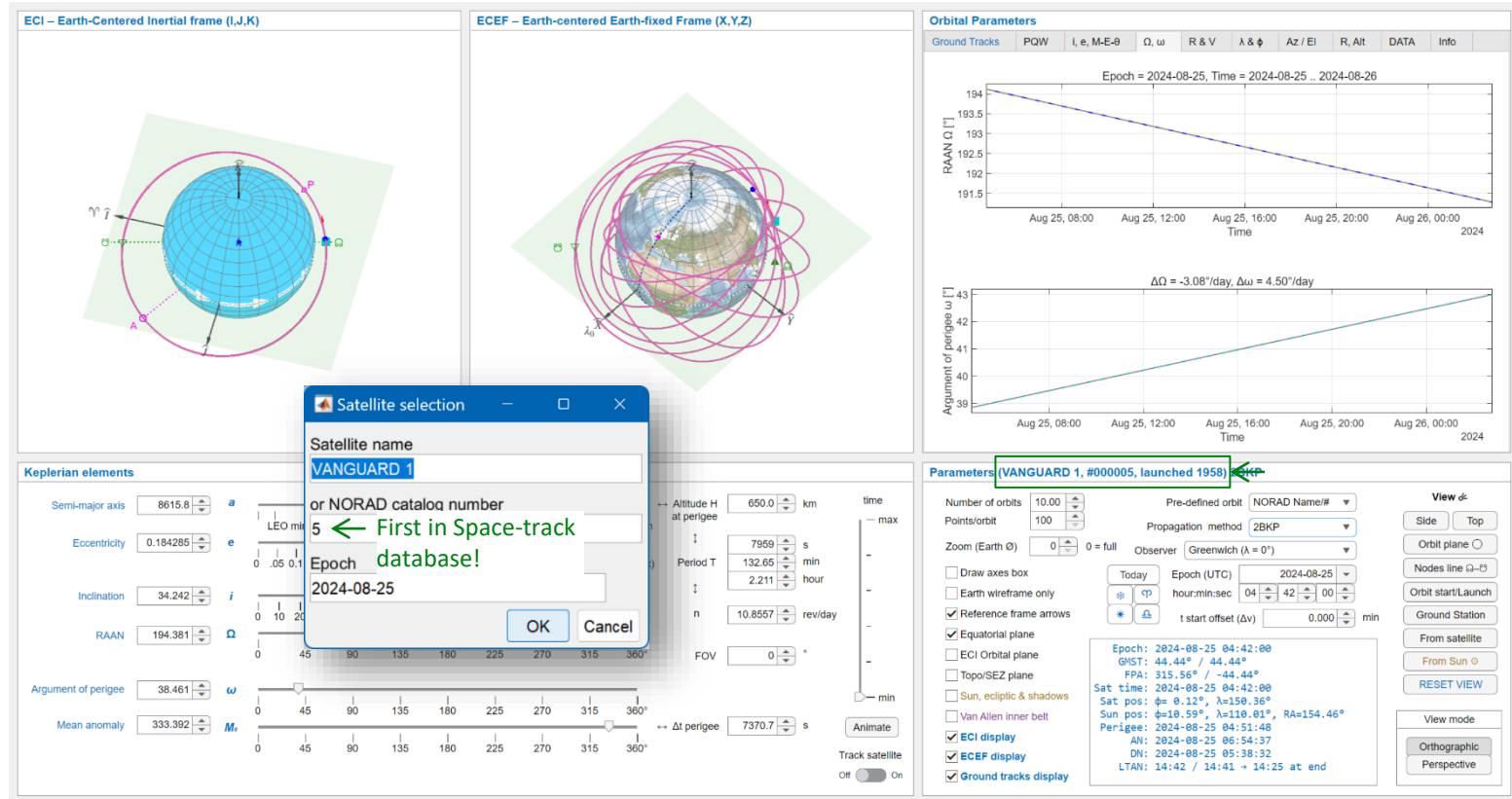
Ephemeris.H_km = Ephemeris.a - Earth.r1_km; % [km] Minimum radius at perigee

Ephemeris.h = sqrt(Ephemeris.a * Earth.mu * (1 - Ephemeris.e^2)); % [km^2/s]

% Computes required inclination to be in SSO orbit
SSO_i = SSO_incl(Ephemeris.a, Ephemeris.e, Earth);
if abs(SSO_i - Ephemeris.i) < 0.4 % ← criteria TBD; to tag Metop-A
 Ephemeris.SSO = true;
 Ephemeris.SSO_i = SSO_i;
else
 Ephemeris.SSO = false;
 Ephemeris.SSO_i = NaN;
end
Ephemeris.LTAN = LTAN_from_RAAN(Ephemeris.Epoch, Ephemeris.RAAN); % [hr] ← indicative; meaningful only for SSO orbits

```

# Vanguard – 1, after 68 years



# IMP: Note on TLE “Mean Elements” & SGP4

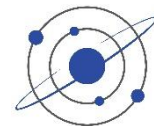


- TLE data is disseminated as [General Perturbations](#) (GP) orbital data following the [Simplified General Perturbations 4](#) ([SGP4](#)) propagation model
  - Such GP orbital data has traditionally been transmitted in [Two-Line Element Sets](#) (TLEs)
    - Although in May 2020 it started being published in CCSDS Orbit Mean-Elements Message (OMM) format as well
  - The advantage of [SGP4](#) is that it's an analytical model that includes the most important natural perturbations, and it's much faster than an equivalent numerical model.
    - Therefore, it's good for initial orbit design based on low-precision requirements
- However, it turns out that GP data in general, and TLEs in particular, are *poorly understood* even by professionals
- The core issue is that TLEs and OMMs contain [Brouwer mean elements](#) (representing values that are averaged over time rather than instantaneous values) which cannot be directly translated to osculating elements

The most important point to be noted is that not just any prediction model will suffice

- The NORAD element sets are “mean” values obtained by removing periodic variations in a particular way
- In order to obtain good predictions, these periodic variations must be reconstructed (by the prediction model) in exactly the same way they were removed by NORAD

# IMP: Note on TLE “Mean Elements” (continued)



- Mean elements are defined as quantities averaged over a suitable interval (or over an angular variable such as true anomaly), which **removes short-period oscillations** and **yields parameters that vary smoothly over time**
  - Notice that mean elements **depend on some unspecified averaging interval of the time**; the true, eccentric, or mean anomaly; or the longitude. Because there are many kinds of mean elements, it's important to understand how they're defined and used. Mean elements are most useful for long-range mission planning because they approximate the satellite's long-term behavior
  - Therefore, **TLE are not considered true orbits representation**
    - This is because TLE are in fact a merge between orbital state and a propagation model
    - The state is only meaningful with respect to the associated SGP4/SDP4 propagation model, and **cannot be used in any other model**
  - The elements in the two-line element sets are mean elements calculated to fit a set of observations using a specific model — the SGP4/SDP4 orbital model
    - Just as you shouldn't expect the arithmetic and geometric means of a set of data to have the same value, you shouldn't expect mean elements from different element sets, calculated using different orbital models, to have the same value
- Data cannot simply be reformatted unless you are willing to accept predictions with unpredictable errors

# TLE validity period

---



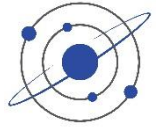
- TLEs can only be used to determine the position and velocity of an RSO for a short window around the time it is referenced to (its epoch)
- As the time between the epoch and the time that the position estimation is desired grows, the estimation quickly becomes very inaccurate
- TLEs can only be used to determine positions of RSOs **within approximately a week of their epoch** and therefore **the TLE of an satellite require constant updating** to provide accurate estimations of its position.
  - Because real-world satellite orbits are dynamic and influenced by various forces and operational activities, making precise long-term predictions challenging without fresh data (→ see next slides)

→ See [ISS Orbit Study](#)



# TLE data needs to be updated frequently

---



TLE data needs to be updated frequently for the following reasons:

## 1. Perturbations in the orbit (see Section [Orbital perturbations](#))

- Satellites are influenced by forces other than gravity from the central body (Earth), such as:
  - **Atmospheric Drag:** Low Earth Orbit (LEO) satellites encounter air molecules that slow them down, causing their orbits to decay gradually
  - **Gravitational Perturbations:** Earth's gravity is not perfectly uniform due to its oblate shape (flattened at the poles) and local mass concentrations (mass anomalies)
  - **Third-Body Effects:** The gravitational pull of the Moon, Sun, and other planets slightly alters the satellite's orbit
  - **Solar Radiation Pressure:** The force exerted by photons from the Sun can change a satellite's velocity and trajectory

These factors cause deviations from the idealized orbit described in the TLE

## 2. TLE is a Simplified model

- TLE uses a simplified propagator model (SGP, SGP4, or SGP8) to approximate the satellite's orbit
- Over time, small errors in the model's assumptions (e.g., neglecting some perturbations or modeling them imprecisely) accumulate, reducing accuracy

# TLE data needs to be updated frequently

---



## 3. Numerical Propagation Errors

- Orbital propagation involves numerical computations based on initial conditions (the TLE data). These computations introduce small errors due to finite precision
- Over time, these errors amplify, leading to discrepancies between the predicted and actual positions

## 4. Orbital Decay and Maneuvers

- Satellites in LEO gradually lose altitude due to drag, changing their orbital period and trajectory
- Geostationary and other satellites may perform [station-keeping maneuvers](#) to maintain their assigned orbital slots. For example the ISS needing periodic reboost maneuvers to counteract atmospheric drag and prevent orbital decay leading to reentry.
- These maneuvers change the orbit, rendering old TLE data obsolete

## 5. Time Sensitivity of TLE

- TLE data is most accurate near its epoch (the reference time specified in the TLE). As time passes, the accuracy decreases due to the cumulative effects of the above factors

# Consequences of outdated TLE Data

---



- If TLE data isn't updated, predictions of a satellite's position become less accurate, which can lead to:
  - Errors in tracking and ground station communication
  - Increased risk of orbital collisions due to inaccurate conjunction assessments
  - Inefficiencies in planning satellite operations or observations

## How TLE Updates Work

- Organizations like U.S. Space Command, NORAD or private companies track satellites [using radar and optical systems](#)
- They calculate updated orbital parameters based on observations and publish new TLEs periodically
- Frequent updates ensure that the TLE remains accurate enough for operational use

# COE exercise



- Given the following COEs, which of the following apply to the satellite's orbit and position (check all that apply)  
 $a = 10000 \text{ km}$ ,  $e = 0 / 0.2$ ,  $i = 30^\circ$ ,  $\Omega = 180^\circ$ ,  $\omega = 90^\circ$ ,  $\nu = 90^\circ$
- |                                                                                    |                                                                                       |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <input type="checkbox"/> Orbit is circular                                         | → when $e = 0$                                                                        |
| <input type="checkbox"/> Orbit is elliptical                                       | → when $e > 0$                                                                        |
| <input type="checkbox"/> Orbit is equatorial                                       | → when $i = 0^\circ$                                                                  |
| <input type="checkbox"/> Orbit is polar                                            | → when $i = 90^\circ$                                                                 |
| <input type="checkbox"/> Orbit is prograde                                         | → when $0 < i \leq 90^\circ$ (ambiguous for $i = 90^\circ$ ; not defined)             |
| <input type="checkbox"/> Orbit is retrograde                                       | → when $i > 90^\circ$                                                                 |
| <input type="checkbox"/> Satellite is at Perigee                                   | → when $\nu = 0^\circ$                                                                |
| <input type="checkbox"/> Satellite is at Apogee                                    | → when $\nu = 180^\circ$                                                              |
| <input type="checkbox"/> Satellite is at Ascending Node $\oslash$                  | → when $\omega + \nu = 0^\circ$ with $e = 0$ (or numerical solution)                  |
| <input type="checkbox"/> Satellite is at Descending Node $\oslash$                 | → when $\omega + \nu = 180^\circ$ with $e = 0$                                        |
| <input type="checkbox"/> Satellite is along $\hat{I}$ axis in the + or – direction | → when $\Omega = 0^\circ/180^\circ$ , $\omega + \nu = 0^\circ/180^\circ$ with $e = 0$ |
| <input type="checkbox"/> Satellite is along $\hat{J}$ axis in the + or – direction | → when $\Omega = \pm 90^\circ$ , $\omega + \nu = 0^\circ/180^\circ$ with $e = 0$      |
| <input type="checkbox"/> Satellite is along $\hat{K}$ axis in the + or – direction | → when $e = 0$ , $i = 0^\circ$ , $\omega + \nu = \pm 90^\circ$                        |

# COE exercise (continued)

---



- Given the following COEs, which of the following apply to the satellite's orbit and position (check all that apply)  
 $a = 10000 \text{ km}$ ,  $e = 0 / 0.2$ ,  $i = 30^\circ$ ,  $\Omega = 180^\circ$ ,  $\omega = 90^\circ$ ,  $\nu = 90^\circ$ 
  - ☐ Satellite is over the Equator  $\rightarrow$  when  $e = 0$ ,  $\omega + \nu = 0^\circ$  or  $180^\circ$ ,  $\forall i$
  - ☐ Satellite is over North Pole  $\rightarrow$  when  $e = 0$ ,  $i = 90^\circ$ ,  $\omega + \nu = 90^\circ$ ,  $\forall \Omega$
  - ☐ Satellite is over South Pole  $\rightarrow$  when  $e = 0$ ,  $i = 90^\circ$ ,  $\omega + \nu = 270^\circ$  or  $-90^\circ$ ,  $\forall \Omega$
  - ☐ Satellite is gaining potential energy  $\rightarrow$  when satellite going towards perigee ( $e \neq 0$ )
  - ☐ Satellite is losing potential energy  $\rightarrow$  when satellite going towards apogee ( $e \neq 0$ )